UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement High Gain 3.3V MOSFET SPICE Model Document

Version 0.2 Phase 1
2011/8/23

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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2_P1	2010/05/11	Shih-Hsin Yeh	Original release.

1.2 General information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

3.3V n-ch MOSFET
3.3V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of the compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24 , there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Model usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNSP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow the description for net-list of LOD effect, and these steps to invoke the model file during simulation.

Example:

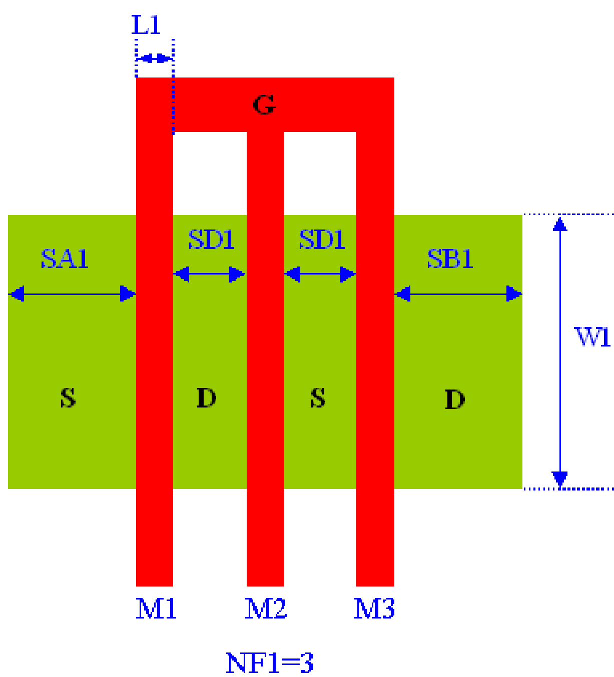


Figure 1-1 Layout example for LOD effect

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mx' statement;

Example : (Refer to Figure 1-1)

```
M1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1
+ SB=SD1*2+L1*2+SB1
+ PS='2*(SA1+W1)'
+ PD='2*(SD1*2+L1*2+SB1+W1)'
+ AS='SA1*W1'
+ AD='(SD1*2+L1*2+SB1)*W1'

M2 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*1+L1*1 SB=SD1*1+L1*1+SB1
+ PS='2*(SA1+SD1*1+L1*1+W1)'
+ PD='2*(SB1+SD1*1+L1*1+W1)'
+ AS='(SA1+SD1*1+L1*1)*W1'
+ AD='(SB1+SD1*1+L1*1)*W1'

M3 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*2+L1*2 SB=SB1
+ PS='2*(SD1*2+L1*2+SA1+W1)'
+ PD='2*(SB1+W1)'
+ AS='(SD1*2+L1*2+SA1+W1)*W1'
+ AD='SB1*W1'
```

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mxx' statement;

Example : (Refer to Figure 1-1)

```
M1 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1
+ sb=SD1*2+L1*2+SB1
+ ps='2*(SA1+W1)'
```

```
+ pd='2*(SD1*2+L1*2+SB1+W1)'
+ as='SA1*W1'
+ ad='(SD1*2+L1*2+SB1)*W1'


M2 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1+SD1*1+L1*1  sb=SD1*1+L1*1+SB1
+ ps='2*(SA1+SD1*1+L1*1+W1)'
+ pd='2*(SB1+SD1*1+L1*1+W1)'
+ as='(SA1+SD1*1+L1*1)*W1'
+ ad='(SB1+SD1*1+L1*1)*W1'

M3 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1+SD1*2+L1*2  sb=SB1
+ ps='2*(SD1*2+L1*2+SA1+W1)'
+ pd='2*(SB1+W1)'
+ as='(SD1*2+L1*2+SA1+W1)*W1'
+ ad='SB1*W1'
```

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.Option  scale=0.9
.lib "./ModelFileName.lib.eldo" CaseName
```

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

- Define device condition by 'Mx' statement;

Example : (Refer to Figure 1-1)

```
M1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1
+ SB=SD1*2+L1*2+SB1
+ PS='2*(SA1+W1)'
+ PD='2*(SD1*2+L1*2+SB1+W1)'
+ AS='SA1*W1'
+ AD='(SD1*2+L1*2+SB1)*W1'

M2 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*1+L1*1  SB=SD1*1+L1*1+SB1
+ PS='2*(SA1+SD1*1+L1*1+W1)'
+ PD='2*(SB1+SD1*1+L1*1+W1)'
+ AS='(SA1+SD1*1+L1*1)*W1'
+ AD='(SB1+SD1*1+L1*1)*W1'

M3 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*2+L1*2  SB=SB1
+ PS='2*(SD1*2+L1*2+SA1+W1)'
```

+ PD='2*(SB1+W1)'
+ AS='(SD1*2+L1*2+SA1+W1)*W1'
+ AD='SB1*W1'

ModelFileName for 3.3V MOSFET: l110ae_hg33_v021
DeviceName for 3.3V n-ch MOSFET: n_33_hgl110e
DeviceName for 3.3V p-ch MOSFET: p_33_hgl110e

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range (After 90% shrink):
For both 3.3V n-ch and p-ch MOSFET:

$0.324\ \mu m \leqq L_{DES} \leqq 50\ \mu m$

$0.35\ \mu m \leqq W_{DES} \leqq 100\ \mu m$

Adopt multi-finger structures if $W_{DES} > 9\ \mu m$.

$0.324\ \mu m \leqq SA,SB \leqq 1.98\ \mu m$

Note: SAREF=SBREF=1.98μm is the reference dimension of the STI-stress (LOD) effect.

Voltage range:
For 3.3V n-ch MOSFET:

$0V \leqq V_{GS} \leqq 3.3V$ (*1.1)

$0V \leqq V_{DS} \leqq 3.3V$ (*1.1)

$-3.3V \leqq V_{BS} \leqq 0V$

For 3.3V p-ch MOSFET:

$0V \geqq V_{GS} \geqq -3.3V$ (*1.1)

$0V \geqq V_{DS} \geqq -3.3V$ (*1.1)

$3.3V \geqq V_{BS} \geqq 0V$

Temperature range:
For 3.3V n-ch and p-ch MOSFET:

-50 ℃ ~ +125 ℃

UMC L110E is UMC L130E 90% direct shrunk option.The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 12 nm) 9/0.12
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Sloting inserting	metal line width>5.556um	mwtal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department

2 3.3 V n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	**MOSFET AC Model**	**Junction DC Model**	**Junction AC Model**
Mask ID	BSW0WA	BSW0WA	BSW0WA	BSW0WA
Lot ID	D5JQJ	D5JQJ	D5JQJ	D5JQJ
Wafer Number	16	16	16	16

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS, respectively. The dimension is after 90% shrink and 12nm sizing channel length back.
All the normalized current are divided with W_{DES}.

All parameters are given for T=25 ℃, V_{BS}=0 V unless specified otherwise.

Table 2-2 Electrical Parameters for NMOS

All parameters are given for T=25 °C , V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W/L= 10 μ m / 10.02 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.537	0.536	0.547	0.537
I_{ON}	V_{DS} = 3.3 V	μA/ μm	44.37	43.98	43.95	44.38
Wide and short channel device (W/L= 10 μ m / 0.34 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.574	0.576	0.581	0.574
V_{TSAT}	V_{DS} = 3.3 V	V	0.536	0.535	0.548	0.536
Body Effect V_T shift	V_{DS} = 3.3 V V_{BS}= 0 ~ -3.3 V	V	--	0.551	0.575	0.575
DIBL	V_{DS}= 0.1~3.3 V	V	--	0.040	0.033	0.038
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	561	556.44	559.73	561.04
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	5.8E-12	2.3E-12	9.8E-13	5.2E-13
Narrow and long channel device (W/L= 0.35 μ m / 10.02 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.535	0.533	0.539	0.535
V_{TSAT}	V_{DS}= 3.3 V	V	--	0.526	0.523	0.519
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	37.8	37.42	37.01	37.77
Narrow and short channel device (W/L= 0.35 μm / 0.34 μm)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.540	0.538	0.554	0.540
V_{TSAT}	V_{DS}= 3.3 V	V	0.496	0.513	0.529	0.497
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	606.9	606.51	594.29	606.63
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	2.75E-12	2.9E-11	1.9E-11	3.1E-12
Capacitance						
CJ	V_J = 0 V	F/m^2	--	1.04E-03	1.03E-03	1.03E-03
CJSW	V_J = 0 V	F/m	--	1.46E-10	1.41E-10	1.41E-10
CJSWG	V_J = 0 V	F/m	--	2.08E-10	1.97E-10	1.97E-10
C_{OVL}	V_{GS} = -0.5 V	F/m	--	2.16E-10	2.01E-10	2.01E-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below(The dimensions are after 90% shrink and 12nm sizing channel length back.).

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10.02	1.02	0.82	0.7	0.52	0.46	0.36	0.34
10	X	X	X	X	X	X	X	X
1								X
0.8		X		X		X		X
0.6		X		X		X		X
0.43	X							X
0.35	X	X		X		X		X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	3.3	3.2
V_{GS} (V)	-0.5	3.6	0.05
V_{BS} (V)	0	-3.3	-0.825

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	3.6	0.05
V_{GS} (V)	0.7	3.3	0.65
V_{BS} (V)	0	-3.3	-3.3

2.3.3 Model verification plots

The simulation results (solid lines) obtained from the extracted model were plotted together with the measurement results (dotted lines) for comparison. The L and W in the plots refer to (L_{DES}*0.9+12nm and W_{DES}*0.9) respectively.

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

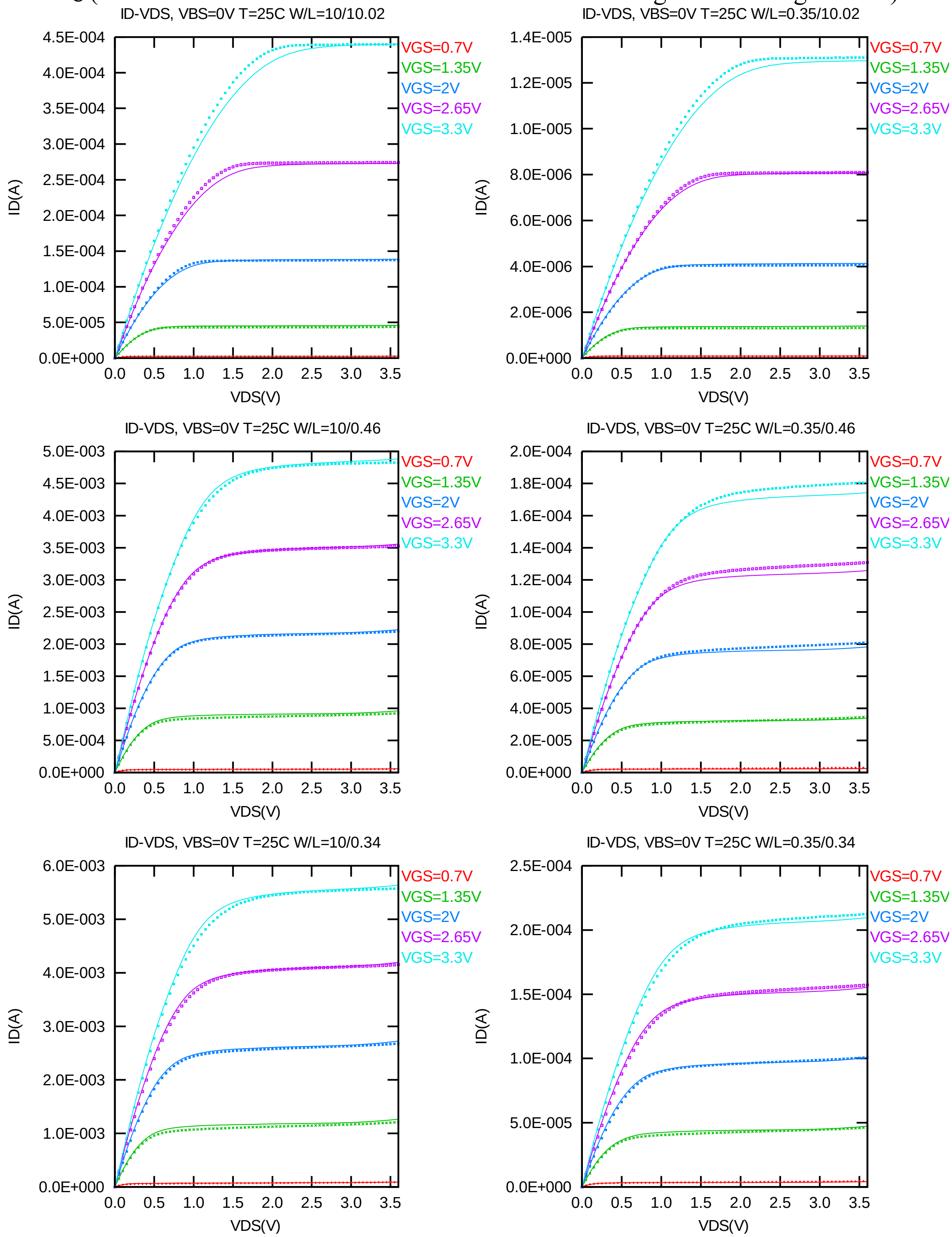


Figure 2-1 I_D vs. V_{DS} at V_{BS} = 0 V for NMOS

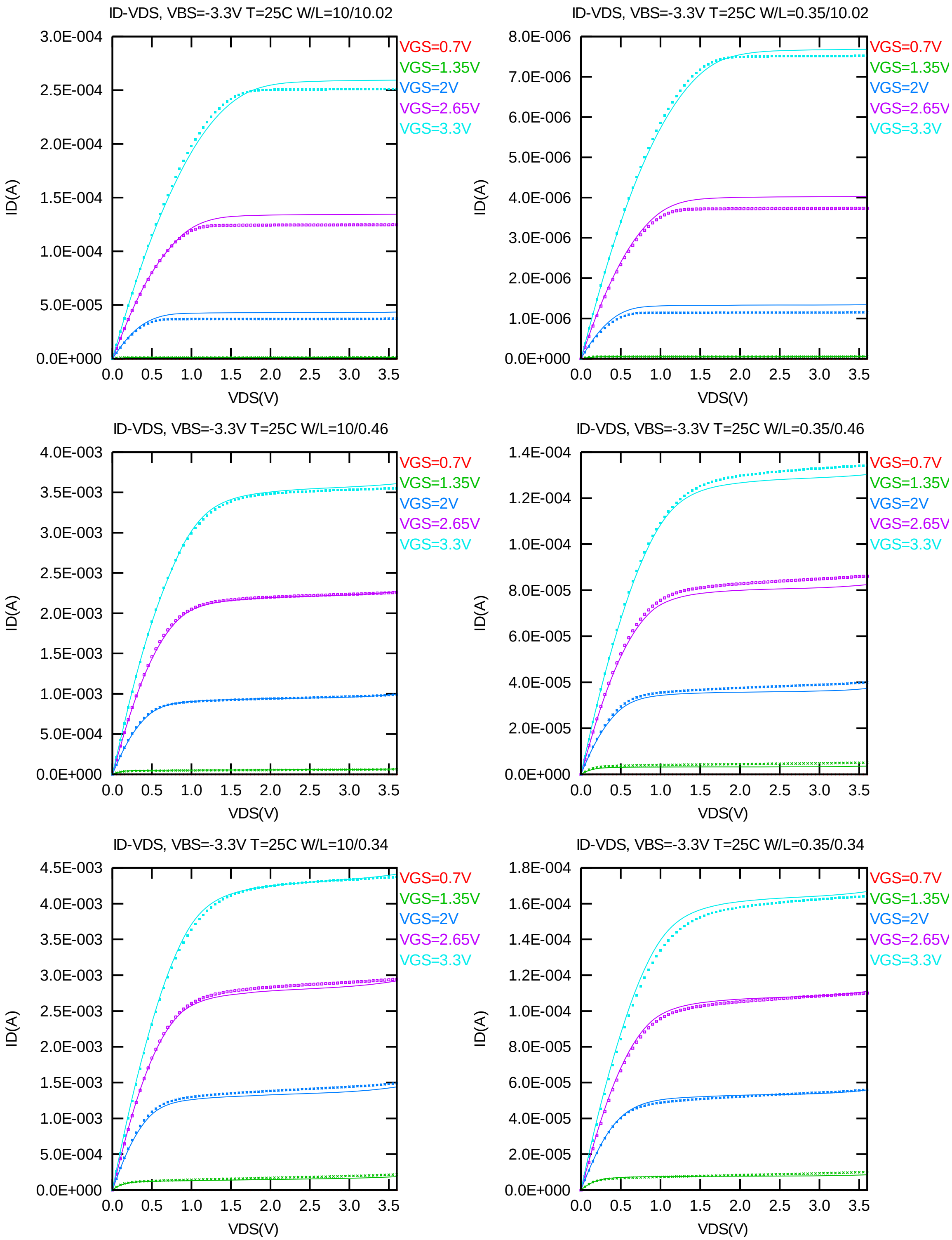


Figure 2-2 I_D vs. V_{DS} at V_{BS}=-3.3 V for NMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

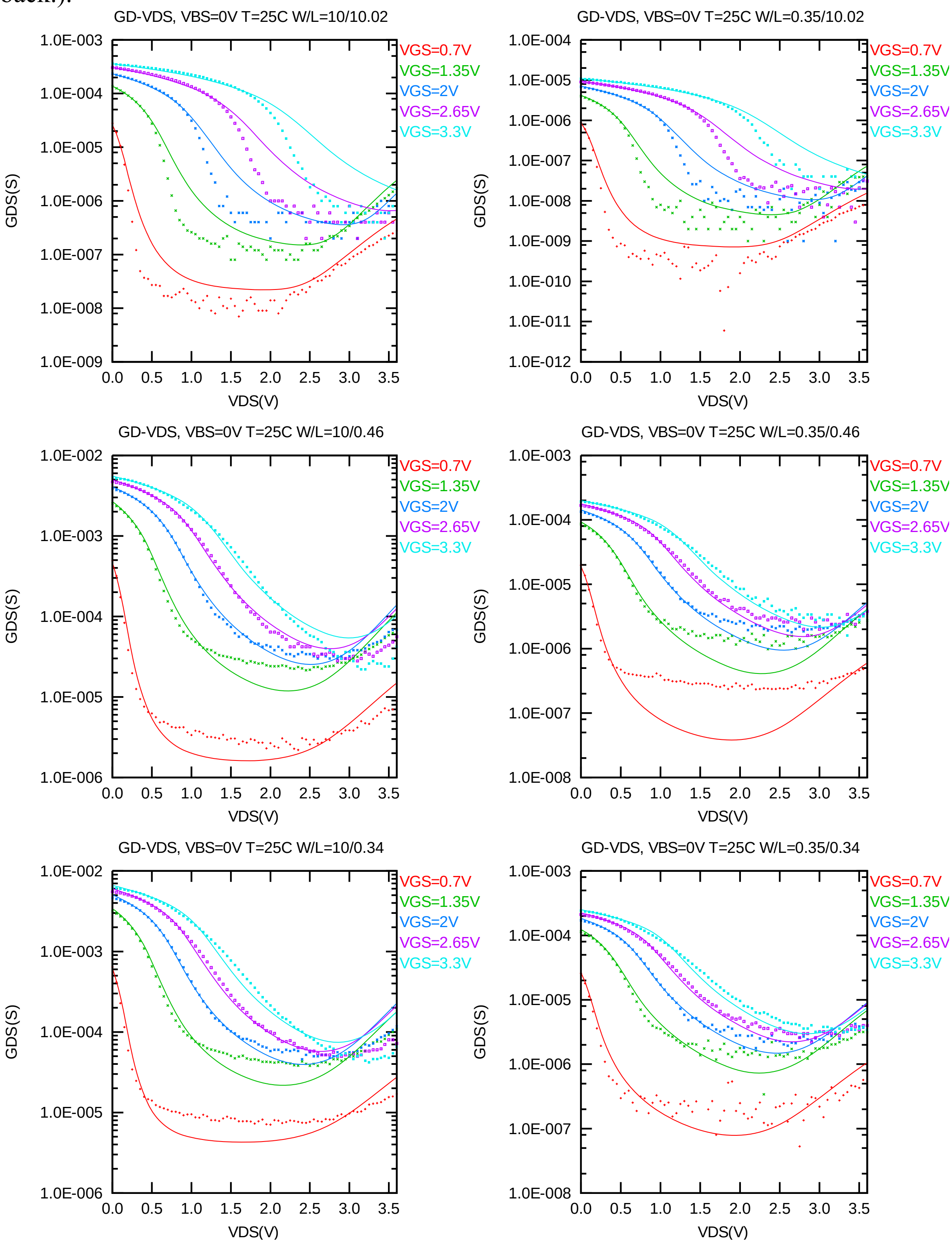


Figure 2-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for NMOS

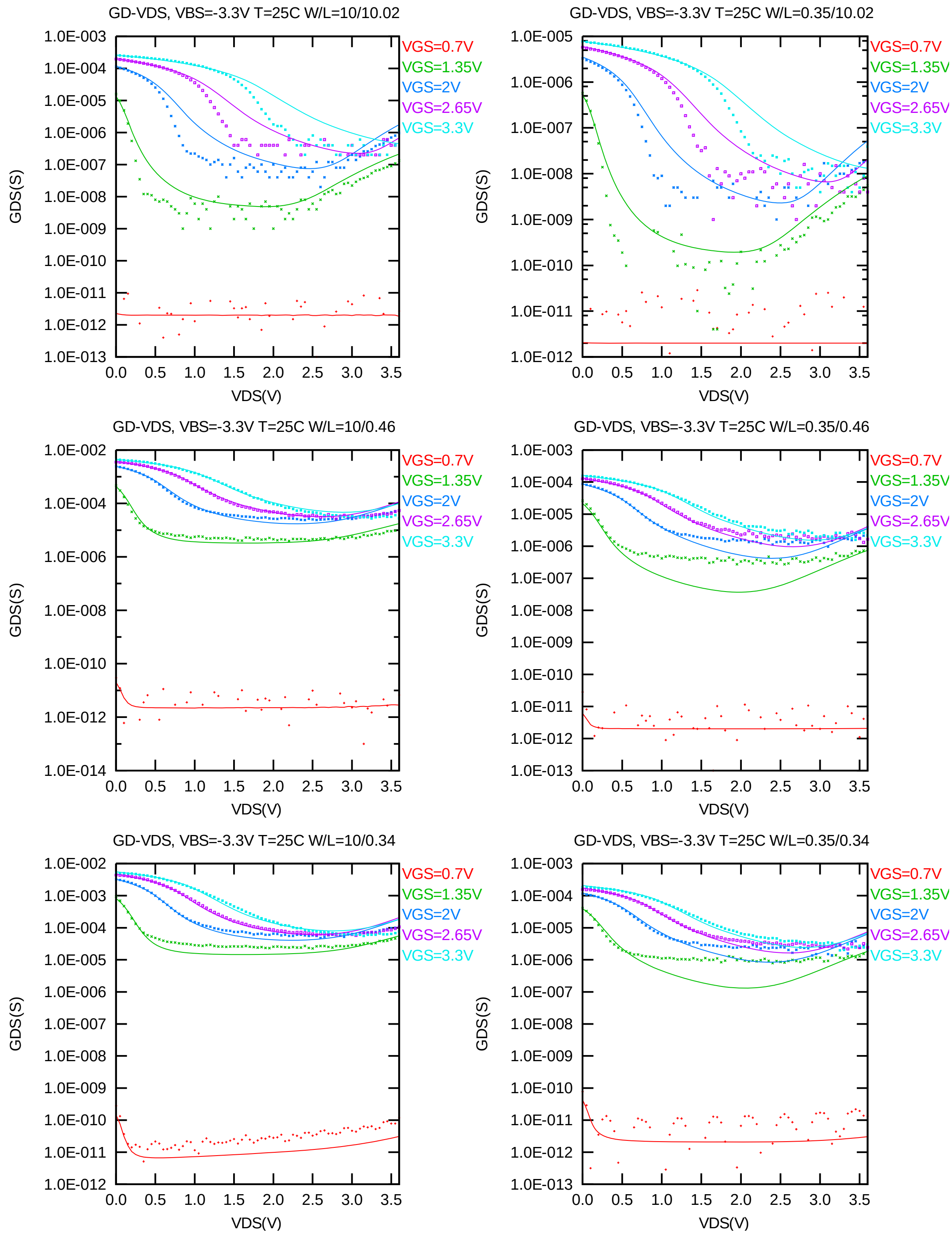


Figure 2-4 G_{DS} vs. V_{DS} at V_{BS} = -3.3 V for NMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

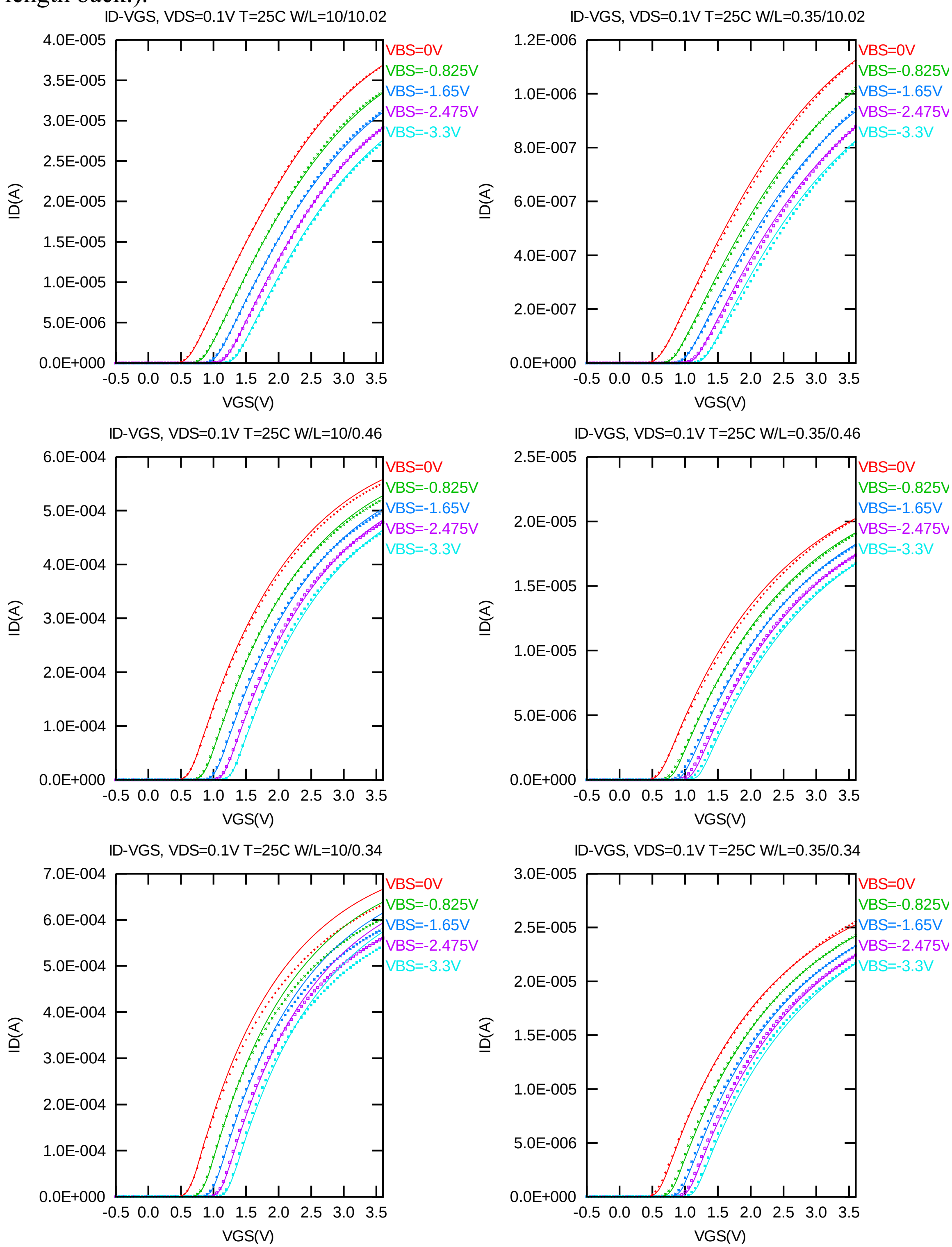


Figure 2-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

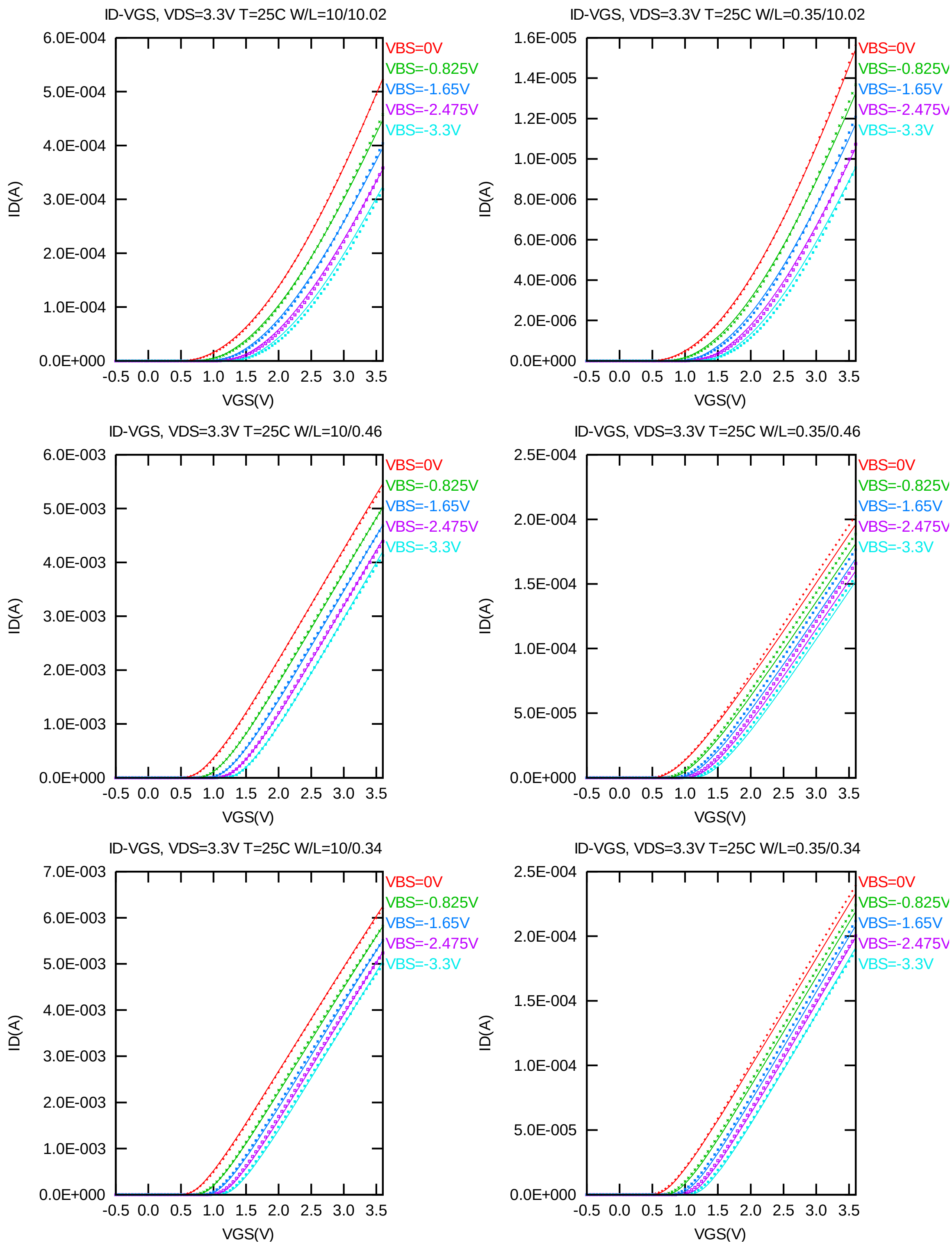


Figure 2-6 I_D vs. V_{GS} at V_{DS} = 3.3 V for NMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

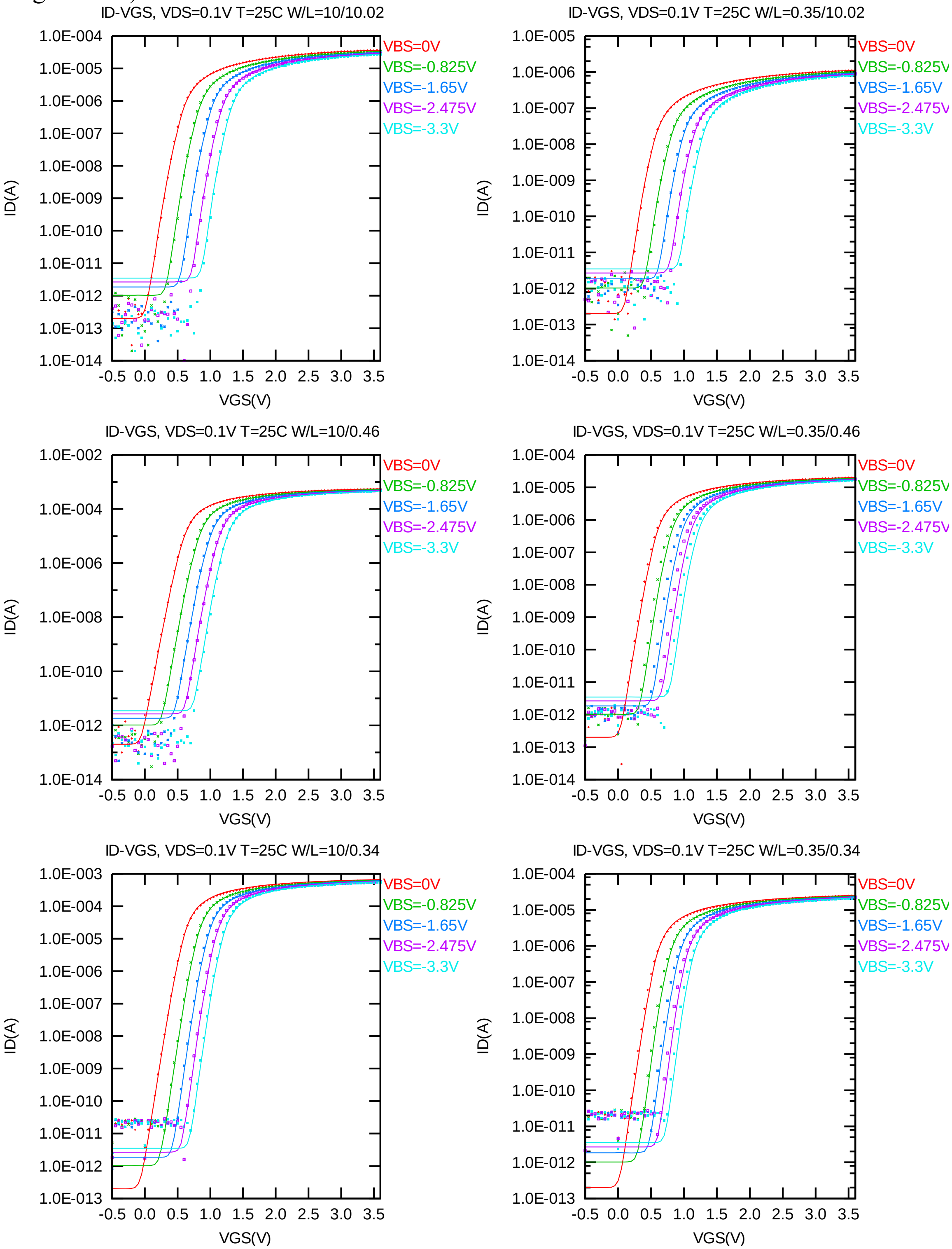


Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

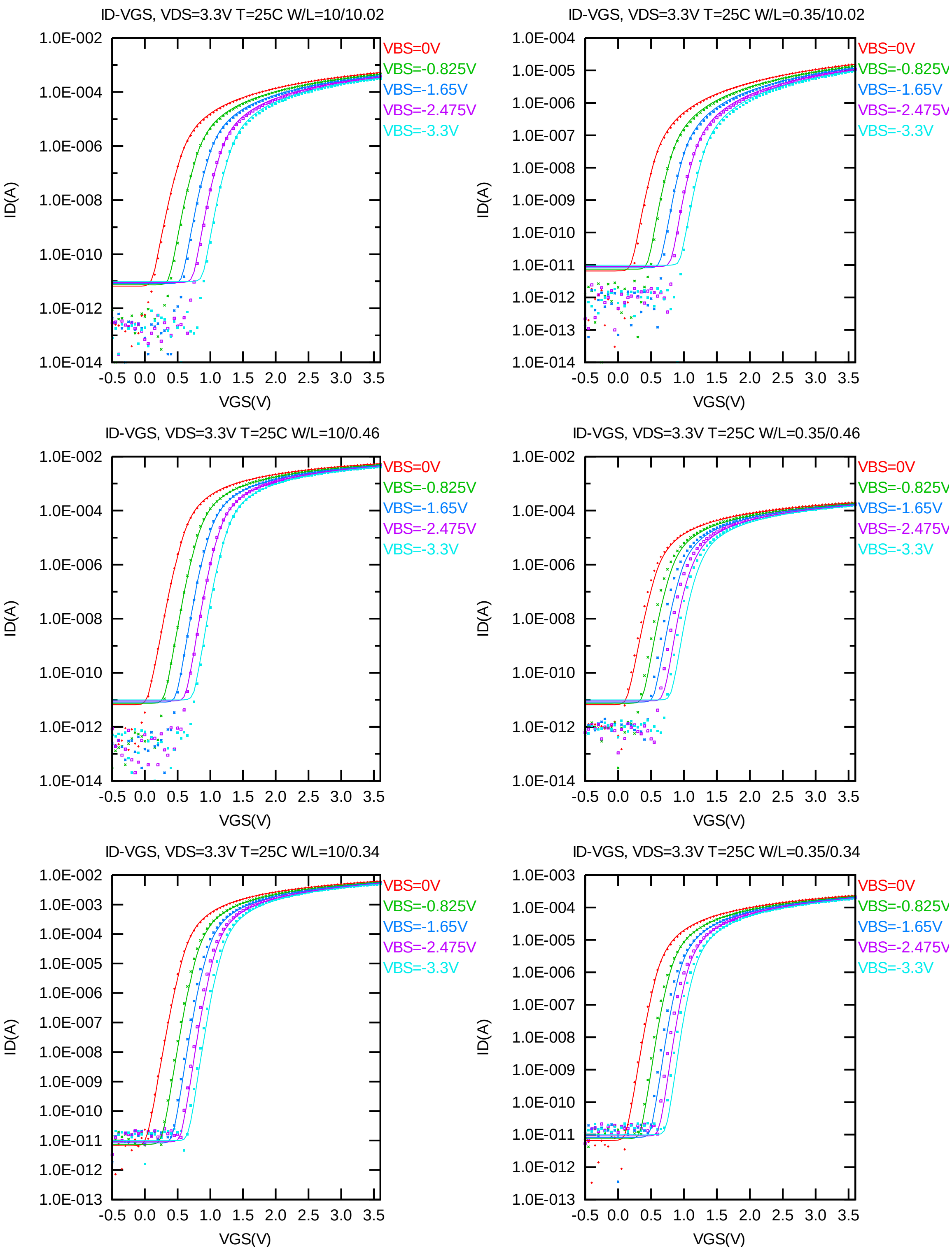


Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS} = 3.3 V for NMOS

G_M vs. V_{GS} Plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

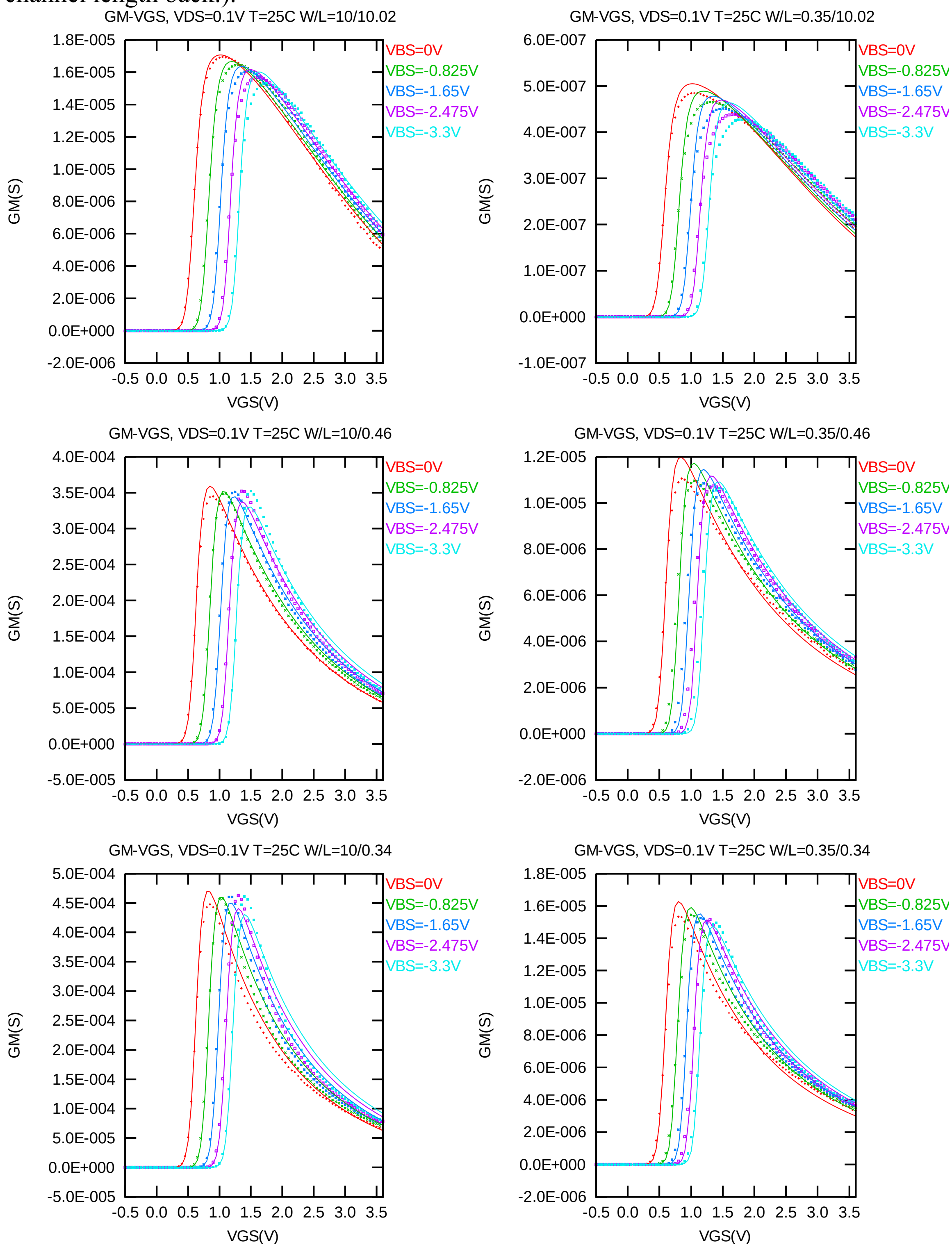


Figure 2-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for NMOS

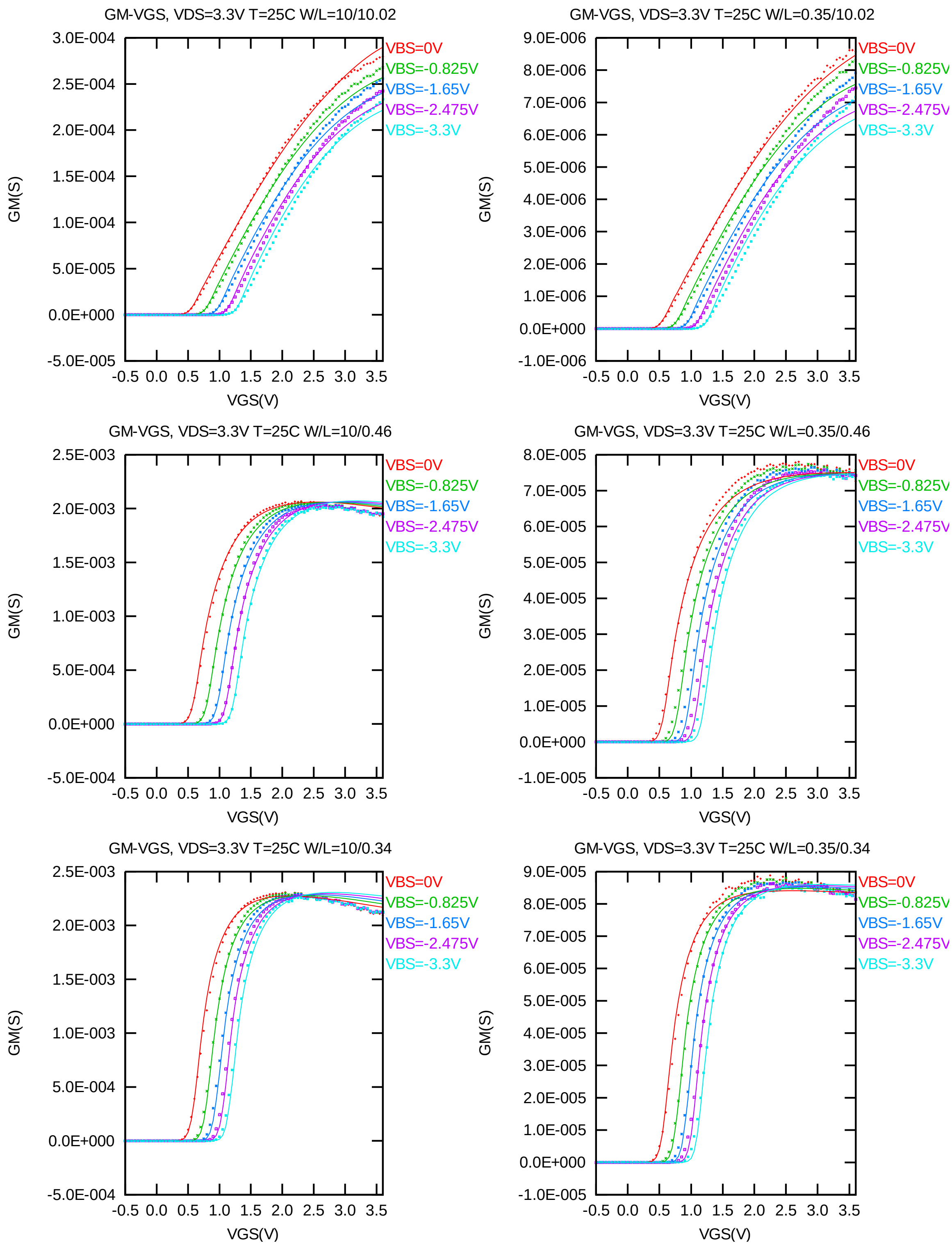
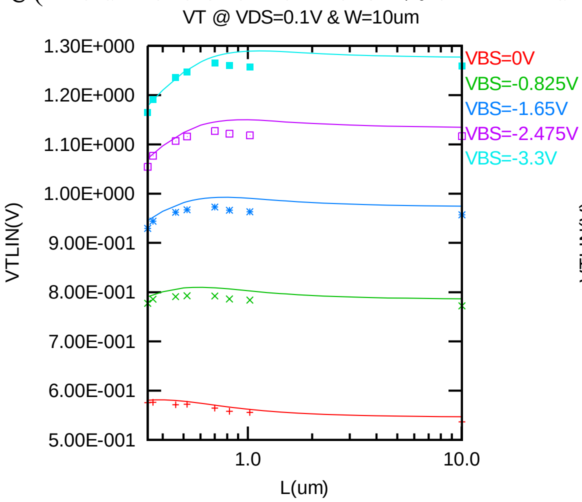
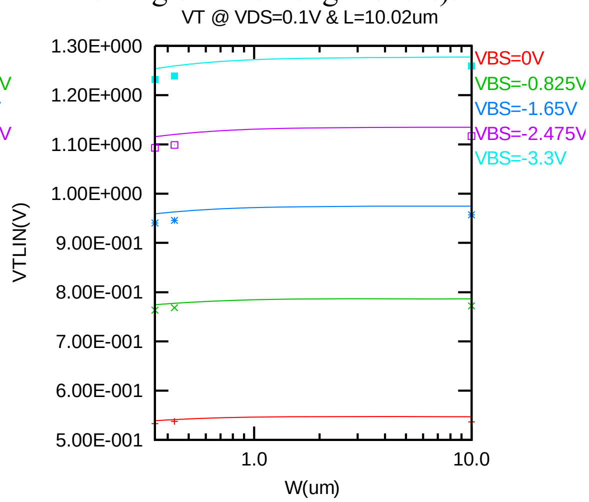


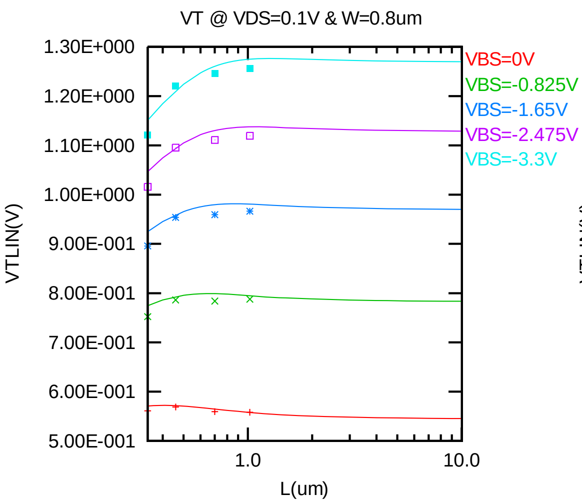
Figure 2-10 G_M vs. V_{GS} at V_{DS} = 3.3 V for NMOS

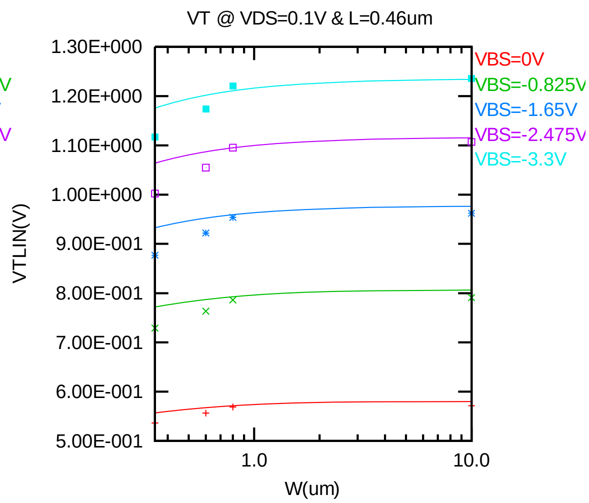
V_{TLIN} & V_{TSAT} vs. L_{DES} & W_{DES}

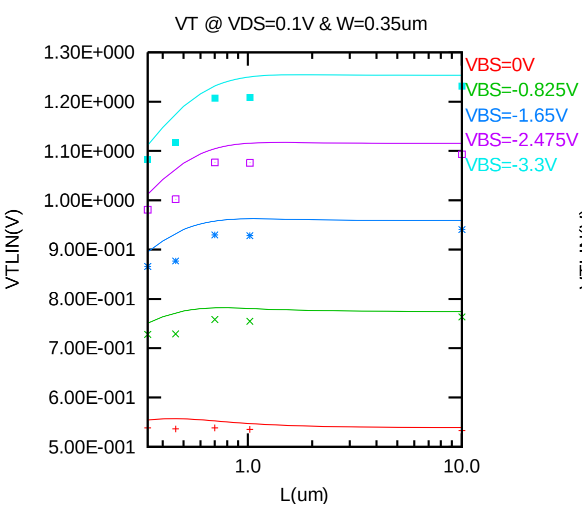
The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).











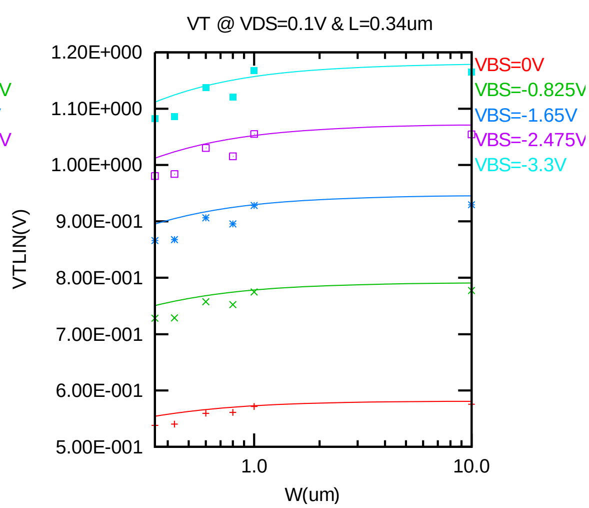


Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for NMOS

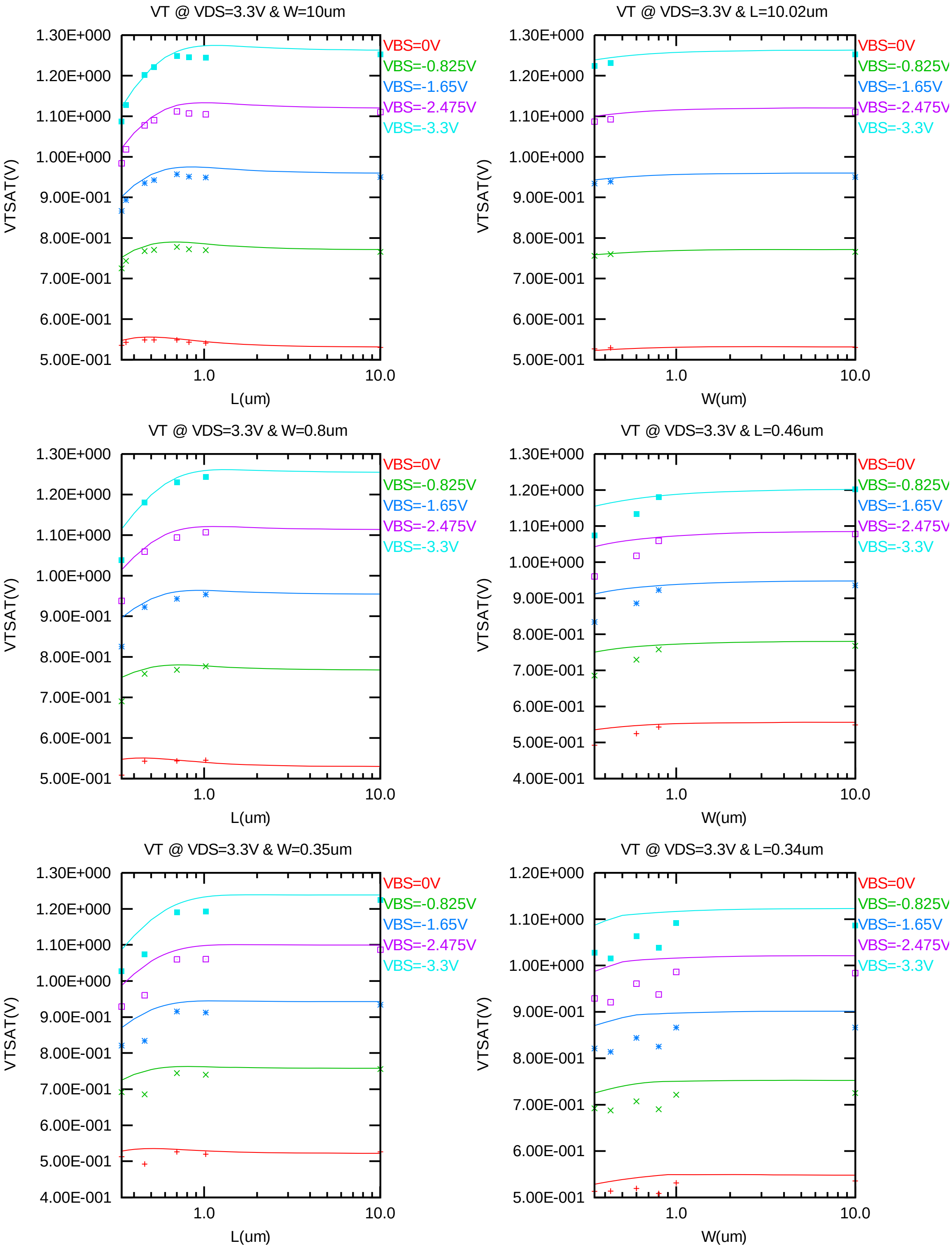


Figure 2-12 V_{TSAT} vs. channel length & channel width for high drain bias (V_{DS} = 3.3 V) for NMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 3.3$ V for $V_{BS} = 0$ V and $V_{BS} = -3.3$ V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

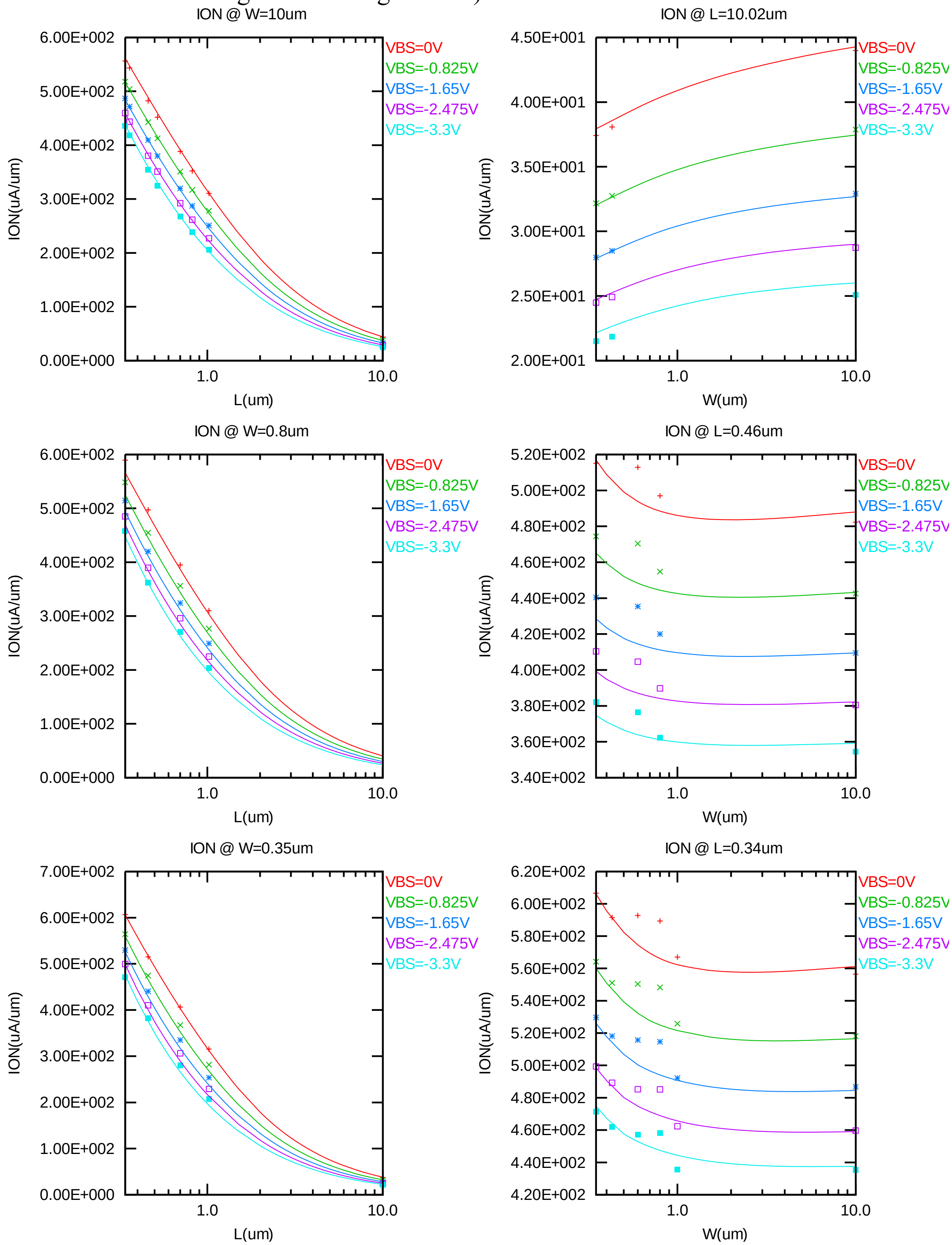


Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

I_{OFF} vs. L_{DES} & I_{OFF} vs. W_{DES}

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0$ V to -3.3V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.). A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

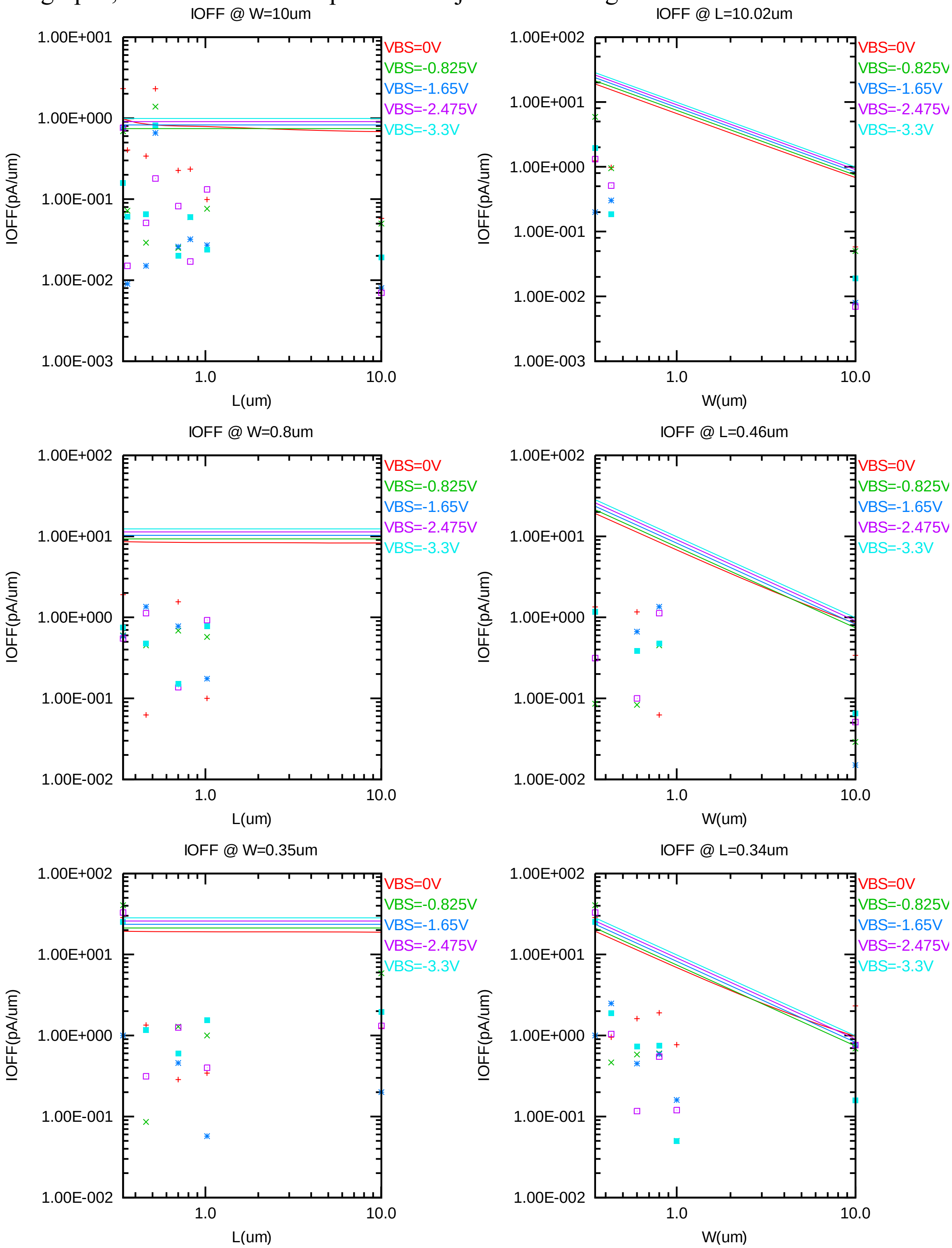


Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} and V_{TSAT} vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at I_D = 300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) (The dimensions are after 90% shrink and 12nm sizing channel length back.).

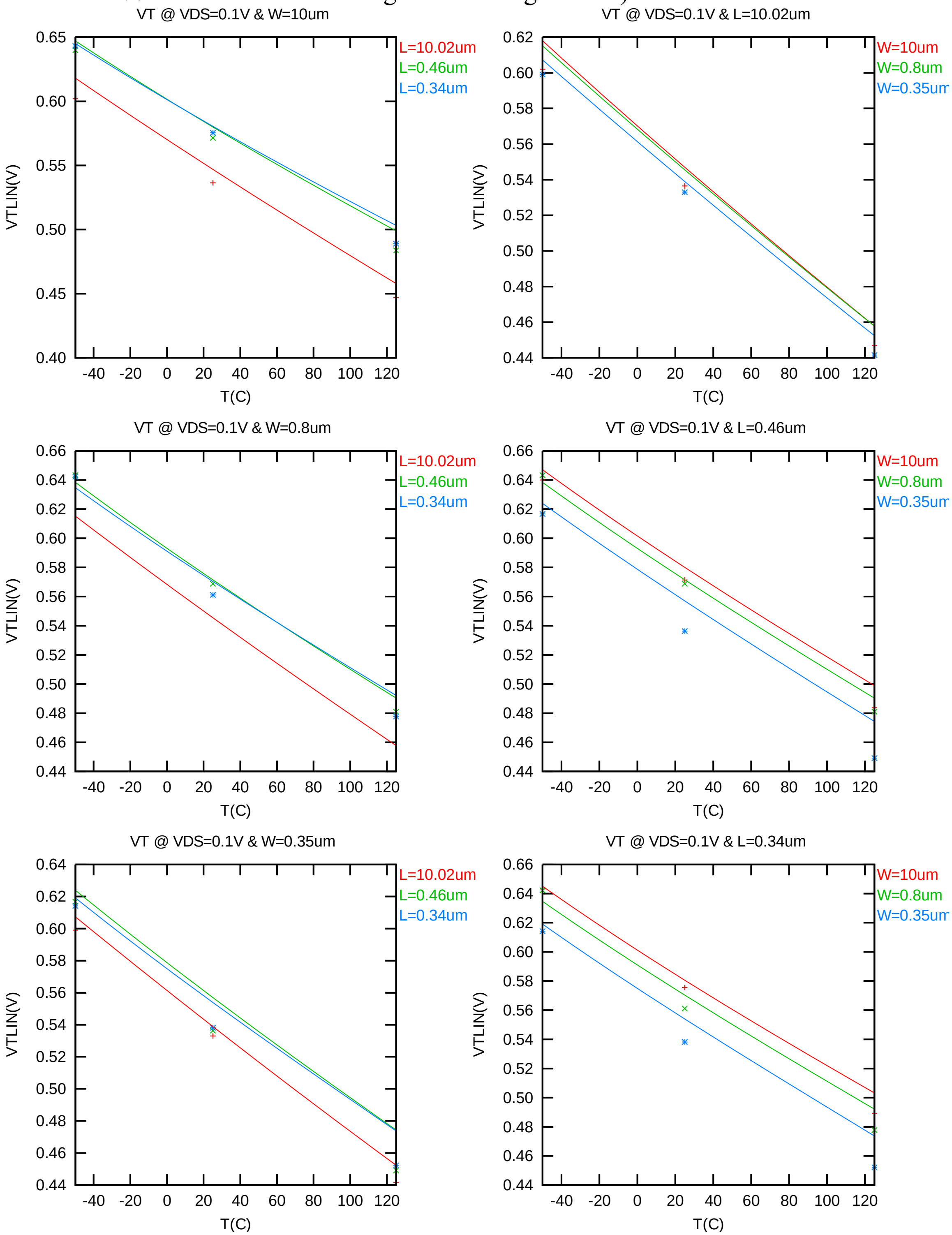


Figure 2-15 V_{TLIN} vs. temperature for NMOS

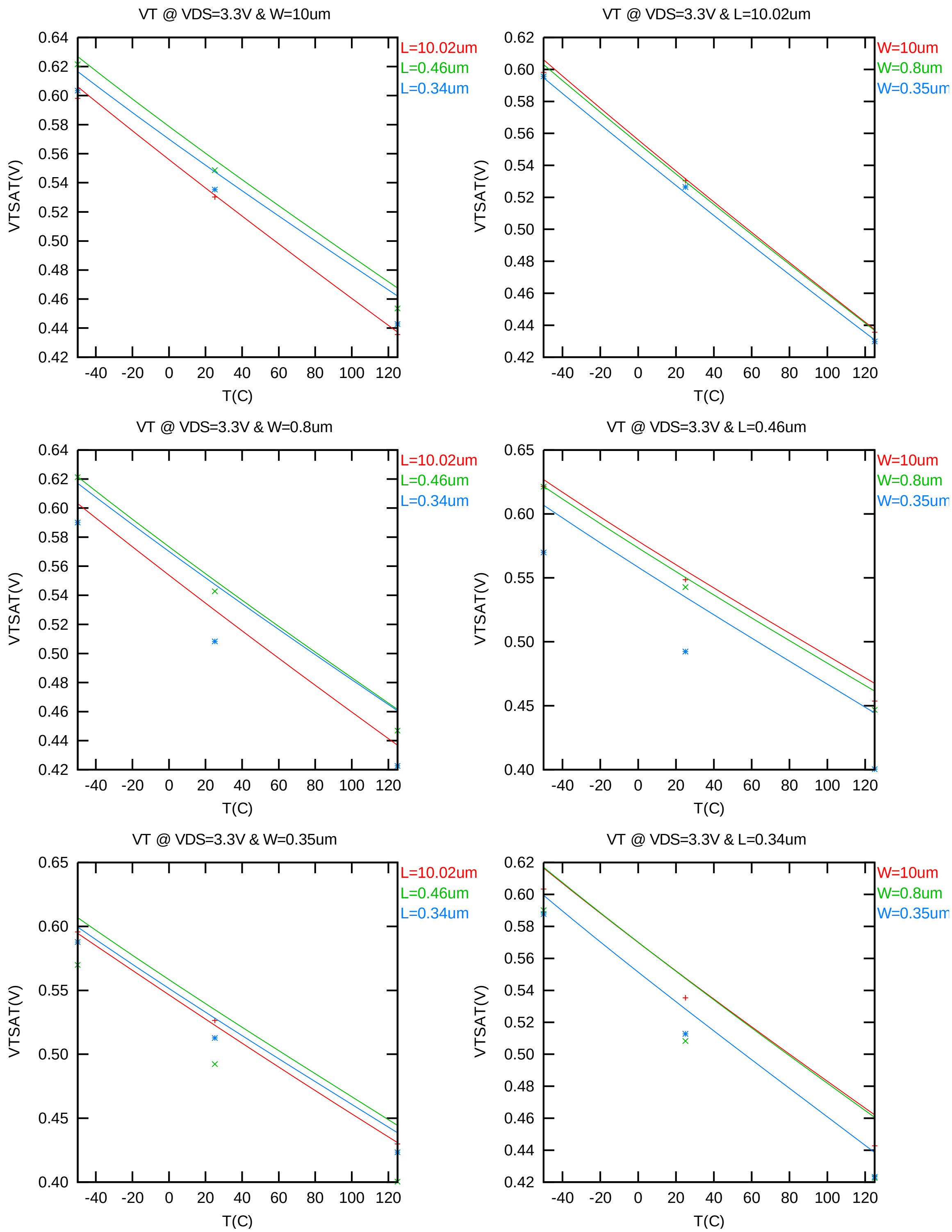


Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs Temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 3.3$ V, $V_{BS} = 0$ V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

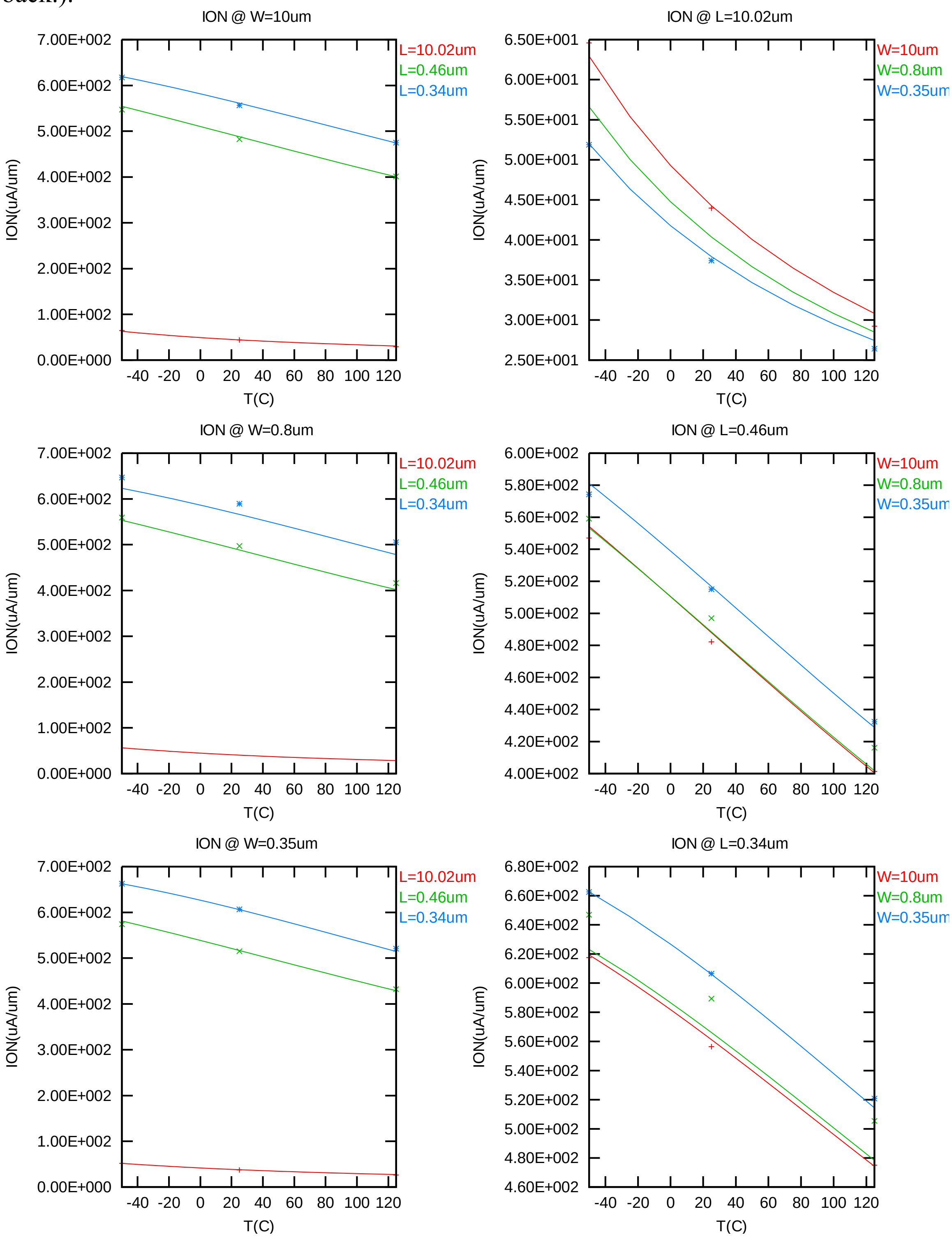


Figure 2-17 Saturation Current vs. temperature for NMOS

I_{OFF} vs Temperature

The following plot shows the temperature dependence of I_{OFF} extracted at V_{DS} = 3.3 V, V_{GS} = V_{BS} = 0 V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

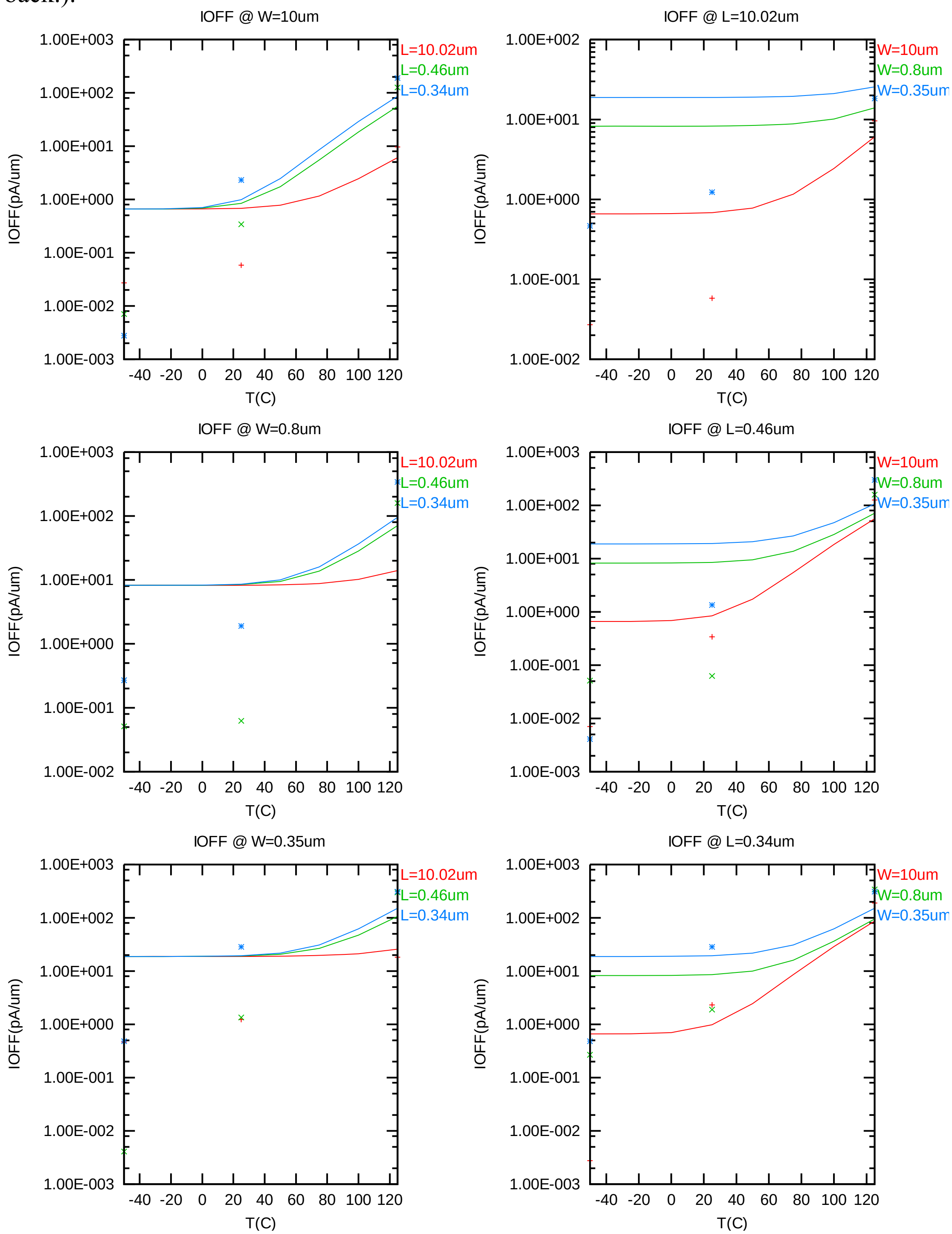


Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
e. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirement for model fit quality

Target of Quality Check	$V_{GS}< V_T$ (sub-threshold)	$V_T\leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS}\leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I_D Accuracy

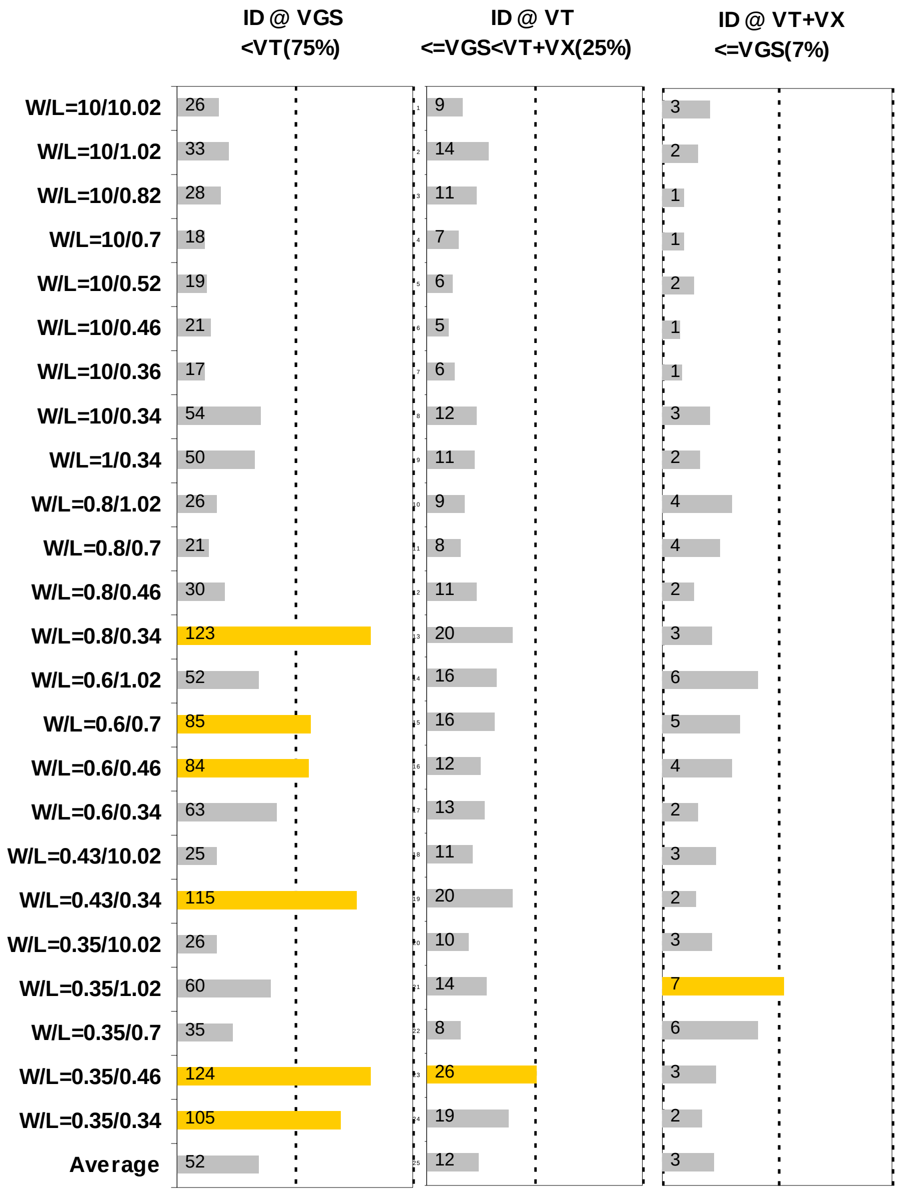


Figure 2-19 I_D accuracy at 25 ℃ for NMOS

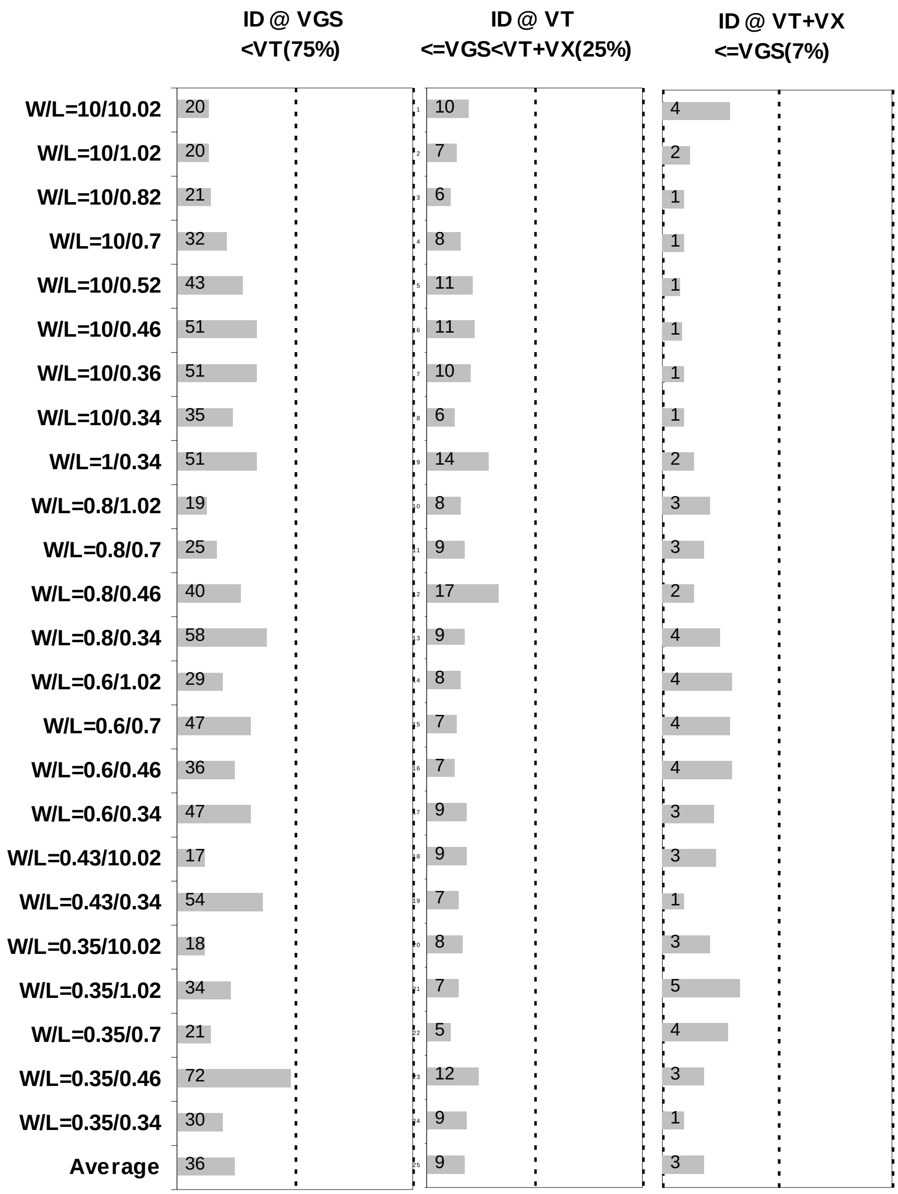


Figure 2-20 I_D accuracy at 125 ℃ for NMOS

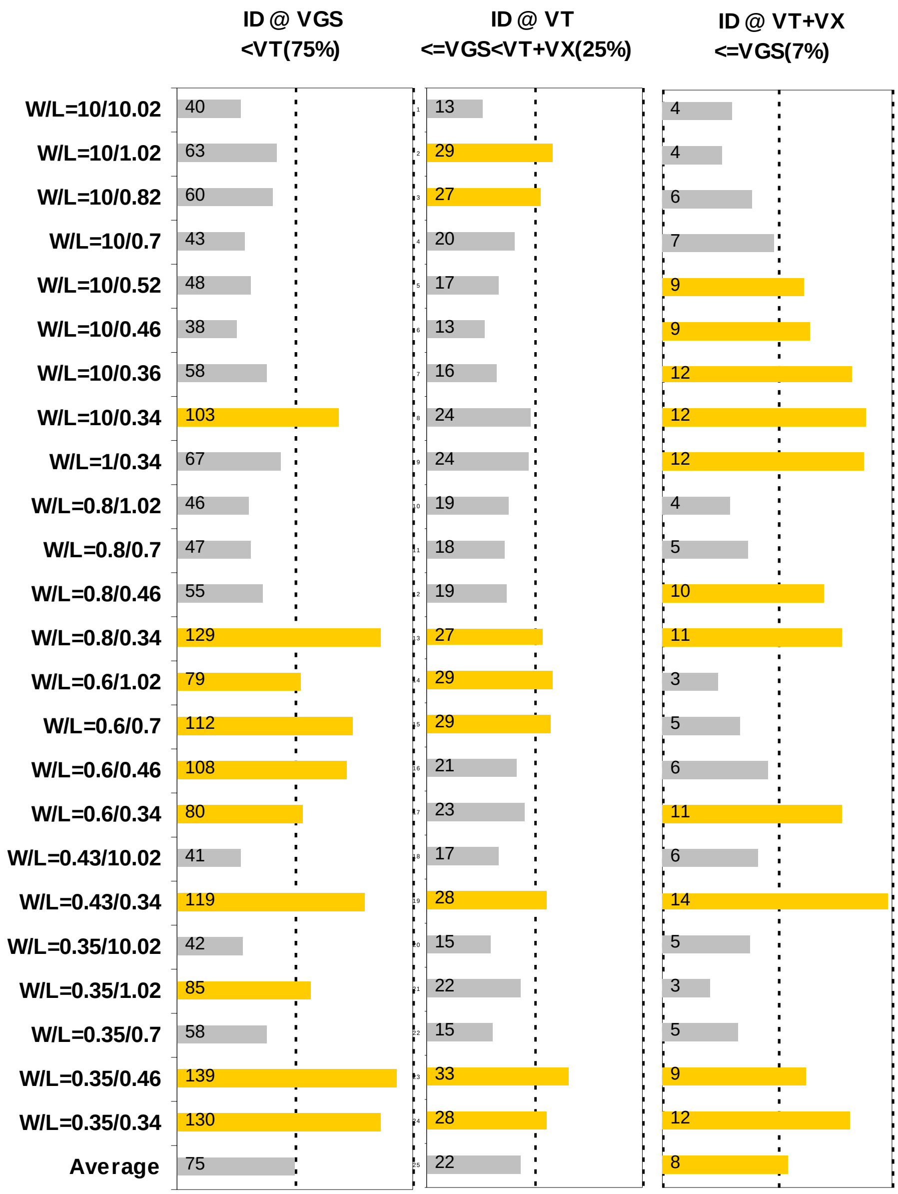


Figure 2-21 I_D accuracy at -50 ℃ for NMOS

G_M and G_{DS} Accuracy

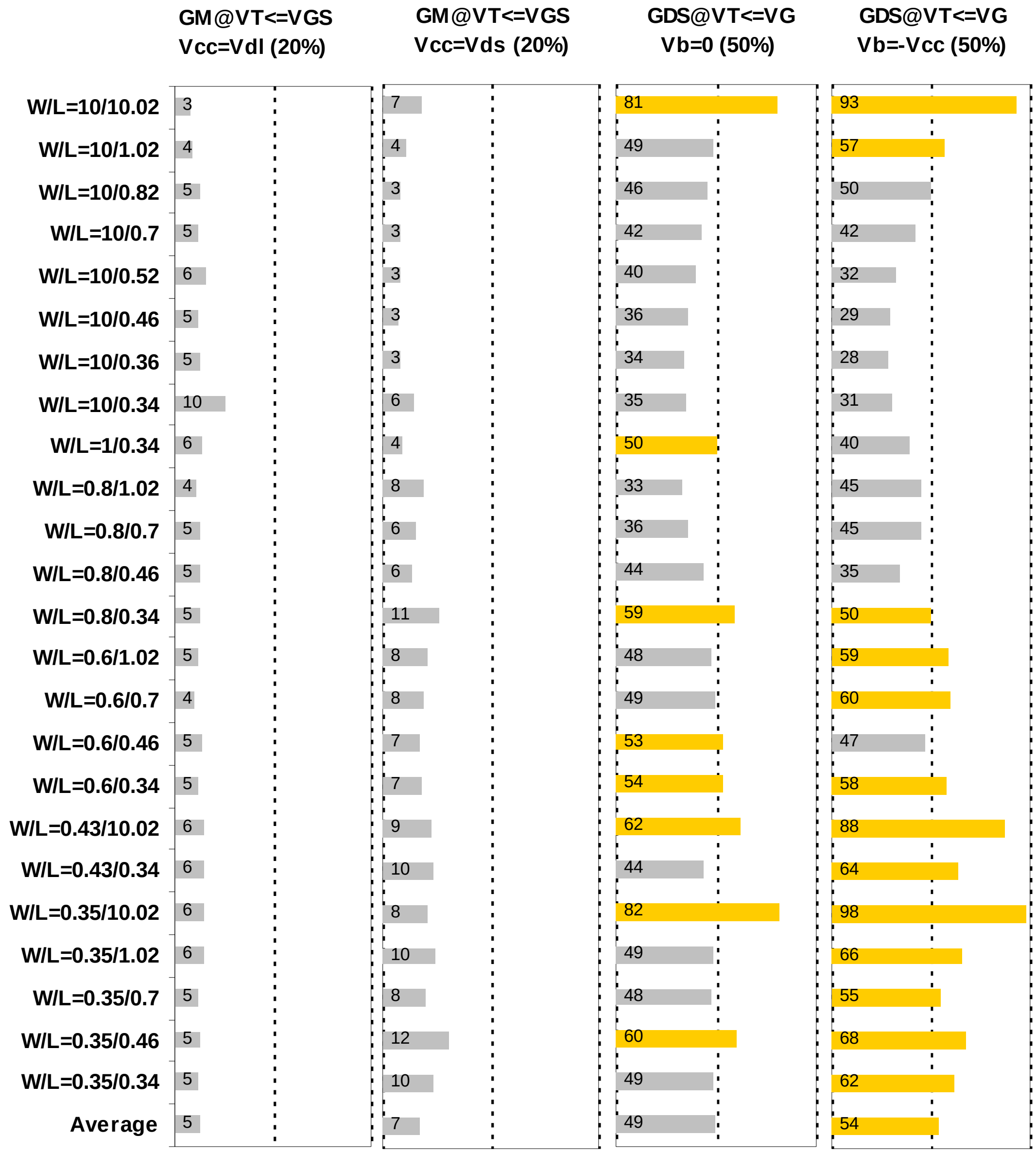


Figure 2-22 G_M and G_{DS} accuracy at 25 ℃ for NMOS

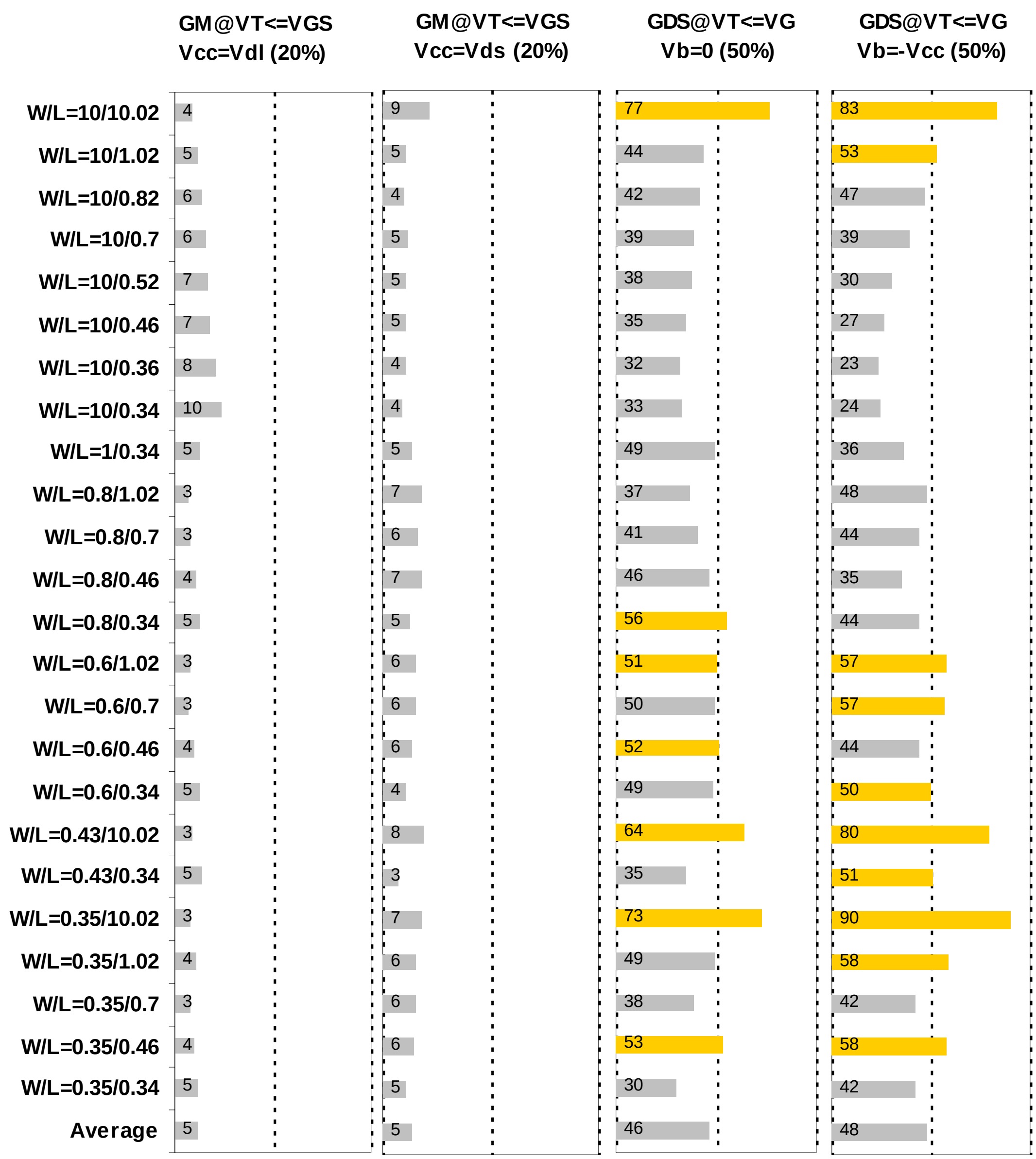


Figure 2-23 G_M and G_{DS} accuracy at 125 ℃ for NMOS

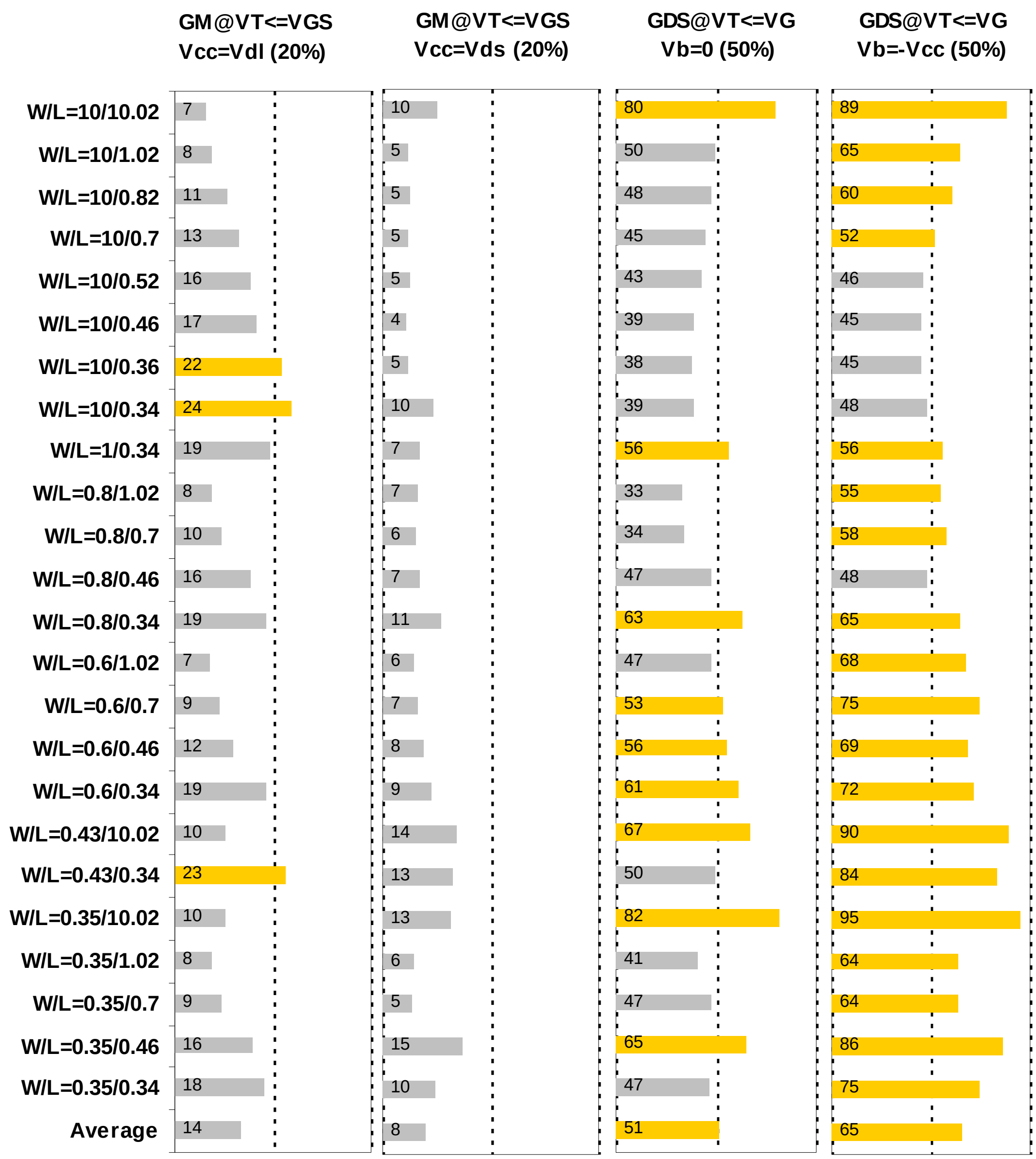


Figure 2-24 G_M and G_{DS} accuracy at -50 ℃ for NMOS

2.4 MOSFET capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS Cgg capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components, C_{GDS}	2.02	9000

Table 2-7 Large area measurement structure for NMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	2.02	100
	0.52	100
	0.34	100

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized as below.

Table 2-8 Measurement condition for NMOS capacitance model

V_{BIAS} [V]	From -3.65 to 3.65 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

The oxide thickness model parameter "TOX" was extracted to be 74.9 Å.

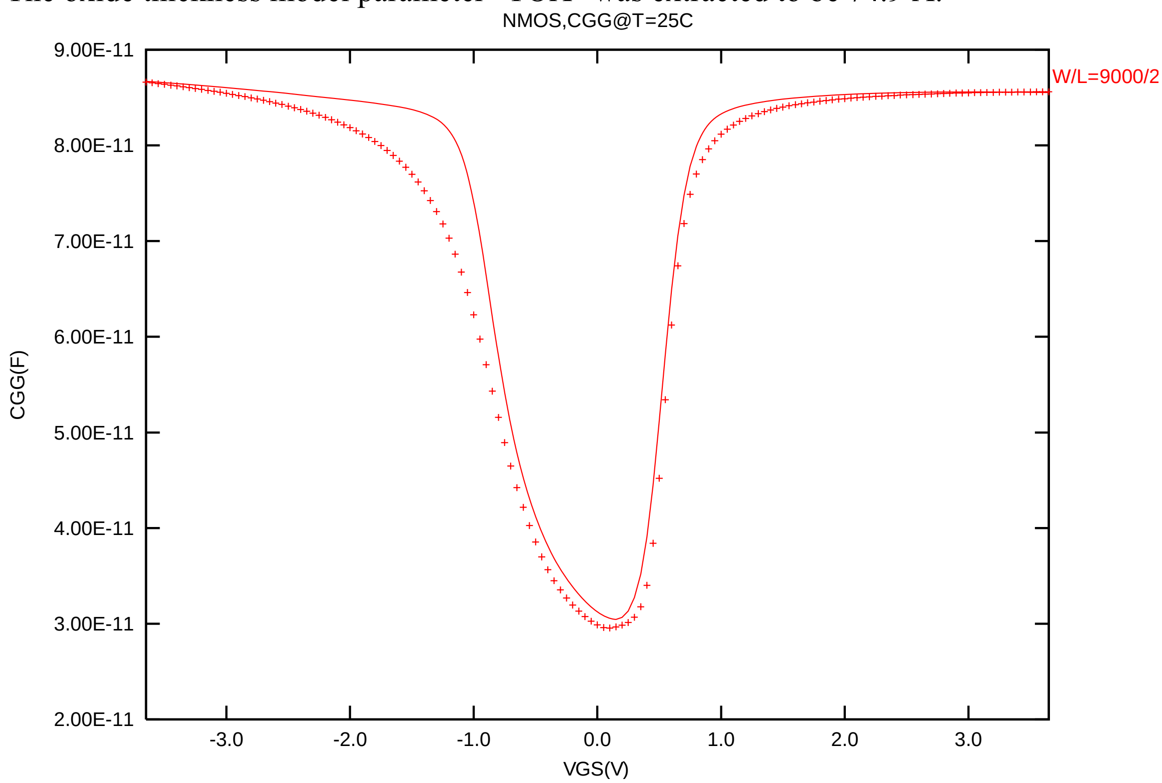


Figure 2-25 Measured and Simulated C_{GG} curves

2.4.4 C_{GC} vs. V_{GS} Plots

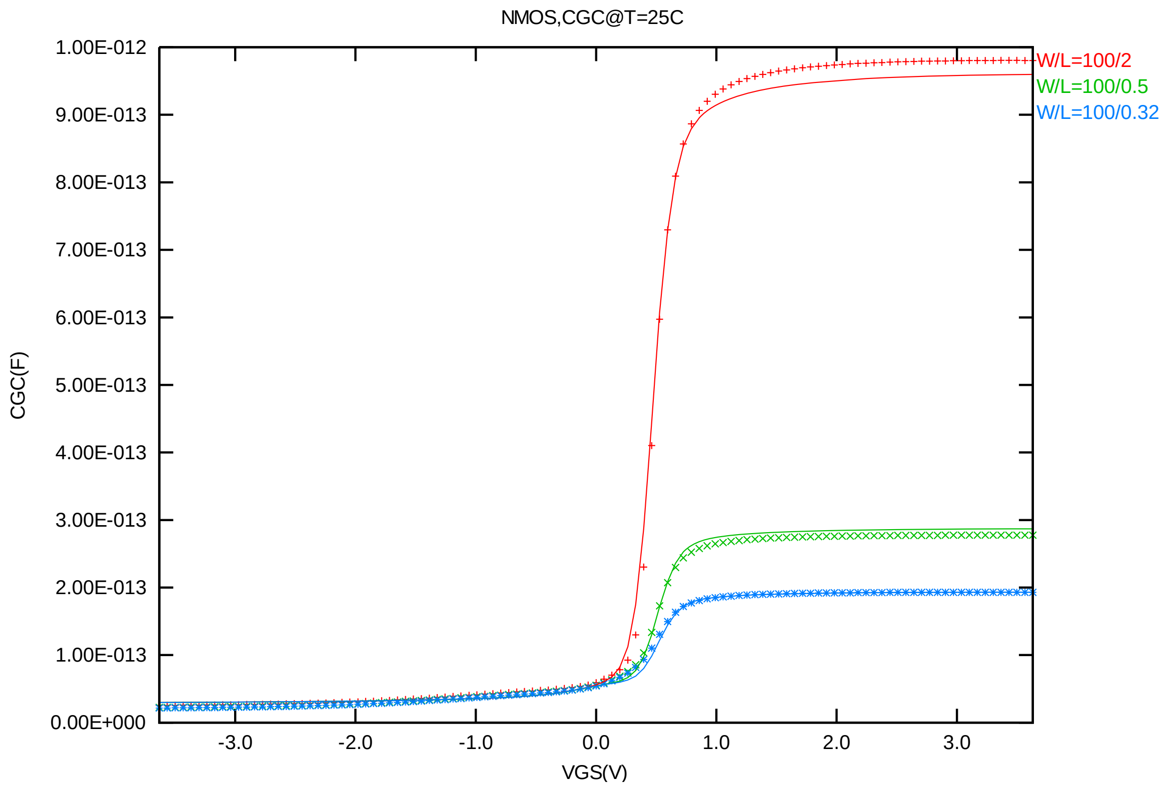


Figure 2-26 Measured and Simulated C_{GC} curves for large area measurement structure for NMOS

2.5 Parasitic source / drain junction capacitance model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	**Area[m^2]**	**Perimeter[m]**	**Description**
Bottom area junction capacitance	5.0E-08	2.0E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.6E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.74E-08	1.833E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.6 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

2.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-11 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	1.03E-03	CJSW	F/m	1.41E-10	CJSWG	F/m	1.97E-10
PB	V	7.19E-01	PBSW	V	6.56E-01	PBSWG	V	6.56E-01
MJ	--	3.40E-01	MJSW	--	2.03E-01	MJSWG	--	3.65E-01
TCJ	1/K	7.99E-04	TCJSW	1/K	9.14E-05	TCJSWG	1/K	1.00E-02
TPB	V/K	1.83E-03	TPBSW	V/K	8.11E-04	TPBSWG	V/K	0.00E+00

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

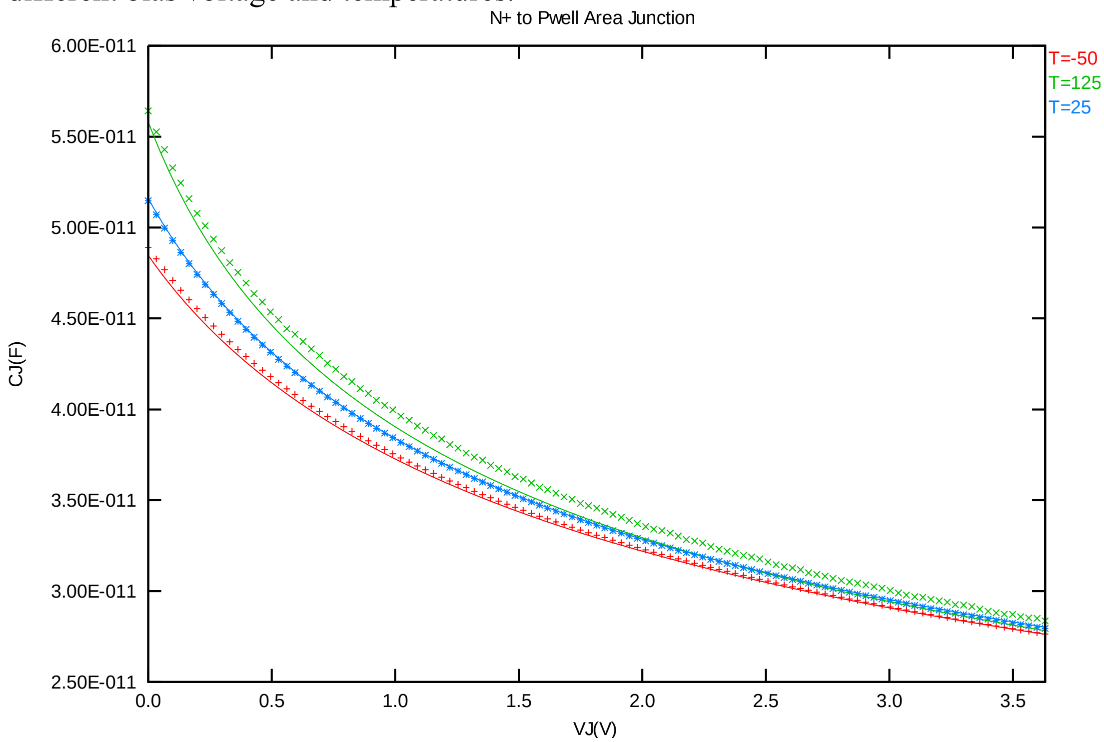


Figure 2-27 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

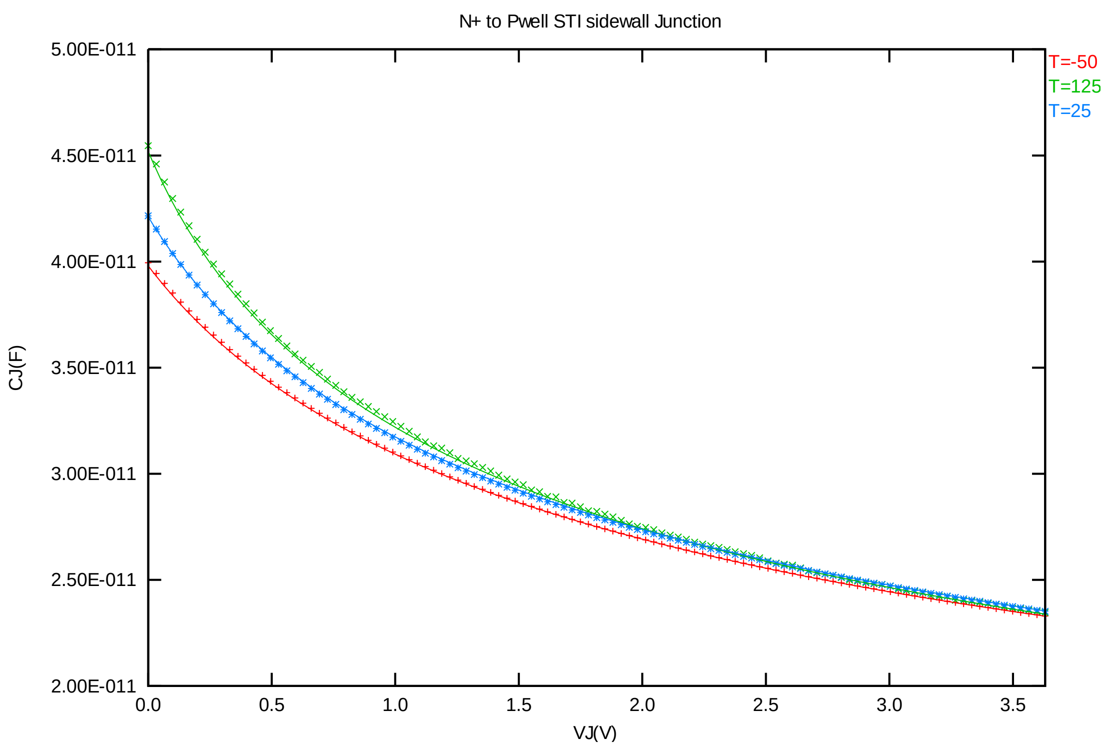


Figure 2-28 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS 3.3 V MOSFET

3 3.3 V p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW0WA	BSW0WA	BSW0WA	BSW0WA
Lot ID	D5JQJ	D5JQJ	D5JQJ	D5JQJ
Wafer Number	16	16	16	16

3.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS, respectively. The dimension is after 90% shrink and 12nm sizing channel length back.
All the normalized current are divided with W_{DES}.

All parameters are given for T=25 ℃,V_{BS}=0 V unless specified otherwise.

Table 3-2 Electrical Parameters for PMOS

All parameters are given for T=25 ^oC ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W/L= 10 μ m / 10.02 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.644	-0.642	-0.654	-0.644
I_{ON}	V_{DS} = -3.3 V	μA/ μm	-8.12	-8.10	-8.18	-8.13
Wide and short channel device (W/L= 10 μ m / 0.34 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.581	-0.581	-0.583	-0.581
V_{TSAT}	V_{DS} = -3.3 V	V	-0.554	-0.552	-0.531	-0.554
Body Effect V_T shift	V_{DS} = -3.3 V V_{BS}= 0 ~ --3.3 V	V	--	-0.805	-0.797	-0.801
DIBL	V_{DS}= -0.1~-3.3 V	V	--	-0.028	-0.052	-0.027
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-255.50	-253.16	-255.32	-255.50
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/μm	-4.6e-13	-3.6E-13	-2.3E-12	-5.0E-13
Narrow and long channel device (W/L= 0.35 μ m / 10.02 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.642	-0.641	-0.661	-0.642
V_{TSAT}	V_{DS}= -3.3 V	V	--	-0.638	-0.656	-0.637
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-7.9	-7.89	-7.81	-7.89
Narrow and short channel device (W/L= 0.35 μm / 0.34 μm)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.589	-0.582	-0.613	-0.590
V_{TSAT}	V_{DS}= -3.3 V	V	-0.559	-0.545	-0.573	-0.556
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-228.4	-225.83	-222.12	-228.37
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/μm	-1.8e-12	5.1E-13	-1.9E-11	-2.2E-12
Capacitance						
CJ	V_J = 0 V	F/m^2	--	9.66E-04	9.68E-04	9.68E-04
CJSW	V_J = 0 V	F/m	--	8.82E-11	8.81E-11	8.81E-11
CJSWG	V_J = 0 V	F/m	--	2.46E-10	2.29E-10	2.29E-10
C_{OVL}	V_{GS} = 0.5 V	F/m	--	3.19E-10	2.97E-10	2.97E-10

3.3 MOSFET DC model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below(The dimensions are after 90% shrink and 12nm sizing channel length back.).

Table 3-3 Geometry of PMOS devices measured for parameter extraction

W\L	10.02	1.02	0.82	0.7	0.52	0.46	0.36	0.34
10	X	X	X	X	X	X	X	X
1								X
0.8		X		X		X		X
0.6		X		X		X		X
0.43	X							X
0.35	X	X		X		X		X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-3.3	-3.2
V_{GS} (V)	0.5	-3.6	-0.05
V_{BS} (V)	0	3.3	0.825

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-3.6	-0.05
V_{GS} (V)	-0.7	-3.3	-0.65
V_{BS} (V)	0	3.3	3.3

3.3.3 Model verification plots

The simulation results (solid lines) obtained from the extracted model were plotted together with the measurement results (dotted lines) for comparison. The L and W in the plots refer to (L_{DES}*0.9+12nm and W_{DES}*0.9) respectively.

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

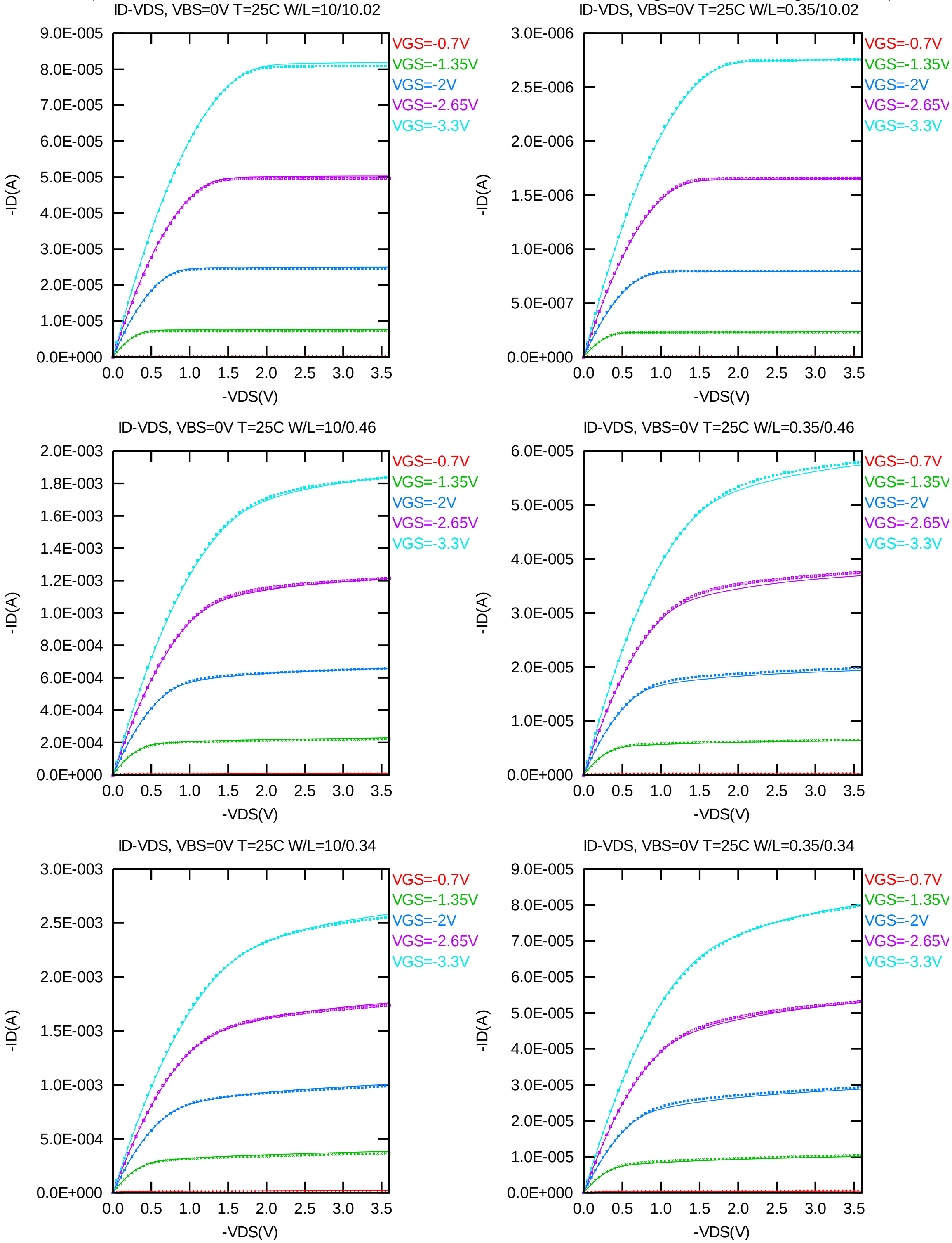


Figure 3-1 I_D vs. V_{DS} at V_{BS} = 0 V for PMOS

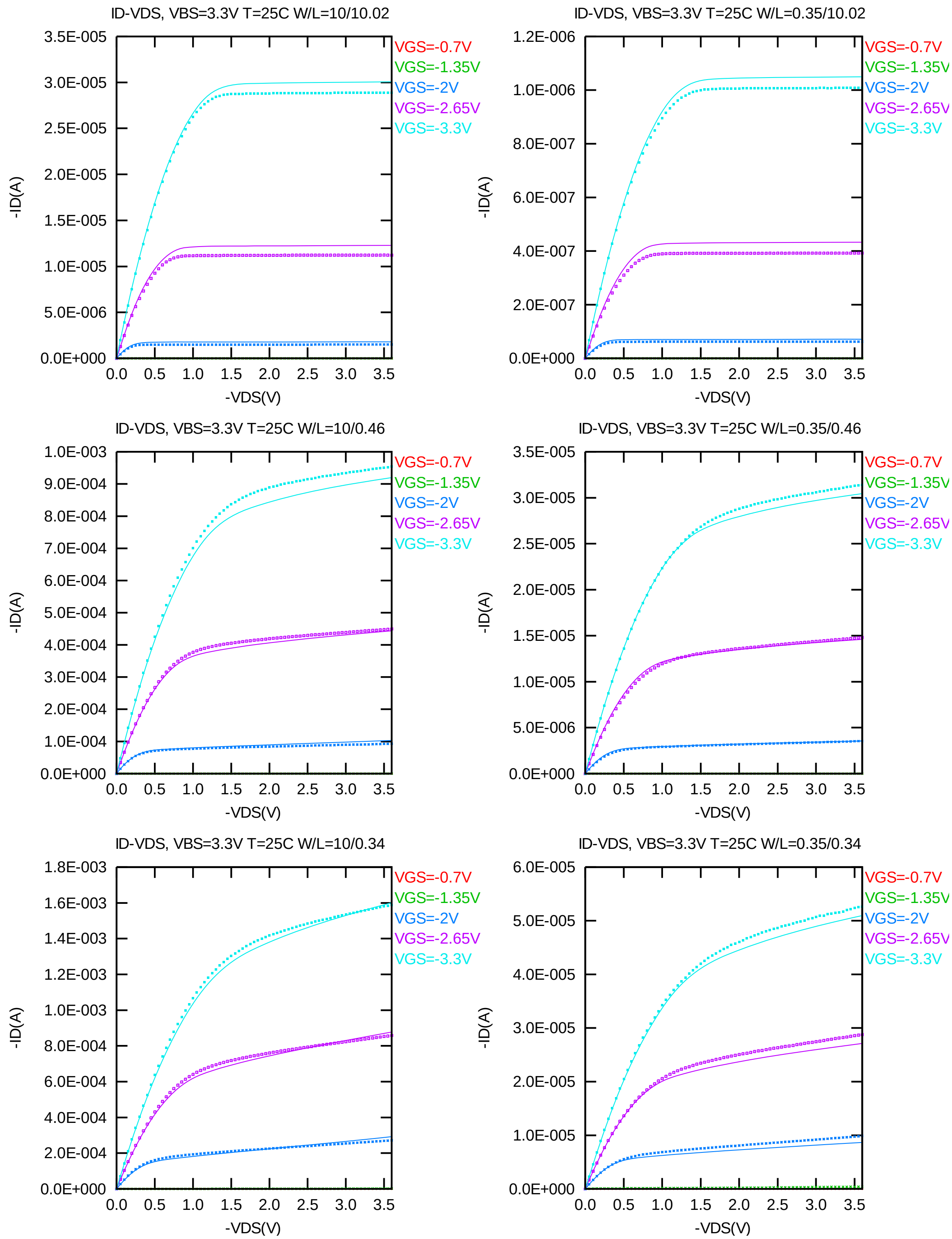


Figure 3-2 I_D vs. V_{DS} at V_{BS}=3.3 V for PMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

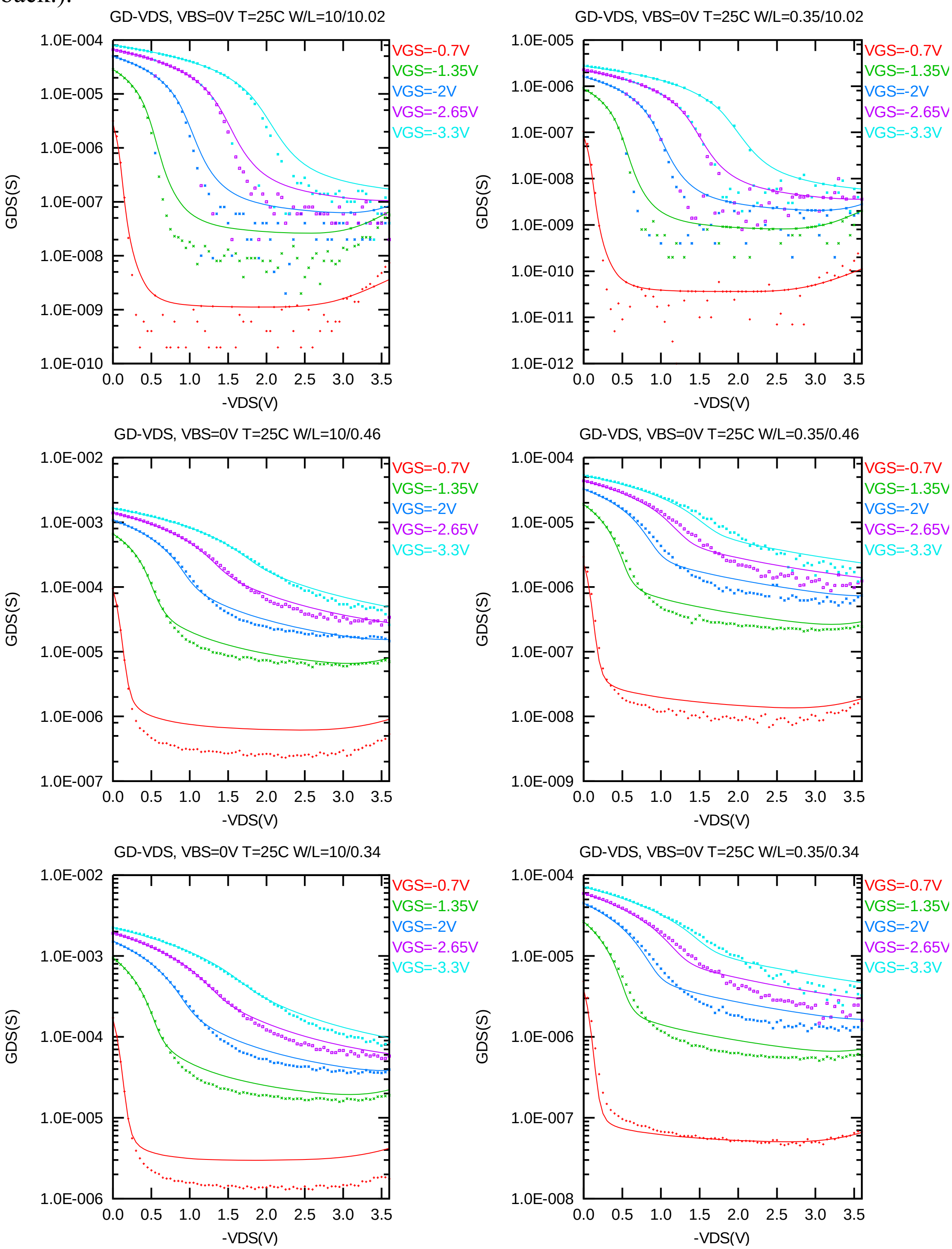


Figure 3-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for PMOS

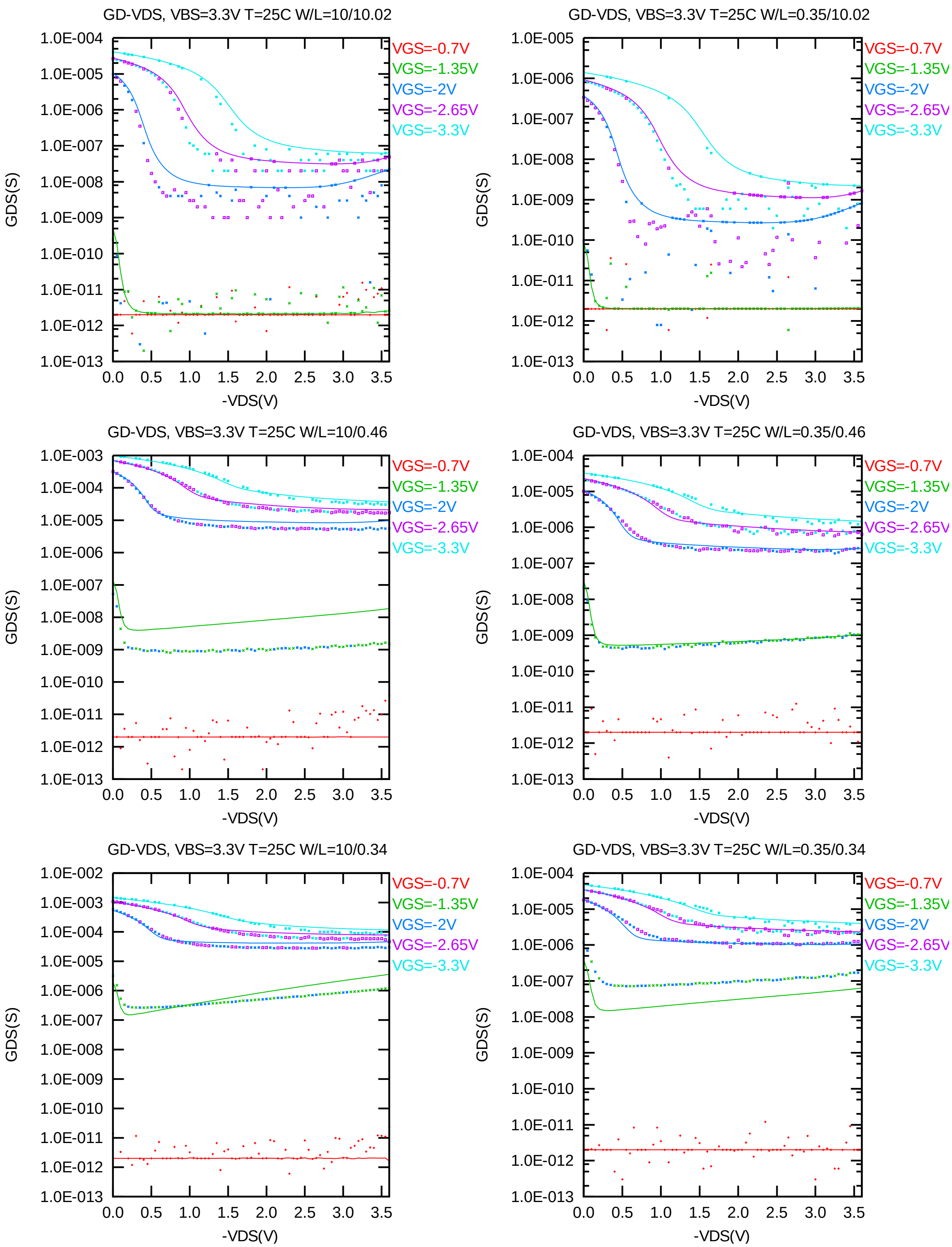


Figure 3-4 G_{DS} vs. V_{DS} at V_{BS} = 3.3V for PMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

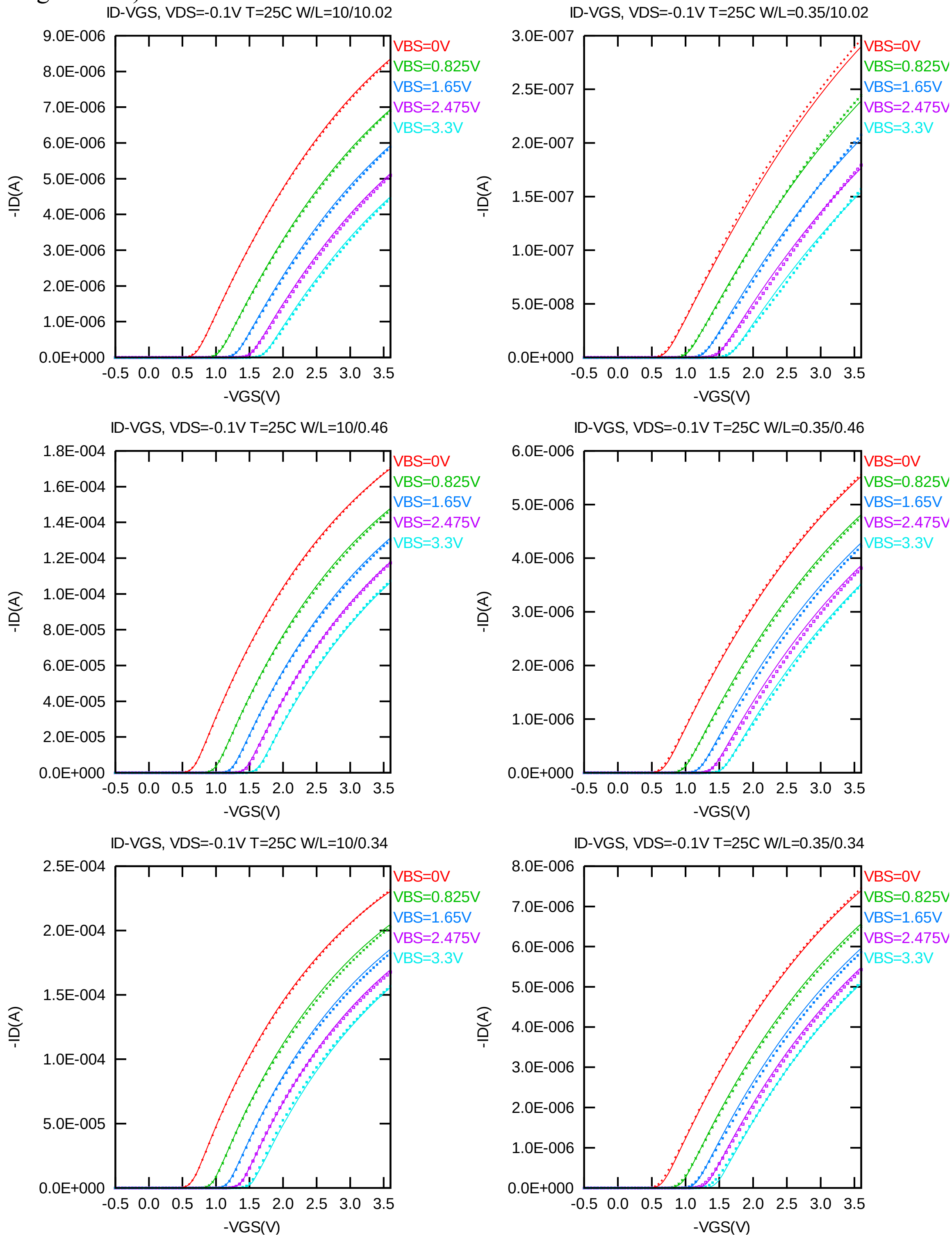


Figure 3-5 I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

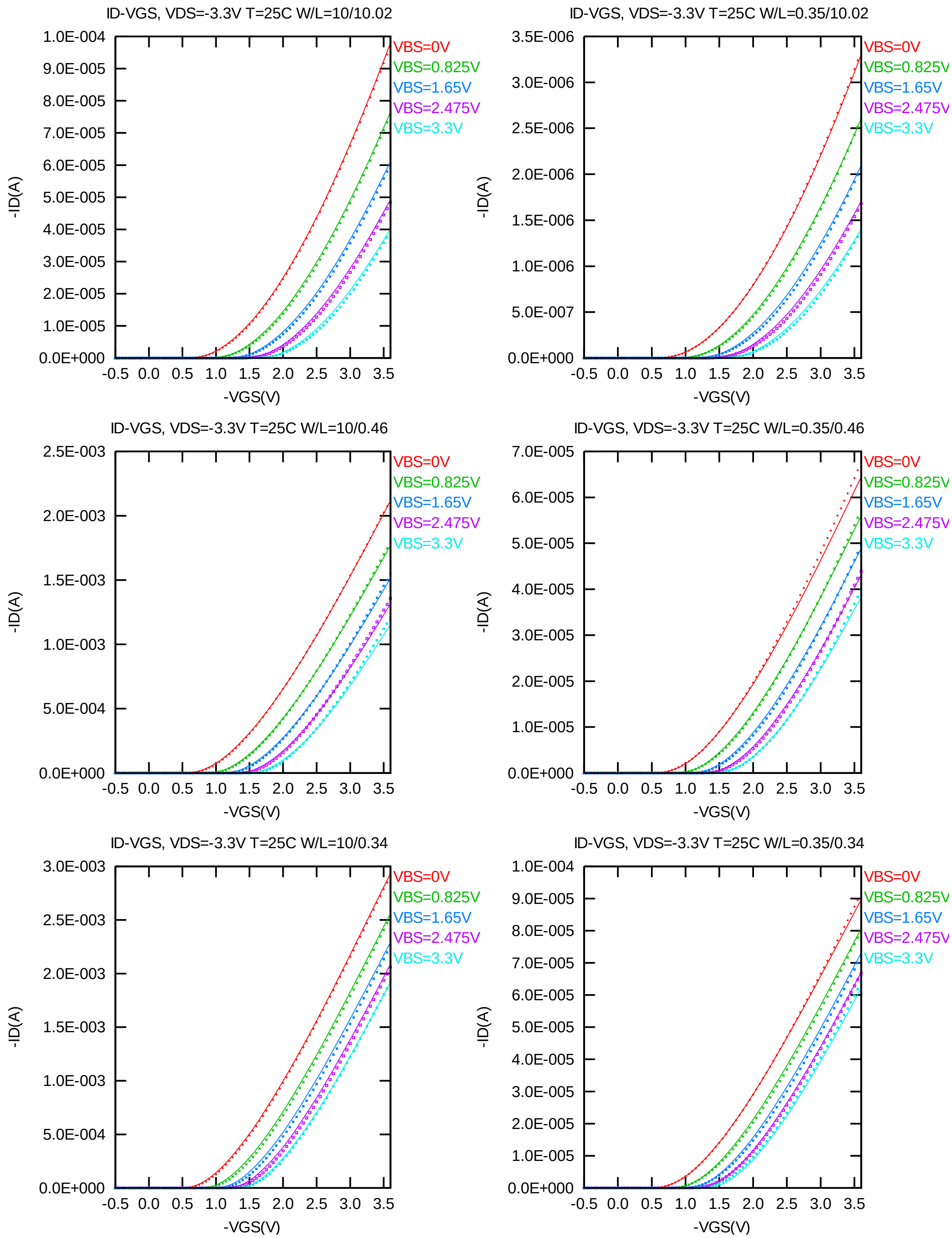


Figure 3-6 I_D vs. V_{GS} at V_{DS} = -3.3 V for PMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

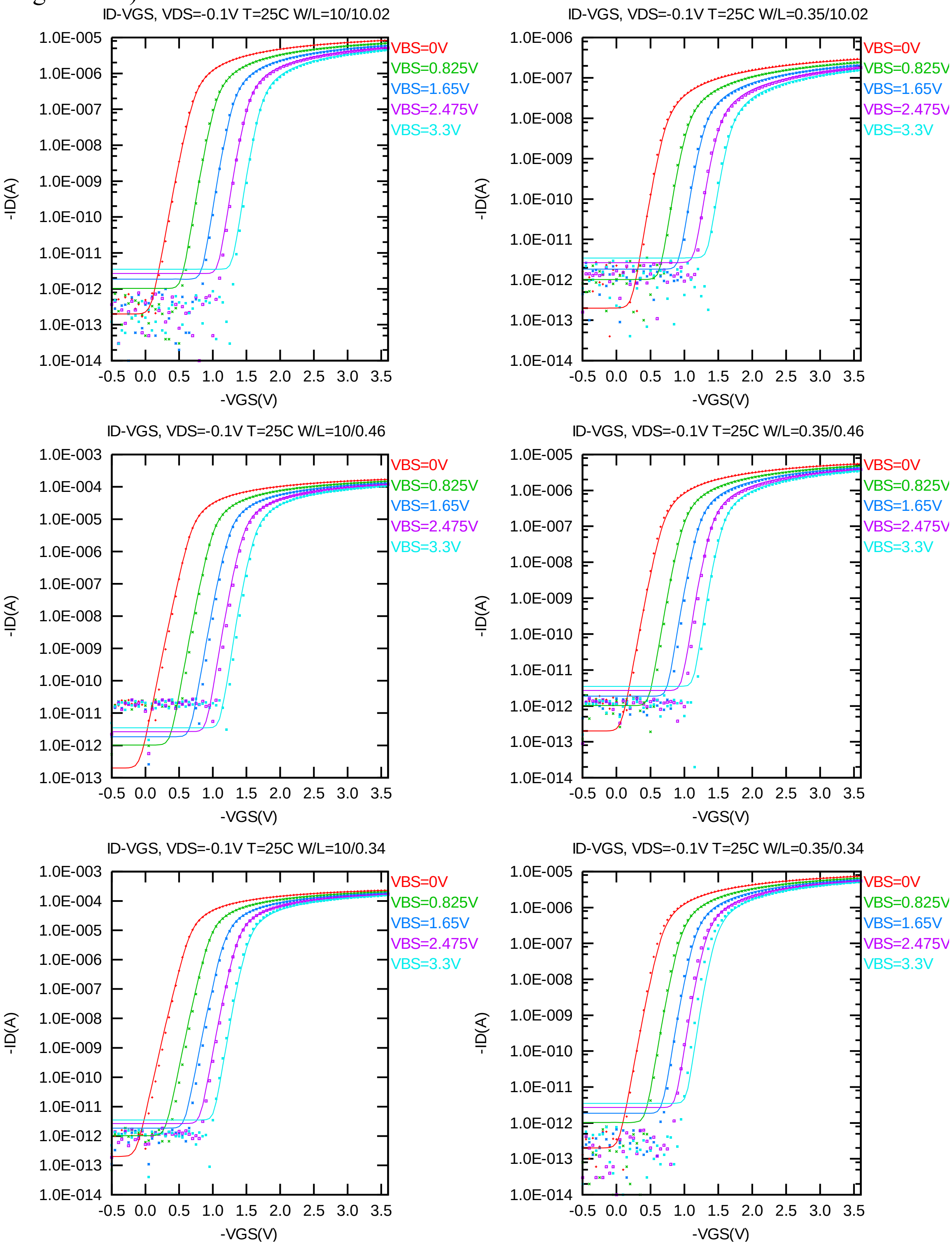


Figure 3-7 Log scaled I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

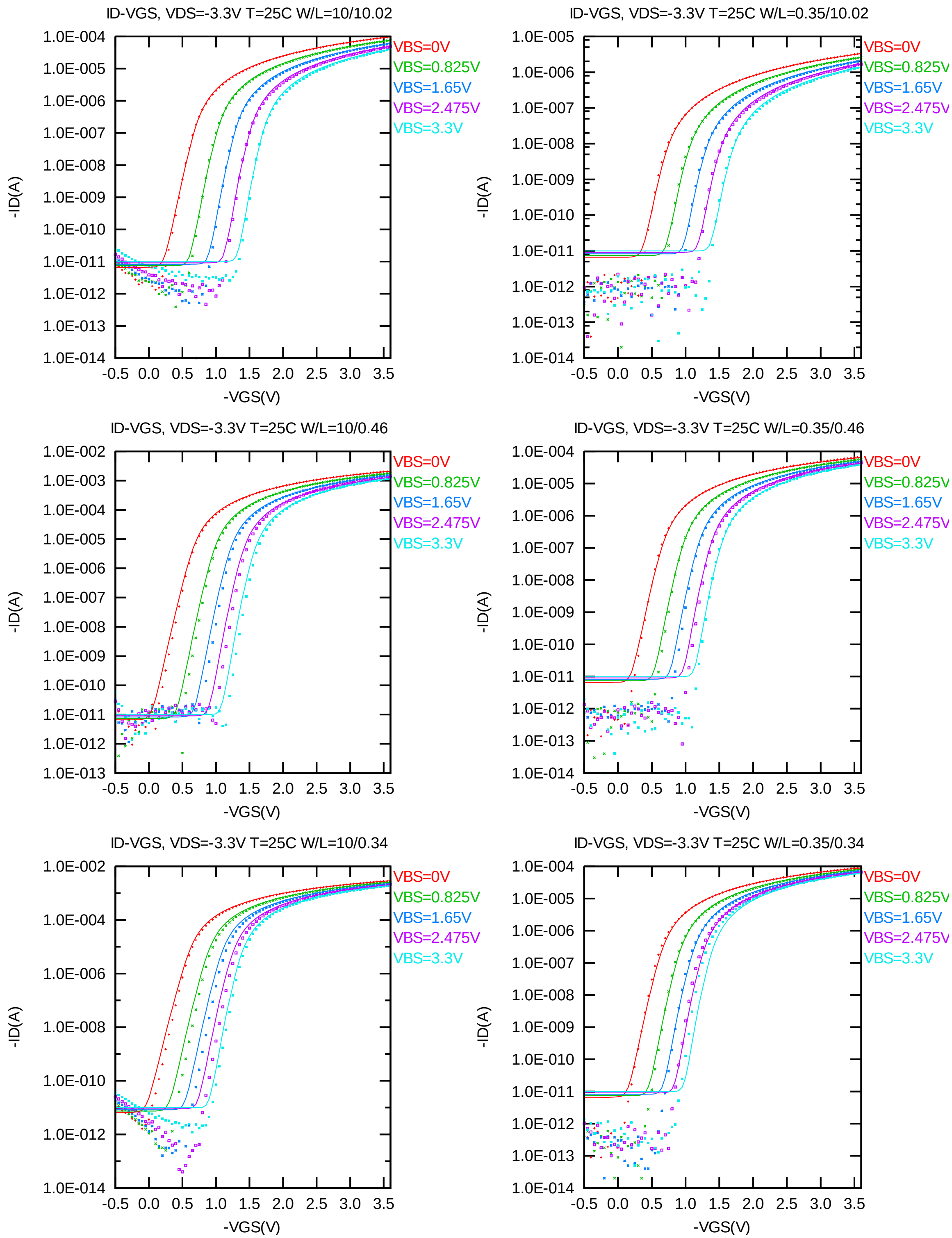


Figure 3-8 Log scaled I_D vs. V_{GS} at V_{DS} = -3.3 V for PMOS

G_M vs. V_{GS} Plots

The following plots show the measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

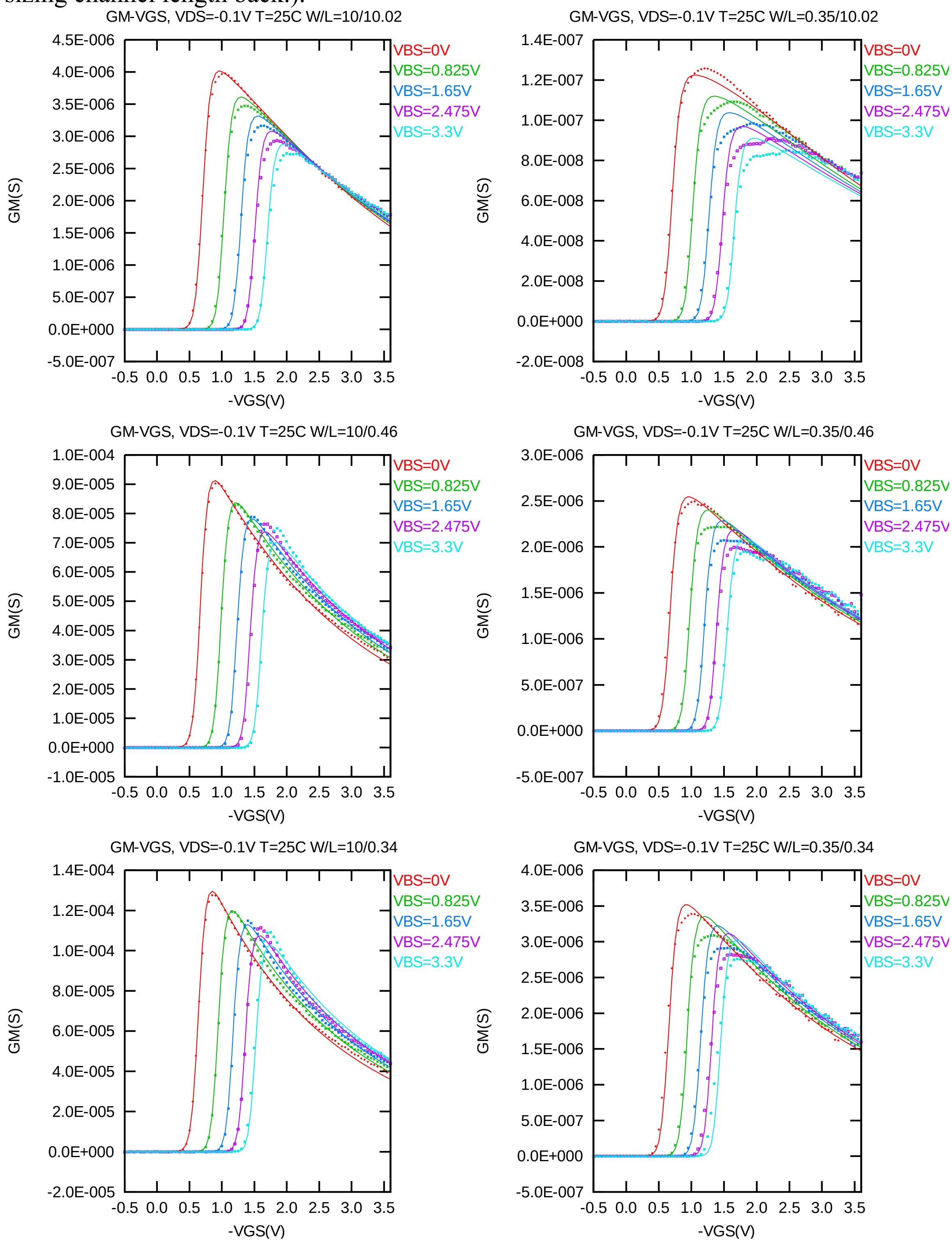


Figure 3-9 G_M vs. V_{GS} at V_{DS} = -0.1 V for PMOS

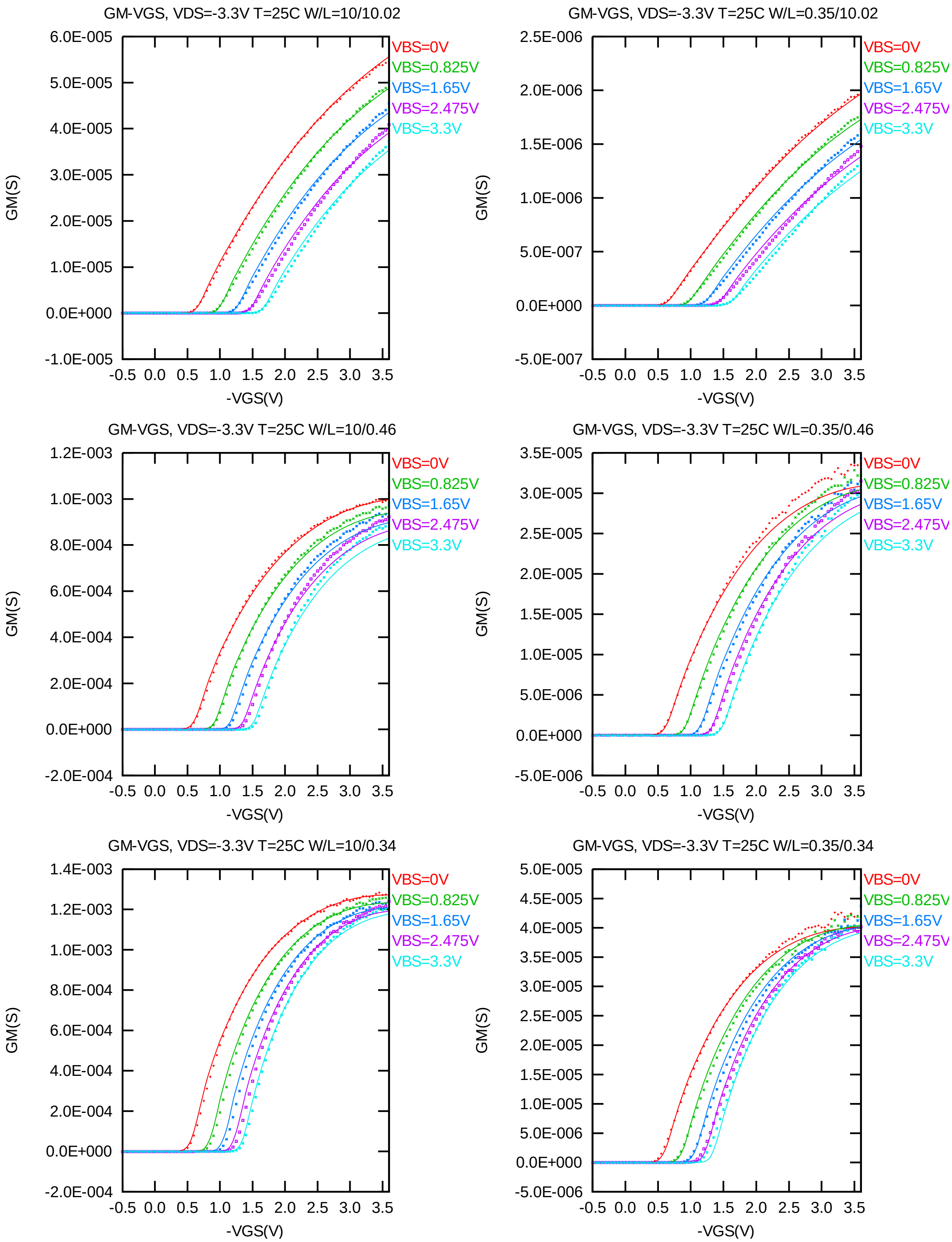


Figure 3-10 G_M vs. V_{GS} at V_{DS} = - 3.3 V for PMOS

V_{TLIN} & V_{TSAT} vs. L_{DES} & W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

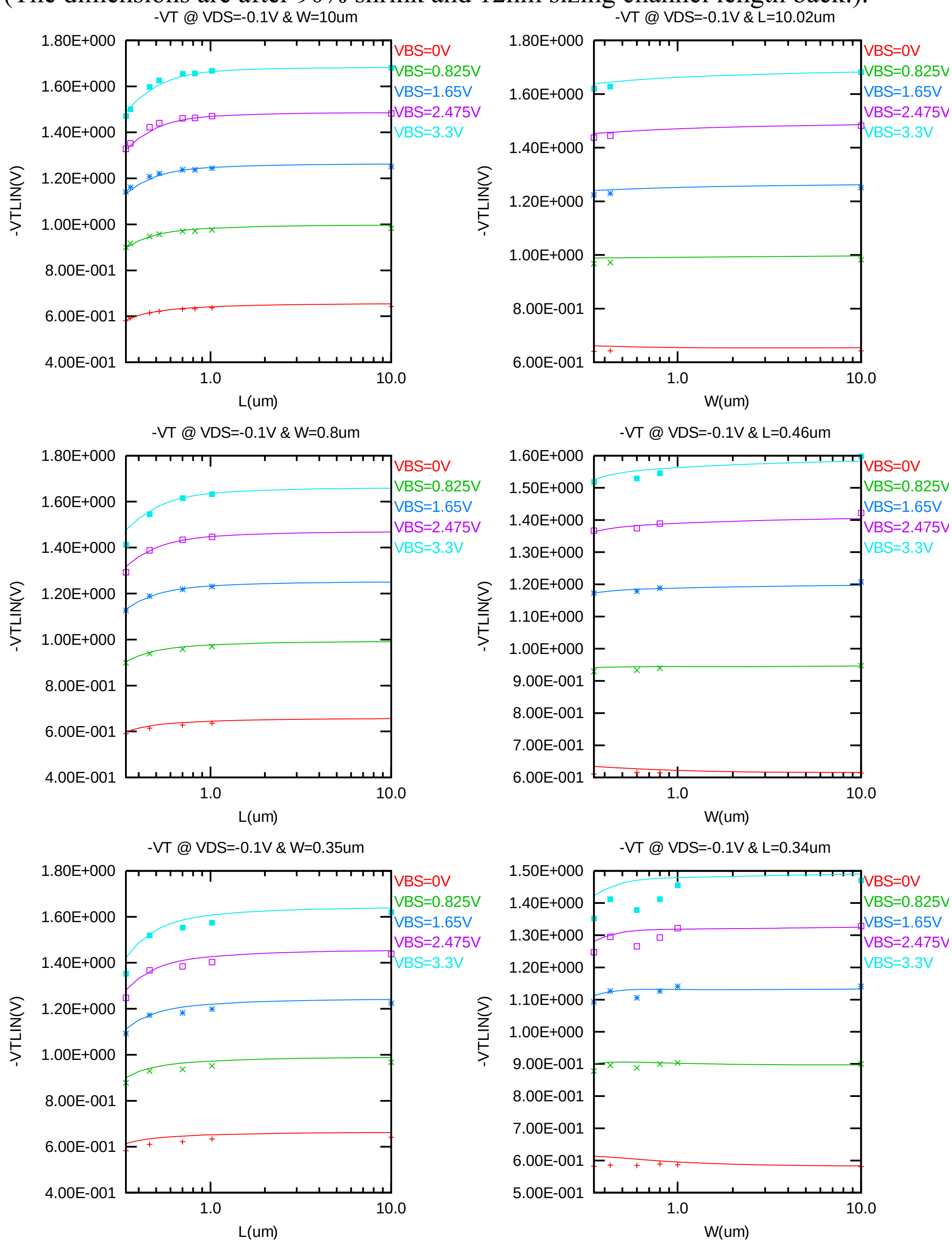


Figure 3-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = -0.1 V) for PMOS

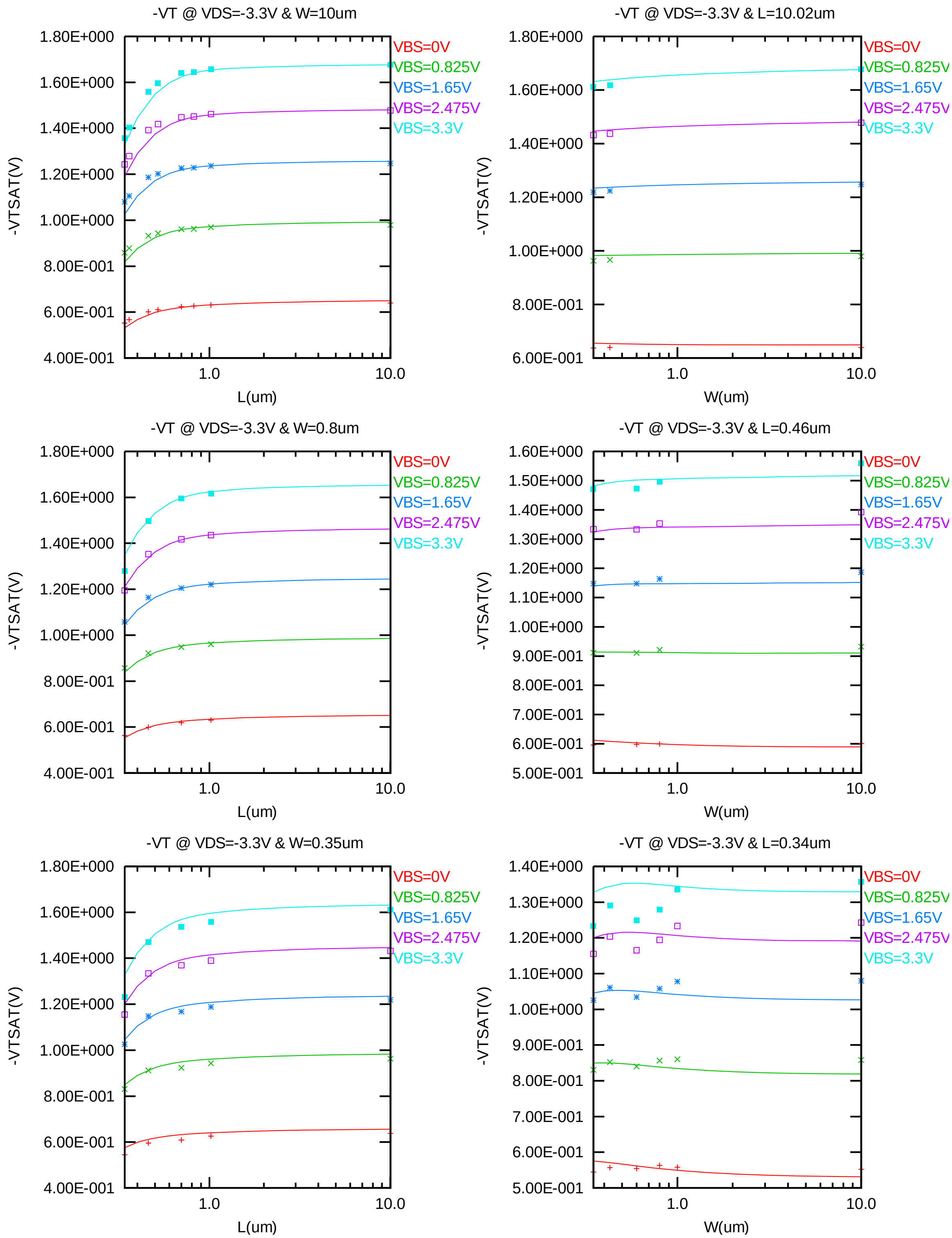


Figure 3-12 V_{TSAT} vs. channel length & channel width for high drain bias (V_{DS} = -3.3 V) for PMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -V_{CC}$ for $V_{BS} = 0$ V and $V_{BS} = 3.3$V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

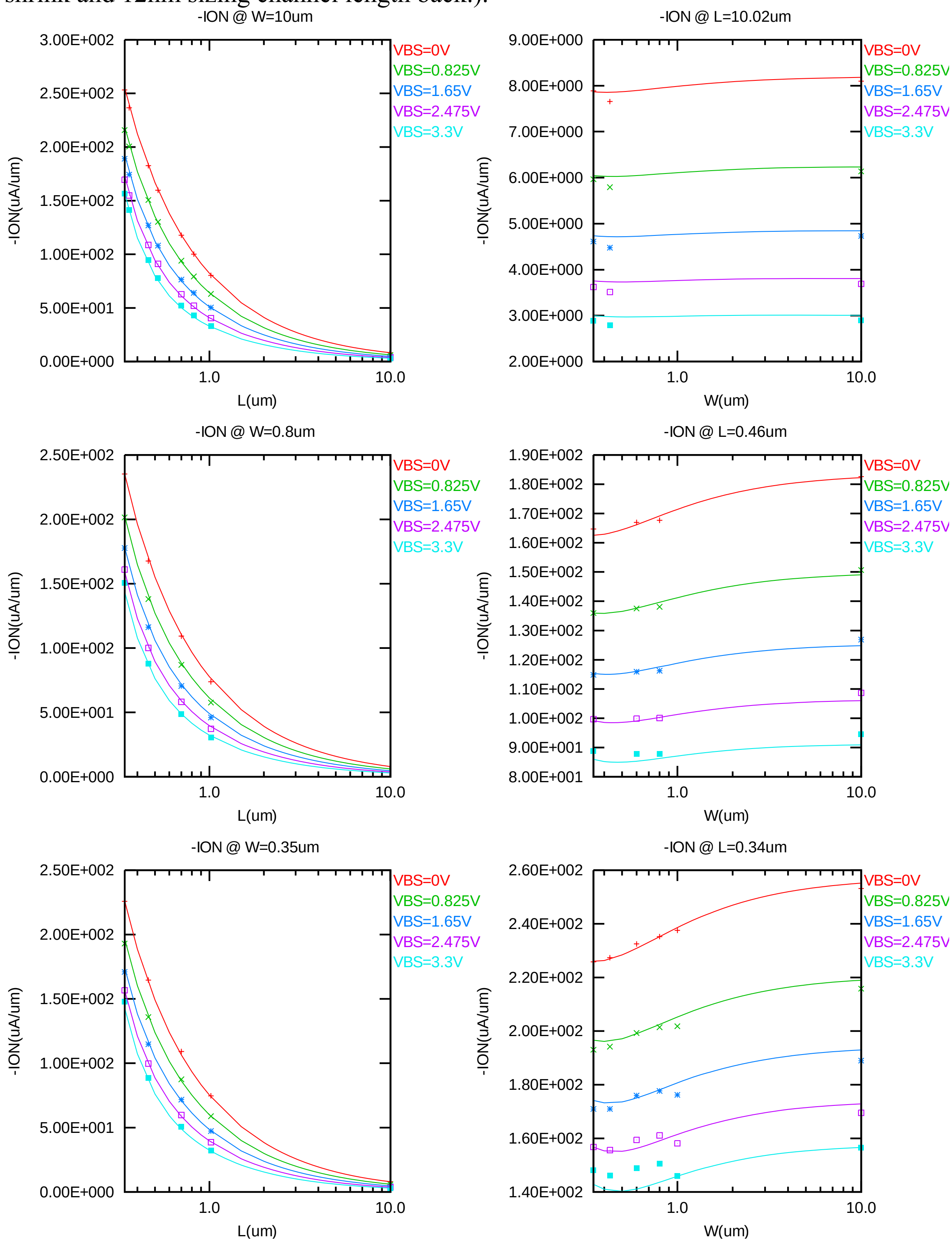


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

I_{OFF} *vs.* L_{DES} & I_{OFF} *vs.* W_{DES}

The following plots show the channel length and width dependence of I_{OFF} extracted at V_{DS} = -V_{CC} and V_{GS} = 0V for V_{BS} = 0 V to 3.3V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.). A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

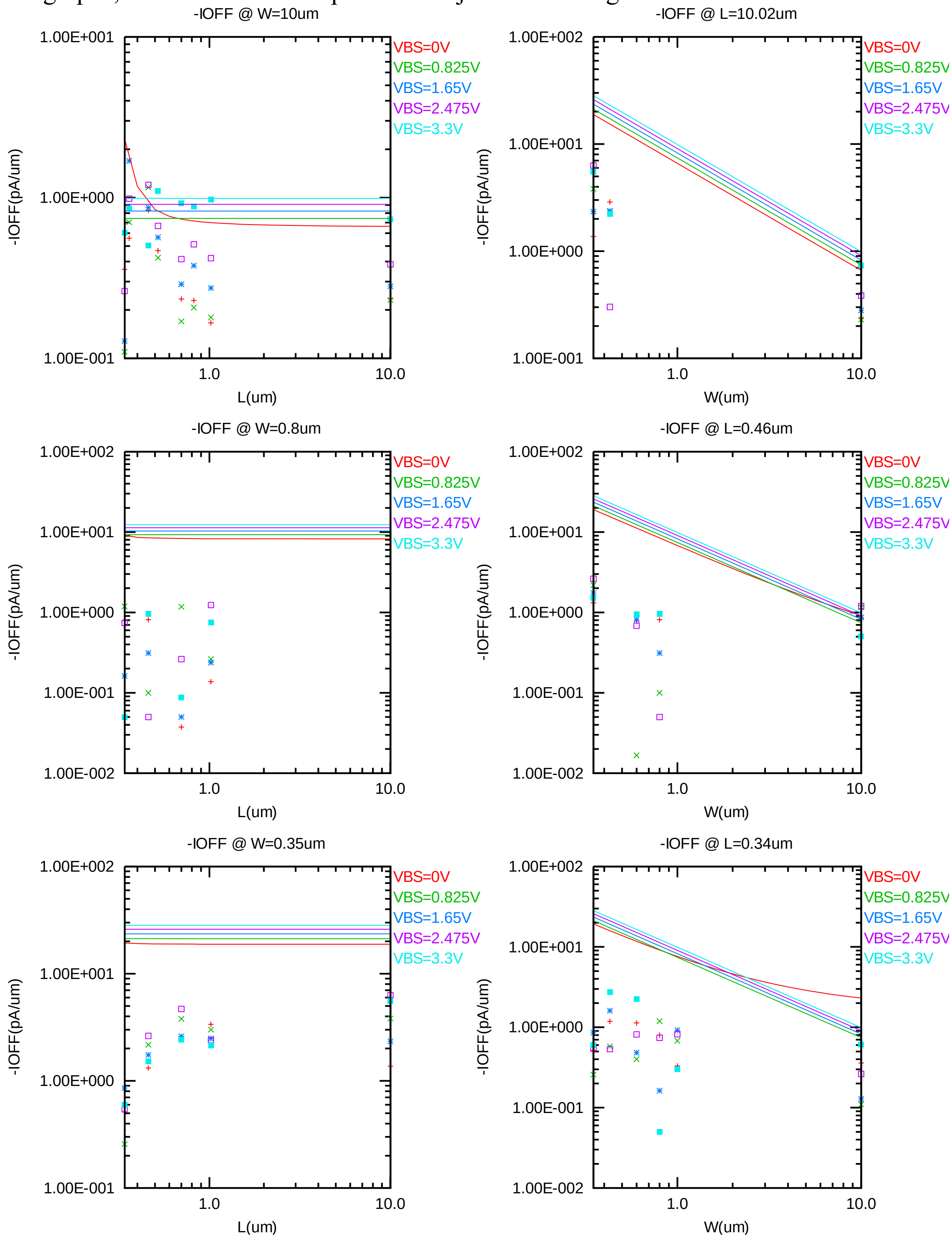


Figure 3-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} & V_{TSAT} vs. Temperature

The following plots give the temperature dependence of the linear and saturation threshold voltage extracted at I_D = -70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm)(The dimensions are after 90% shrink and 12nm sizing channel length back.).

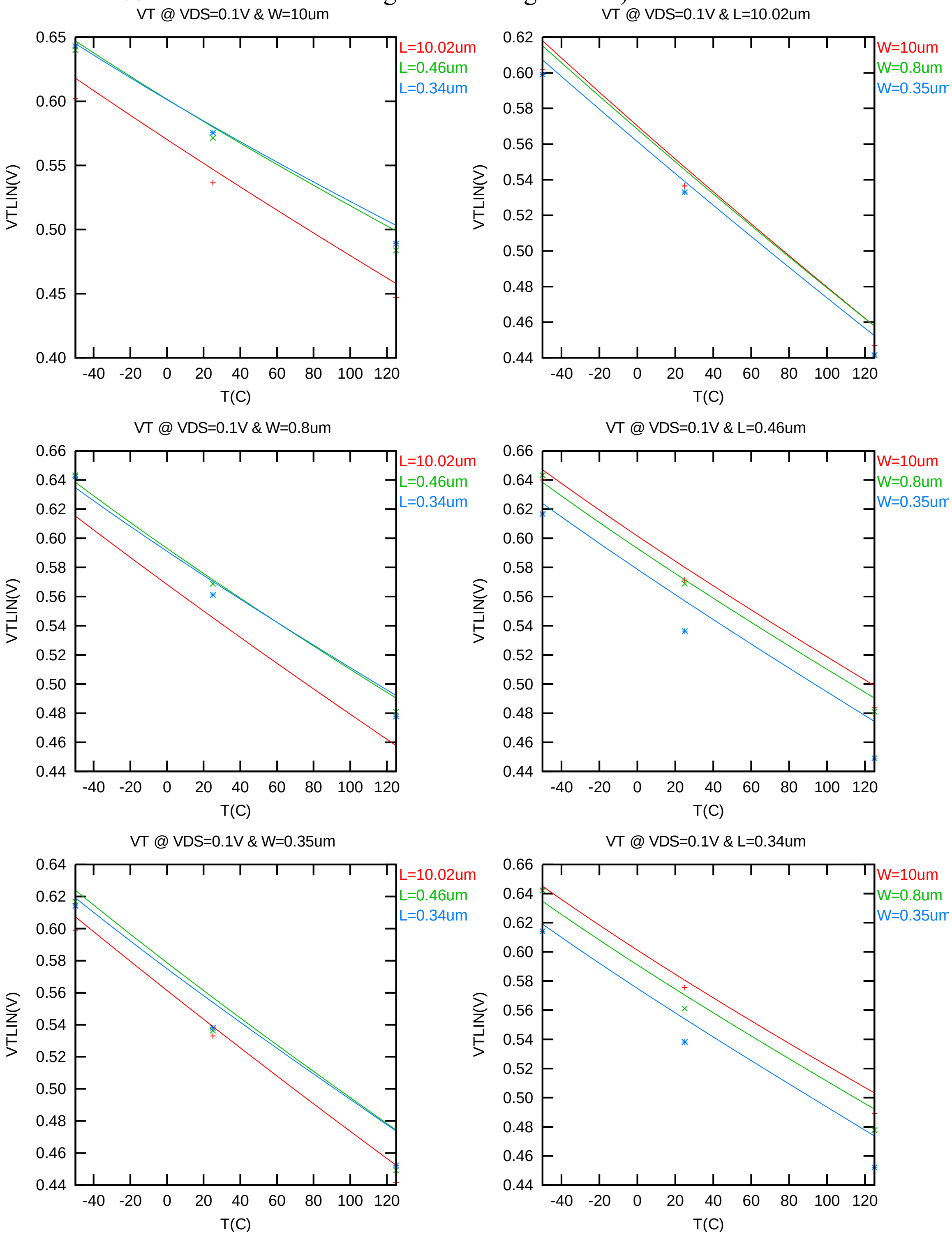


Figure 3-15 V_{TLIN} vs. temperature for PMOS

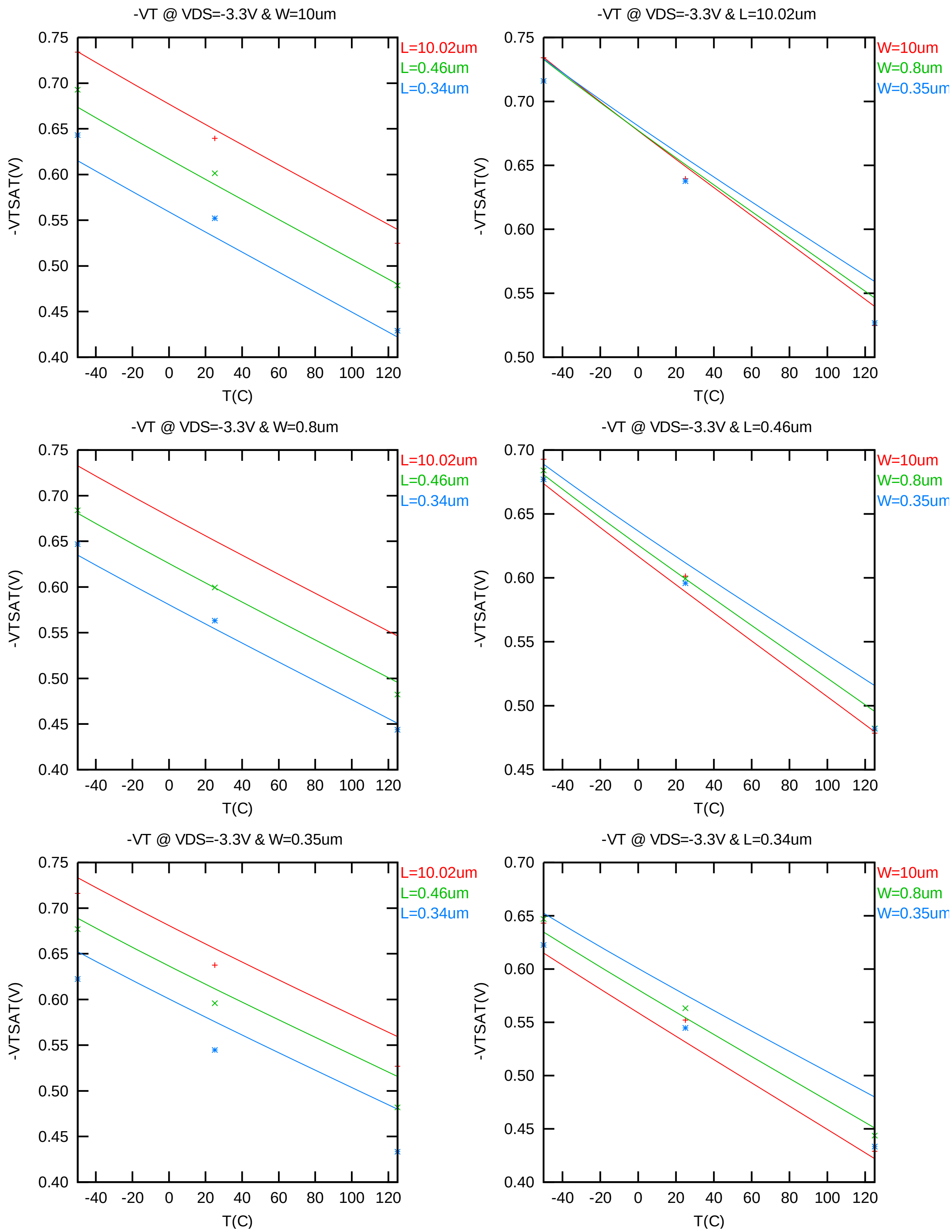


Figure 3-16 V_{TSAT} vs. temperature for PMOS

I_{ON} vs Temperature

The following plot shows the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = -3.3$ V, $V_{BS} = 0$ V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

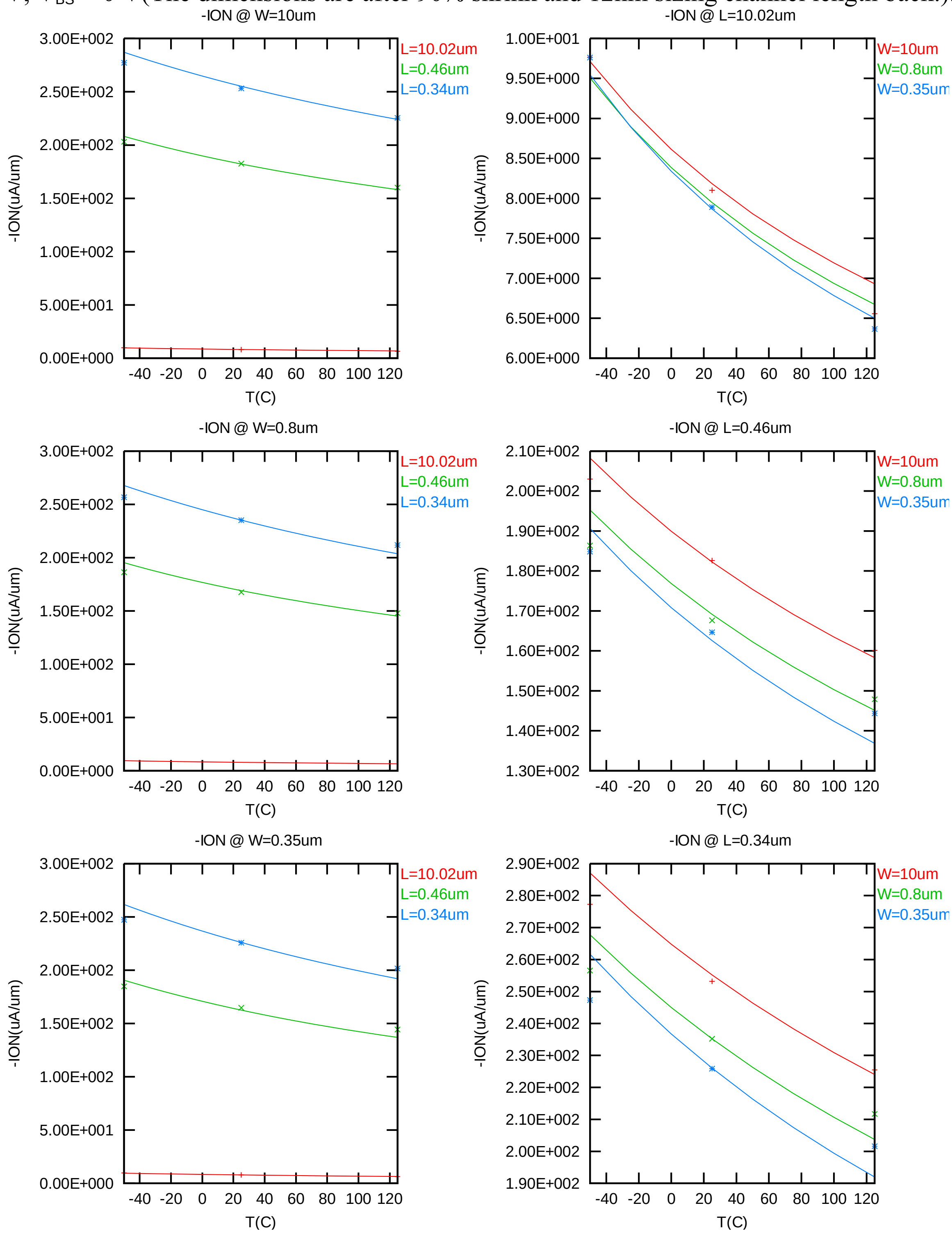


Figure 3-17 Saturation Current vs. temperature for PMOS

I_{OFF} vs Temperature

The following plot shows the temperature dependence of I_{OFF} extracted at V_{DS}= -3.3 V, V_{GS} = V_{BS} = 0 V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

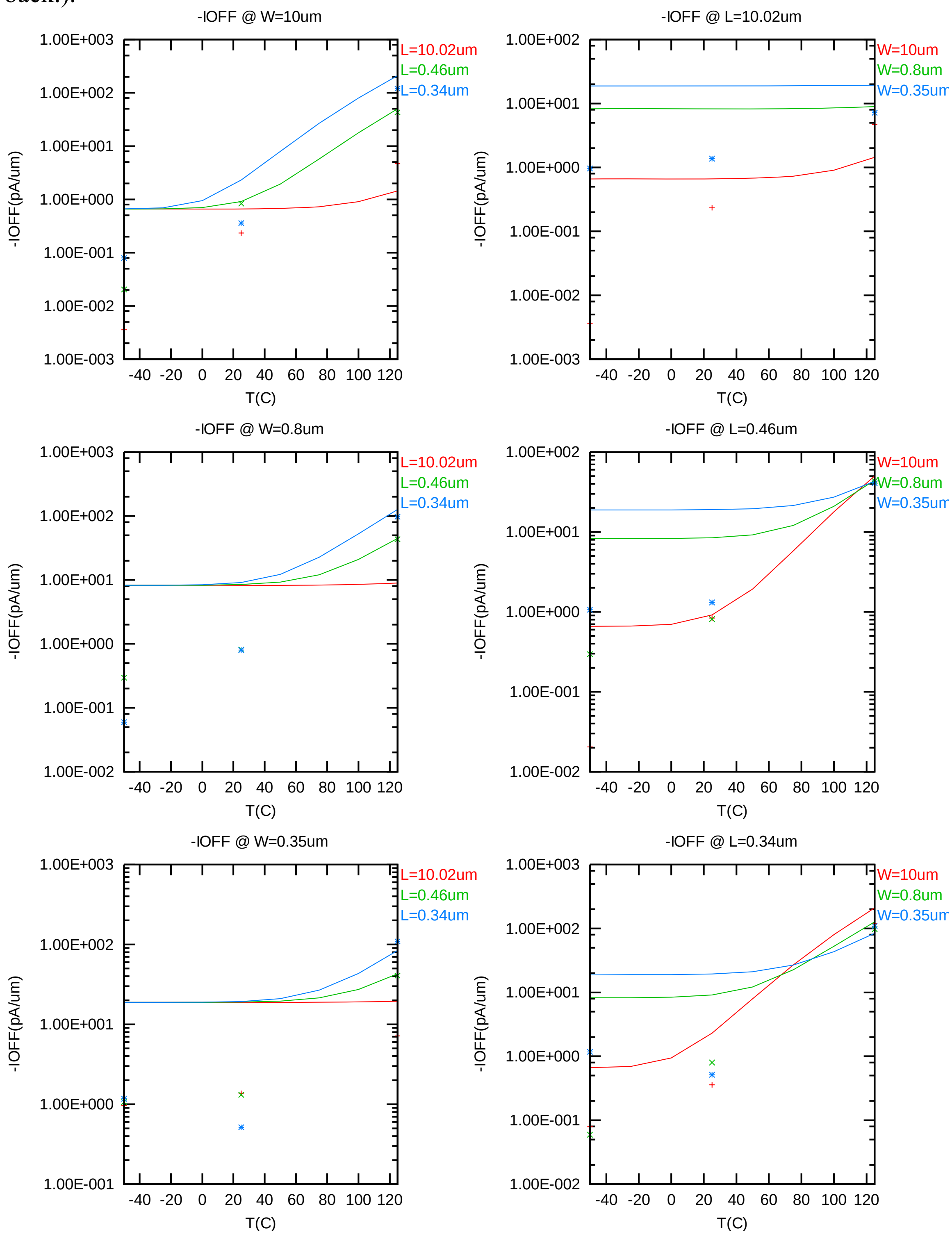


Figure 3-18 I_{OFF} vs. temperature for PMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
g. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
h. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = -3.3$ V

2) G_M and G_{DS} accuracy criteria

i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
j. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 3-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I$_D$ *Accuracy*

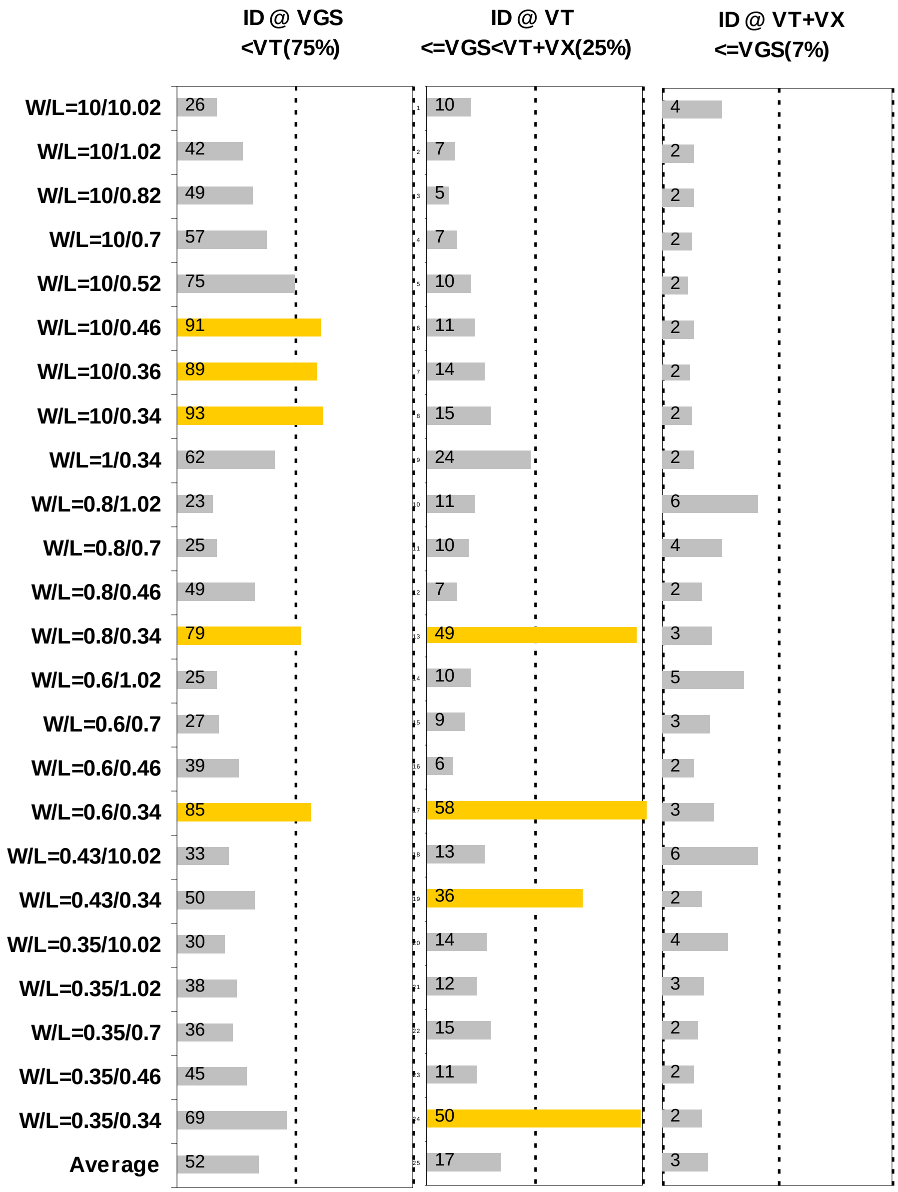


Figure 3-19 I$_D$ accuracy at 25 ℃ for PMOS

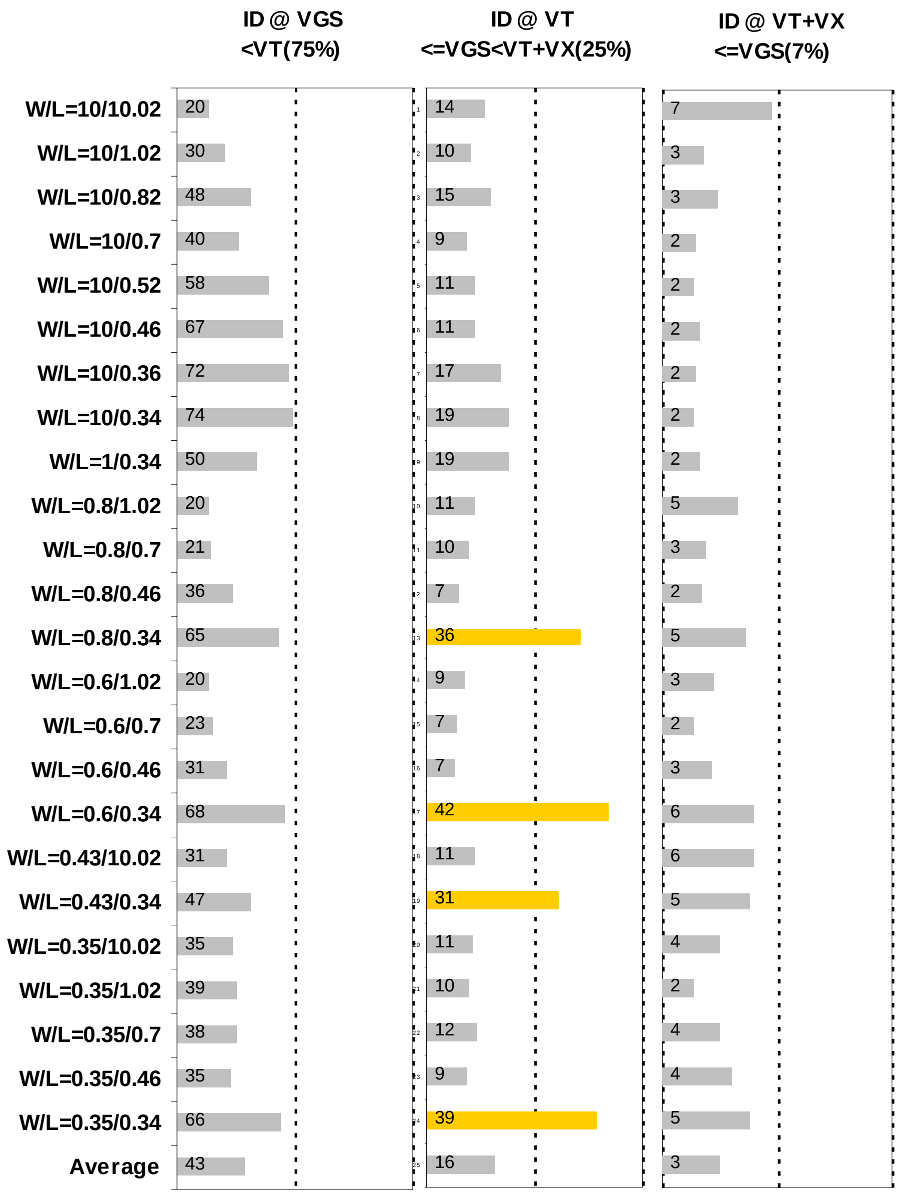


Figure 3-20 I_D accuracy at 125 ℃ for PMOS

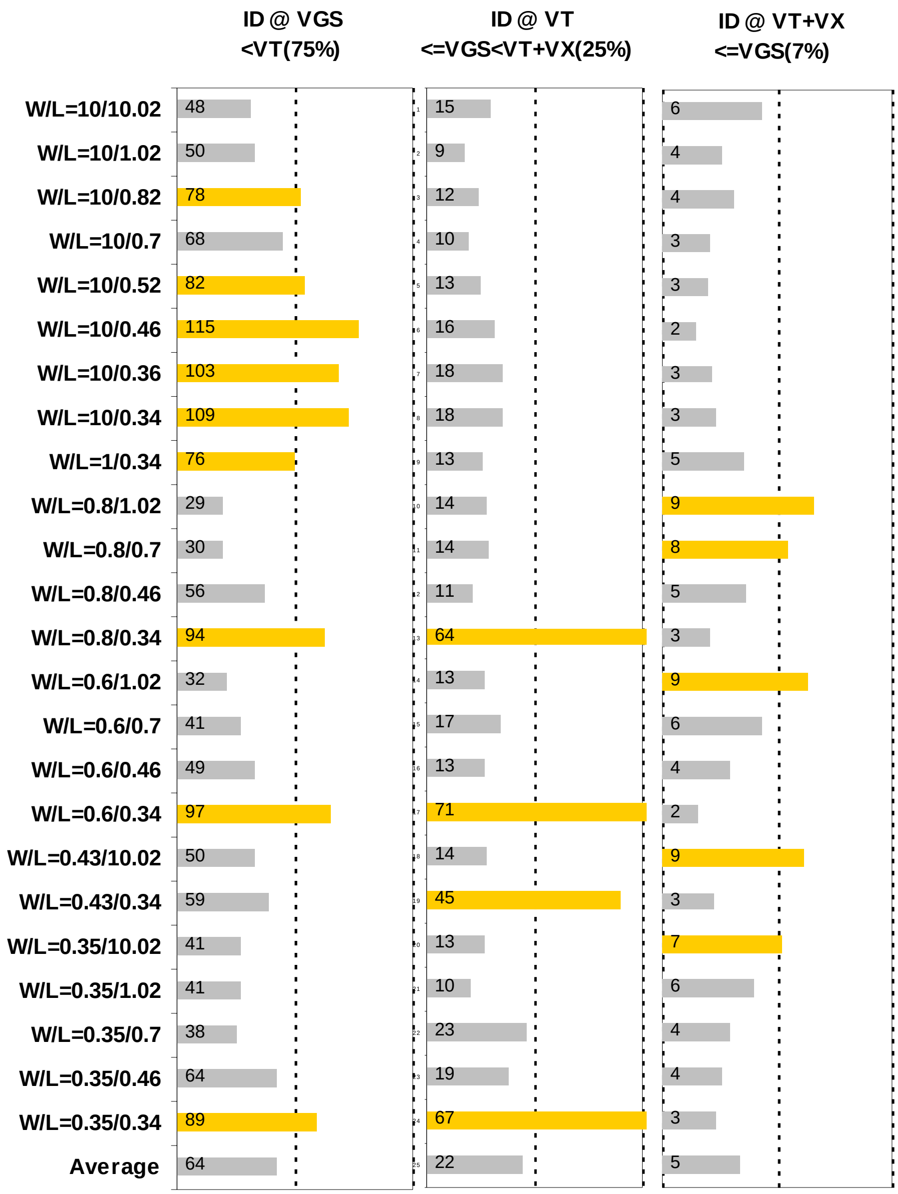


Figure 3-21 I_D accuracy at -40 ℃ for PMOS

G_M and G_{DS} Accuracy

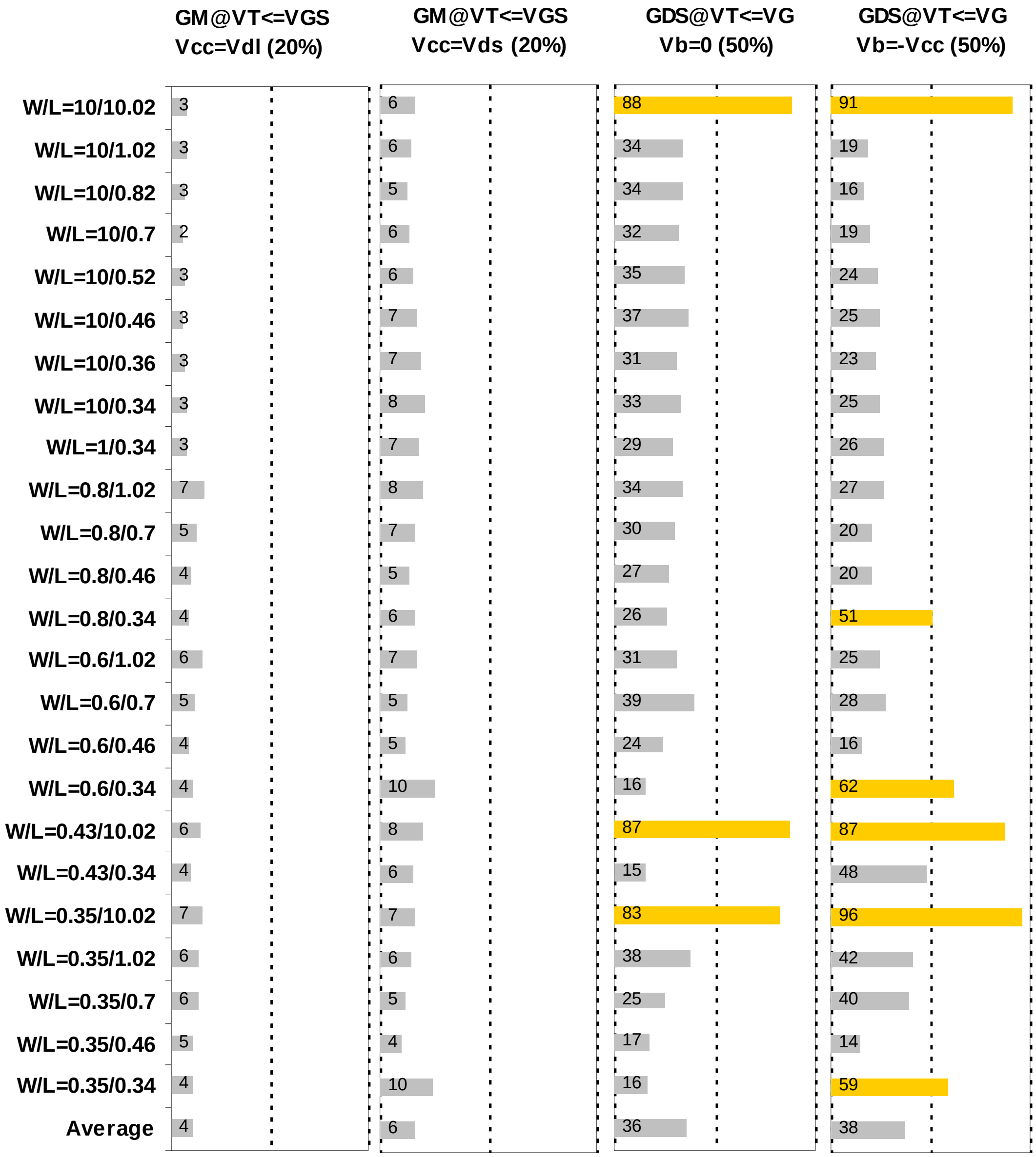


Figure 3-22 G_M and G_{DS} accuracy at 25 ℃ for PMOS

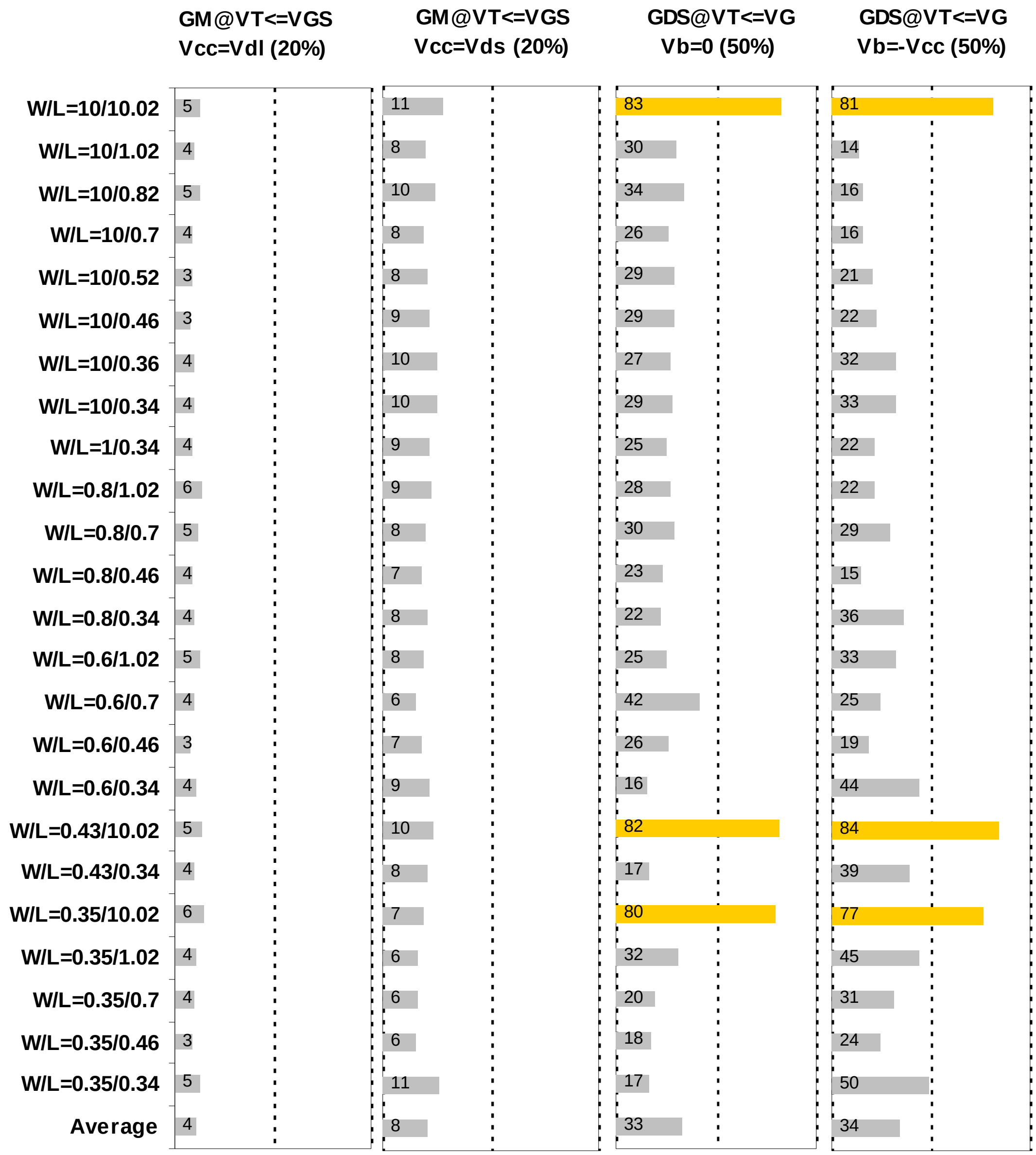


Figure 3-23 G_M and G_{DS} accuracy at 125 ℃ for PMOS

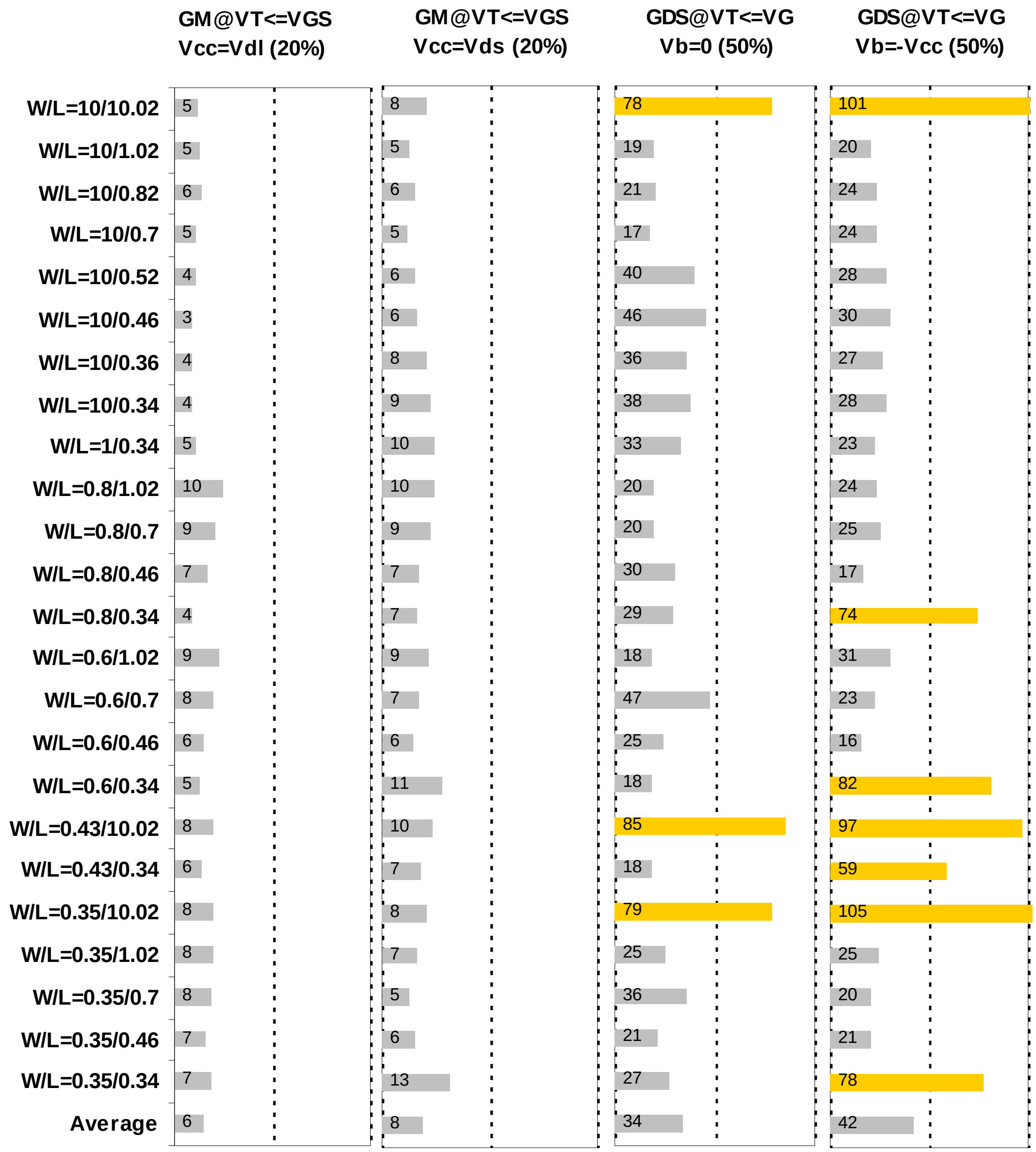


Figure 3-24 G_M and G_{DS} accuracy at -40 ℃ for PMOS

3.4 MOSFET capacitance model

3.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Measurement structure for PMOS Cgg capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components, C_{GDS}	2	9000

Table 3-7 Large area measurement structure for PMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	2.02	100
	0.52	100
	0.34	100

3.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized as below.

Table 3-8 Meaurement condition for PMOS capacitance model

V_{BIAS} [V]	From -3.65 to 3.65 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

3.4.3 T_{OX} extraction

The oxide thickness model parameter "TOX" was extracted to be 76.8 Å.

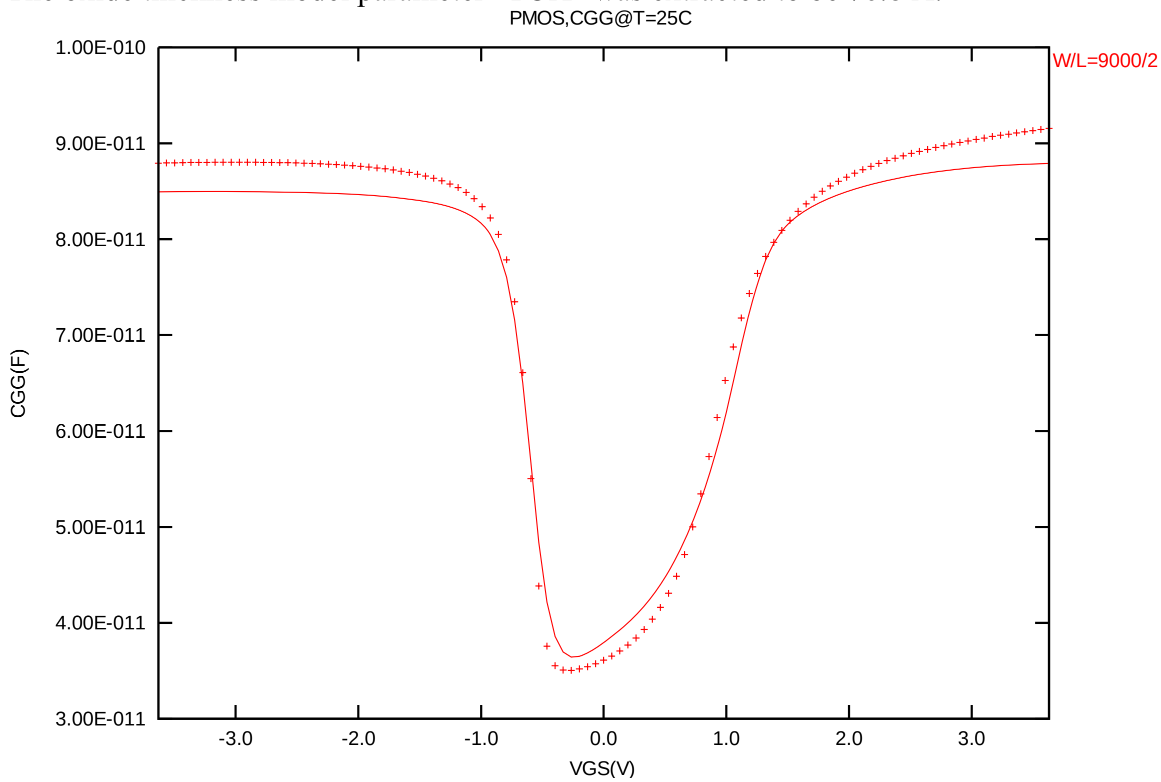


Figure 3-25 Measured and Simulated C_{GG} curves for PMOS

3.4.4 C_{GC} vs. V_{GS} Plots

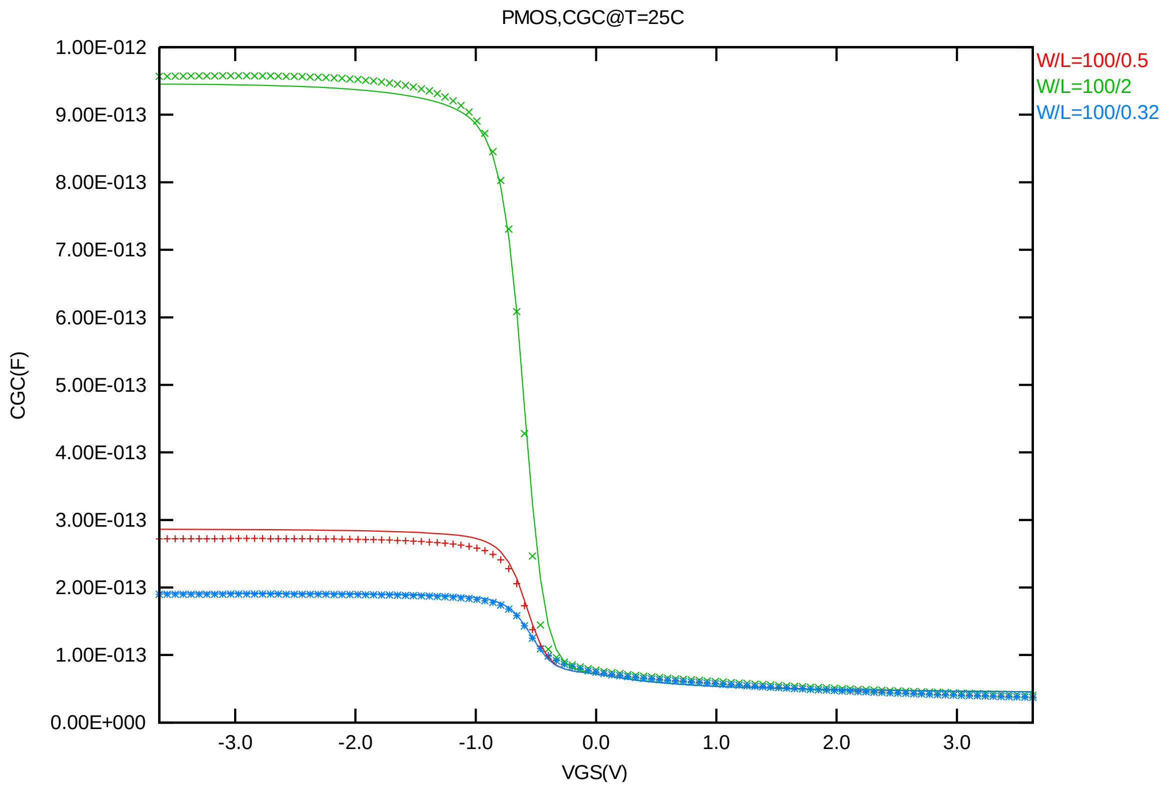


Figure 3-26 Measured and Simulated C_{GC} curves for large area measurement structure for PMOS

3.5 Parasitic source / drain junction capacitance model

3.5.1 Test devices

The following test structures were used for parameter extraction.

Table 3-9 Junction capacitor structures for PMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	5.0E-08	2.0E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.6E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.74E-08	1.833E-02	Poly finger structure

3.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-10 Junction capacitor measurement conditions for PMOS

V_{BIAS} [V]	VH: nwell & VL: p+ From 0 to 3.6 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

3.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 3-11 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	9.68E-04	CJSW	F/m	8.81E-11	CJSWG	F/m	2.29E-10
PB	V	6.80E-01	PBSW	V	4.04E-01	PBSWG	V	6.06E-01
MJ	--	3.03E-01	MJSW	--	3.21E-02	MJSWG	--	3.71E-01
TCJ	1/K	8.81E-04	TCJSW	1/K	9.14E-05	TCJSWG	1/K	1.42E-03
TPB	V/K	1.59E-03	TPBSW	V/K	3.65E-03	TPBSWG	V/K	2.58E-03

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

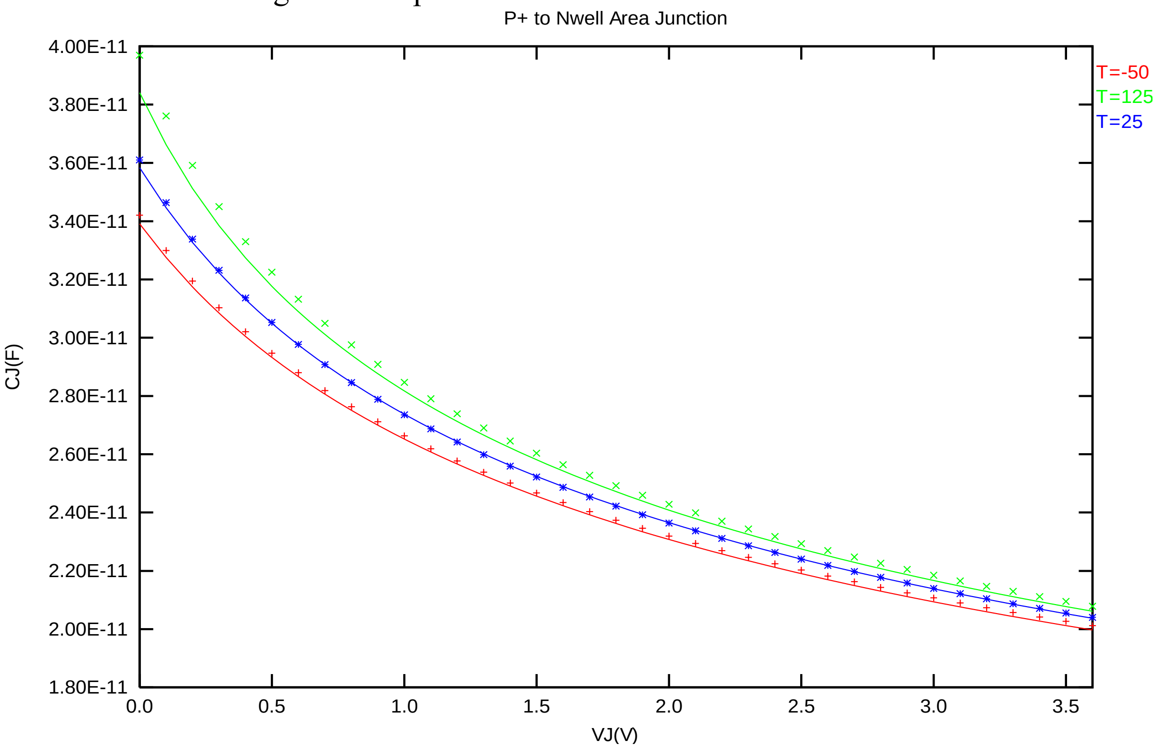


Figure 3-27 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

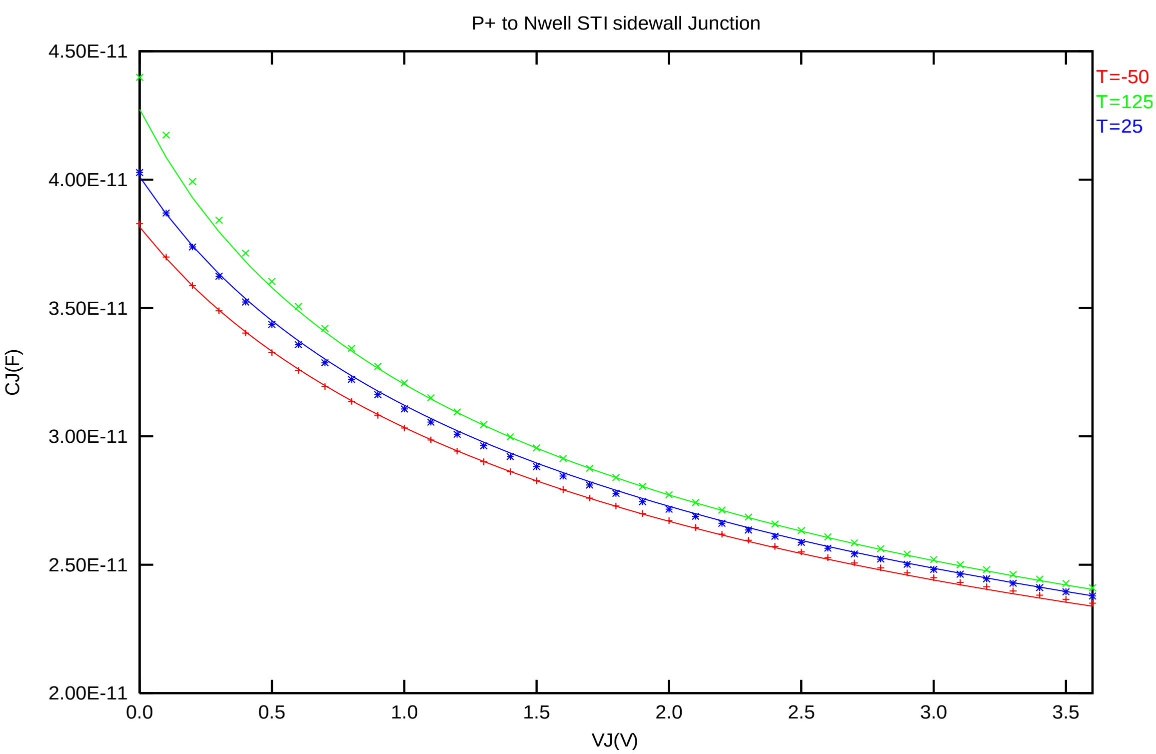


Figure 3-28 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

4 Worst case model

4.1 Parameter Variations

Table 4-1 Corner parameters

	n_33_hgl110e	TT	SS	FF	SNFP	FNSP
1	tox	7.49E-09	+2.25E-10	-2.25E-10	+0.00E+00	+0.00E+00
2	wwl	1.97E-21	-3.77E-21	+3.26E-21	-1.41E-21	+1.32E-21
3	xl	1.10E-08	+3.40E-08	-3.40E-08	+1.00E-08	-1.00E-08
4	xw	-4.07E-10	-1.22E-08	+7.99E-09	-5.43E-09	+4.42E-09
5	vth0	5.07E-01	+1.22E-02	-1.30E-02	+1.65E-02	-1.67E-02
6	lvth0	2.72E-08	+1.25E-08	-1.01E-08	+5.84E-09	-5.50E-09
7	pvth0	-3.38E-15	+5.27E-16	-2.35E-16	+2.28E-16	-1.59E-16
8	k3	-5.98E-01	+1.60E+00	-1.85E+00	+8.44E-01	-8.77E-01
9	u0	3.91E-02	-2.65E-03	+2.53E-03	-1.46E-03	+1.44E-03
10	vsat	1.30E+05	-1.61E+04	+1.16E+04	-1.03E+04	+9.48E+03
11	pvsat	-2.30E-10	-8.34E-12	+3.72E-11	-7.13E-13	+2.49E-24
12	ags	3.00E-01	+7.33E-03	-7.09E-03	+0.00E+00	-2.90E-03
13	b0	1.69E-06	-5.46E-07	+5.94E-07	-3.38E-07	+2.77E-07
14	rdsw	7.30E+02	+1.80E+02	-9.47E+01	+9.27E+01	-6.98E+01
15	cgso	4.60E-11	-10.0%	+10.0%	+0.0%	+0.0%
16	cgdo	4.60E-11	-10.0%	+10.0%	+0.0%	+0.0%
17	cgsl	1.80E-10	-10.0%	+10.0%	+0.0%	+0.0%
18	cgdl	1.80E-10	-10.0%	+10.0%	+0.0%	+0.0%
19	cj	1.03E-03	+10.0%	-10.0%	+10.0%	-10.0%
20	cjsw	1.41E-10	+10.0%	-10.0%	+10.0%	-10.0%

21	cjswg	1.97E-10	+10.0%	-10.0%	+10.0%	-10.0%

	p_33_hgl110e	**TT**	**SS**	**FF**	**SNFP**	**FNSP**
1	tox	7.68E-09	+2.25E-10	-2.25E-10	+0.00E+00	+0.00E+00
2	wwl	-7.64E-22	+6.31E-22	-4.59E-22	-2.68E-22	+3.81E-22
3	xl	1.10E-08	+3.88E-08	-2.72E-08	-6.61E-09	+6.50E-09
4	xw	4.21E-09	-1.95E-08	+1.41E-08	+7.15E-09	-8.28E-09
5	vth0	-6.18E-01	-1.60E-02	+1.64E-02	+2.20E-02	-2.18E-02
6	lvth0	9.80E-09	-1.90E-09	+1.94E-09	+2.36E-09	-2.60E-09
7	pvth0	1.35E-15	-3.00E-15	+3.15E-15	+1.55E-15	-1.46E-15
8	k3	-1.68E+00	+9.45E+00	-1.07E+01	-5.03E+00	+4.90E+00
9	u0	1.14E-02	-6.88E-04	+6.30E-04	+3.84E-04	-3.94E-04
10	vsat	9.17E+04	+7.05E+03	-4.84E+03	+5.34E+03	-7.17E+03
11	ags	1.42E-01	+2.50E-02	-2.59E-02	-9.66E-03	+9.68E-03
12	pags	-4.85E-14	+0.00E+00	-2.42E-14	-1.39E-15	-7.25E-16
13	b0	3.80E-08	-1.04E-08	+5.51E-09	+4.51E-09	-4.78E-09
14	rdsw	1.10E+03	+1.52E+02	-8.76E+01	-1.49E+02	+2.12E+02
15	cgso	1.06E-10	-10.0%	+10.0%	+0.0%	+0.0%
16	cgdo	1.06E-10	-10.0%	+10.0%	+0.0%	+0.0%
17	cgsl	2.02E-10	-10.0%	+10.0%	+0.0%	+0.0%
18	cgdl	2.02E-10	-10.0%	+10.0%	+0.0%	+0.0%
19	cj	9.68E-04	+10.0%	-10.0%	-10.0%	+10.0%
20	cjsw	8.81E-11	+10.0%	-10.0%	-10.0%	+10.0%
21	cjswg	2.29E-10	+10.0%	-10.0%	-10.0%	+10.0%

4.2 Wafer Data vs Corner graph

Threshold voltage in linear region V_{TLIN}

The threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

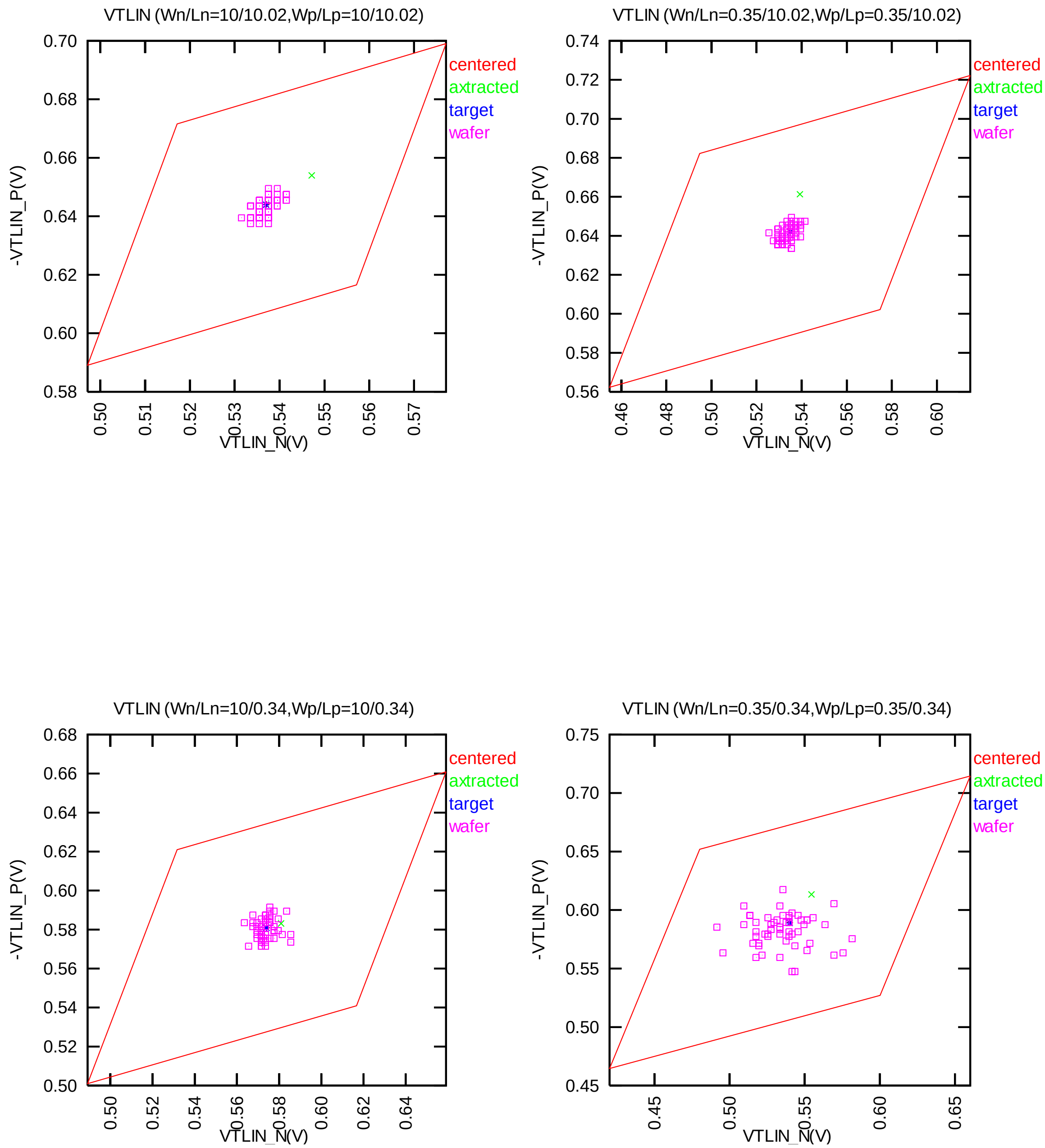


Figure 4-1 Corner graph of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

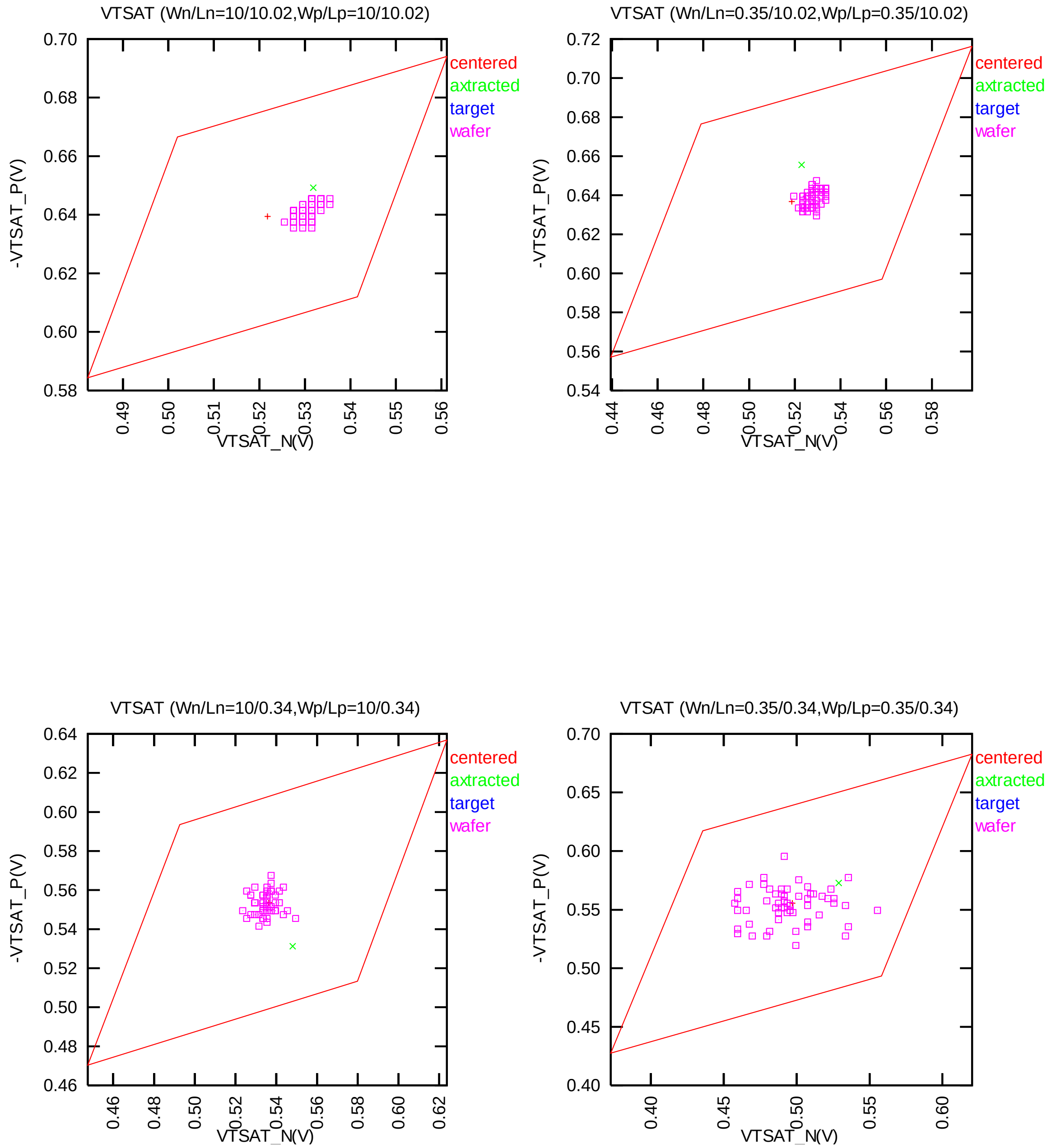


Figure 4-2 Corner graph of threshold voltage at high drain bias

Saturation Current I_{ON}

The dimensions are after 90% shrink and 12nm sizing channel length back.

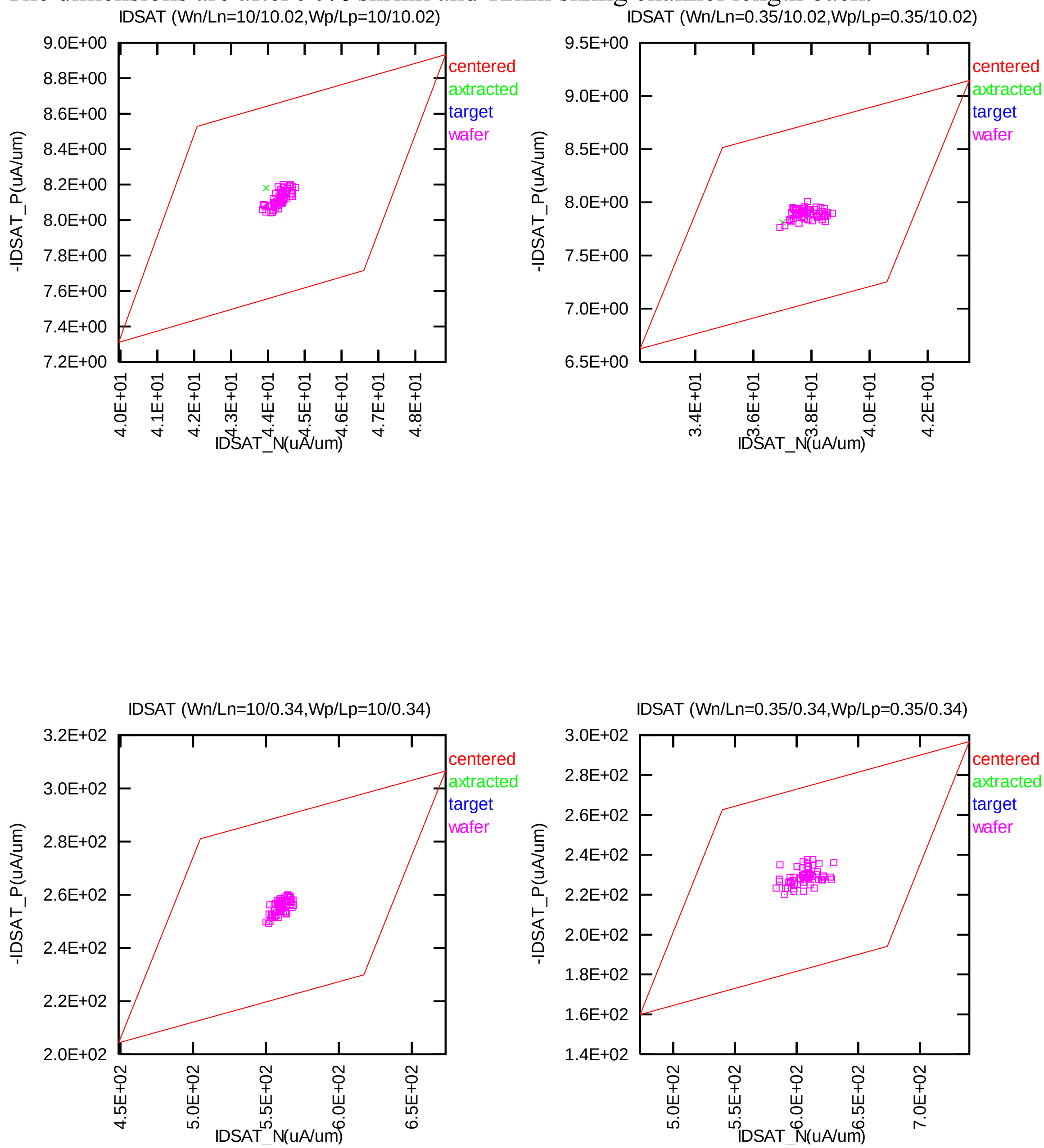


Figure 4-3 Corner graph of on-current

4.3 Device performance reference tables

Note: The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The real silicon size of W/L=9µm/0.324µm would be the same as the model simulation results in size of W/L=9µm/0.336µm as below table (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 4-2 NMOS ID current at VGS=VDS= Vcc/2 and VGS=VDS =Vcc.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (µA)	VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	113.79	625.91	83.14	440.88	61.91	307.19
SS	9/9	96.39	563.55	70.81	396.54	53.05	276.05
FF	9/9	132.96	687.05	96.70	485.11	71.61	338.68
SNFP	9/9	105.89	592.18	77.56	417.92	57.90	291.71
FNSP	9/9	121.95	658.49	88.90	462.98	66.03	322.03
TT	9/0.324	1869.60	5593.10	1656.60	5072.00	1372.10	4288.80
SS	9/0.324	1399.70	4518.30	1227.00	4055.20	1011.70	3395.80
FF	9/0.324	2383.10	6630.90	2146.00	6089.10	1798.80	5218.20
SNFP	9/0.324	1636.20	5050.50	1446.20	4564.00	1197.90	3848.90
FNSP	9/0.324	2113.30	6128.70	1879.50	5580.30	1559.50	4735.10
TT	0.35/9	3.63	20.06	2.76	14.66	2.14	10.61
SS	0.35/9	2.78	17.00	2.14	12.45	1.68	9.04
FF	0.35/9	4.61	23.10	3.47	16.86	2.67	12.19
SNFP	0.35/9	3.23	18.48	2.47	13.55	1.92	9.85
FNSP	0.35/9	4.06	21.64	3.08	15.76	2.38	11.37
TT	0.35/0.324	79.01	233.73	70.34	213.86	58.38	181.22
SS	0.35/0.324	56.20	184.80	49.41	166.73	40.89	139.50
FF	0.35/0.324	104.34	280.55	94.72	261.00	79.50	224.83
SNFP	0.35/0.324	67.67	209.07	60.03	190.26	49.81	160.45
FNSP	0.35/0.324	90.97	258.02	81.40	237.49	67.69	202.33

Table 4-3 NMOS ID current density at VGS=VDS= Vcc/2 and VGS=VDS =Vcc.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (µA/um)	VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	12.64	69.55	9.24	48.99	6.88	34.13
SS	9/9	10.71	62.62	7.87	44.06	5.89	30.67
FF	9/9	14.77	76.34	10.74	53.90	7.96	37.63
SNFP	9/9	11.77	65.80	8.62	46.44	6.43	32.41
FNSP	9/9	13.55	73.17	9.88	51.44	7.34	35.78
TT	9/0.324	207.73	621.46	184.07	563.56	152.46	476.53
SS	9/0.324	155.52	502.03	136.33	450.58	112.41	377.31
FF	9/0.324	264.79	736.77	238.44	676.57	199.87	579.80
SNFP	9/0.324	181.80	561.17	160.69	507.11	133.10	427.66
FNSP	9/0.324	234.81	680.97	208.83	620.03	173.28	526.12
TT	0.35/9	10.38	57.31	7.89	41.87	6.13	30.33
SS	0.35/9	7.95	48.57	6.11	35.57	4.80	25.84
FF	0.35/9	13.17	65.99	9.93	48.18	7.63	34.83
SNFP	0.35/9	9.22	52.81	7.05	38.72	5.50	28.14
FNSP	0.35/9	11.61	61.83	8.79	45.03	6.79	32.50
TT	0.35/0.324	225.75	667.80	200.98	611.03	166.80	517.77
SS	0.35/0.324	160.59	528.00	141.19	476.37	116.83	398.57
FF	0.35/0.324	298.11	801.57	270.64	745.71	227.15	642.37
SNFP	0.35/0.324	193.34	597.34	171.52	543.60	142.33	458.43
FNSP	0.35/0.324	259.90	737.20	232.57	678.54	193.40	578.09

Table 4-4 NMOS VTLIN and VTSAT.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	0.609	0.596	0.537	0.522	0.449	0.428
SS	9/9	0.647	0.635	0.578	0.562	0.491	0.469
FF	9/9	0.570	0.558	0.497	0.482	0.407	0.387
SNFP	9/9	0.628	0.616	0.558	0.542	0.470	0.448
FNSP	9/9	0.589	0.577	0.517	0.502	0.428	0.407
TT	9/0.324	0.640	0.606	0.576	0.537	0.500	0.453
SS	9/0.324	0.724	0.693	0.662	0.626	0.588	0.543
FF	9/0.324	0.555	0.517	0.490	0.448	0.412	0.361
SNFP	9/0.324	0.682	0.649	0.619	0.581	0.544	0.497
FNSP	9/0.324	0.598	0.562	0.533	0.493	0.456	0.408
TT	0.35/9	0.603	0.591	0.535	0.519	0.449	0.427
SS	0.35/9	0.681	0.668	0.615	0.598	0.531	0.508
FF	0.35/9	0.525	0.513	0.455	0.439	0.366	0.346
SNFP	0.35/9	0.642	0.630	0.575	0.558	0.490	0.467
FNSP	0.35/9	0.564	0.552	0.495	0.479	0.407	0.386
TT	0.35/0.324	0.606	0.569	0.542	0.498	0.462	0.408
SS	0.35/0.324	0.725	0.691	0.662	0.622	0.586	0.535
FF	0.35/0.324	0.487	0.445	0.421	0.372	0.338	0.280
SNFP	0.35/0.324	0.666	0.629	0.602	0.559	0.524	0.471
FNSP	0.35/0.324	0.547	0.508	0.481	0.436	0.400	0.345

Table 4-5 NMOS ID current at VGS=0 and VDS=0.1V or VDS=Vcc(off current).

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	2.00E-14	6.57E-13	2.46E-13	9.45E-13	5.35E-11	6.81E-11
SS	9/9	1.99E-14	6.57E-13	1.00E-13	7.64E-13	2.41E-11	3.10E-11
FF	9/9	2.05E-14	6.57E-13	6.73E-13	1.49E-12	1.21E-10	1.53E-10
SNFP	9/9	2.00E-14	6.54E-13	1.46E-13	8.21E-13	3.45E-11	4.41E-11
FNSP	9/9	2.03E-14	6.57E-13	4.26E-13	1.18E-12	8.32E-11	1.05E-10
TT	9/0.324	2.09E-14	6.68E-13	1.76E-12	4.88E-12	4.82E-10	1.03E-09
SS	9/0.324	1.99E-14	6.57E-13	2.01E-13	1.05E-12	8.53E-11	1.67E-10
FF	9/0.324	3.90E-14	7.25E-13	1.76E-11	4.99E-11	2.80E-09	6.66E-09
SNFP	9/0.324	2.01E-14	6.54E-13	5.36E-13	1.86E-12	1.92E-10	3.97E-10
FNSP	9/0.324	2.44E-14	6.79E-13	5.92E-12	1.56E-11	1.21E-09	2.67E-09
TT	0.35/9	2.00E-14	6.60E-13	3.12E-14	6.74E-13	2.31E-12	3.55E-12
SS	0.35/9	2.00E-14	6.60E-13	2.12E-14	6.61E-13	4.58E-13	1.21E-12
FF	0.35/9	2.02E-14	6.61E-13	1.28E-13	7.97E-13	1.26E-11	1.65E-11
SNFP	0.35/9	2.00E-14	6.60E-13	2.35E-14	6.65E-13	9.73E-13	1.86E-12
FNSP	0.35/9	2.00E-14	6.60E-13	5.67E-14	7.06E-13	5.56E-12	7.64E-12
TT	0.35/0.324	2.01E-14	6.60E-13	1.87E-13	1.11E-12	3.99E-11	9.64E-11
SS	0.35/0.324	2.00E-14	6.60E-13	2.62E-14	6.75E-13	3.26E-12	7.63E-12
FF	0.35/0.324	2.89E-14	6.95E-13	4.74E-12	1.57E-11	5.01E-10	1.37E-09
SNFP	0.35/0.324	2.00E-14	6.60E-13	4.96E-14	7.36E-13	1.08E-11	2.56E-11
FNSP	0.35/0.324	2.11E-14	6.64E-13	9.63E-13	3.34E-12	1.47E-10	3.67E-10

Table 4-6 PMOS ID current at VGS=VDS= -Vcc/2 and VGS=VDS = -Vcc.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA)	-VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	15.99	96.68	14.66	81.35	13.80	68.81
SS	9/9	13.05	86.89	12.09	73.12	11.51	61.89
FF	9/9	19.31	106.43	17.54	89.57	16.33	75.72
SNFP	9/9	17.47	101.76	15.94	85.46	14.92	72.12
FNSP	9/9	14.58	91.61	13.44	77.24	12.72	65.48
TT	9/0.324	588.71	2636.10	536.00	2329.30	494.64	2032.60
SS	9/0.324	417.70	2125.10	382.53	1859.00	356.41	1610.10
FF	9/0.324	787.74	3136.80	715.26	2800.30	655.70	2463.60
SNFP	9/0.324	672.50	2910.40	608.86	2564.50	558.47	2230.50
FNSP	9/0.324	511.17	2359.70	468.66	2094.30	435.57	1836.10
TT	0.35/9	0.58	3.72	0.52	3.07	0.47	2.53
SS	0.35/9	0.43	3.13	0.39	2.57	0.36	2.12
FF	0.35/9	0.76	4.32	0.67	3.56	0.61	2.93
SNFP	0.35/9	0.66	4.03	0.58	3.31	0.53	2.73
FNSP	0.35/9	0.50	3.42	0.45	2.82	0.42	2.33
TT	0.35/0.324	19.46	95.54	17.00	81.09	15.06	67.66
SS	0.35/0.324	11.64	67.47	10.23	56.61	9.16	46.82
FF	0.35/0.324	29.50	123.35	25.64	105.65	22.51	88.68
SNFP	0.35/0.324	23.68	110.24	20.58	93.32	18.14	77.67
FNSP	0.35/0.324	15.67	80.85	13.75	68.86	12.25	57.63

Table 4-7 PMOS ID current density at VGS=VDS= -Vcc/2 and VGS=VDS = -Vcc.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA/um)	-VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	1.78	10.74	1.63	9.04	1.53	7.64
SS	9/9	1.45	9.65	1.34	8.12	1.28	6.88
FF	9/9	2.15	11.83	1.95	9.95	1.81	8.41
SNFP	9/9	1.94	11.31	1.77	9.50	1.66	8.01
FNSP	9/9	1.62	10.18	1.49	8.58	1.41	7.28
TT	9/0.324	65.41	292.90	59.56	258.81	54.96	225.84
SS	9/0.324	46.41	236.12	42.50	206.56	39.60	178.90
FF	9/0.324	87.53	348.53	79.47	311.14	72.86	273.73
SNFP	9/0.324	74.72	323.38	67.65	284.94	62.05	247.83
FNSP	9/0.324	56.80	262.19	52.07	232.70	48.40	204.01
TT	0.35/9	1.65	10.64	1.47	8.76	1.35	7.22
SS	0.35/9	1.22	8.94	1.10	7.35	1.02	6.06
FF	0.35/9	2.16	12.34	1.91	10.16	1.73	8.38
SNFP	0.35/9	1.88	11.52	1.67	9.46	1.52	7.79
FNSP	0.35/9	1.44	9.77	1.29	8.05	1.19	6.66
TT	0.35/0.324	55.61	272.98	48.56	231.68	43.04	193.30
SS	0.35/0.324	33.26	192.77	29.23	161.75	26.18	133.77
FF	0.35/0.324	84.29	352.43	73.25	301.86	64.32	253.38
SNFP	0.35/0.324	67.65	314.97	58.80	266.64	51.84	221.92
FNSP	0.35/0.324	44.76	230.99	39.28	196.75	35.00	164.67

Table 4-8 PMOS VTLIN and VTSAT.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-VTH (V)	-VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	0.731	0.727	0.644	0.639	0.534	0.527
SS	9/9	0.785	0.781	0.700	0.695	0.591	0.584
FF	9/9	0.676	0.672	0.588	0.583	0.477	0.470
SNFP	9/9	0.703	0.700	0.616	0.611	0.505	0.499
FNSP	9/9	0.758	0.754	0.672	0.667	0.562	0.555
TT	9/0.324	0.665	0.639	0.581	0.553	0.474	0.443
SS	9/0.324	0.745	0.723	0.663	0.638	0.558	0.530
FF	9/0.324	0.585	0.555	0.500	0.468	0.390	0.355
SNFP	9/0.324	0.625	0.599	0.541	0.512	0.432	0.401
FNSP	9/0.324	0.705	0.679	0.622	0.594	0.515	0.484
TT	0.35/9	0.721	0.717	0.642	0.637	0.545	0.537
SS	0.35/9	0.799	0.794	0.722	0.716	0.628	0.619
FF	0.35/9	0.643	0.638	0.562	0.557	0.462	0.455
SNFP	0.35/9	0.682	0.677	0.602	0.597	0.503	0.496
FNSP	0.35/9	0.760	0.755	0.682	0.676	0.586	0.578
TT	0.35/0.324	0.667	0.635	0.590	0.556	0.495	0.457
SS	0.35/0.324	0.789	0.760	0.716	0.684	0.627	0.591
FF	0.35/0.324	0.544	0.508	0.464	0.426	0.365	0.324
SNFP	0.35/0.324	0.606	0.573	0.528	0.493	0.431	0.393
FNSP	0.35/0.324	0.729	0.695	0.653	0.618	0.561	0.521

Table 4-9 PMOS ID current at VGS=0 and VDS= -0.1V or VDS= -Vcc(off current).

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IOFF (A)	-VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	2.00E-14	6.57E-13	3.25E-14	6.71E-13	5.15E-12	6.13E-12
SS	9/9	1.99E-14	6.57E-13	2.38E-14	6.61E-13	2.06E-12	2.84E-12
FF	9/9	1.99E-14	6.57E-13	6.46E-14	7.14E-13	1.35E-11	1.51E-11
SNFP	9/9	1.99E-14	6.57E-13	4.54E-14	6.82E-13	8.68E-12	9.88E-12
FNSP	9/9	1.99E-14	6.57E-13	2.63E-14	6.64E-13	3.09E-12	3.94E-12
TT	9/0.324	2.19E-14	6.64E-13	2.47E-12	4.94E-12	5.38E-10	8.68E-10
SS	9/0.324	2.00E-14	6.57E-13	3.73E-13	1.22E-12	1.21E-10	1.81E-10
FF	9/0.324	4.52E-14	7.18E-13	1.73E-11	3.38E-11	2.39E-09	4.17E-09
SNFP	9/0.324	2.73E-14	6.75E-13	6.80E-12	1.26E-11	1.16E-09	1.89E-09
FNSP	9/0.324	2.05E-14	6.57E-13	9.03E-13	2.20E-12	2.49E-10	4.01E-10
TT	0.35/9	2.00E-14	6.60E-13	2.05E-14	6.61E-13	1.83E-13	8.34E-13
SS	0.35/9	2.00E-14	6.60E-13	2.01E-14	6.60E-13	6.37E-14	7.06E-13
FF	0.35/9	2.00E-14	6.60E-13	2.36E-14	6.64E-13	7.15E-13	1.40E-12
SNFP	0.35/9	2.00E-14	6.60E-13	2.15E-14	6.62E-13	3.68E-13	1.03E-12
FNSP	0.35/9	2.00E-14	6.60E-13	2.02E-14	6.60E-13	9.86E-14	7.44E-13
TT	0.35/0.324	2.01E-14	6.60E-13	8.65E-14	7.88E-13	1.25E-11	2.23E-11
SS	0.35/0.324	2.00E-14	6.60E-13	2.33E-14	6.66E-13	1.22E-12	2.55E-12
FF	0.35/0.324	2.35E-14	6.69E-13	1.45E-12	3.72E-12	1.32E-10	2.53E-10
SNFP	0.35/0.324	2.05E-14	6.61E-13	3.45E-13	1.29E-12	4.20E-11	7.41E-11
FNSP	0.35/0.324	2.00E-14	6.60E-13	3.37E-14	6.86E-13	3.69E-12	7.05E-12

5 Shallow Trench Isolation (STI) or LOD Stress Effect

5.1 Simulation results for NMOS at SA=0.36µm (SAREF=2.2µm)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm). The dimension is after 90% shrink and 12nm sizing channel length back.
The simulation results for the NMOS characterization are summarized in the table below.

Table 5-1 Simulation results for NMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W/L= 10 μ m / 10.02 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.538
I_{ON}	V_{DS} = 3.3 V	μA/ μm	43.83
Wide and short channel device (W/L= 10 μ m / 0.34 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.593
V_{TSAT}	V_{DS} = 3.3 V	V	0.555
Body Effect V_T shift	V_{DS} = 3.3 V V_{BS}= 0 ~ -3.3 V	V	0.581
DIBL	V_{DS}= 0.1~3.3 V	V	0.038
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	538.04
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	3.3E-13
Narrow and long channel device (W/L= 0.35 μ m / 10.02 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.536
V_{TSAT}	V_{DS}= 3.3 V	V	0.520
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	37.29
Narrow and short channel device (W/L= 0.35 μm / 0.34 μm)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.559
V_{TSAT}	V_{DS}= 3.3 V	V	0.516
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	582.23
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	2.6E-12

5.1.1 Model verification plots for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D=300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back.).

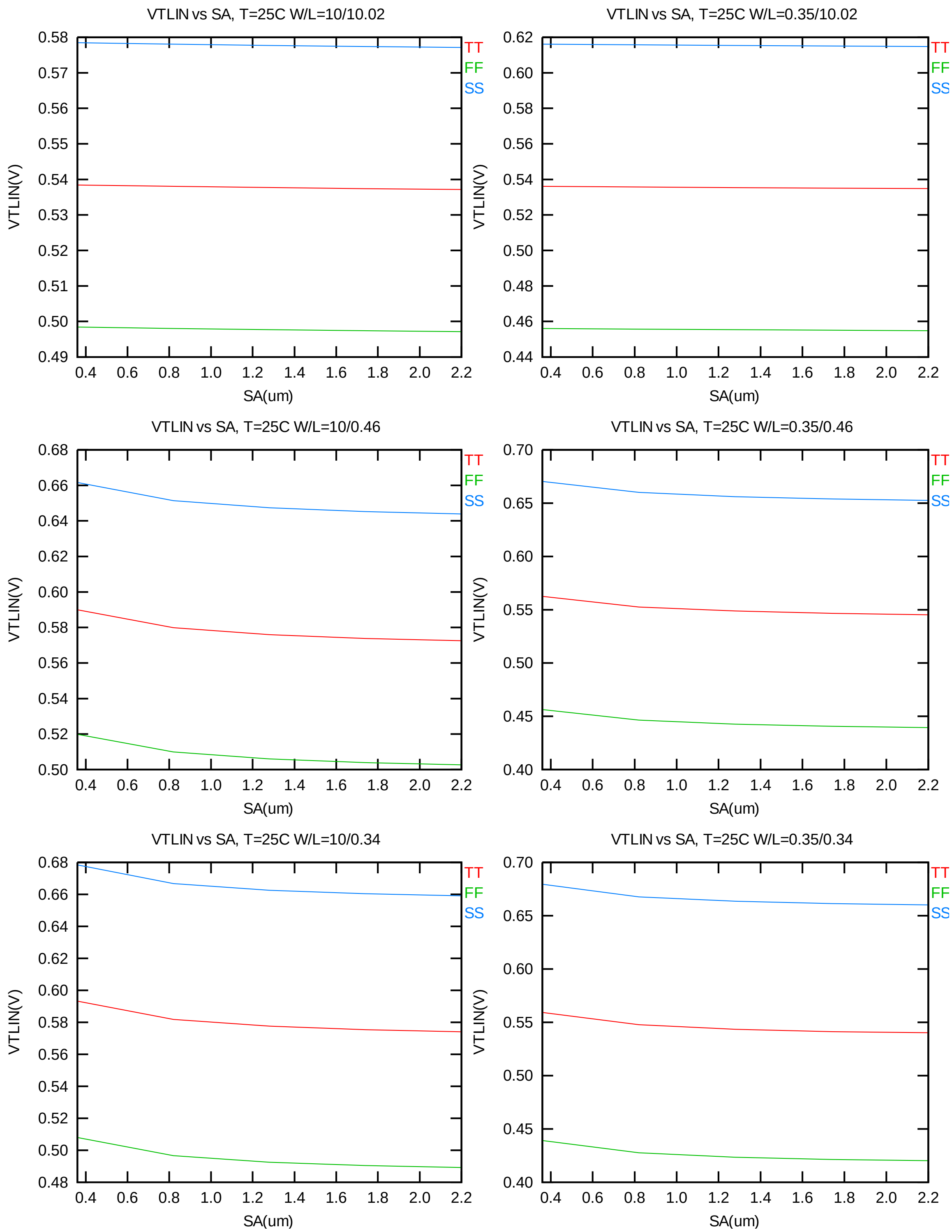


Figure 5-1 V_{TLIN} vs. SA for low drain bias (V_{DS} = 0.1 V) for NMOS

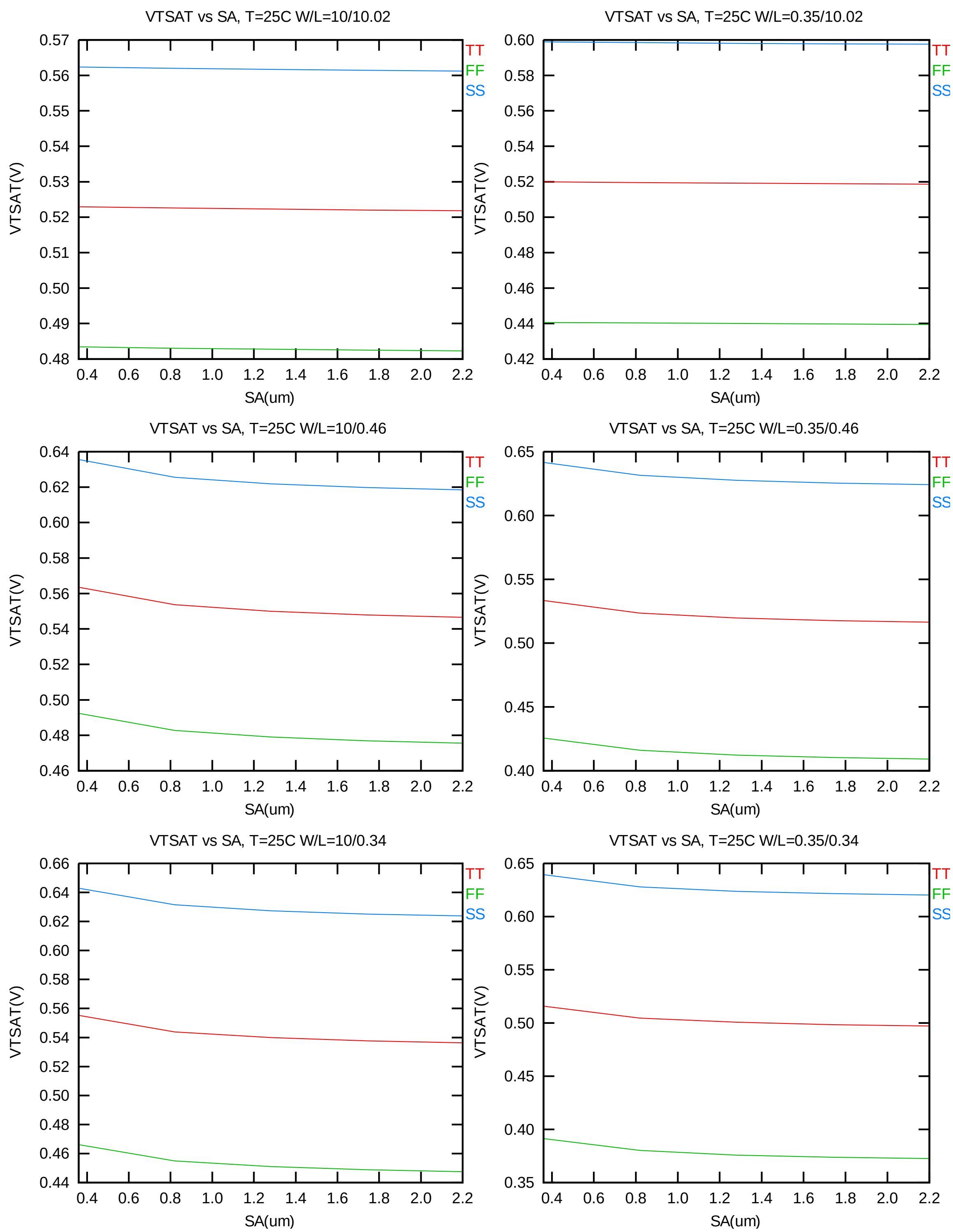


Figure 5-2 V_{TSAT} vs. SA for high drain bias (V_{DS} = 3.3 V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}=3.3 V for V_{BS}=0 at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back.).

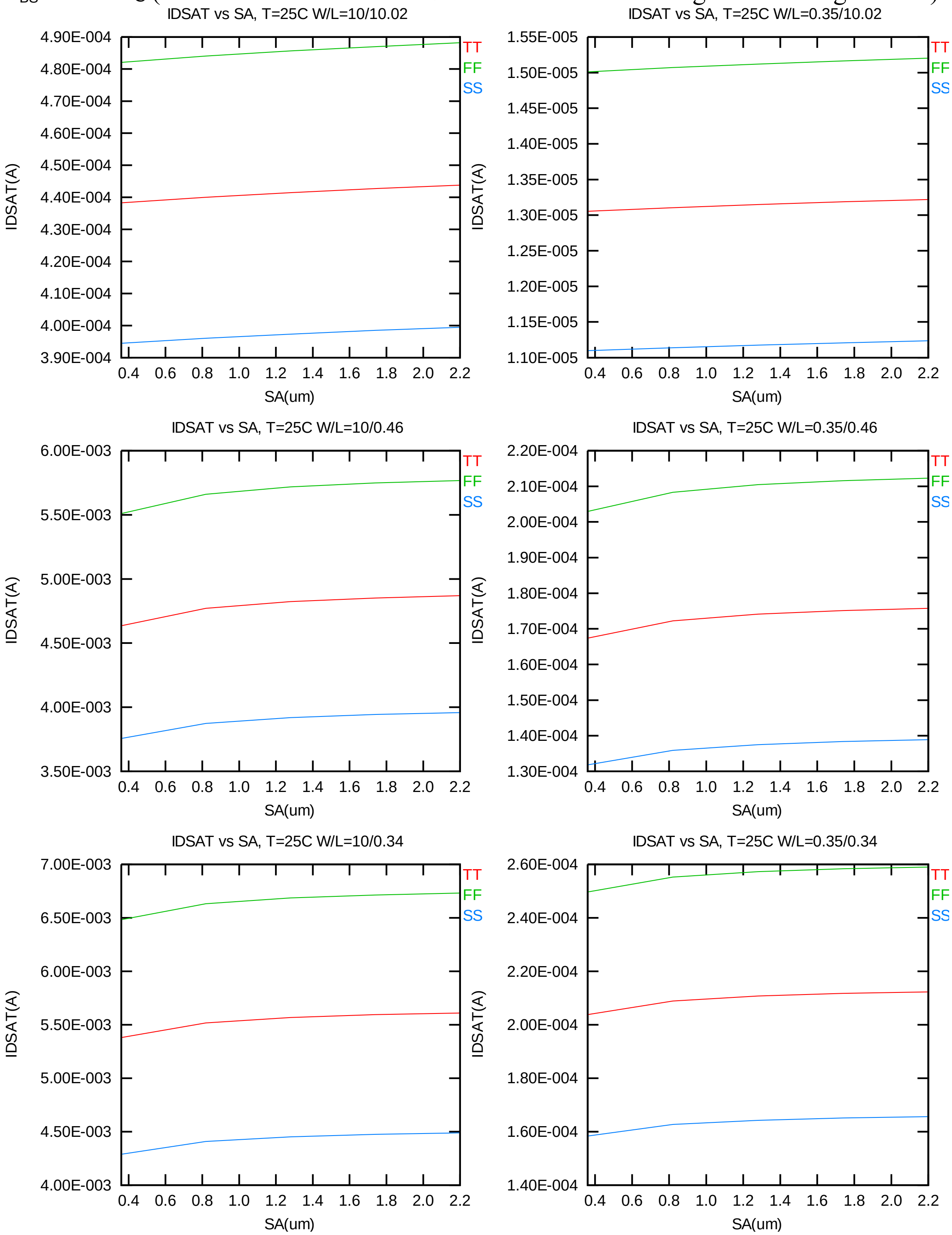


Figure 5-3 Saturation current I_{ON} vs. SA for NMOS

5.2 Simulation results for PMOS at SA=0.36µm (SAREF=2.2µm)

The simulation results for the PMOS characterization are summarized in the table below.

Table 5-2 Simulation results for PMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W/L= 10 µ m / 10.02 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.644
I_{ON}	V_{DS} = -3.3 V	µA/ µm	-8.20
Wide and short channel device (W/L= 10 µ m / 0.34 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.581
V_{TSAT}	V_{DS} = -3.3 V	V	-0.553
Body Effect V_T shift	V_{DS} = -3.3 V V_{BS}= 0 ~ 3.3 V	V	-0.812
DIBL	V_{DS}= -0.1~-3.3 V	V	-0.028
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-269.03
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/µm	-5.0E-13
Narrow and long channel device (W/L= 0.35 µ m / 10.02 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.642
V_{TSAT}	V_{DS}= -3.3 V	V	-0.636
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-7.96
Narrow and short channel device (W/L= 0.35 µm / 0.34 µm)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.589
V_{TSAT}	V_{DS}= -3.3 V	V	-0.555
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-241.88
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/µm	-2.2E-12

5.2.1 Model verification plots for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D=-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back).

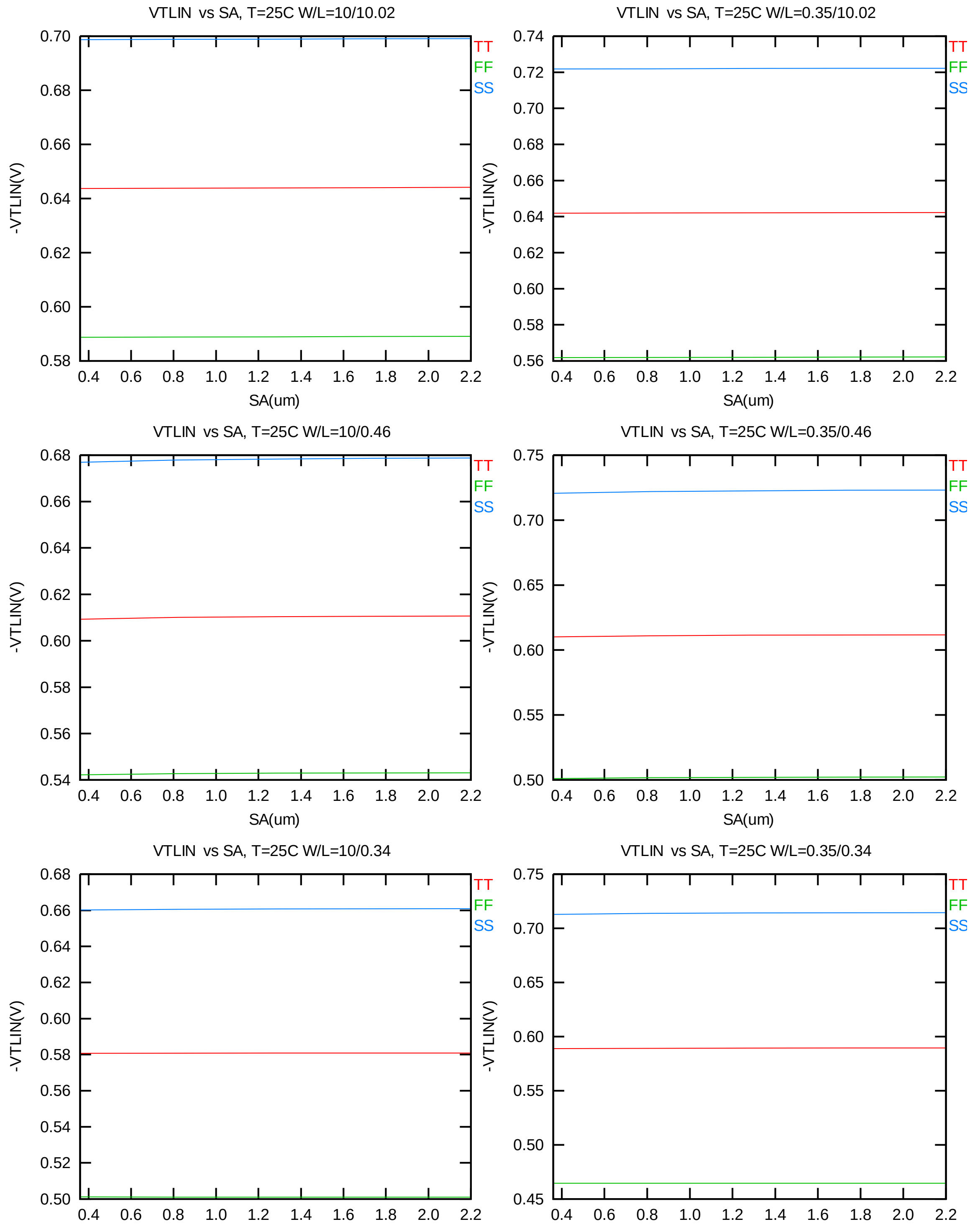


Figure 5-4 V_{TLIN} vs. SA for low drain bias (V_{DS}= -0.1 V) for PMOS

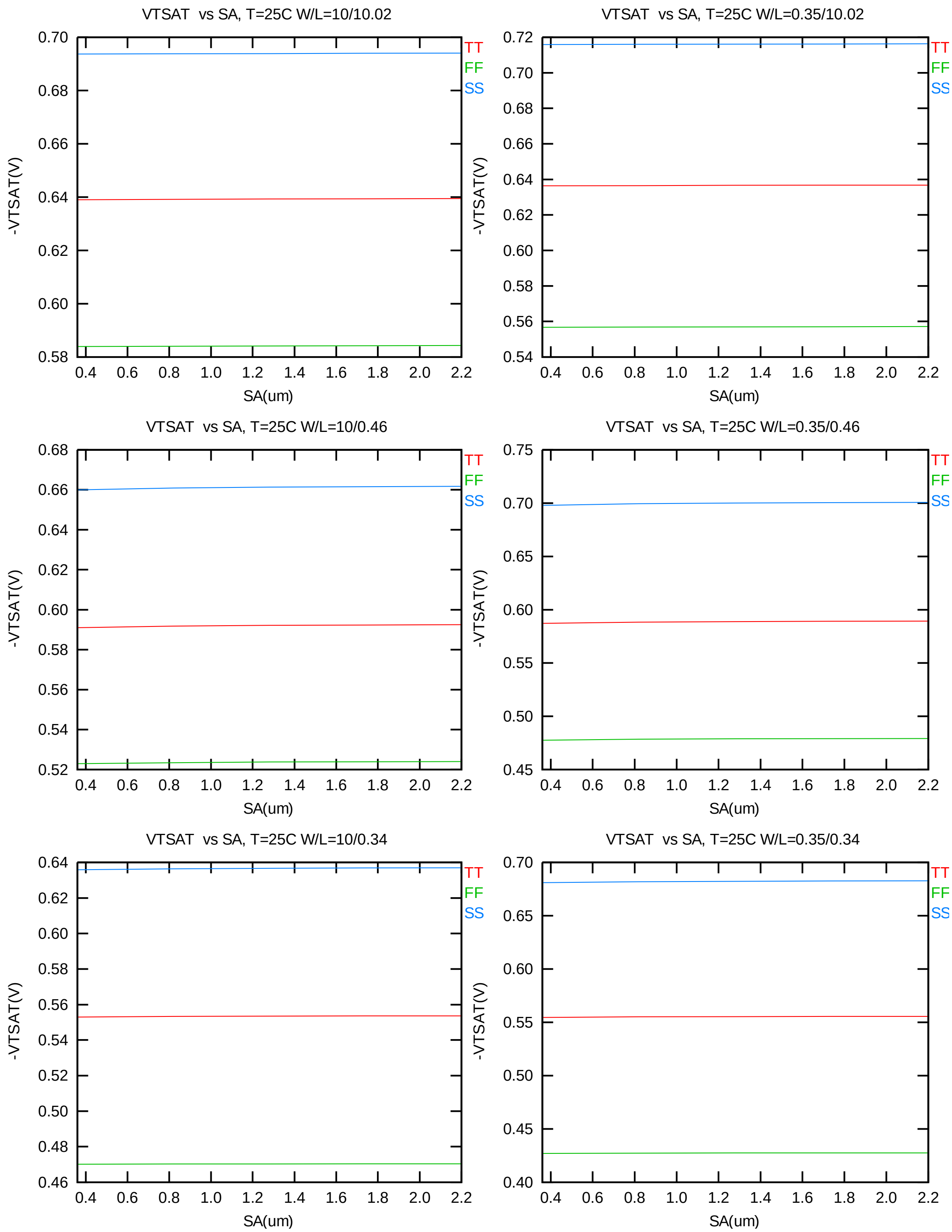


Figure 5-5 V_{TSAT} vs. SA for high drain bias (V_{DS}= -3.3 V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}= -3.3 V for V_{BS}=0 at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back).

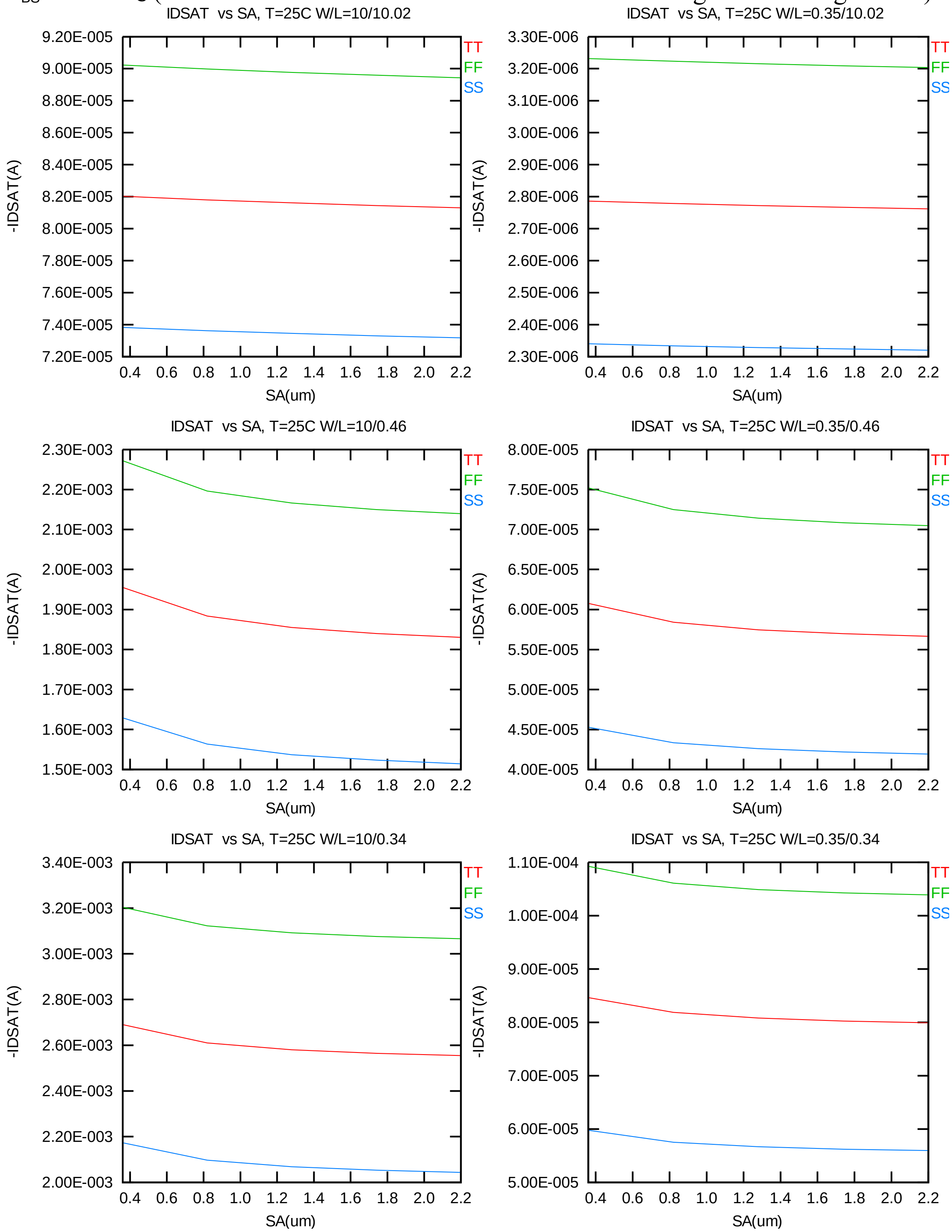


Figure 5-6 Saturation current I_{ON} vs. SA for PMOS

5.3 V_T and I_{ON} variation v. SA for both N/PMOS

Delta V_T is defined as V_T (SA)-V_T (0.36μm) for NMOS and -[V_T (SA)-V_T (0.36μm)] for PMOS. Delta I_{ON} is defined as [I_{ON} (SA)-I_{ON} (0.36μm)]/I_{ON} (0.36μm)*100%.

Delta V_T and I_{ON} vs. SA

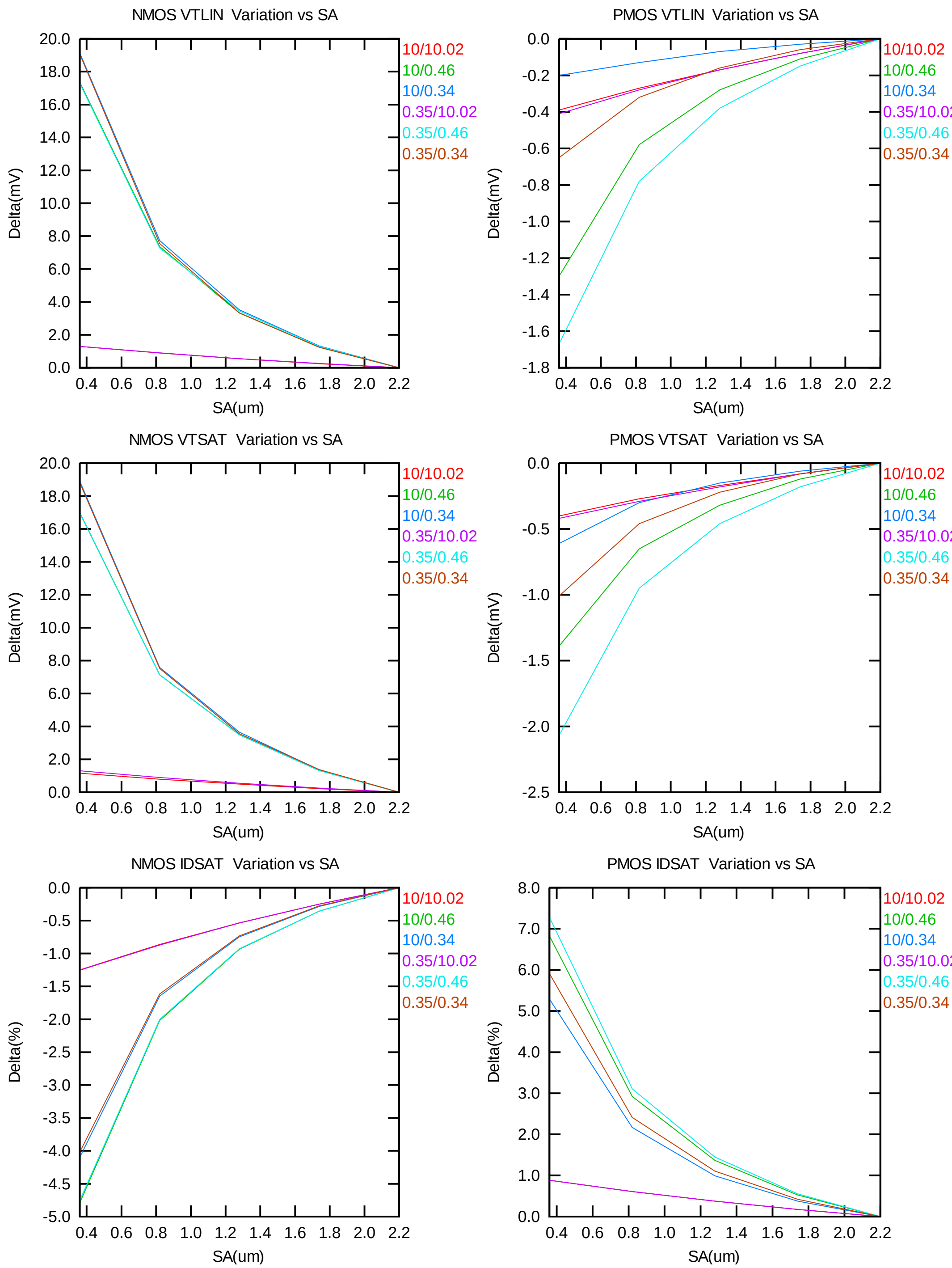


Figure 5-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

6 Dynamic performance

6.1 Ring Oscillator Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configuration is adopted.

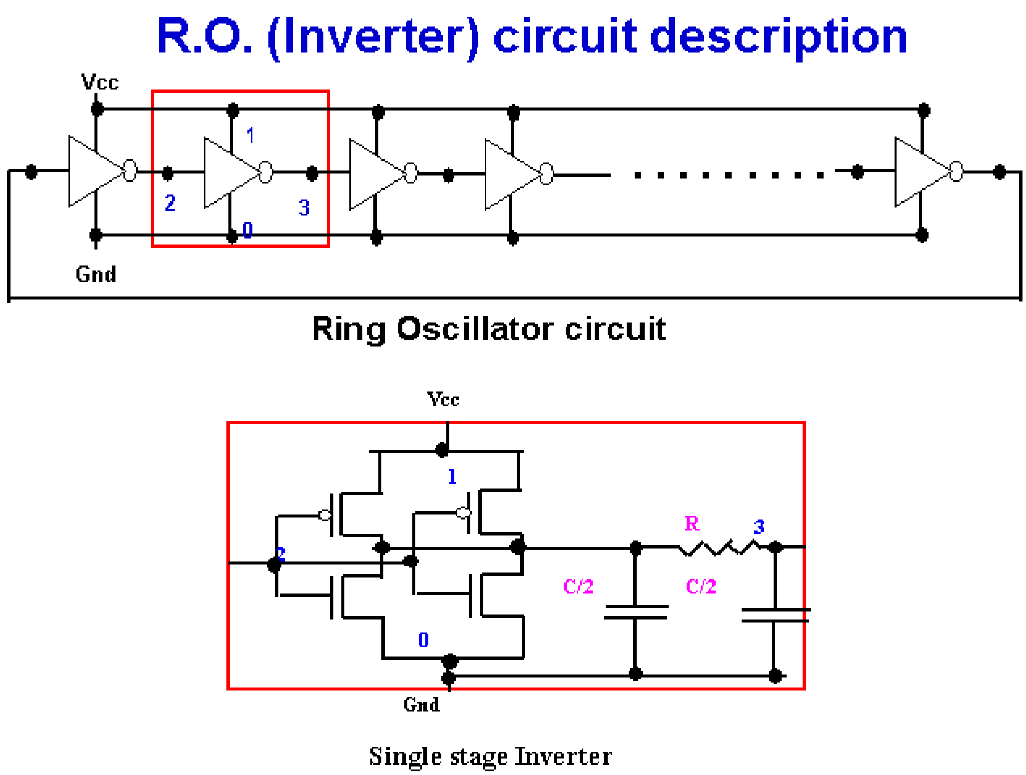


Figure 6-1 Ring Oscillator Netlist Diagram

6.2 Dynamic Performance for SA=0.36μm

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 6-1 Ring oscillator (inverter) structure description for SA=0.36μm

R.O. (Inverter) Structure description		
Contact size	0.16 um X 0.16 um	
Polysilicon gate to contact spacing	0.14 um	
Contact to diffusion edge spacing	0.06 um	
Without RC loading	R = 0ohm	C = 0 F
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	2.5 um X 2	
Channel length for n-ch MOSFET	0.34 um(with 12nm sizing back)	
Channel width for p-ch MOSFET	3.75 um X 2	
Channel Length for p-ch MOSFET	0.34 um(with 12nm sizing back)	

The simulation results at different temperatures and using different worst-case models are listed as below.

Table 6-2 Ring Oscillator simulation results (ps/stage)

Inverter, R=0, C=0f, Fanout=1, Vcc=3.3V			
Temp(C)	**TT**	**FF**	**SS**
-50	35.194	28.503	45.674
0	39.035	31.320	51.040
25	41.114	32.859	53.918
85	46.539	36.925	61.354
125	50.472	39.909	66.677

Inverter, R=0, C=0f, Fanout=1, T=25C			
Vcc(V)	**TT**	**FF**	**SS**
2.40	53.774	41.461	72.981
2.70	48.099	37.649	64.376
3.00	44.091	34.909	58.347
3.30	41.114	32.859	53.918
3.63	38.610	31.119	50.227

The EDR target of Electrical parameters for ring oscillator is listed below. The condition of this EDR target is base on Vcc =3.3V and with extracted RC loading.

Table 6-3 Electrical parameters for ring oscillator of EDR target with RC loading

Parameter	Condition	Unit	Target	Centered Value
T_{pd}	V_{cc}=3.3V	ps/stage	47.4	46.8

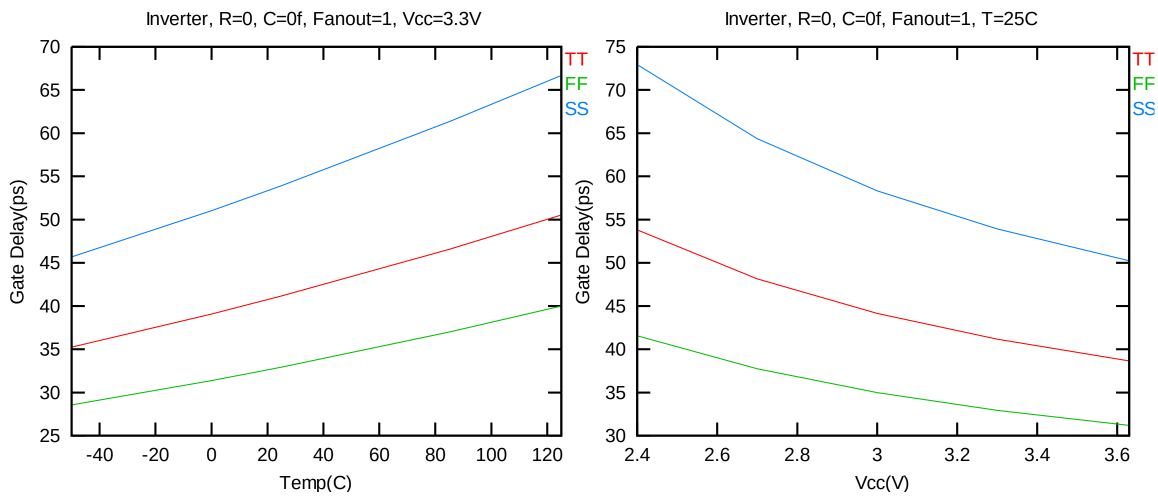


Figure 6-2 The Ring Oscillator (Inverter) Simulation

7 Noise Characteristic

7.1 Verification plots

NMOS, W=10µm L=0.34µm

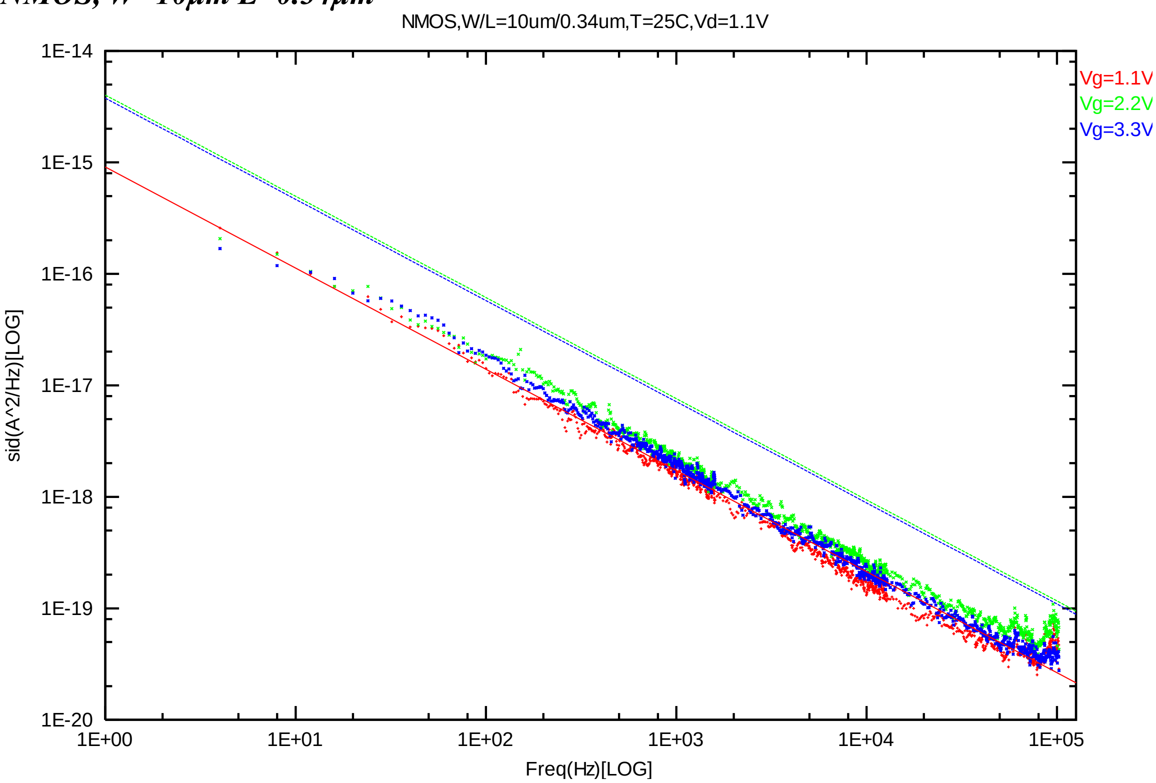


Figure 7-1 NMOS W/L=10µm/0.34µm @ Vd=1.1V

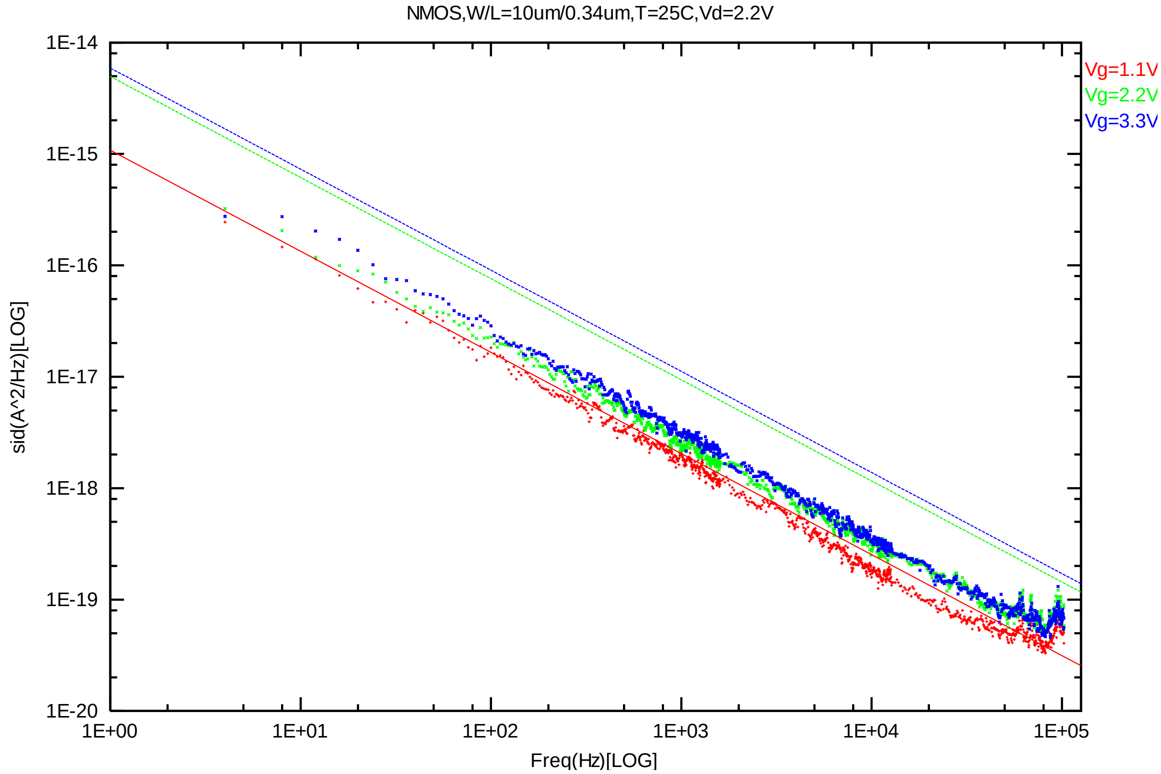


Figure 7-2 NMOS W/L=10µm/0.34µm @ Vd=2.2V

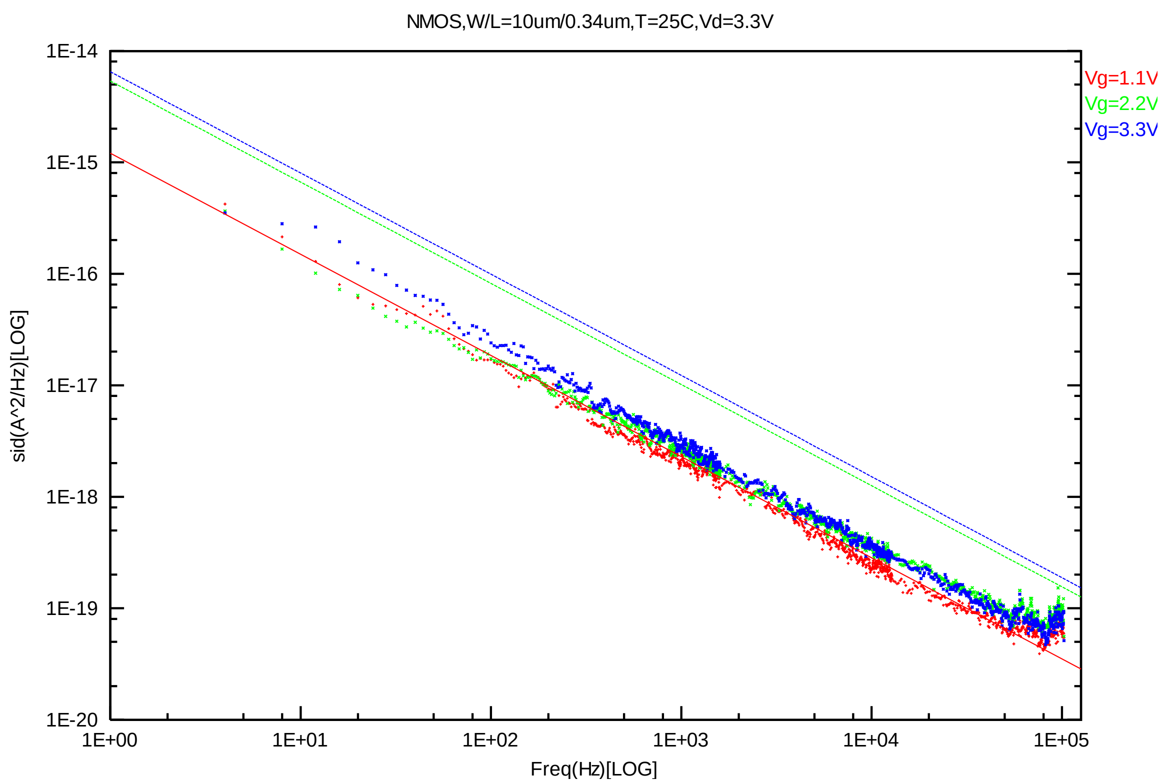


Figure 7-3 NMOS W/L=10µm/0.34µm @ Vd=3.3V

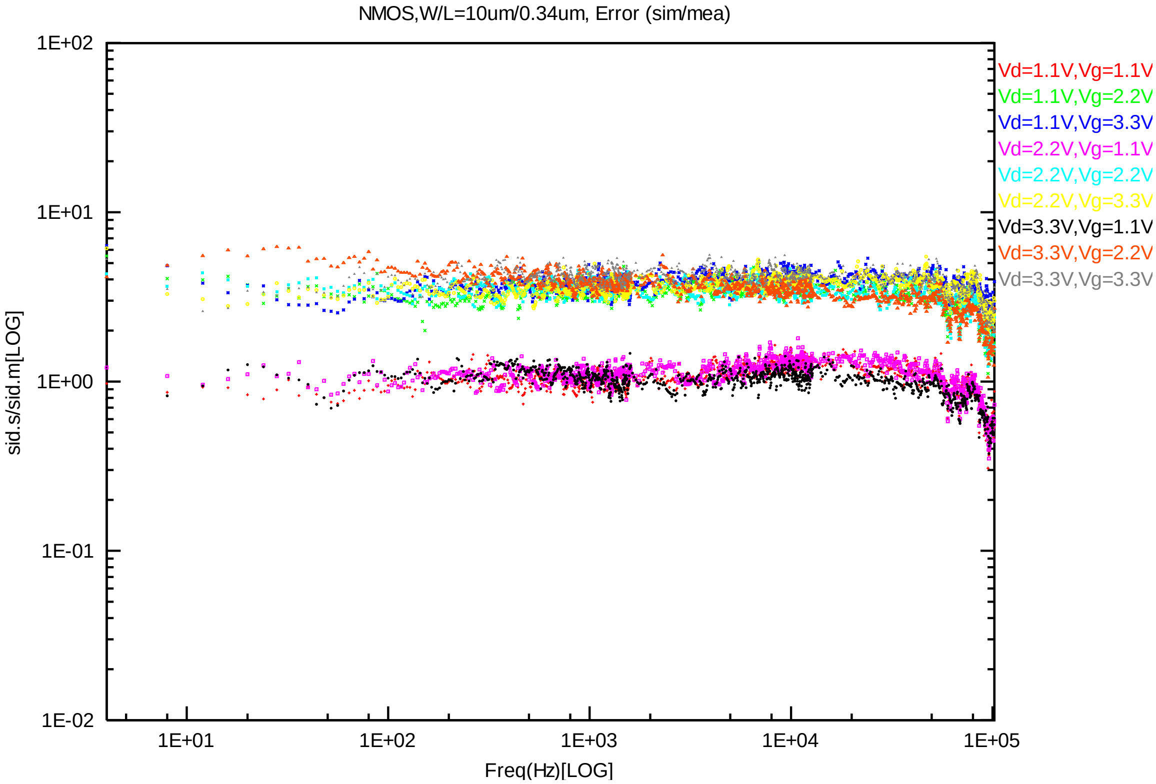


Figure 7-4 NMOS W/L=10µm/0.34µm Error Distribution Plot

NMOS, W=10μm L=0.52μm

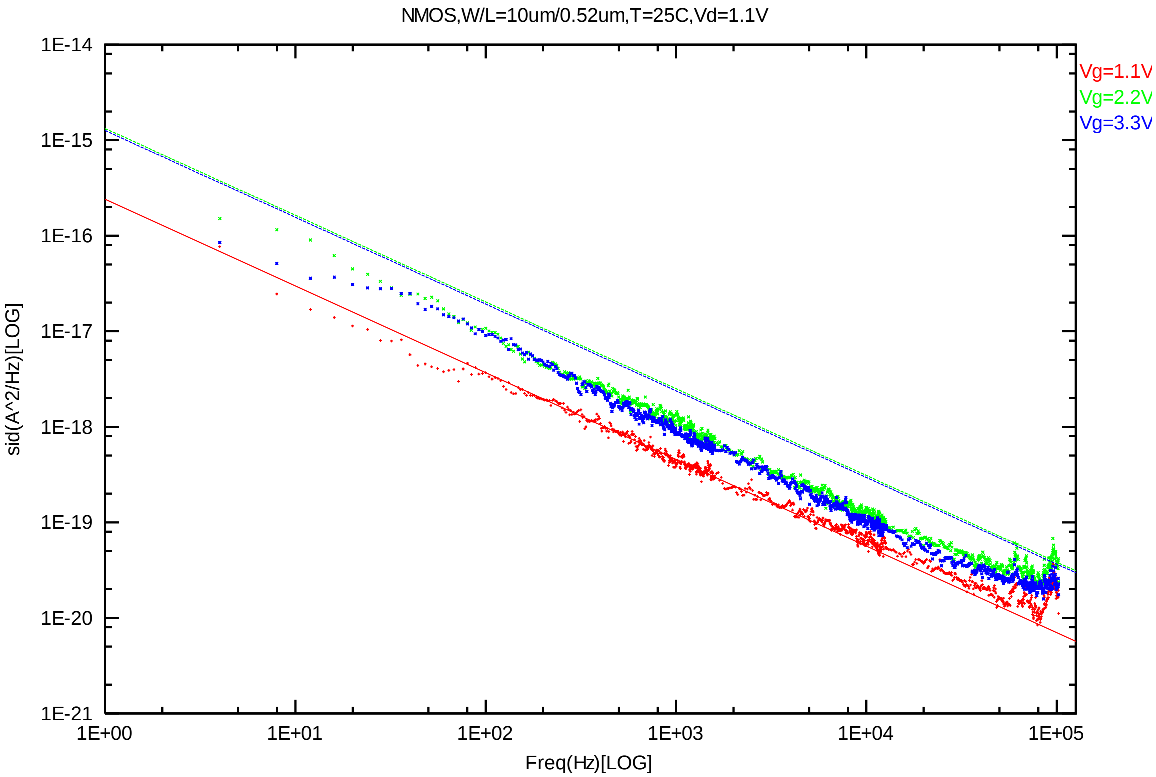


Figure 7-5 NMOS W/L=10μm/0.52μm @ Vd=1.1V

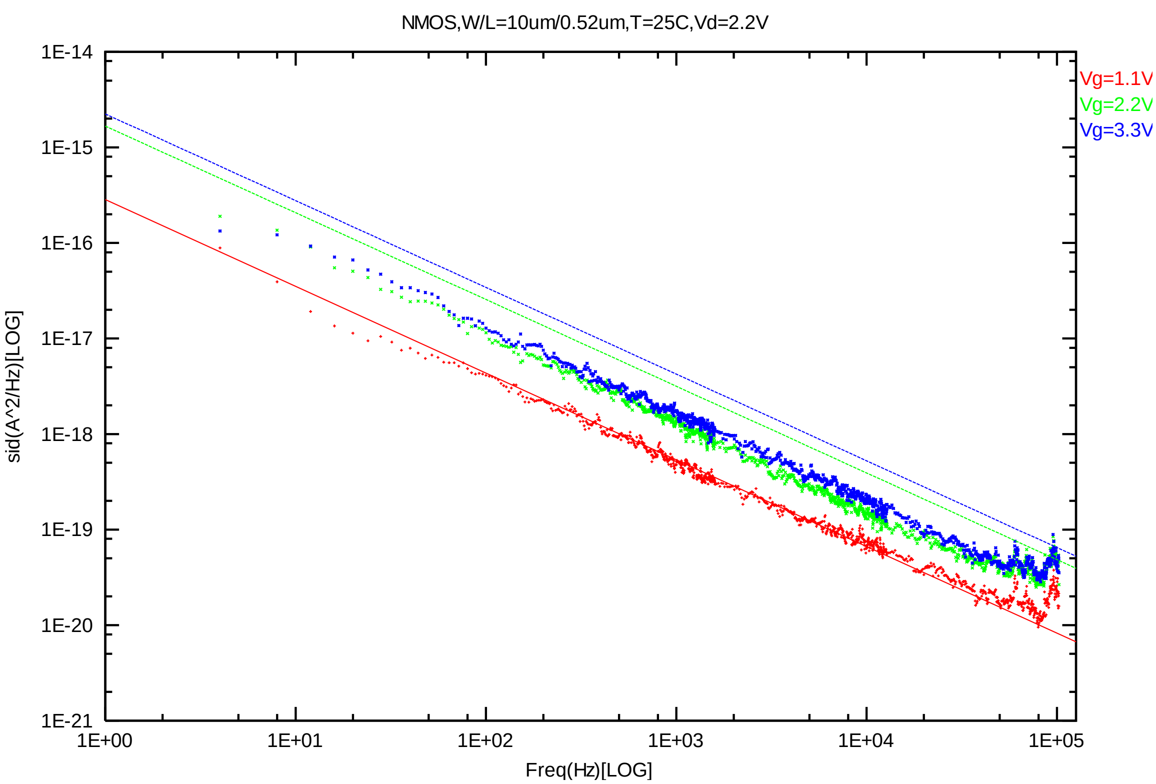


Figure 7-6 NMOS W/L=10μm/0.52μm @ Vd=2.2V

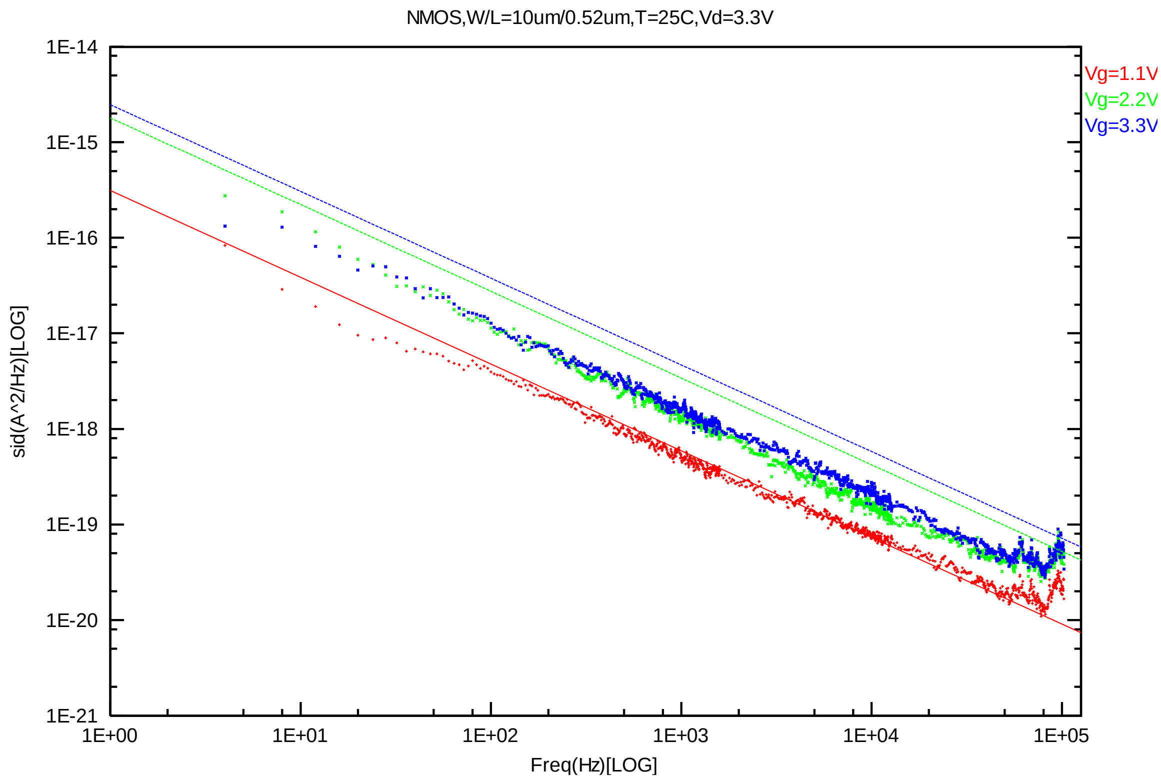


Figure 7-7 NMOS W/L=10µm/0.52µm @ Vd=3.3V

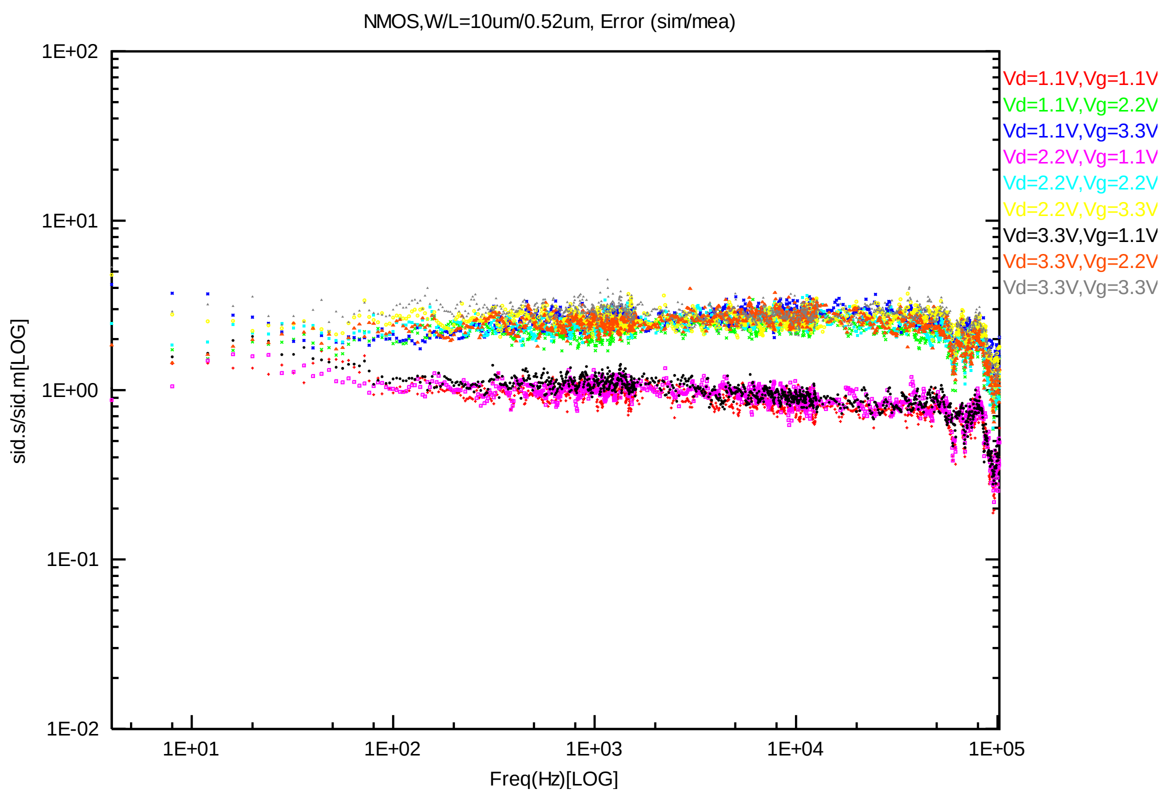


Figure 7-8 NMOS W/L=10µm/0.52µm Error Distribution Plot

NMOS, W=10μm L=1.02μm

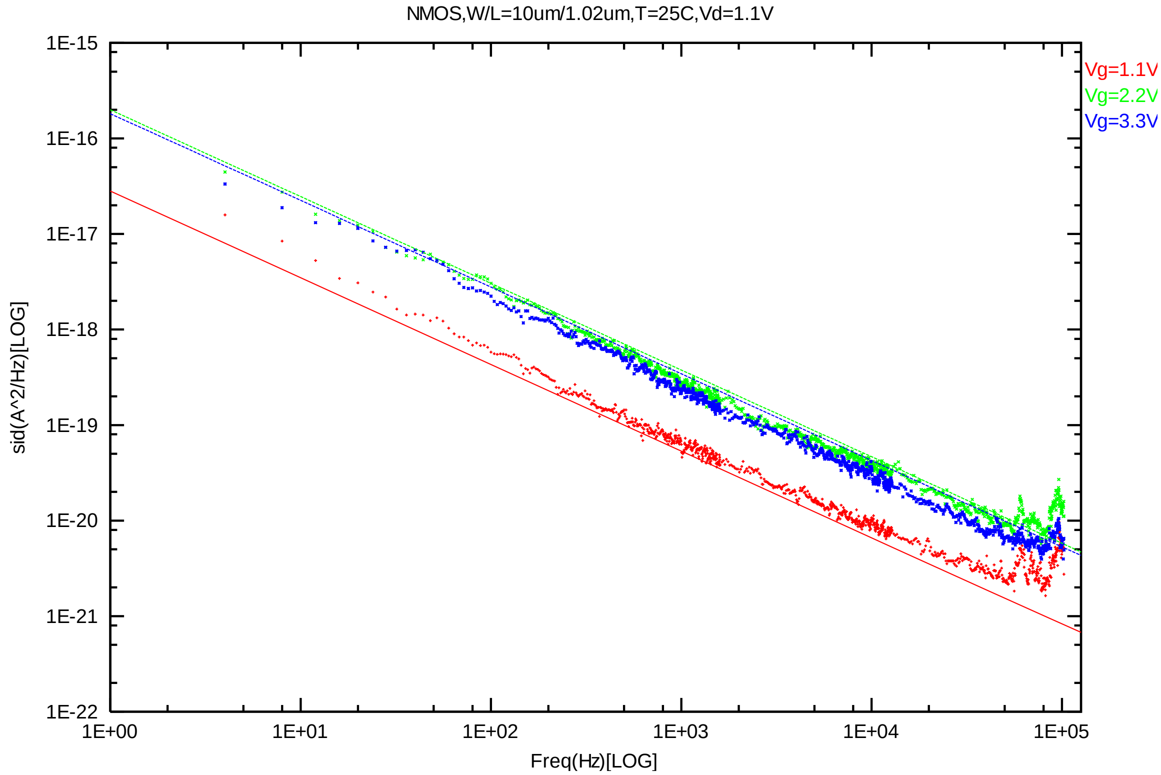


Figure 7-9 NMOS W/L=10μm/1.02μm @ Vd=1.1V

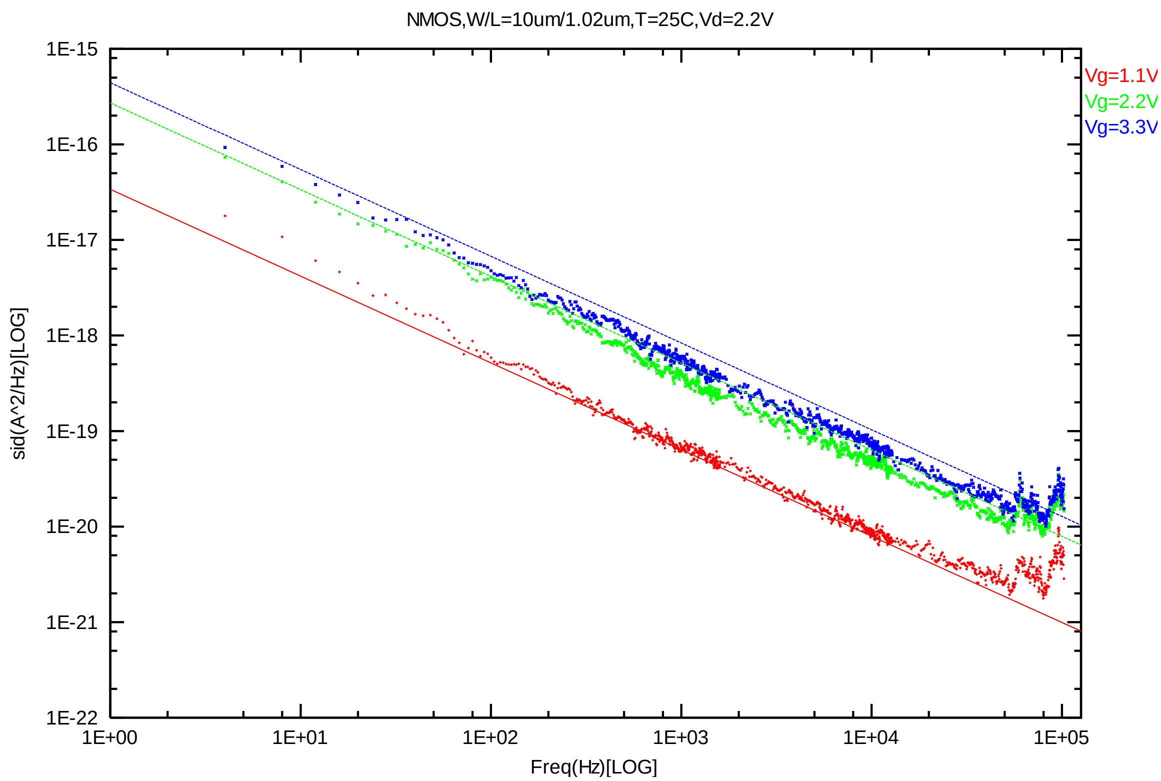


Figure 7-10 NMOS W/L=10μm/1.02μm @ Vd=2.2V

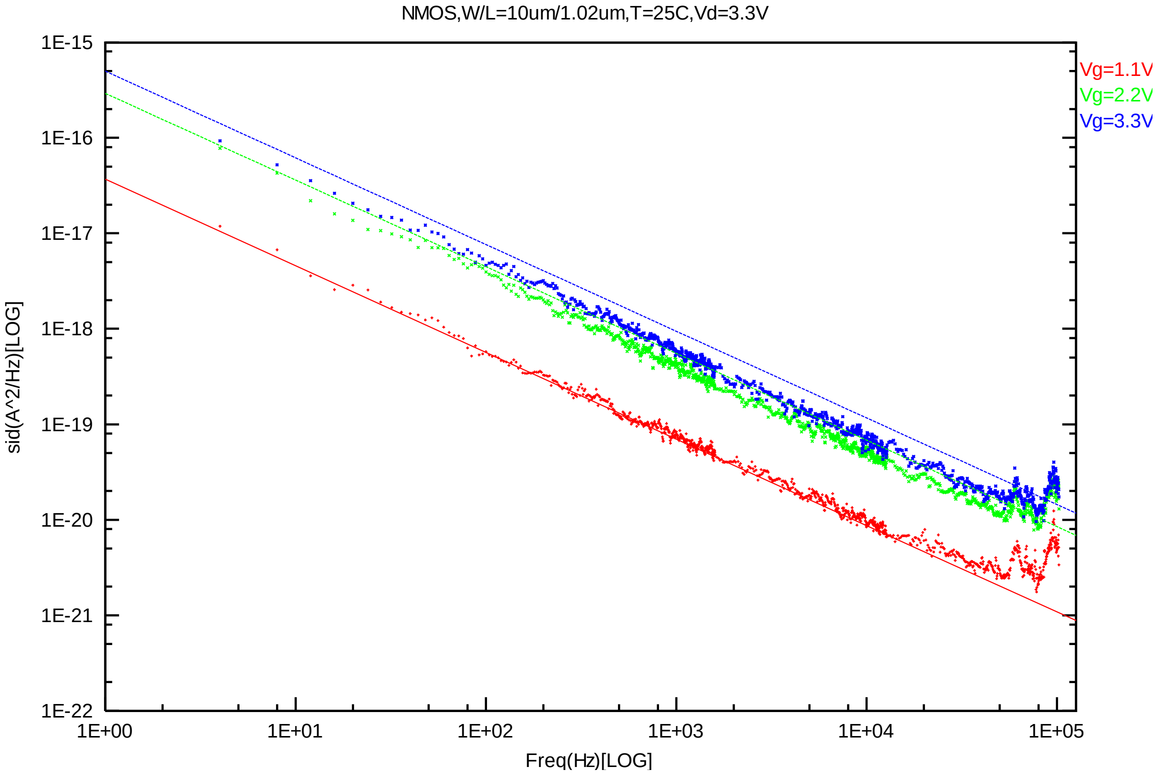


Figure 7-11 NMOS W/L=10µm/1.02µm @ Vd=3.3V

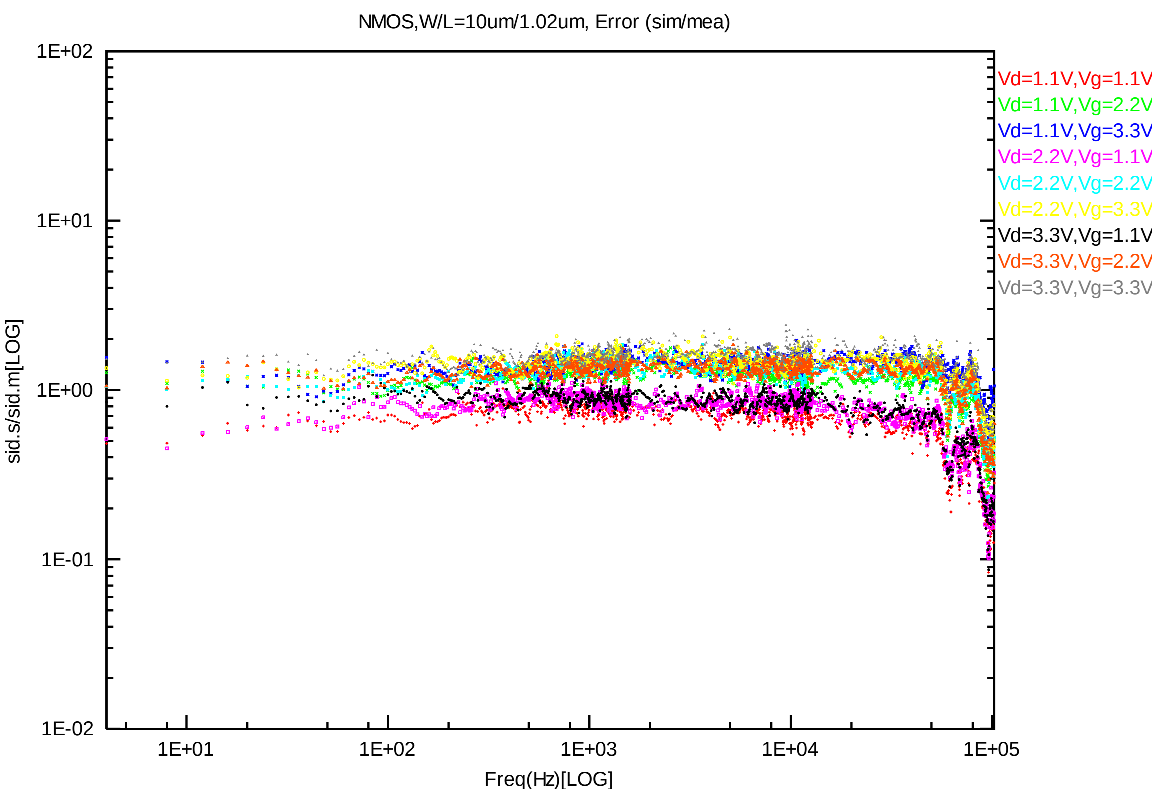


Figure 7-12 NMOS W/L=10µm/1.02µm Error Distribution Plot

NMOS, W=10µm L=10.02µm

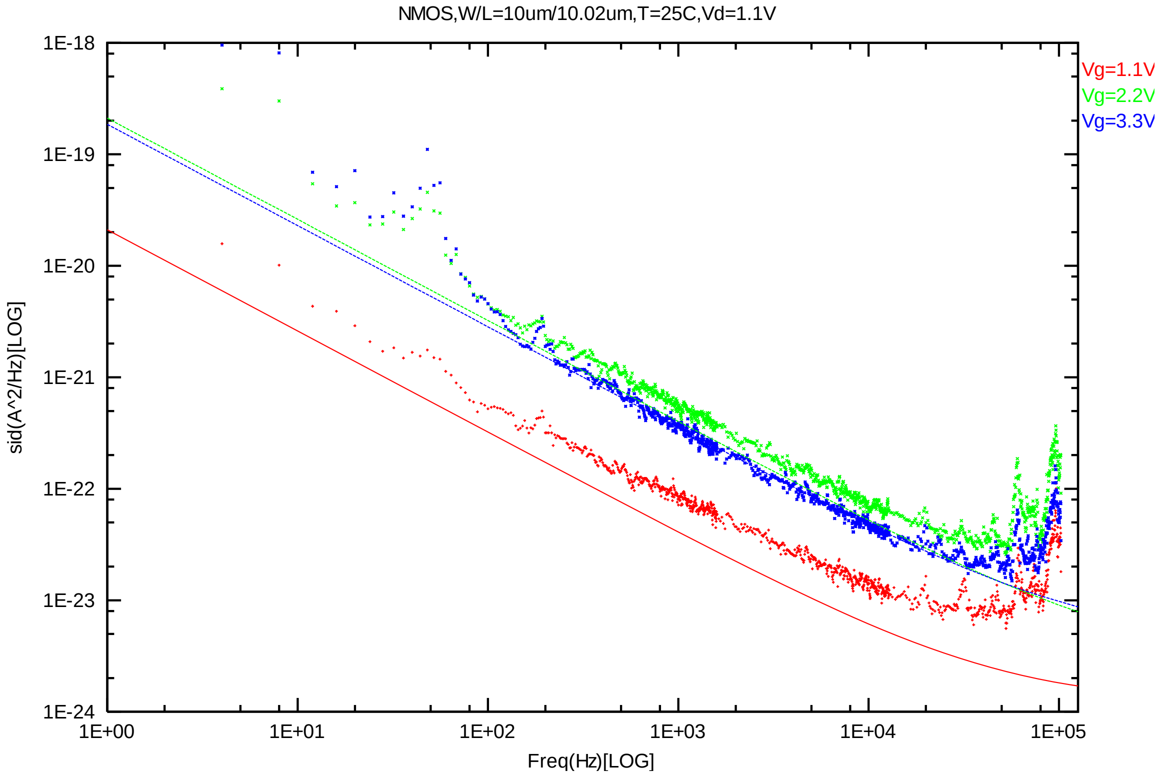


Figure 7-13 NMOS W/L=10µm/10.02µm @ Vd=1.1V

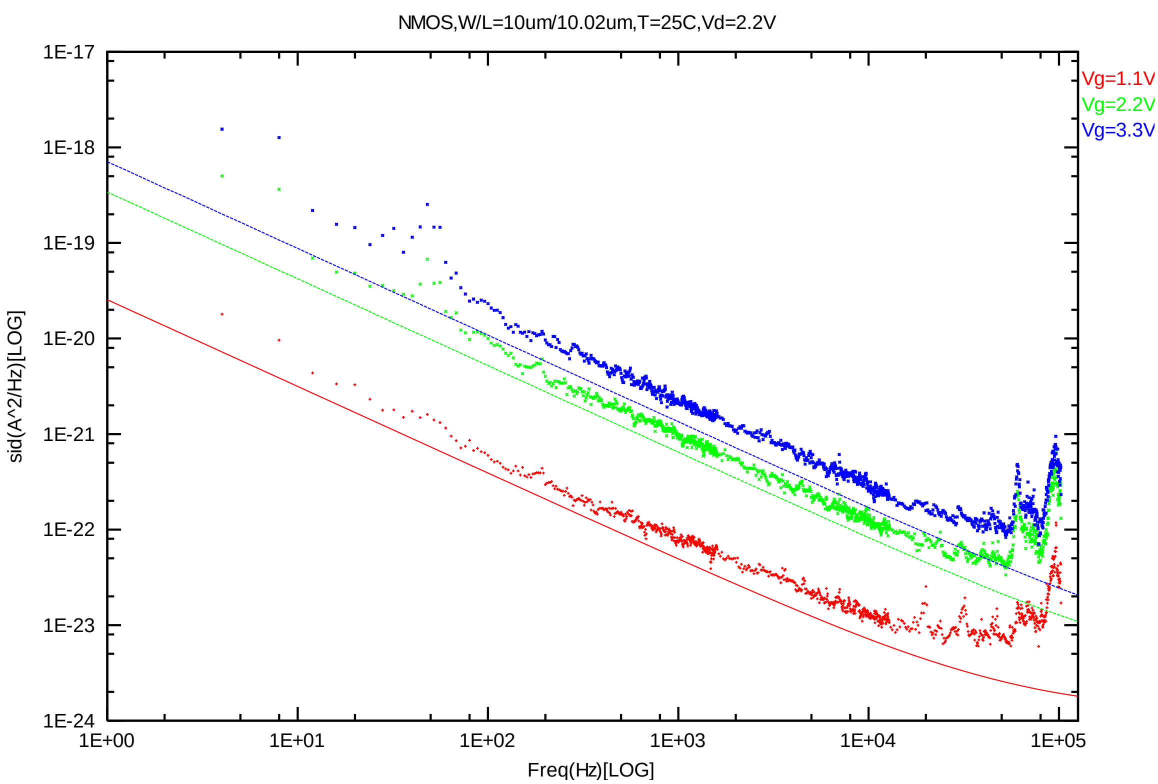


Figure 7-14 NMOS W/L=10µm/10.02µm @ Vd=2.2V

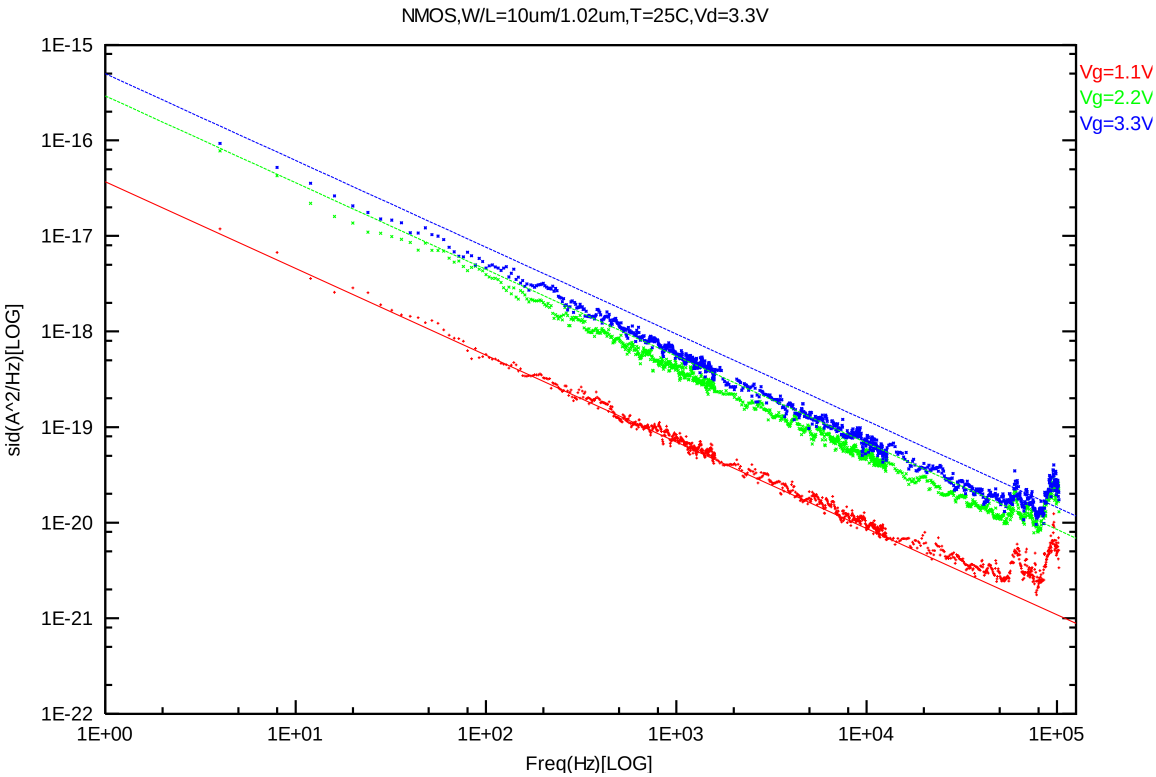


Figure 7-15 NMOS W/L=10µm/10.02µm @ Vd=3.3V

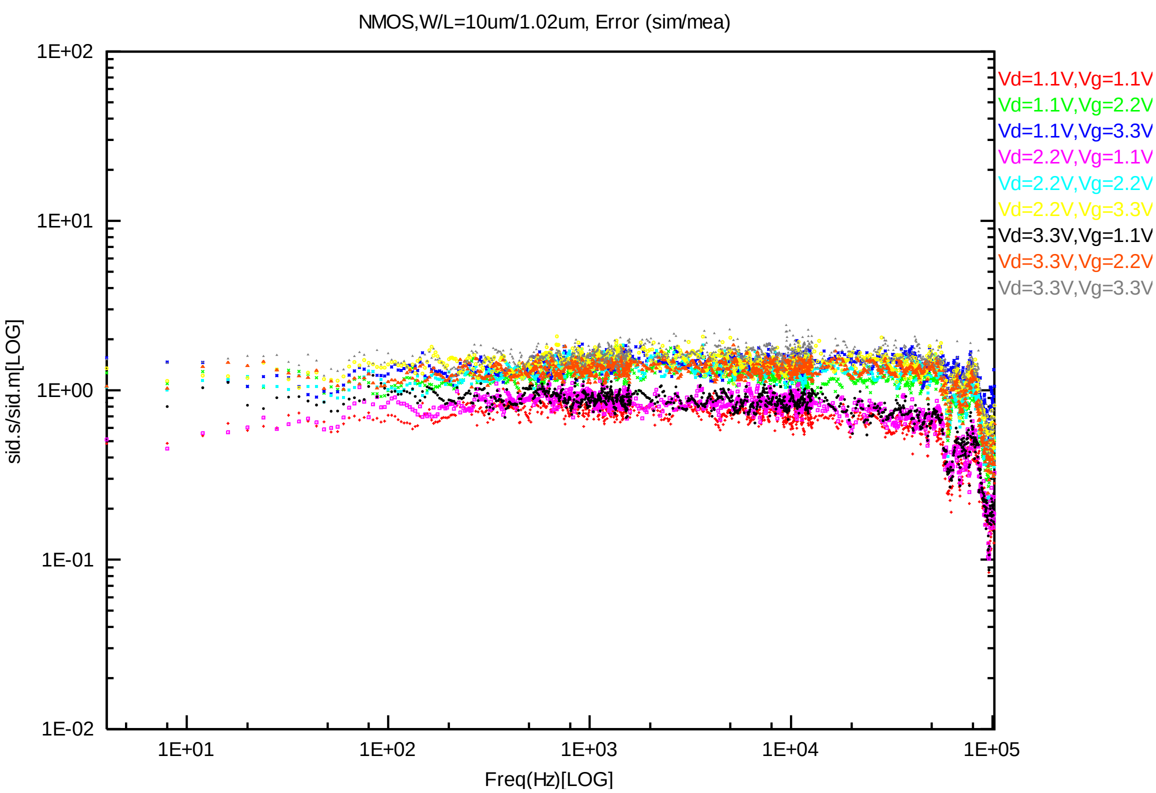


Figure 7-16 NMOS W/L=10µm/10.02µm Error Distribution Plot

PMOS, W=10μm L=0.34μm

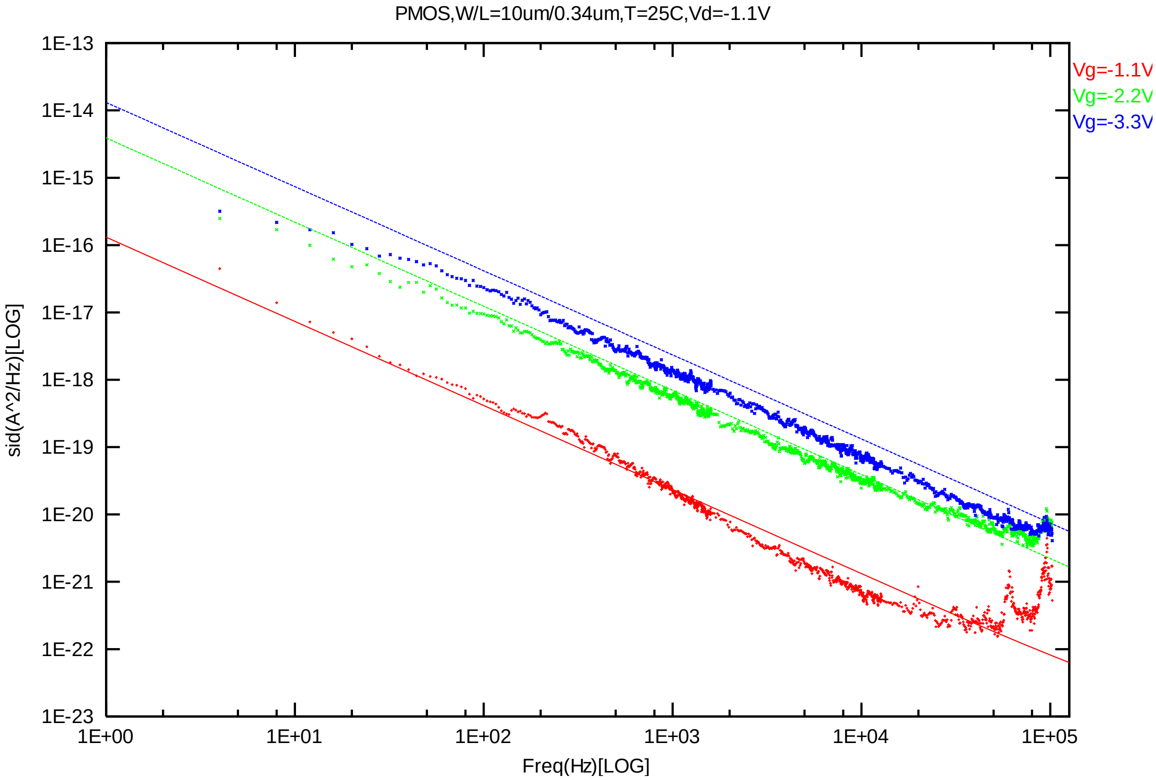


Figure 7-17 PMOS W/L=10μm/0.34μm @ Vd= -1.1V

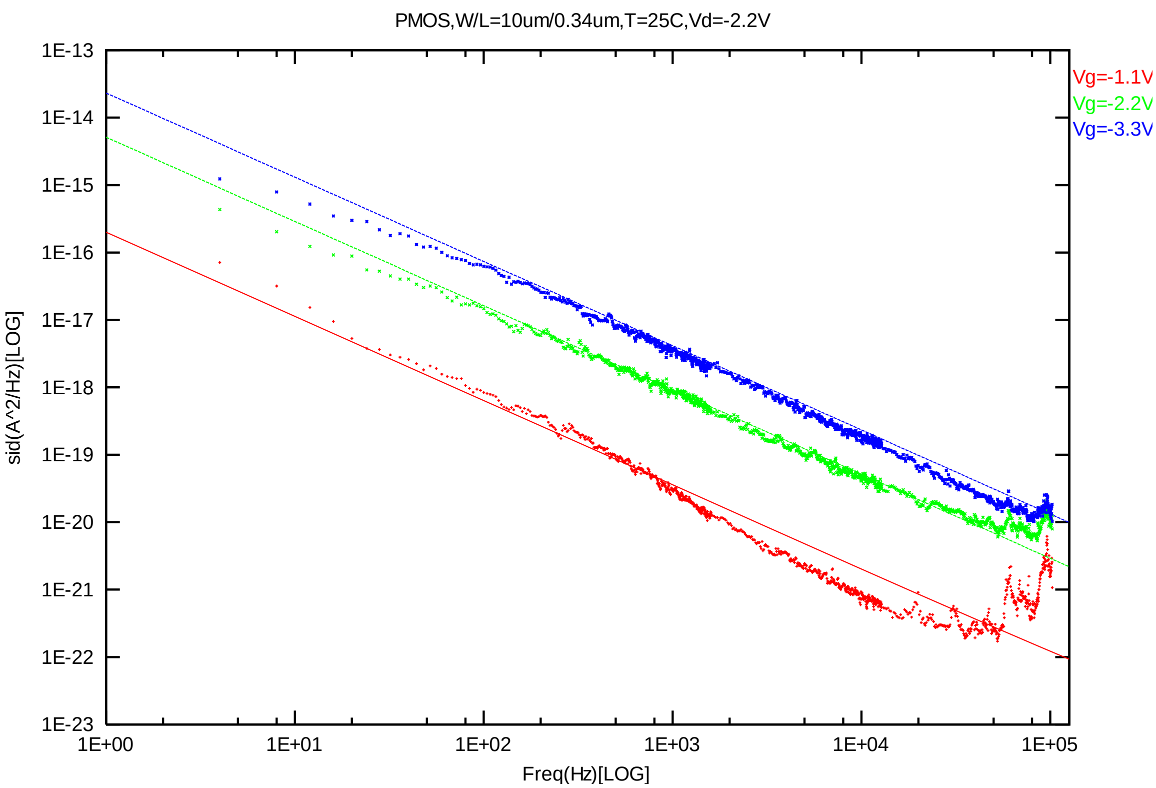


Figure 7-18 PMOS W/L=10μm/0.34μm @ Vd= -2.2V

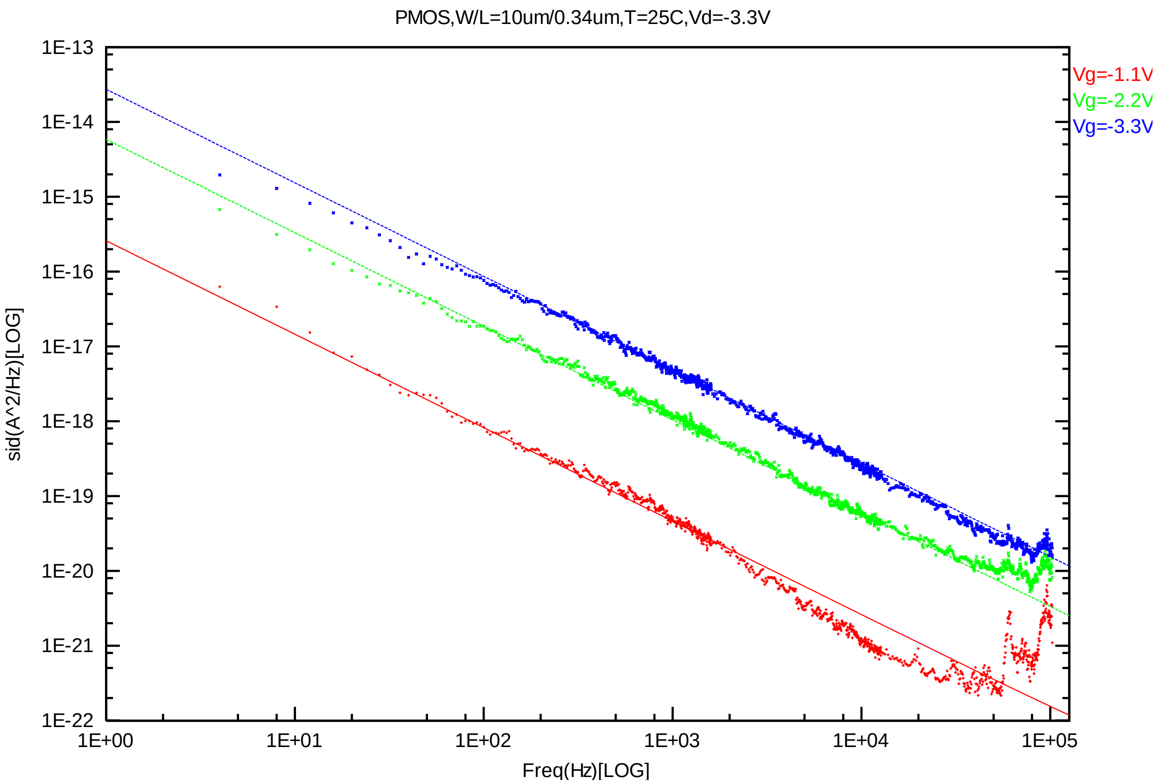


Figure 7-19 PMOS W/L=10µm/0.34µm @ Vd= -3.3V

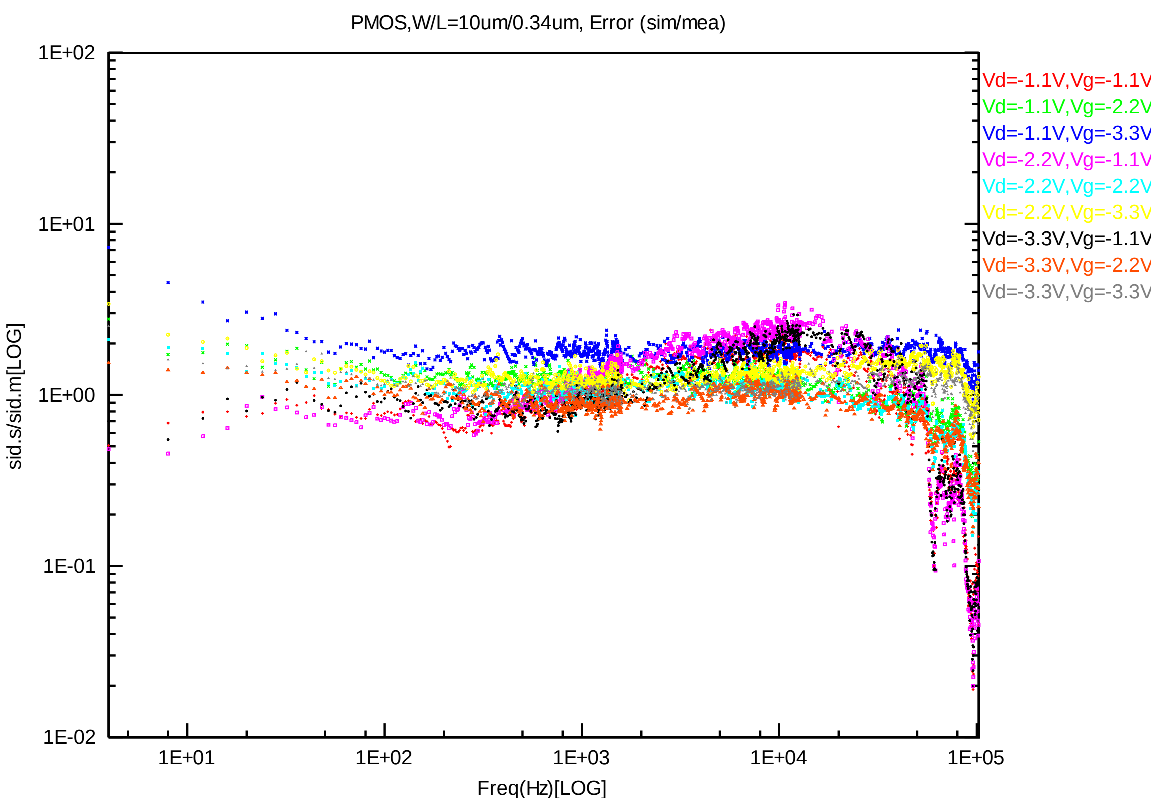


Figure 7-20 PMOS W/L=10µm/0.34µm Error Distribution Plot

PMOS, W=10μm L=0.52μm

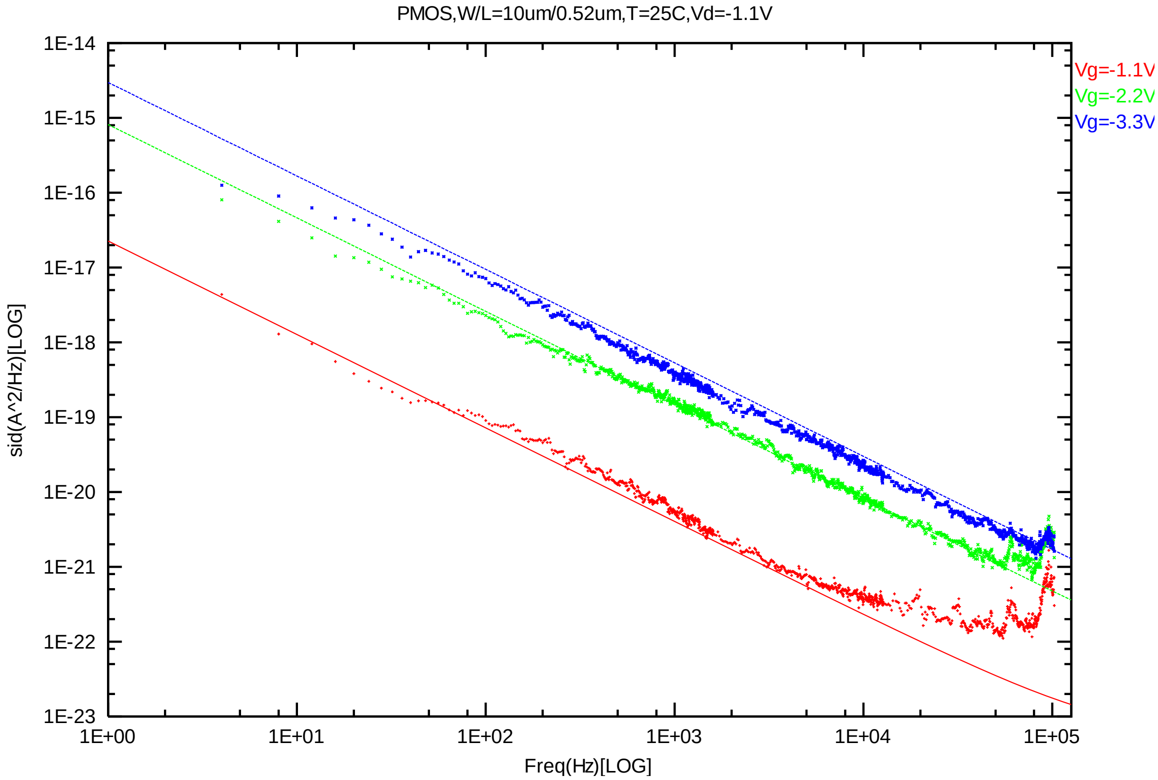


Figure 7-21 PMOS W/L=10μm/0.52μm @ Vd= -1.1V

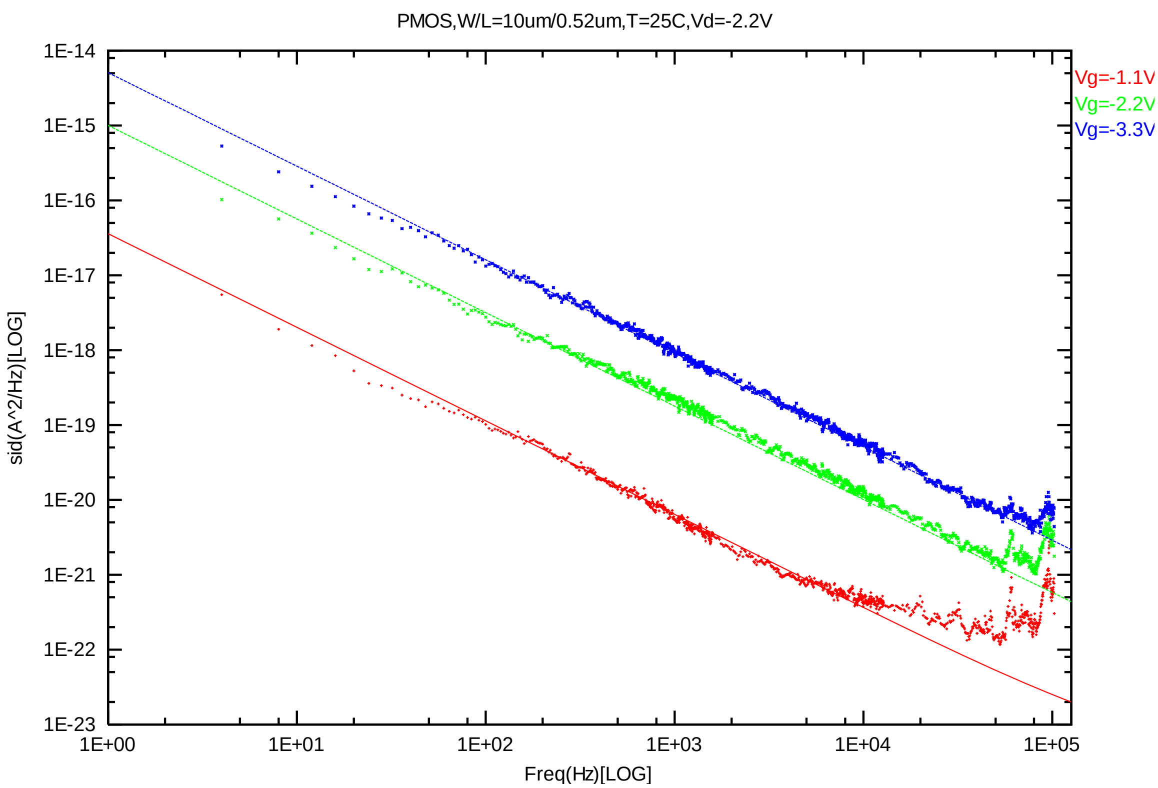


Figure 7-22 PMOS W/L=10μm/0.52μm @ Vd= -2.2V

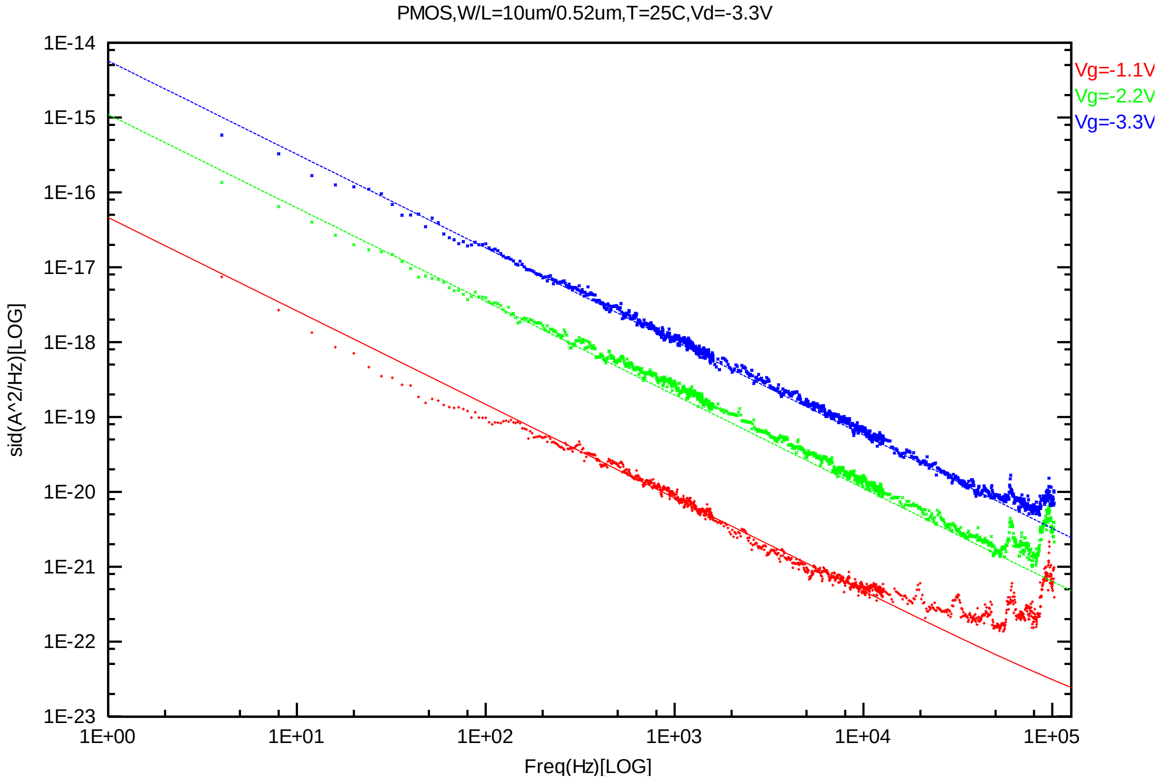


Figure 7-23 PMOS W/L=10µm/0.52µm @ Vd= -3.3V

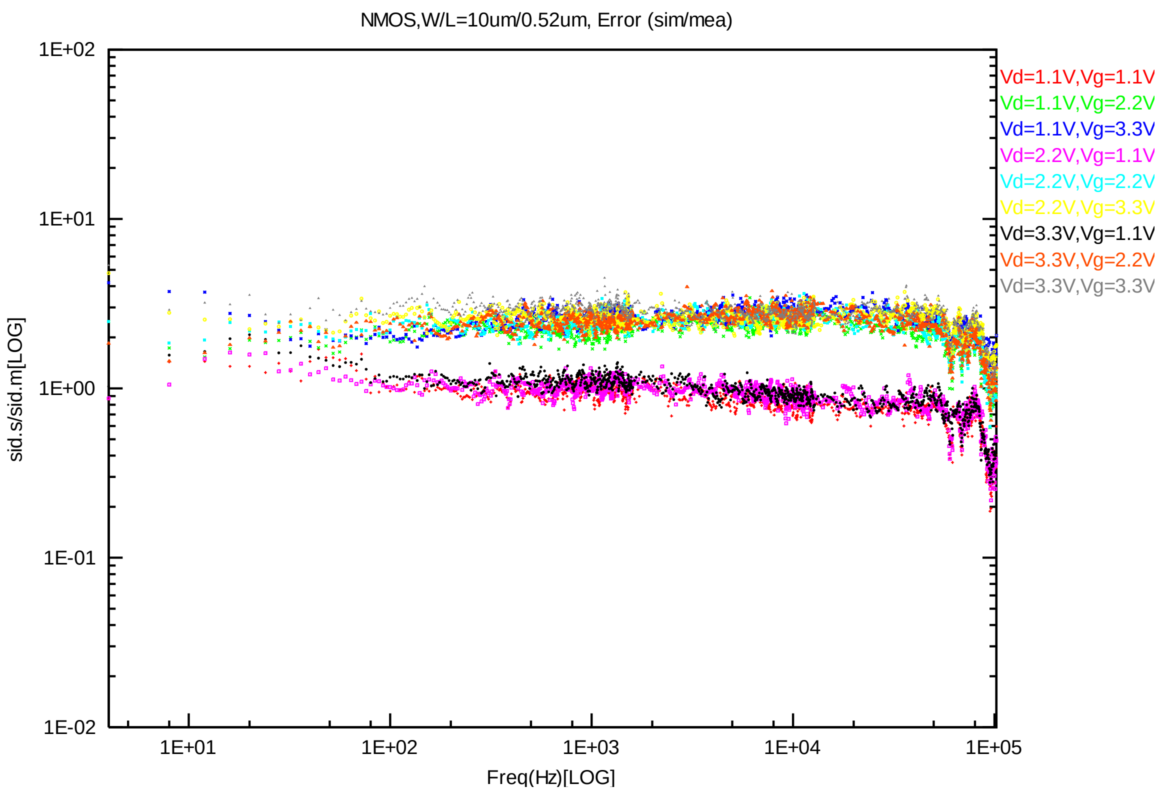


Figure 7-24 PMOS W/L=10µm/0.52µm Error Distribution Plot

PMOS, W=10μm L=1.02μm

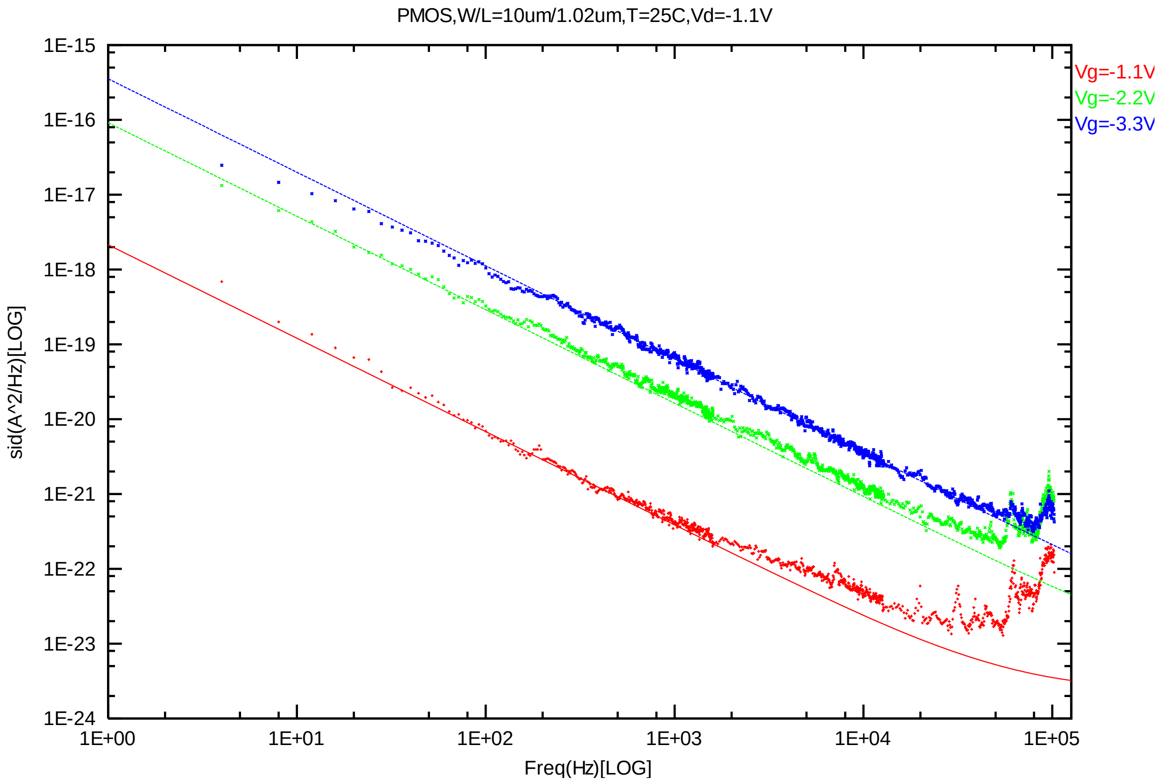


Figure 7-25 PMOS W/L=10μm/1.02μm @ Vd= -1.1V

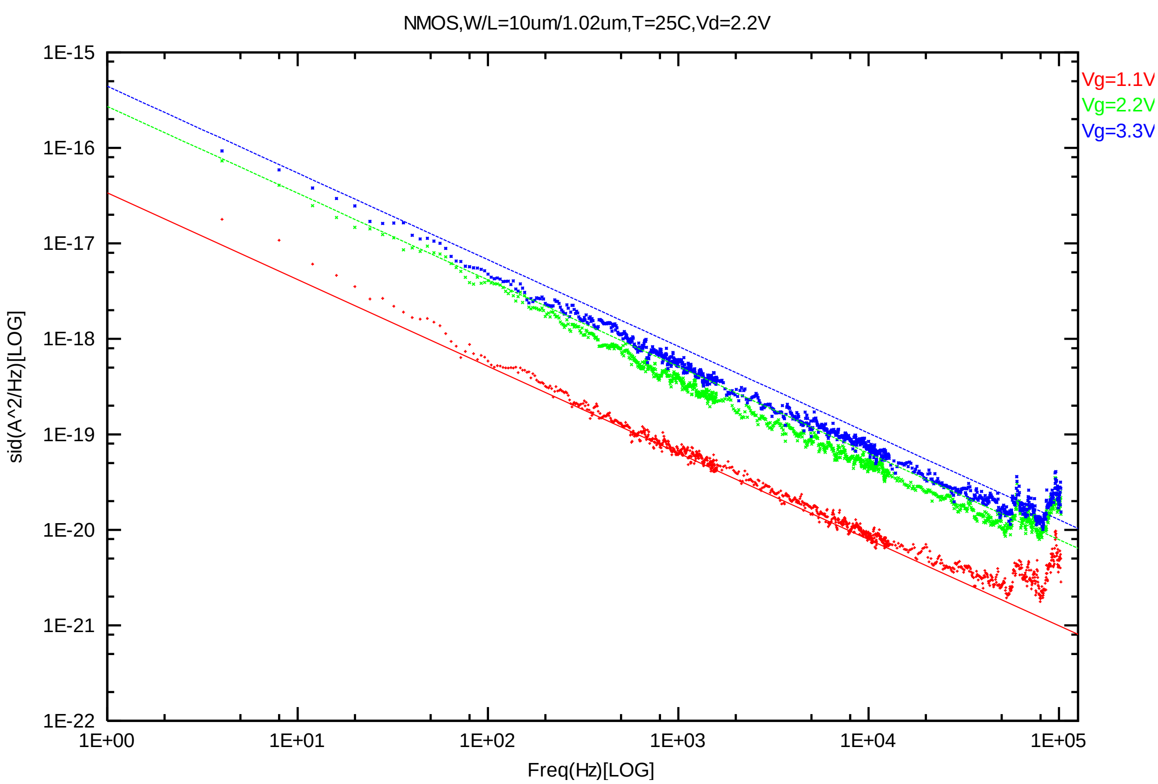


Figure 7-26 PMOS W/L=10μm/1.02μm @ Vd= -2.2V

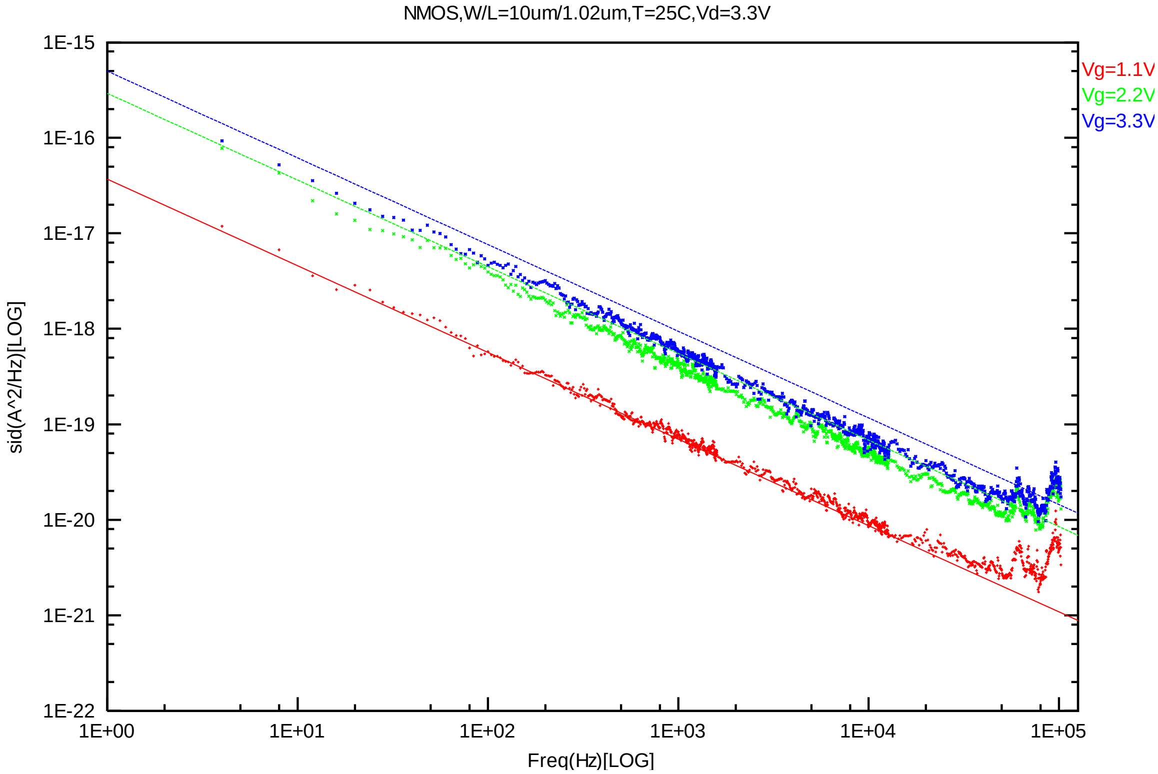


Figure 7-27 PMOS W/L=10µm/1.02µm @ Vd= -3.3V

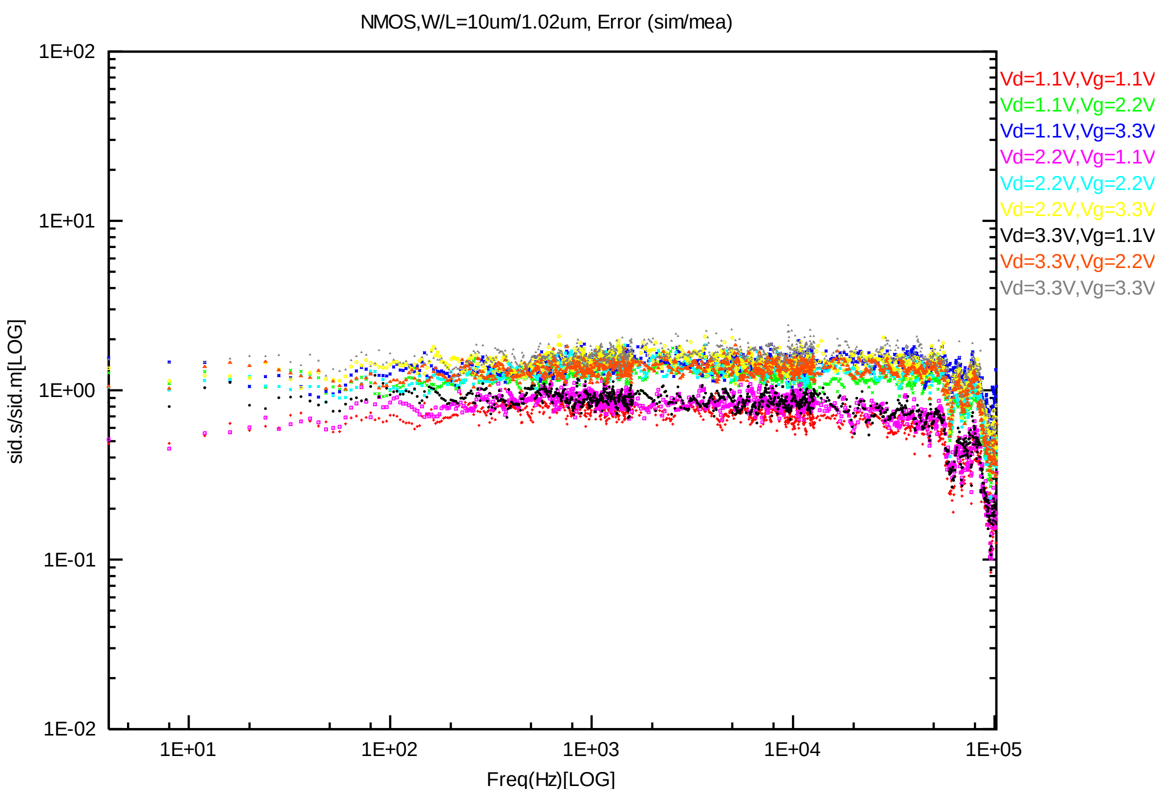


Figure 7-28 PMOS W/L=10µm/1.02µm Error Distribution Plot

PMOS, W=10μm L=10.02μm

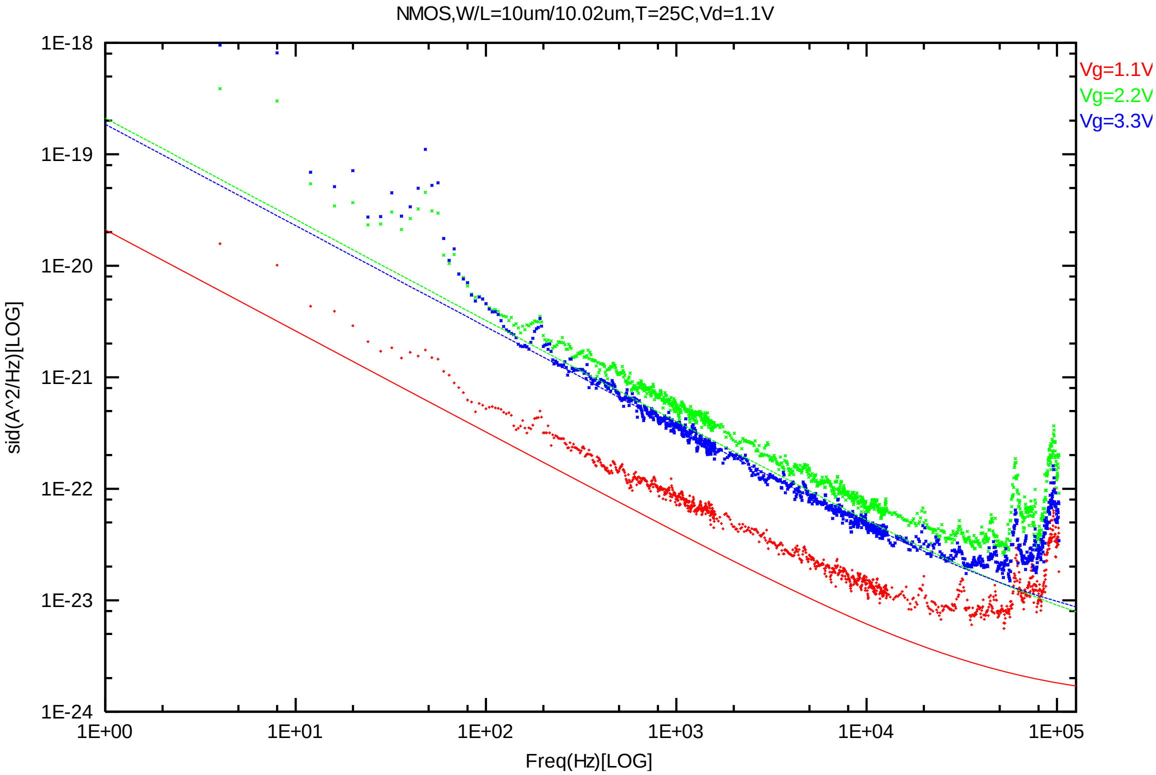


Figure 7-29 PMOS W/L=10μm/10.02μm @ Vd= -1.1V

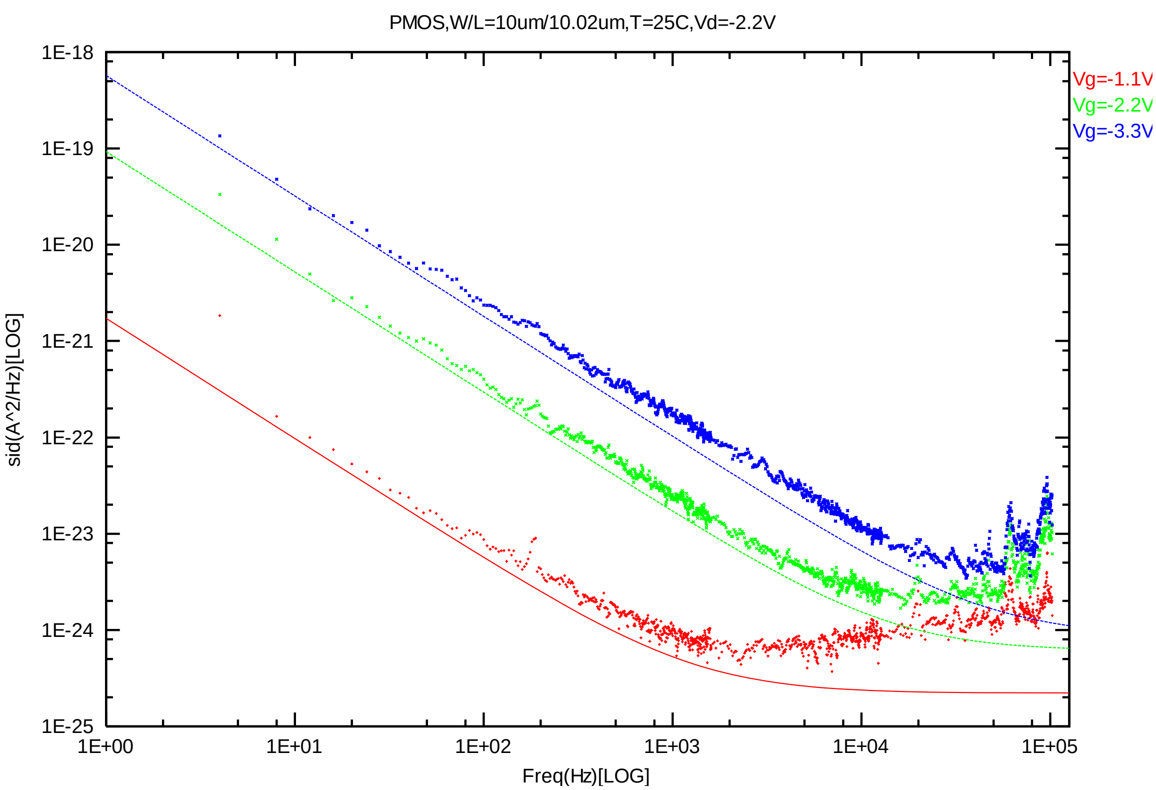


Figure 7-30 PMOS W/L=10μm/10.02μm @ Vd= -2.2V

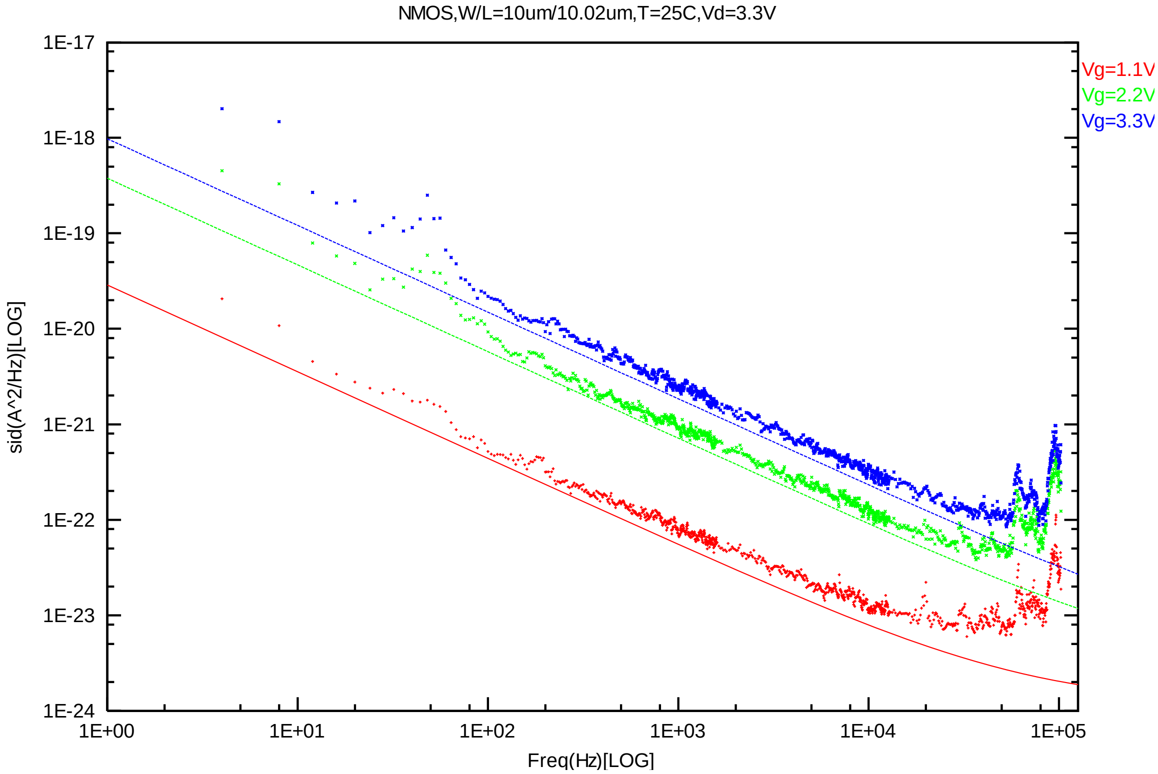


Figure 7-31 PMOS W/L=10μm/10.02μm @ Vd= -3.3V

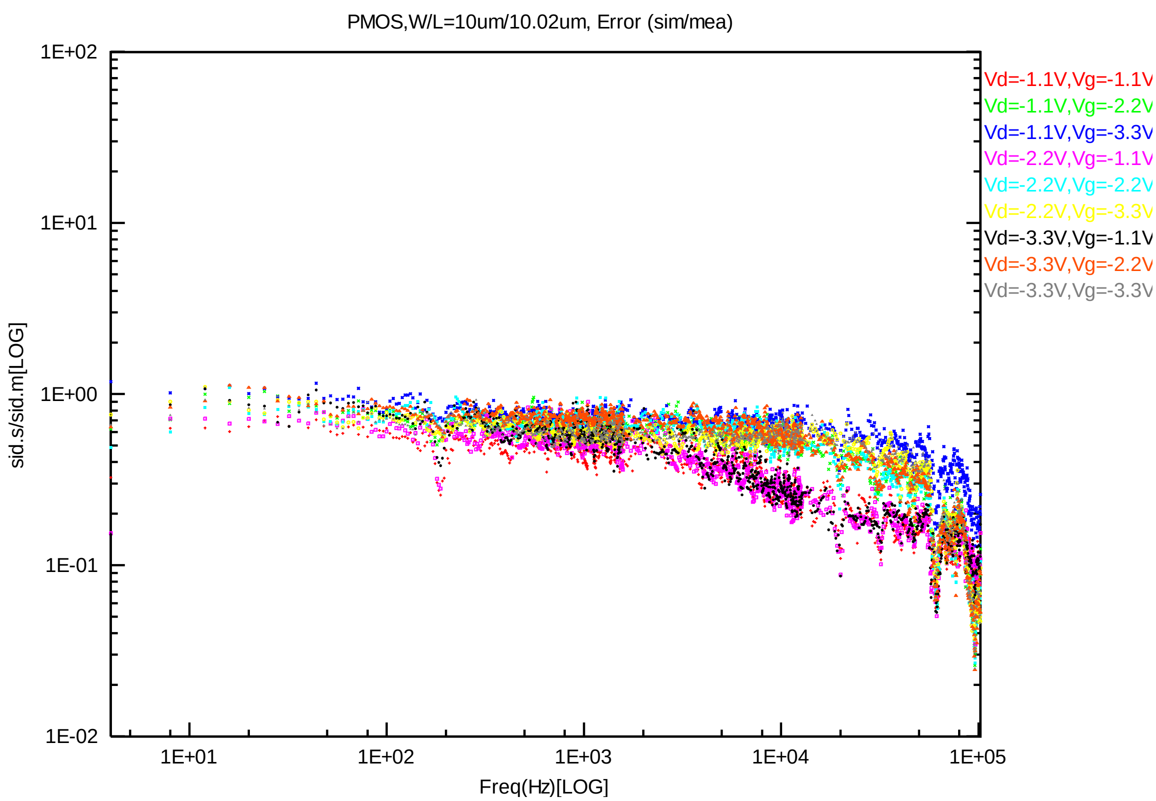


Figure 7-32 PMOS W/L=10μm/10.02μm Error Distribution Plot

7.2 Trend plots at frequency=100Hz

A. NMOS

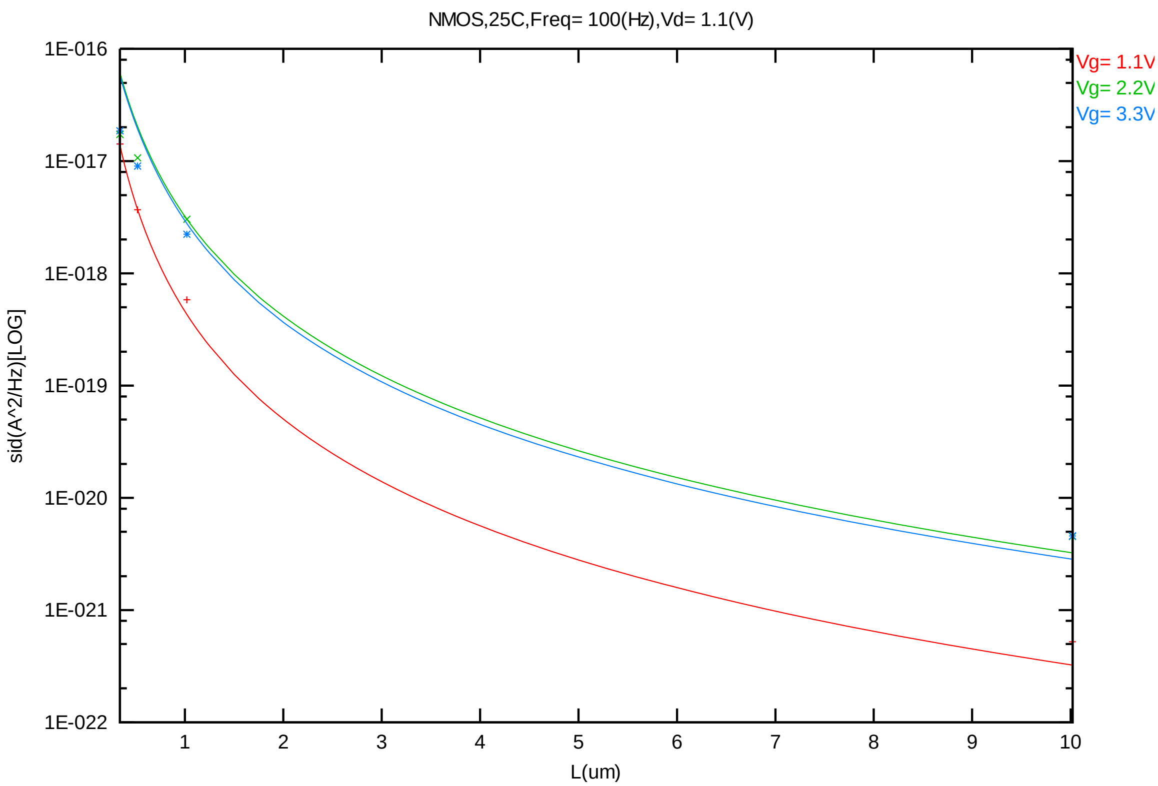


Figure 7-33 Noise vs Length for NMOS @ Vd=1.1V

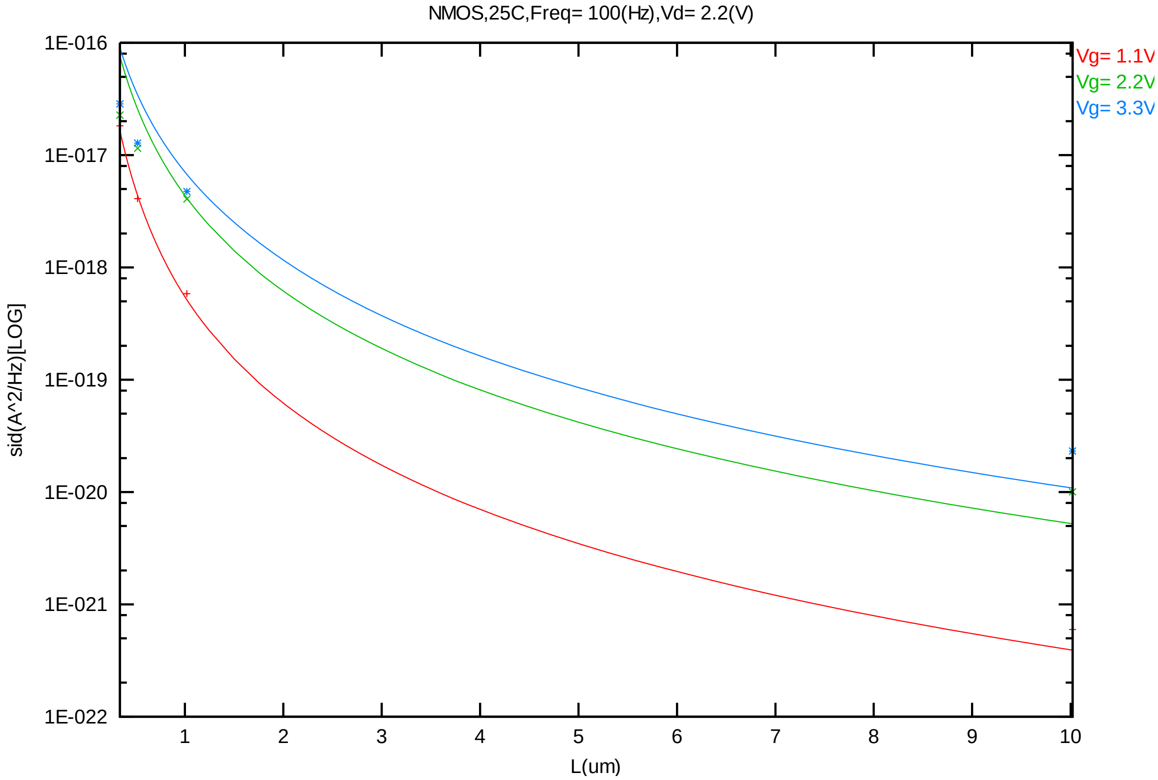


Figure 7-34 Noise vs Length for NMOS @ Vd=2.2V

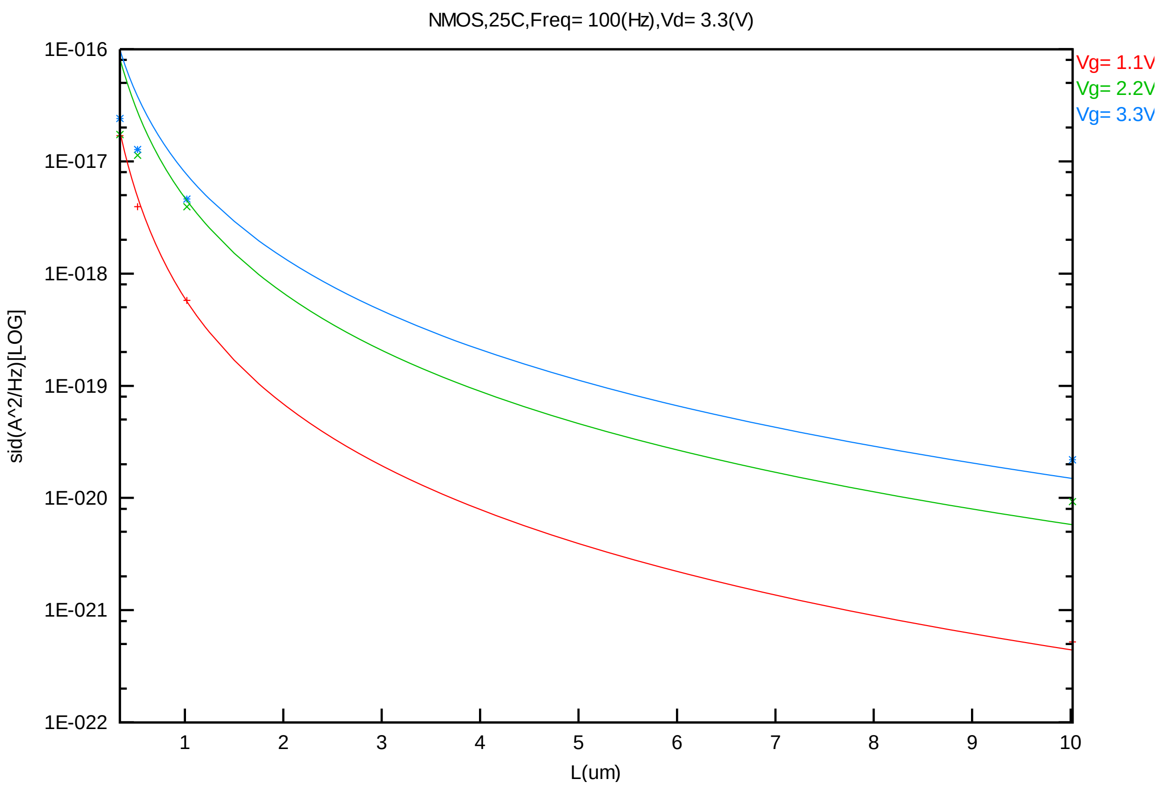


Figure 7-35 Noise vs Length for NMOS @ Vd=3.3V

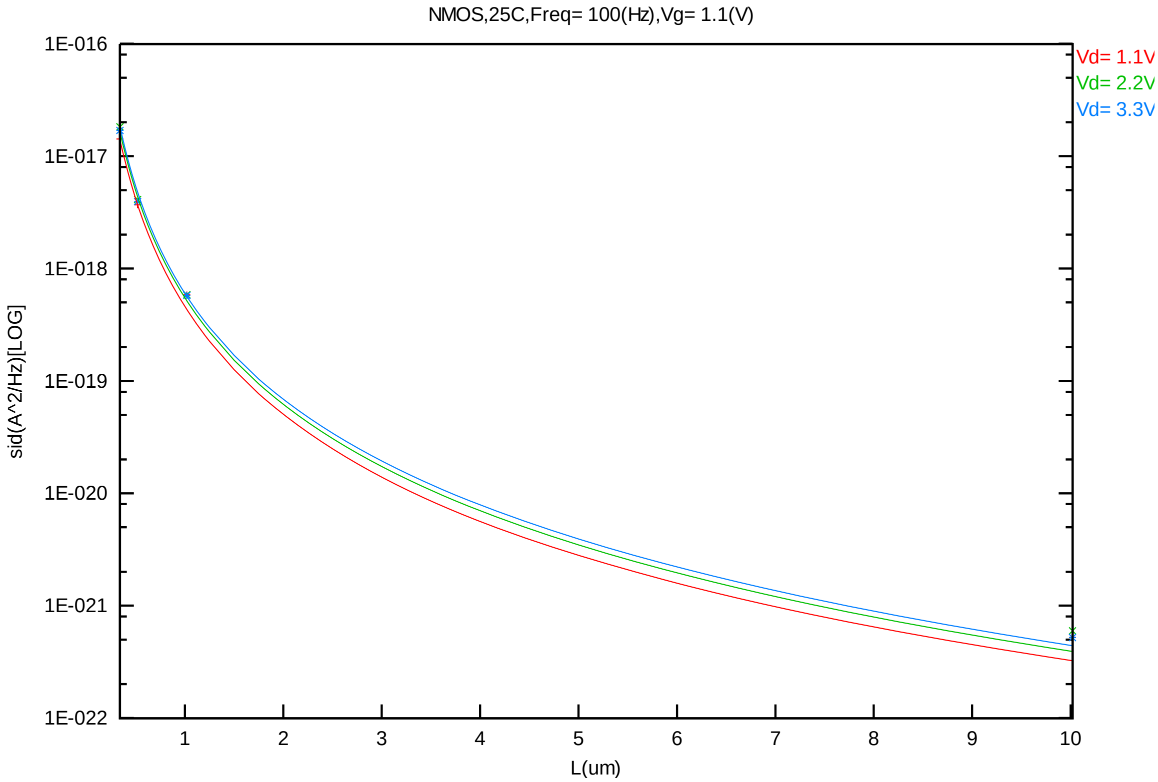


Figure 7-36 Noise vs Length for NMOS @ Vg=1.1V

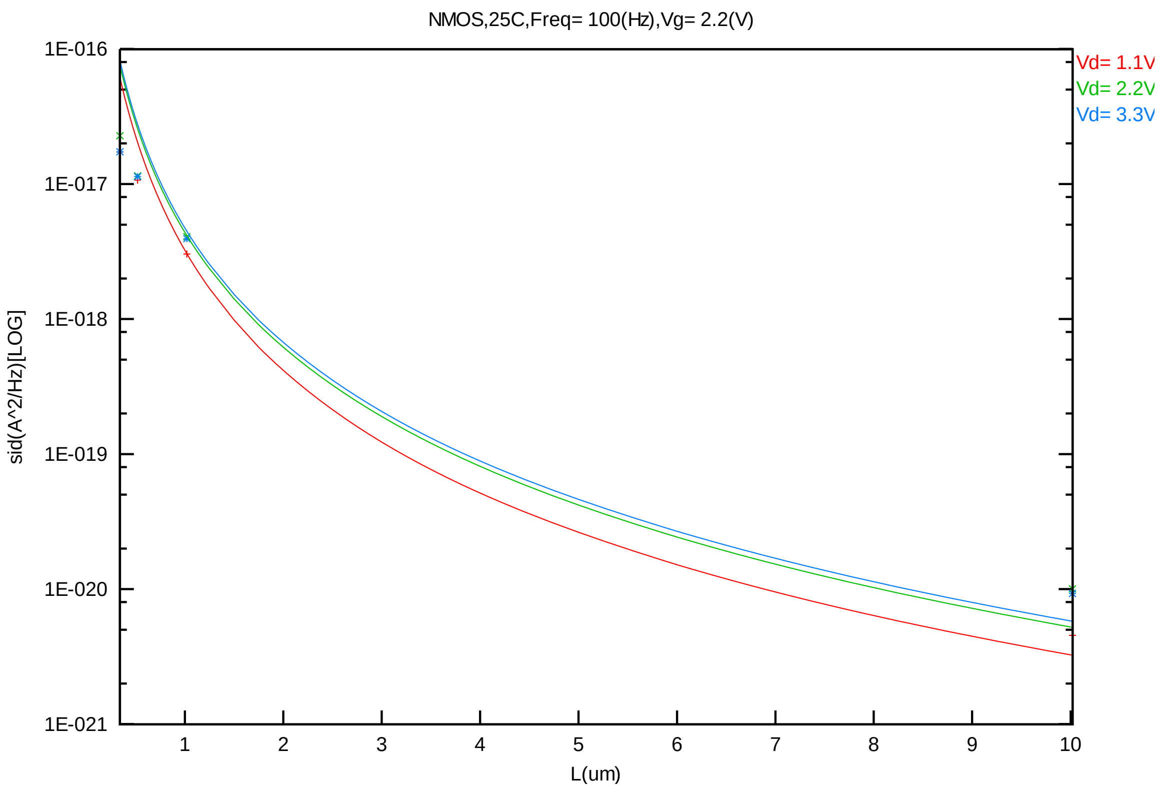


Figure 7-37 Noise vs Length for NMOS @ Vg=2.2V

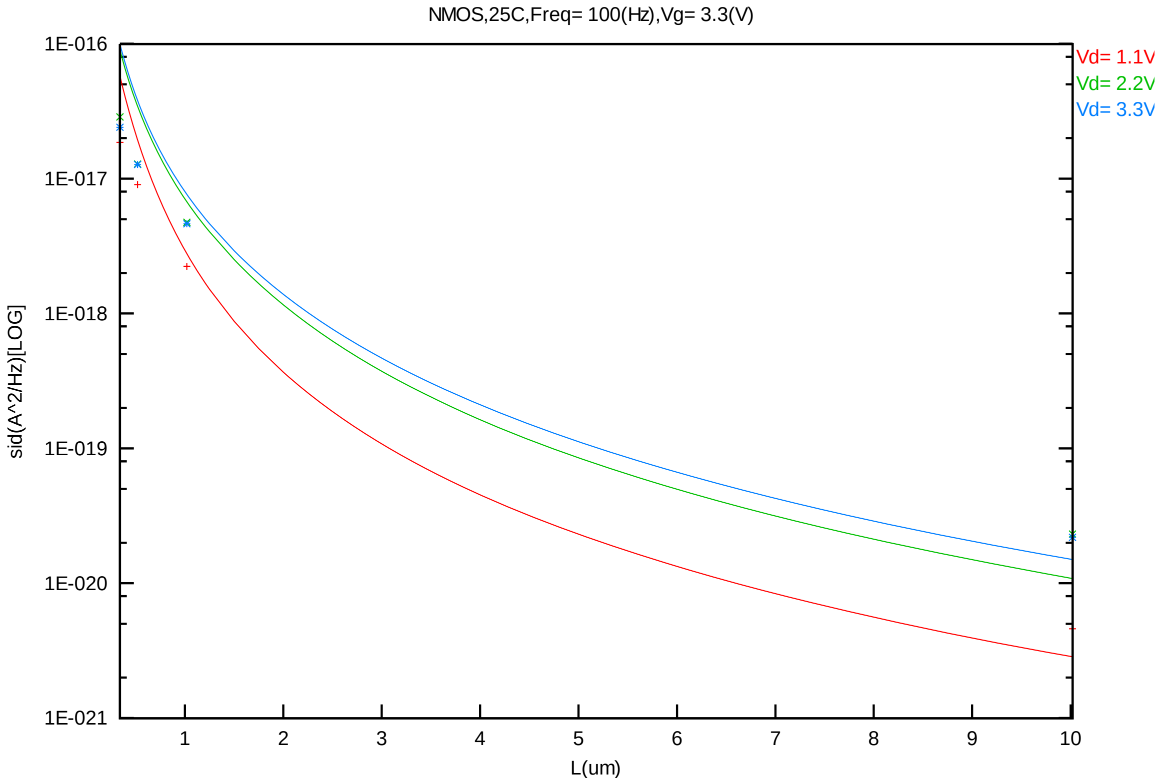


Figure 7-38 Noise vs Length for NMOS @ Vg=3.3V

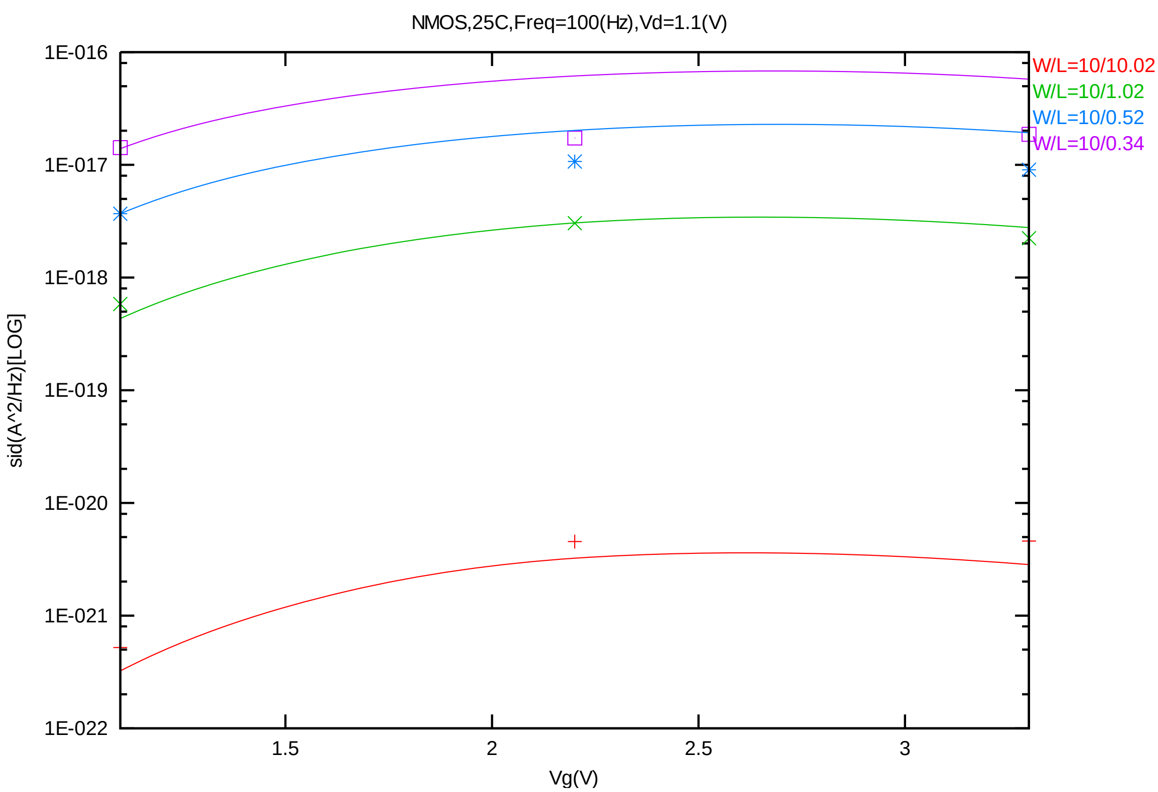


Figure 7-39 Noise vs Vg for NMOS @ Vd=1.1V

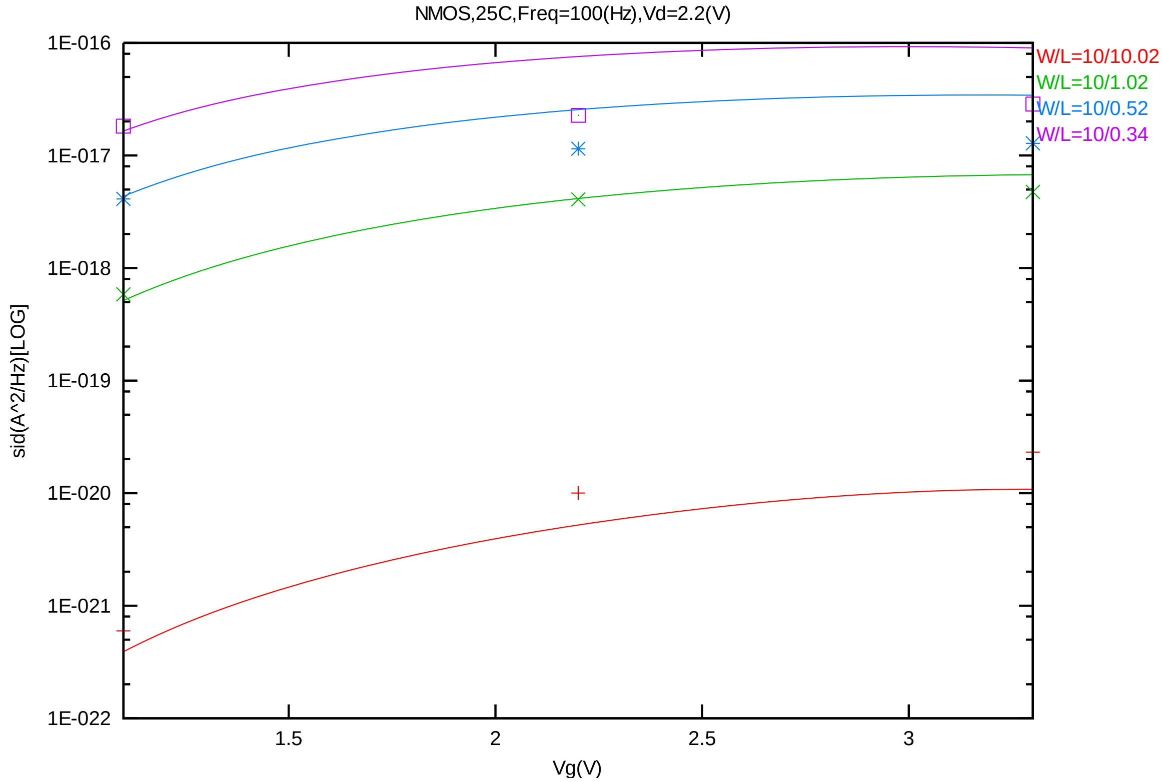


Figure 7-40 Noise vs Vg for NMOS @ Vd=2.2V

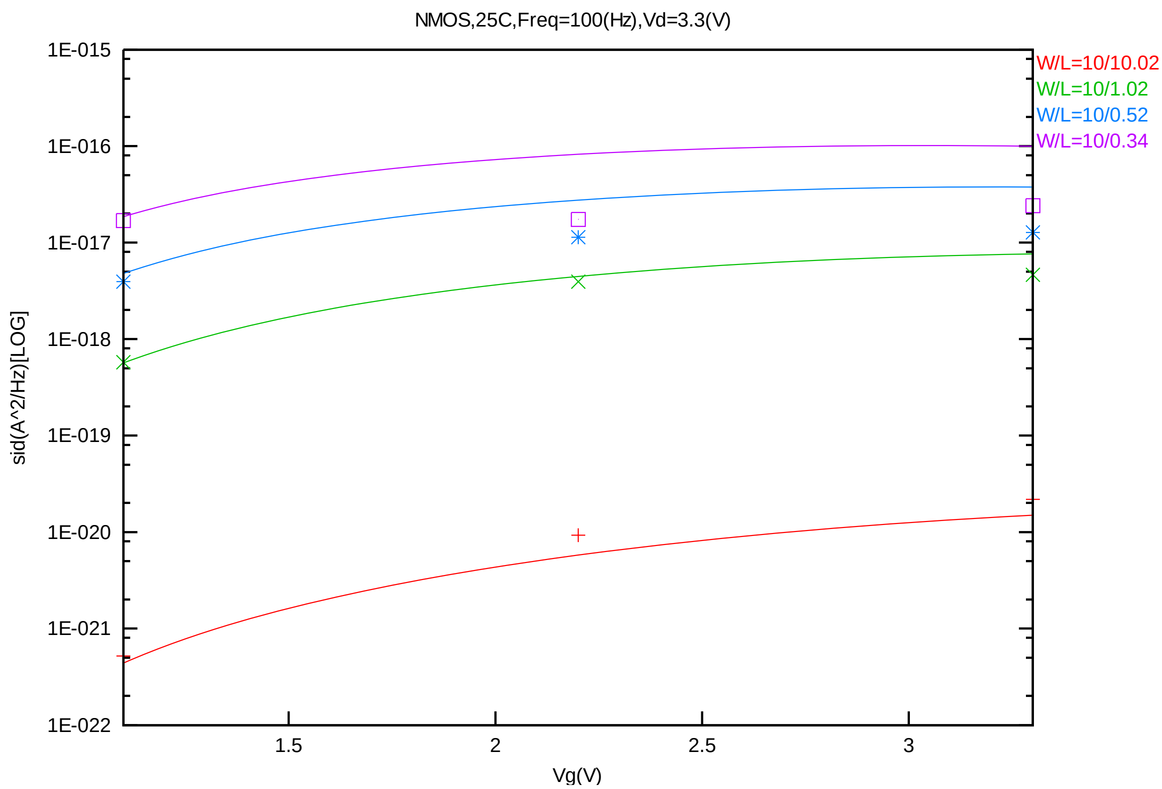


Figure 7-41 Noise vs Vg for NMOS @ Vd=3.3V

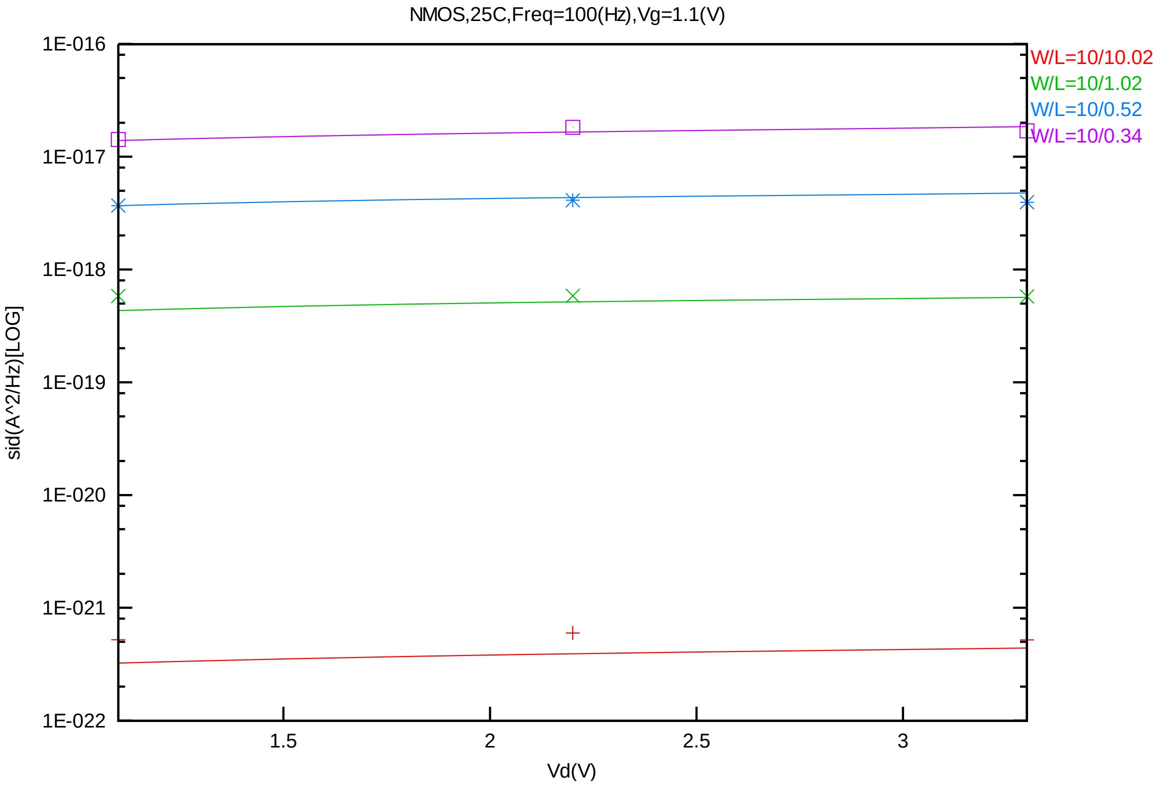


Figure 7-42 Noise vs Vd for NMOS @ Vg=1.1V

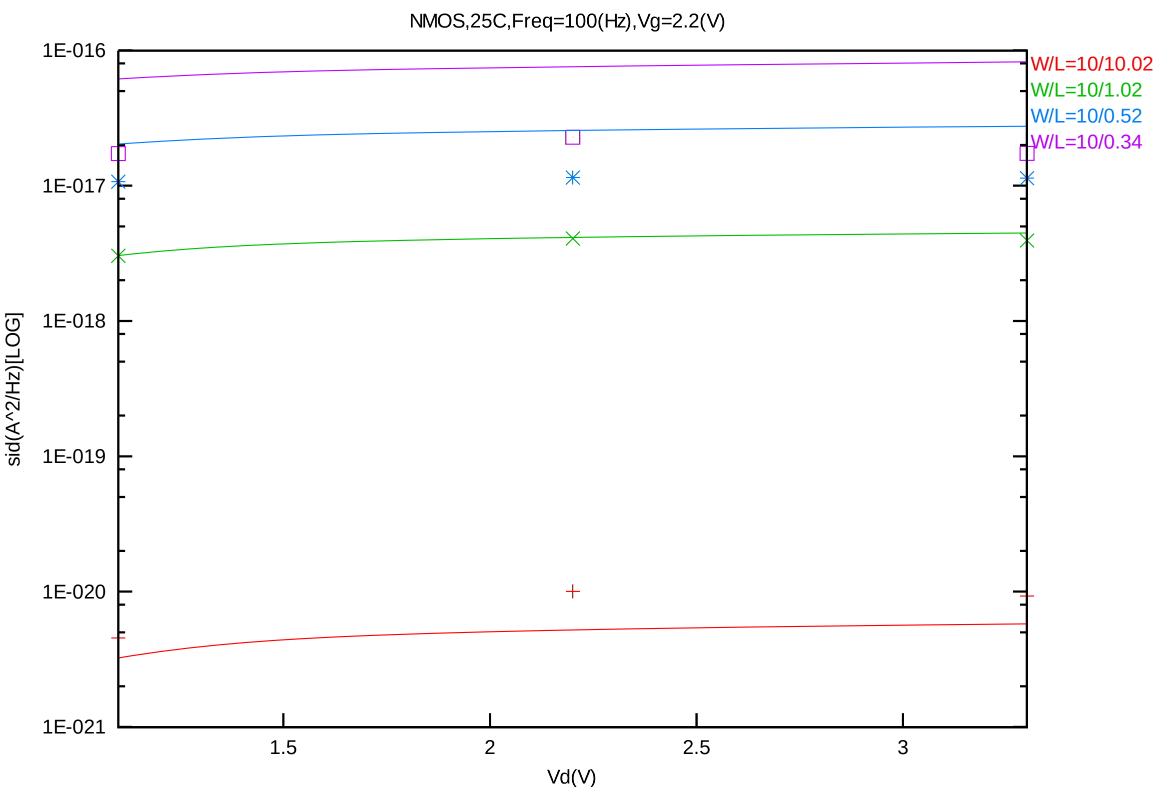


Figure 7-43 Noise vs Vd for NMOS @ Vg=2.2V

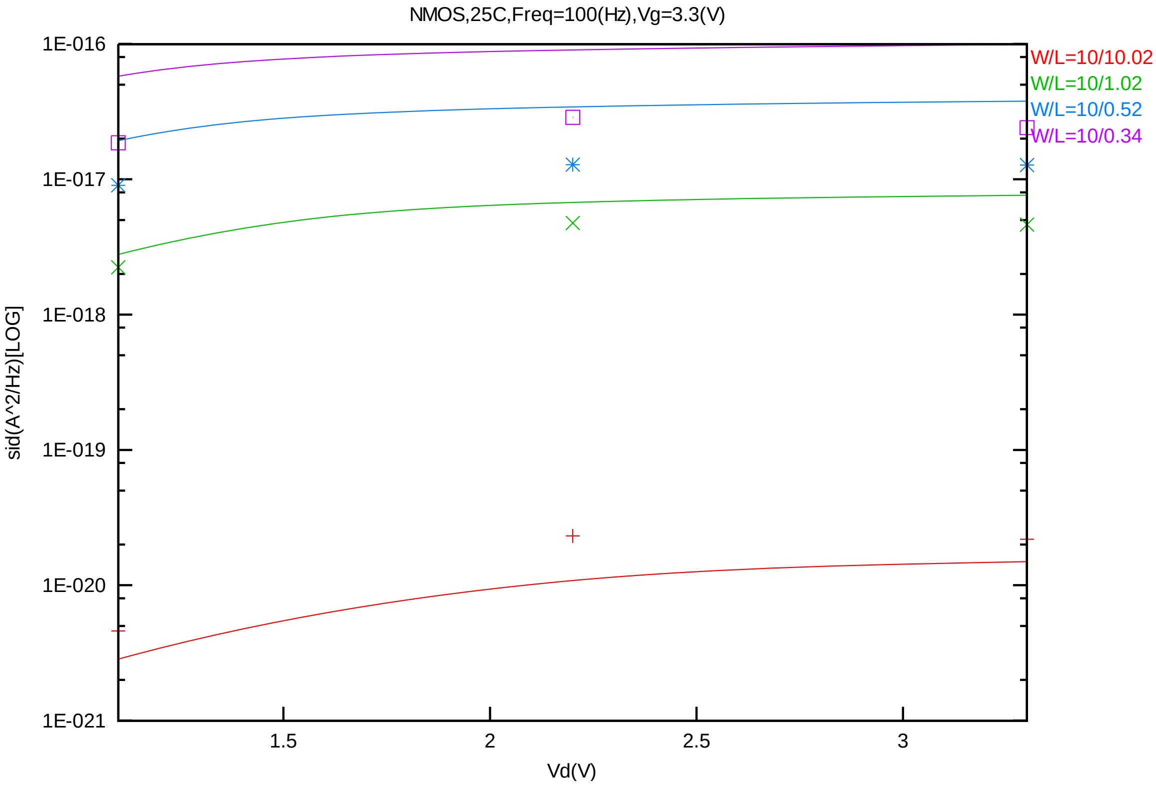


Figure 7-44 Noise vs Vd for NMOS @ Vg=3.3V

B. PMOS

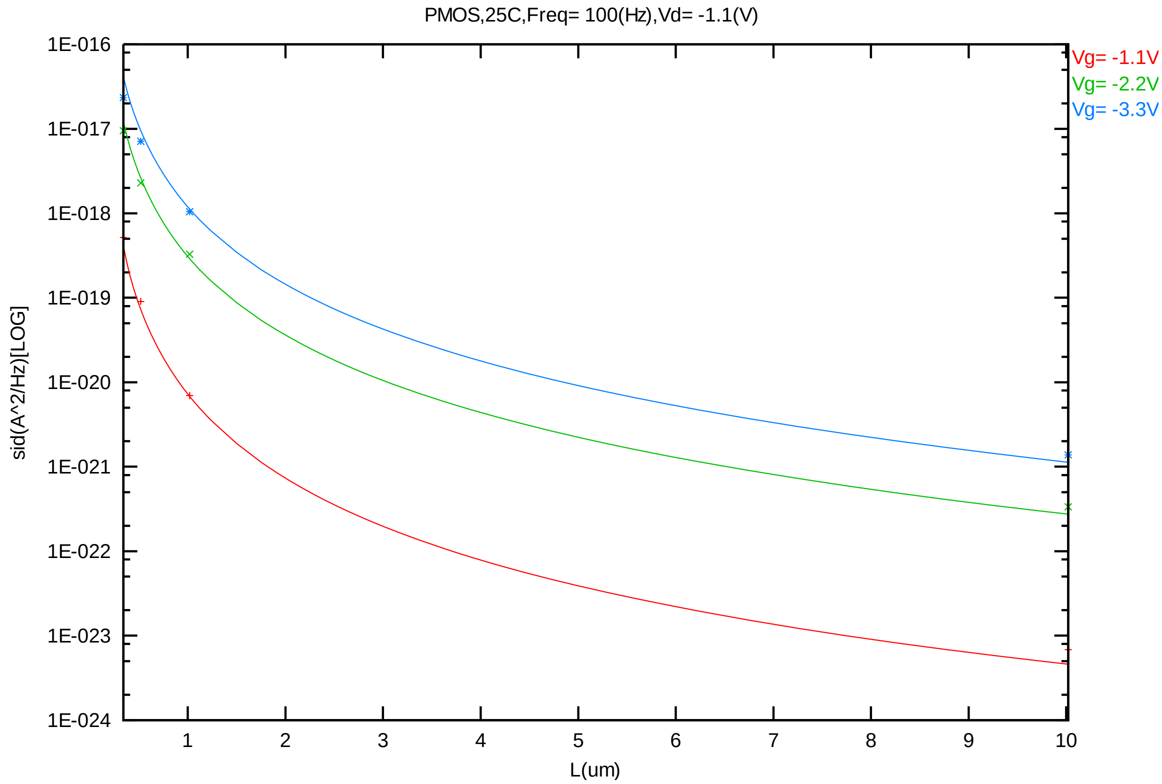


Figure 7-45 Noise vs Length for PMOS @ Vd= -1.1V

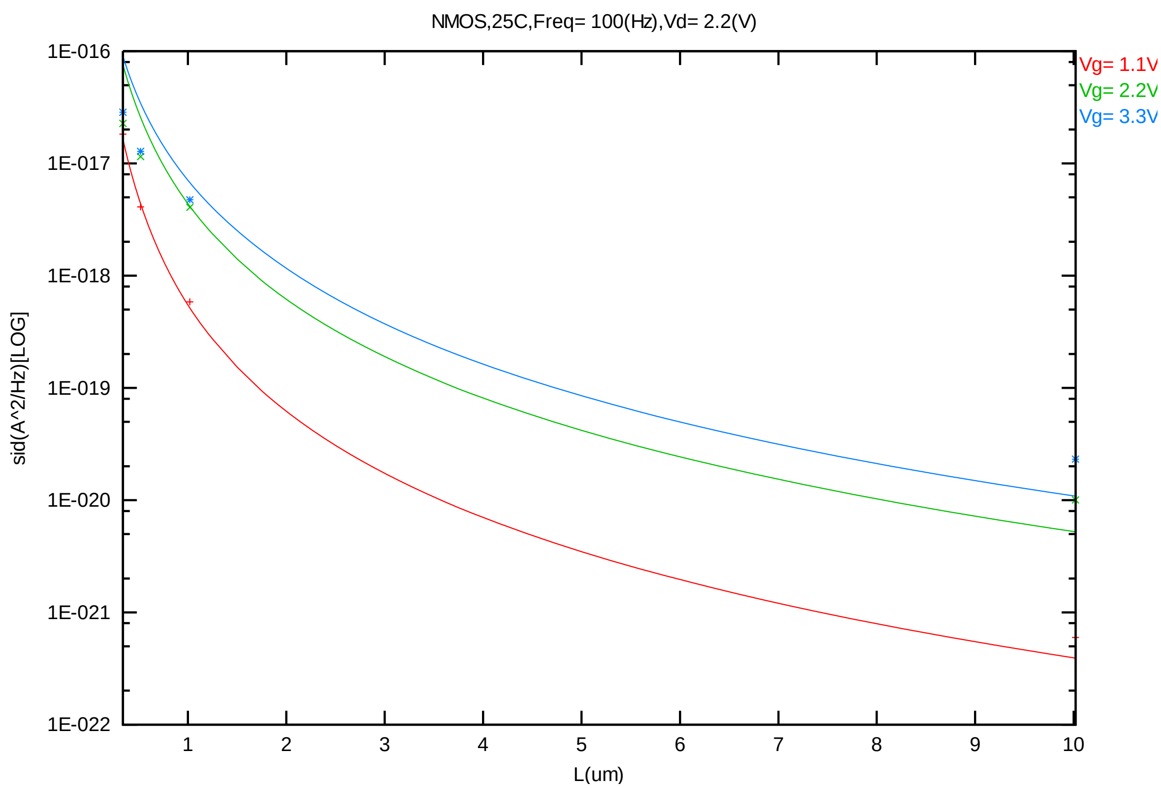


Figure 7-46 Noise vs Length for PMOS @ Vd= -2.2V

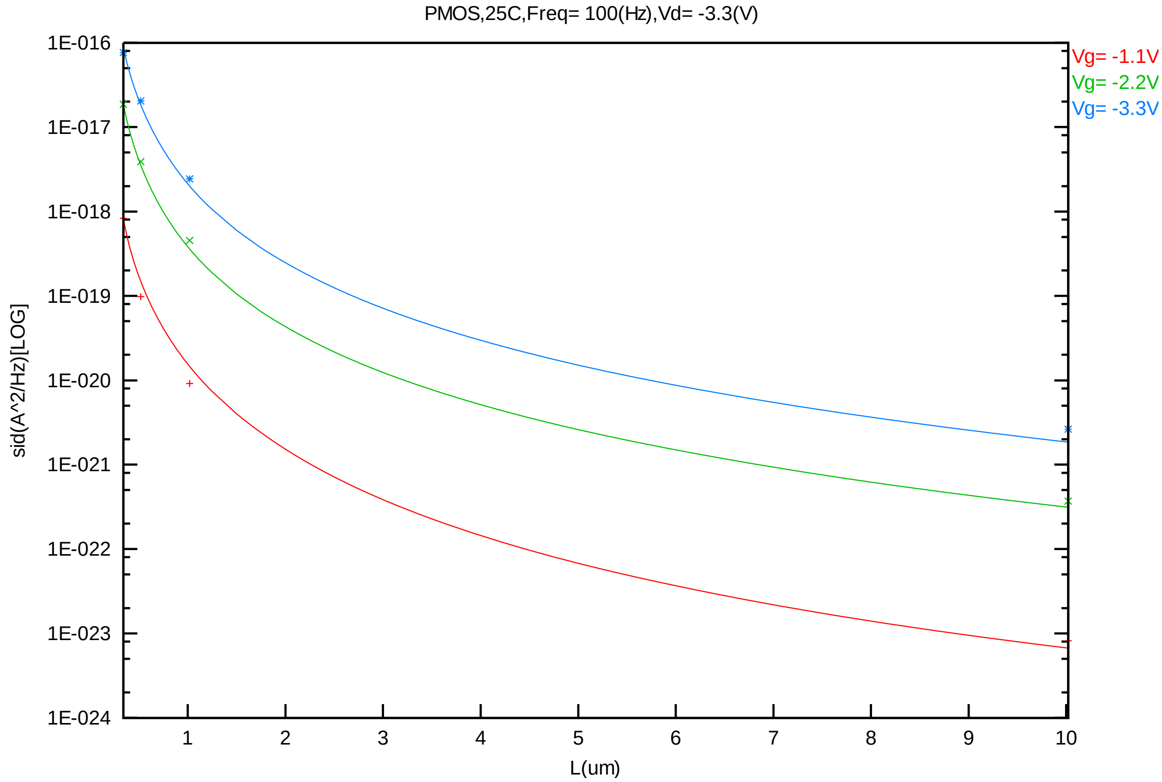


Figure 7-47 Noise vs Length for PMOS @ Vd= -3.3V

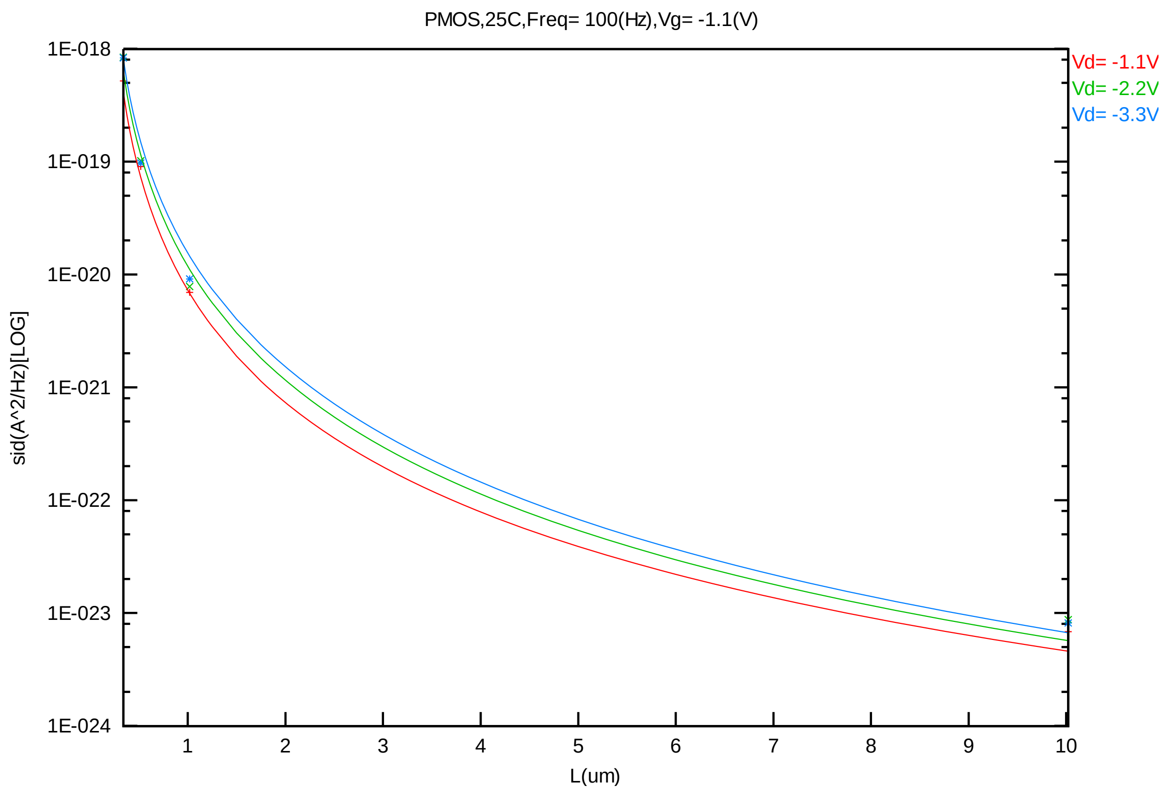


Figure 7-48 Noise vs Length for PMOS @ Vg= -1.1V

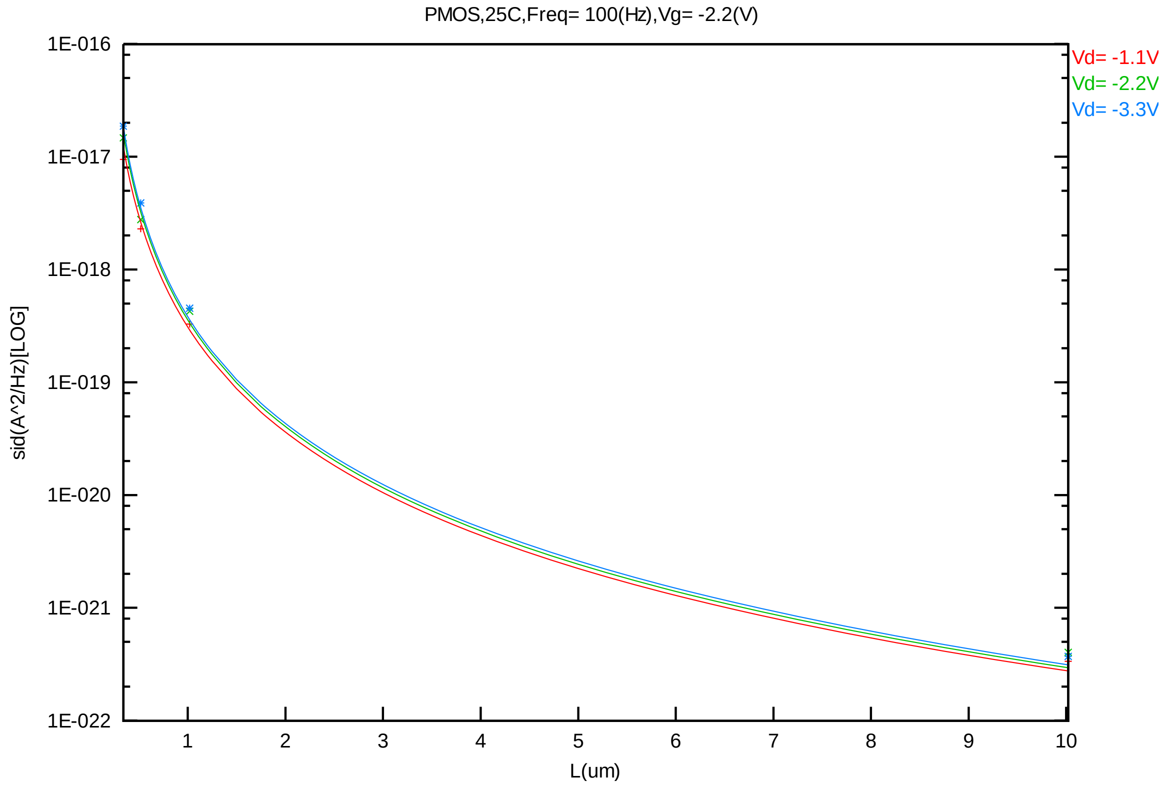


Figure 7-49 Noise vs Length for PMOS @ Vg= -2.2V

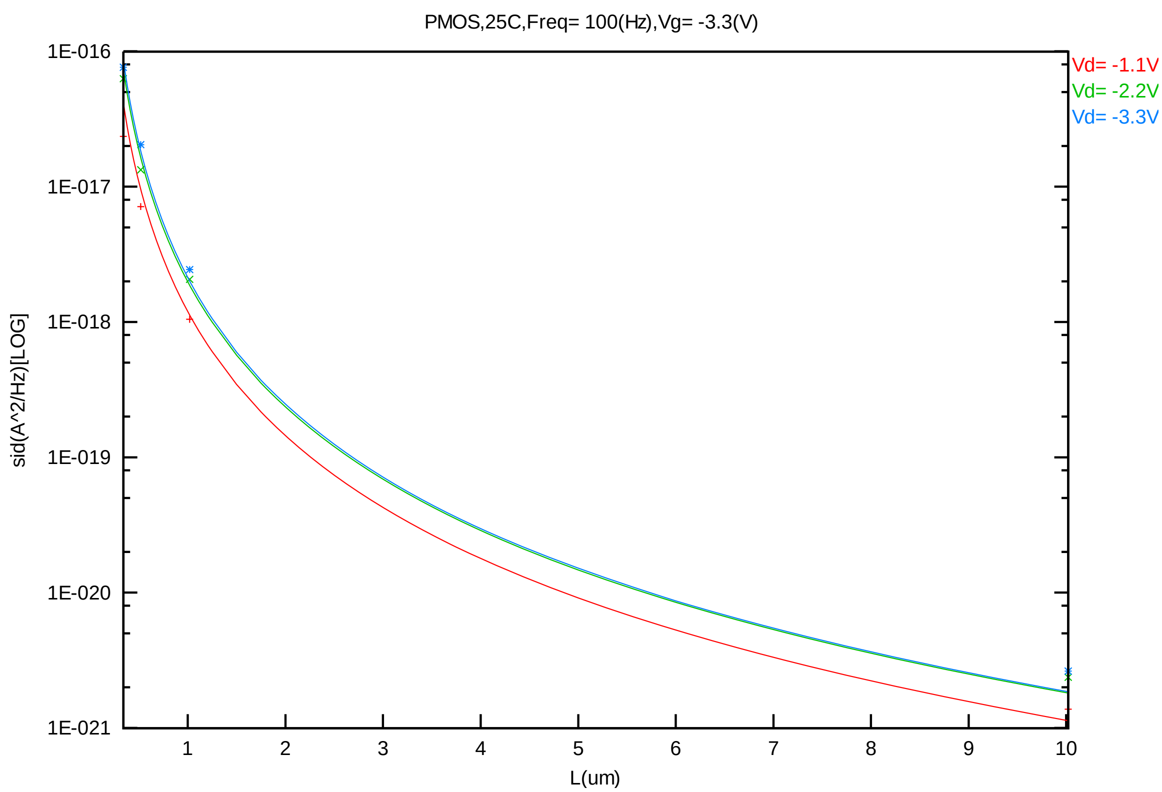


Figure 7-50 Noise vs Length for PMOS @ Vg= -3.3V

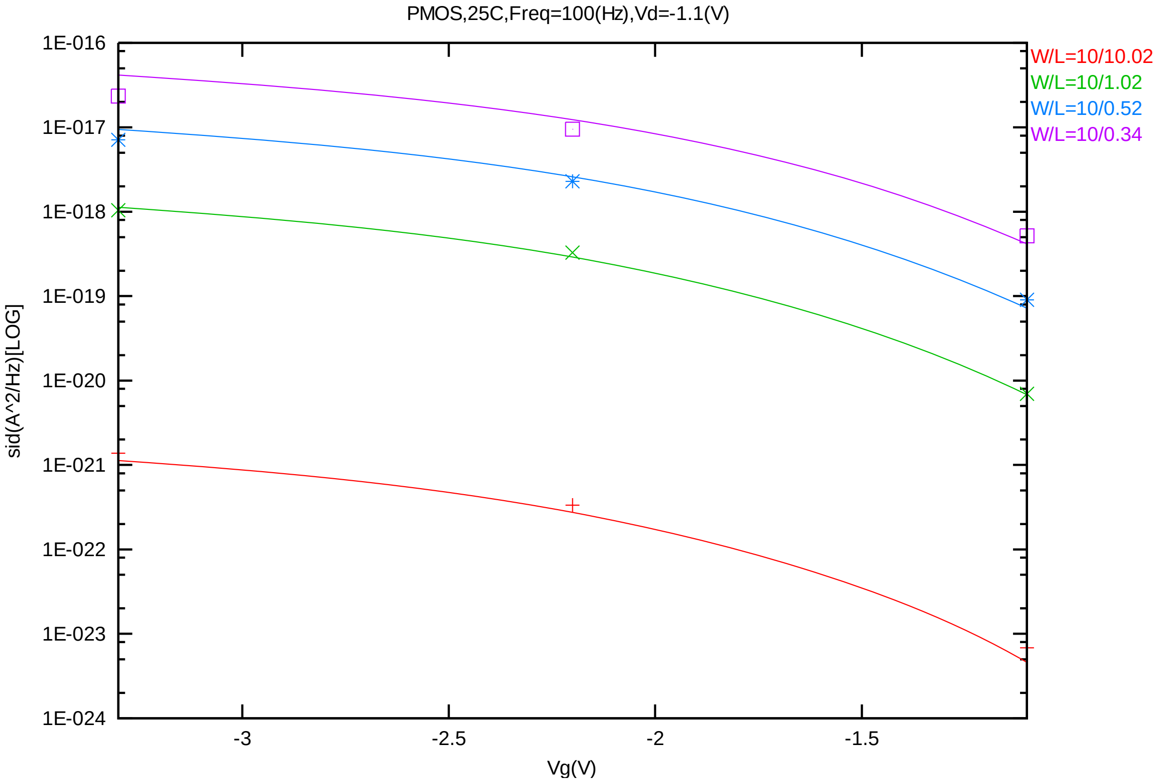


Figure 7-51 Noise vs Vg for PMOS @ Vd= -1.1V

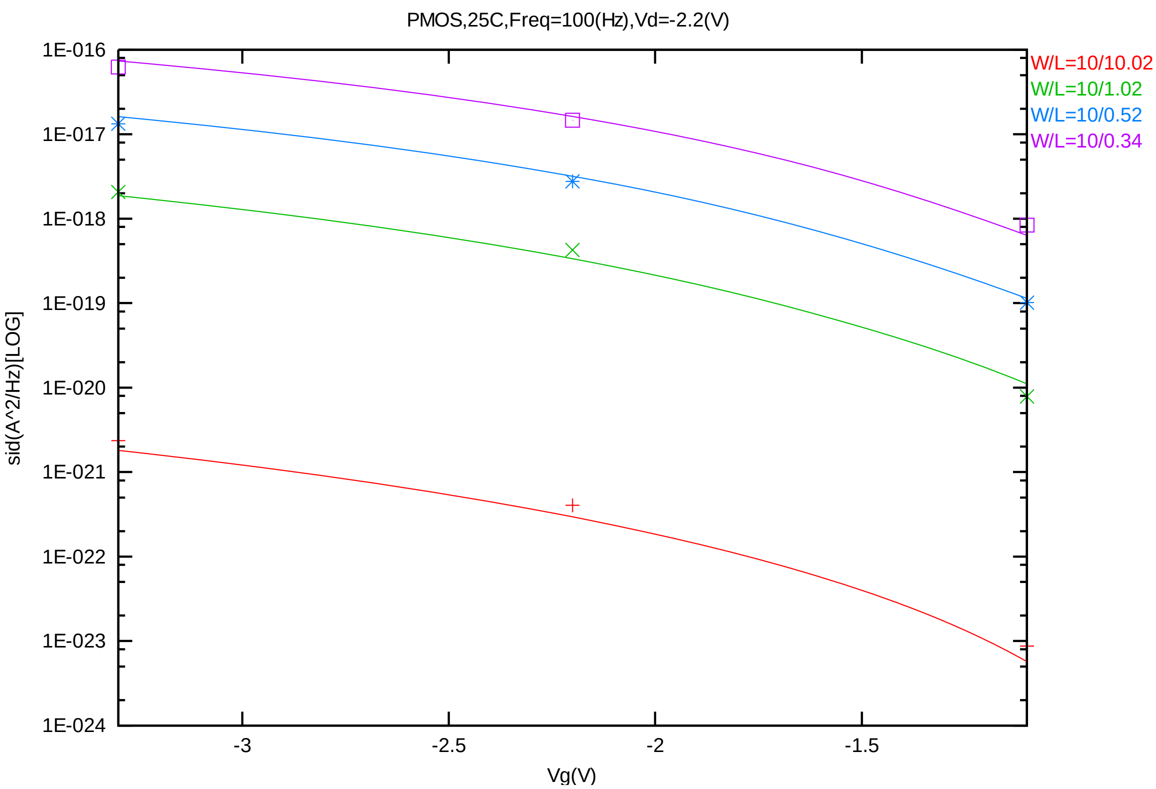


Figure 7-52 Noise vs Vg for PMOS @ Vd= -2.2V

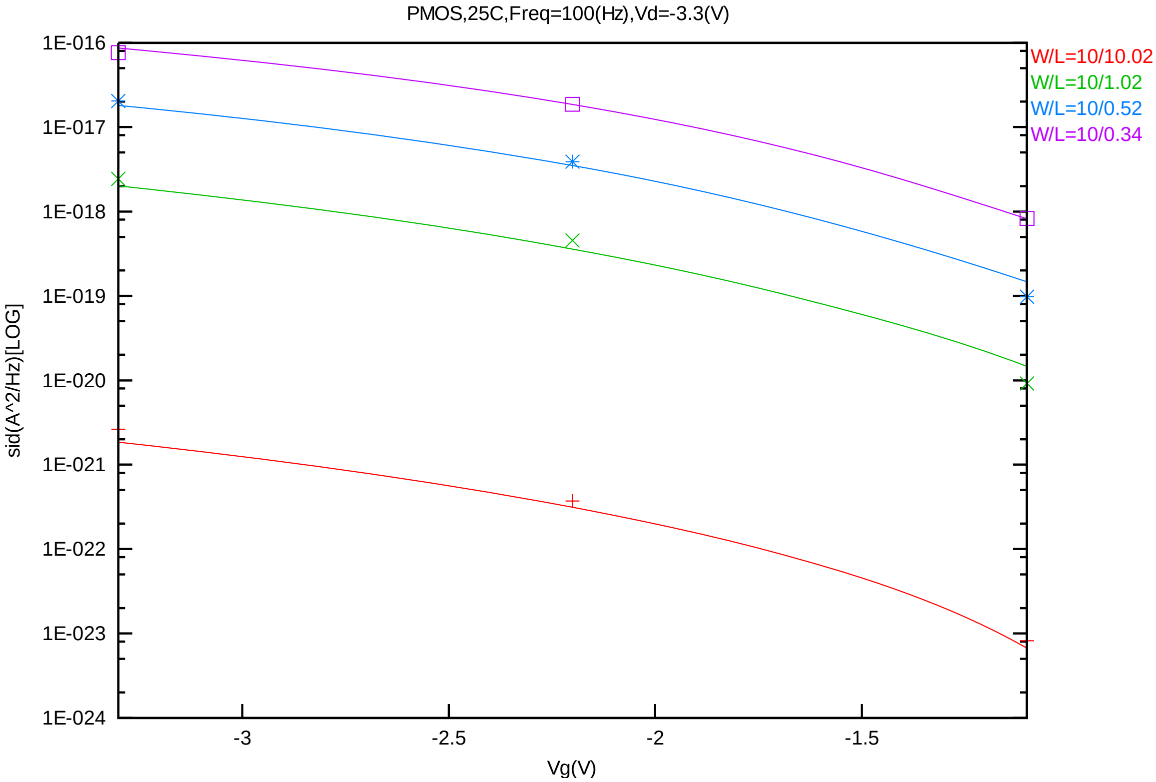


Figure 7-53 Noise vs Vg for PMOS @ Vd= -3.3V

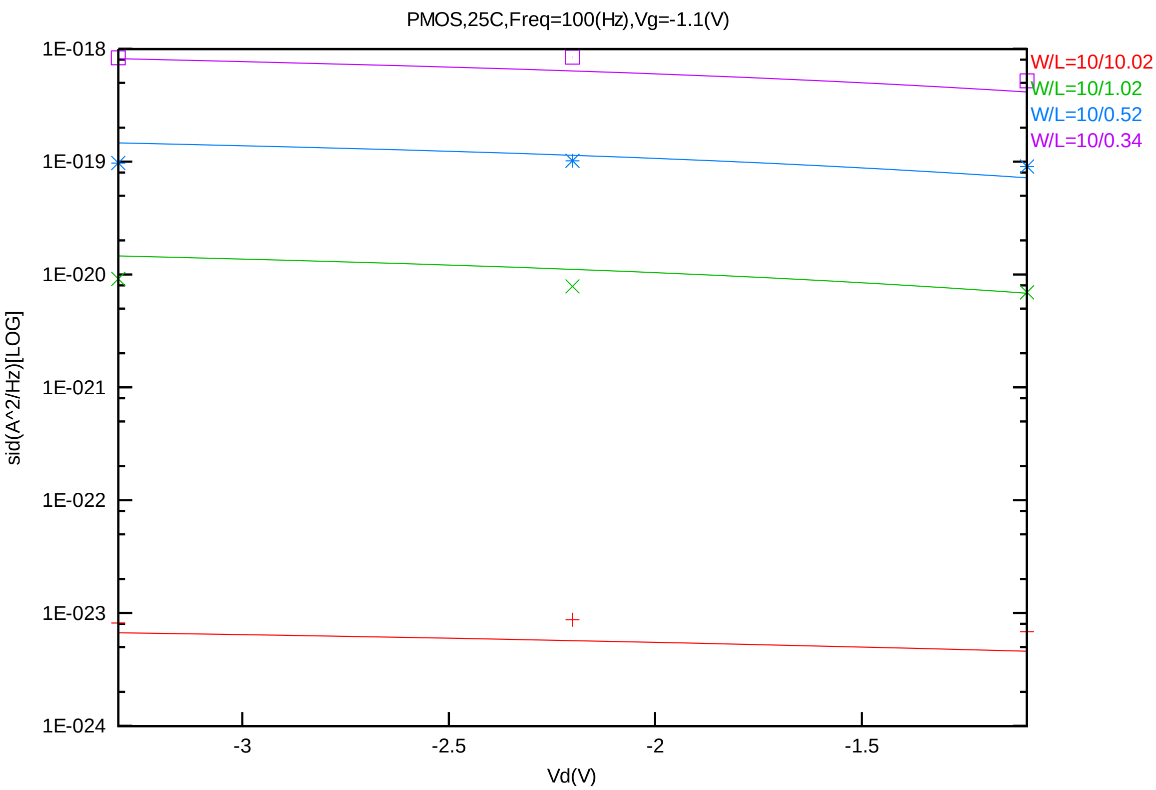


Figure 7-54 Noise vs Vd for PMOS @ Vg= -1.1V

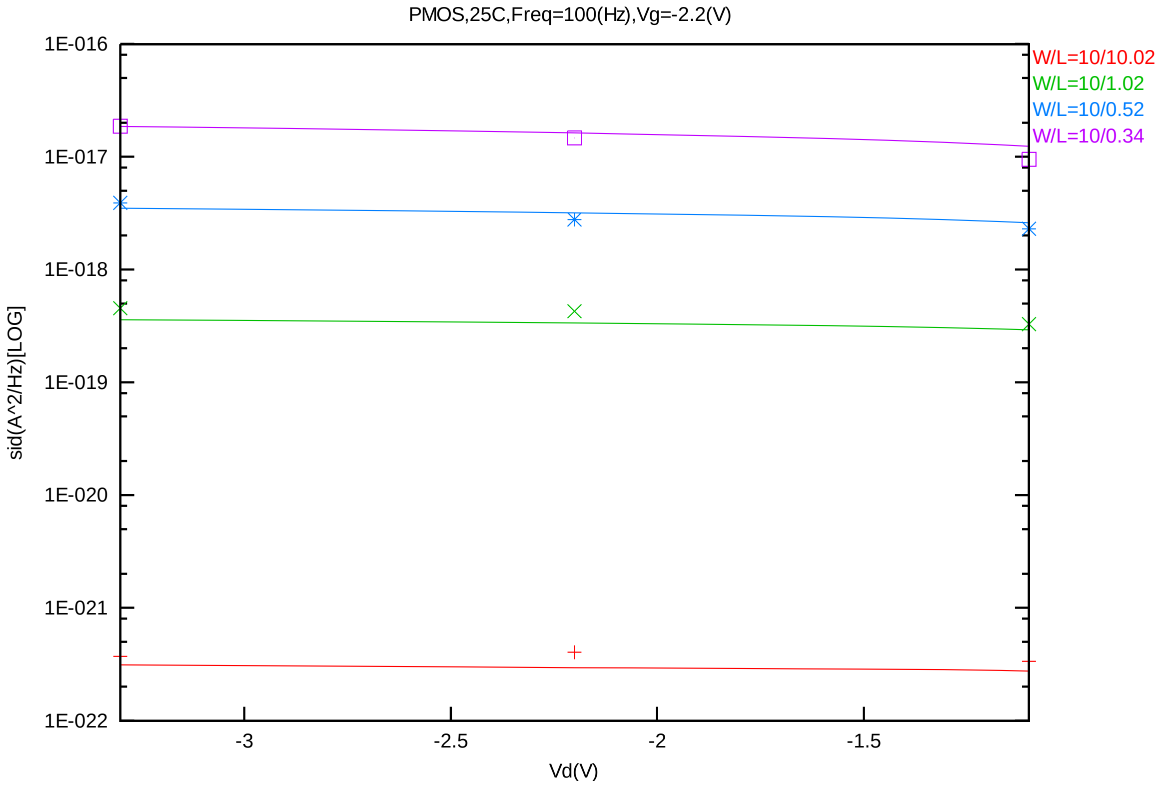


Figure 7-55 Noise vs Vd for PMOS @ Vg= -2.2V

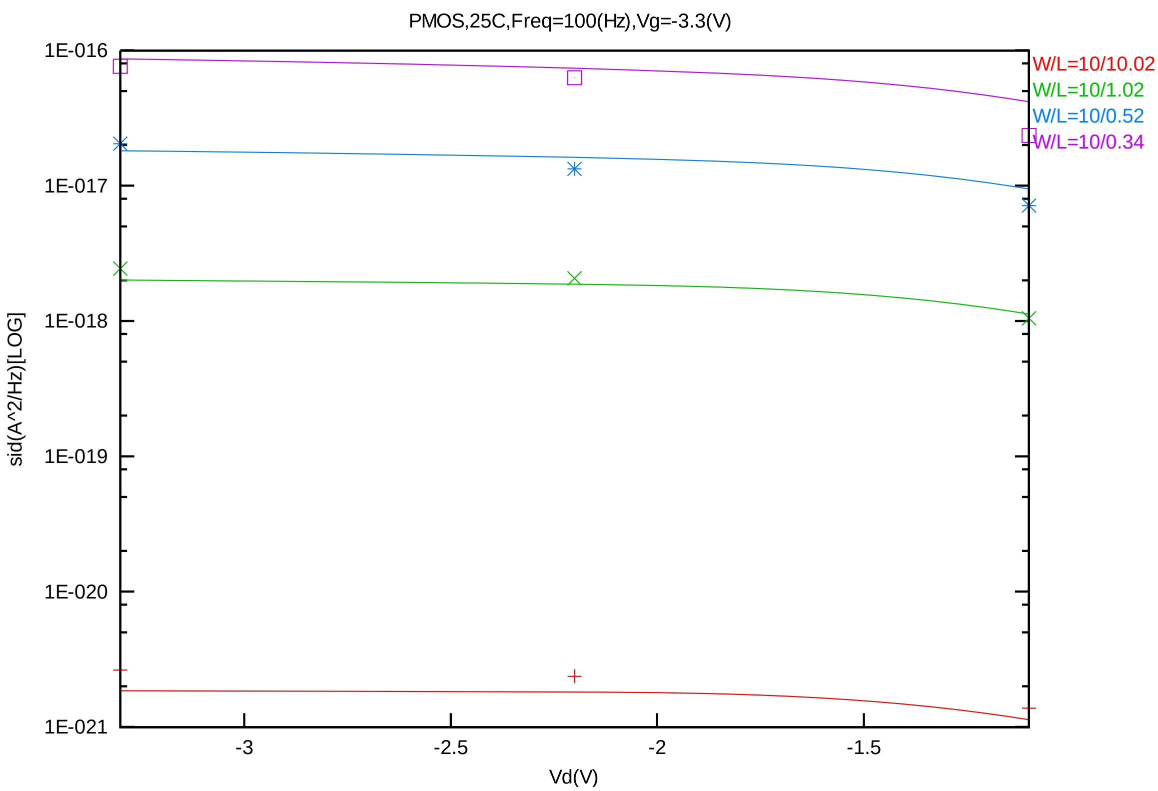


Figure 7-56 Noise vs Vd for PMOS @ Vg= -3.3V

8 Appendix

8.1 HSPICE Warning Message

Table 8-1 Hspice Warning Message

Item	Warning Message	Explanation
1	None	None

8.2 ELDO Warning Message

Table 8-2 Eldo Warning Message

Item	Warning Message	Explanation
1	None	None

8.3 SPECTRE Warning Message

Table 8-3 Spectre Warning Message

Item	Warning Message	Explanation
1	None	None